EGA IV, SECTIONS 16–18: SOURCE-ALIGNED ENGLISH READER

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§16. Differential invariants. Differentially smooth morphisms

In this section we present, in global form, some notions of differential calculus that are particularly useful in algebraic geometry. We pass over many developments classical in differential geometry (connections, infinitesimal transformations associated with a vector field, jets, etc.), although these notions admit particularly natural formulations in the setting of schemes. We likewise pass over here the phenomena special to characteristic $p > 0$ (some of which are studied, in the affine setting, in (**0**, 21)). For further material on the differential formalism for preschemes, the reader may consult Exposés II and VII of [**?**], as well as later chapters of this treatise. **IV-4** | 5

16.1. Normal invariants of an immersion

(16.1.1). Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two ringed spaces, and let $f = (\psi, \theta) : Y \to X$ be a morphism of ringed spaces (**0**, 4.1.1) such that the homomorphism

$$\theta^\# : \psi^*(\mathscr{O}_X) \longrightarrow \mathscr{O}_Y$$

is surjective, so that $\mathscr{O}_Y$ is identified with a sheaf of quotient rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$. We can then endow $\psi^*(\mathscr{O}_X)$ with the $\mathscr{I}_f$-preadic filtration.

Definition (16.1.2). — The $\mathscr{O}_Y$-augmented sheaf of rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ is called the n-th *normal invariant* of f; the ringed space $(Y, \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1})$ is called the n-th *infinitesimal neighbourhood* of Y for the morphism f and is denoted by $Y_f^{(n)}$ or simply $Y^{(n)}$. The sheaf of graded rings associated to the sheaf of filtered rings $\psi^*(\mathscr{O}_X)$

$$\mathscr{G}r_\bullet(f) = \bigoplus_{n \geqslant 0} (\mathscr{I}_f^n / \mathscr{I}_f^{n+1}) \tag{16.1.2.1}$$

is called the sheaf of graded rings *associated with* f. The sheaf $\mathscr{G}r_1(f) = \mathscr{I}_f/\mathscr{I}_f^2$ is called the *conormal sheaf* of f (and is also denoted by $\mathscr{N}_{Y/X}$ when no confusion can result).

It is clear that the $\mathscr{O}_{Y^{(n)}} = \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ (also denoted $\mathscr{O}_{Y_f^{(n)}}$) form a projective system of sheaves of rings on Y; for $n \leqslant m$, the transition homomorphism $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ identifies $\mathscr{O}_{Y^{(n)}}$ with the quotient of $\mathscr{O}_{Y^{(m)}}$ by the power $(\mathscr{I}_f/\mathscr{I}_f^{n+1})^m$ of the *augmentation ideal* of $\mathscr{O}_{Y^{(n)}}$, the kernel of $\phi_{0n} : \mathscr{O}_{Y^{(n)}} \to \mathscr{O}_Y$. The $Y^{(n)}$ therefore form an inductive system of ringed spaces, all having underlying space Y, and we have canonical morphisms of ringed spaces $h_n : Y^{(n)} \to X$ equal to (ψ, θ_n), where $\theta_n^\#$ is the canonical morphism $\psi^*(\mathscr{O}_X) \to \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$. It is clear that the sheaf $\mathscr{G}r_\bullet(f)$ is a sheaf of graded algebras over the sheaf of rings $\mathscr{O}_Y = \mathscr{G}r_0(f)$, and that the $\mathscr{G}r_k(f)$ are $\mathscr{O}_Y$-modules. **IV-4** | 6

As with every sheaf of filtered rings, we have a *canonical surjective homomorphism* of graded $\mathscr{O}_Y$-algebras

$$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_\bullet(f) \tag{16.1.2.2}$$

which coincides in degrees 0 and 1 with the identities.

Examples (16.1.3). —

(i) Suppose that X is a locally ringed space, Y is reduced to a single point y (endowed with a ring $\mathscr{O}_y$) and that, if $x = \psi(y)$, $\theta^\# : \mathscr{O}_x \to \mathscr{O}_y$ is a *surjective* homomorphism of rings having as kernel the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_x$. So the $\mathscr{O}_{Y^{(n)}}$ are identified with the rings $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$ and $\mathscr{G}r_\bullet(f)$ with the graded ring associated with the local ring $\mathscr{O}_x$ endowed with the $\mathfrak{m}_x$-preadic filtration.

(ii) Suppose that Y is a closed subset of an open subspace U of X and that $\mathscr{O}_Y$ is induced on Y by a quotient sheaf $\mathscr{O}_U/\mathscr{I}$, where $\mathscr{I}$ is an ideal of $\mathscr{O}_U$ such that $\mathscr{I}_x = \mathscr{O}_x$ for every $x \notin Y$; if X is a locally ringed space, suppose further that $\mathscr{I}_x \neq \mathscr{O}_x$ for $x \in Y$, so that $(Y, \mathscr{O}_Y)$ is again a locally ringed space.

Let $\psi_0 : Y \to U$ be the canonical injection and denote by $\theta_0 : \mathscr{O}_U \to (\psi_0)_*(\mathscr{O}_Y)$ the homomorphism such that $\theta_0^\#$ is the canonical homomorphism $\psi_0^*(\mathscr{O}_U) = \mathscr{O}_U|Y \to (\mathscr{O}_U/\mathscr{I})|Y$, so that $j_0 = (\psi_0, \theta_0) : Y \to U$ is a morphism of ringed spaces (and of locally ringed spaces if X is a locally ringed space); if $i : U \to X$ is the canonical injection (morphism of ringed spaces), $j = i \circ j_0$ is the morphism (ψ, θ) of Y to X where $\psi : Y \to X$ is the canonical injection and $\theta : \mathscr{O}_X \to \psi_*(\mathscr{O}_Y)$ is the homomorphism such that $\theta^\# = \theta_0^\#$. Since $\theta^\#$ is surjective we can apply the previous definitions; $\mathscr{O}_{Y^{(n)}}$ is equal to $\psi_0^*(\mathscr{O}_U/\mathscr{I}^{n+1})$, and we have $(\psi_0)_*(\mathscr{O}_{Y^{(n)}}) = \mathscr{O}_U/\mathscr{I}^{n+1}$, and $\mathscr{G}r_n(j) = \mathscr{G}r_n(j_0) = \psi_0^*(\mathscr{I}^n/\mathscr{I}^{n+1}) = j_0^*(\mathscr{I}^n/\mathscr{I}^{n+1})$.

(16.1.4). The example (16.1.3, (ii)) shows that in general the $\mathscr{O}_{Y^{(n)}}$ are *not canonically endowed with a structure of an $\mathscr{O}_Y$-module*, or *a fortiori* with a structure of an $\mathscr{O}_Y$-algebra. The data of such a structure is equivalent to the data of a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation morphism ϕ_{0n}; it is also equivalent to the data of a morphism of ringed spaces $(1_Y, \lambda_n) : Y^{(n)} \to Y$ left inverse to the canonical morphism $(1_Y, \phi_{0n}) : Y \to Y^{(n)}$.

Proposition (16.1.5). — *Let $f = (\psi, \theta) : Y \to X$ be an immersion of preschemes. Then:*

(i) *$\mathscr{G}r_\bullet(f)$ is a quasi-coherent graded $\mathscr{O}_Y$-algebra.* **IV-4** | 7

(ii) *The $Y^{(n)}$ are preschemes, canonically isomorphic to subpreschemes of X.*

(iii) *Every homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation homomorphism ϕ_{0n}, makes $\mathscr{O}_{Y^{(n)}}$ and the $\mathscr{O}_{Y^{(k)}}$ for $k \leqslant n$ quasi-coherent $\mathscr{O}_Y$-algebras; the $\mathscr{O}_Y$-module structures induced from the above structures on the $\mathscr{G}r_k(f)$ for $k \leqslant n$ coincide with the ones defined in (16.1.2).*

Proof. (i) Since the question is local on X and Y, we can reduce to the case where Y is a closed subprescheme of X defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$; since $\mathscr{O}_Y$ is the restriction to Y of $\mathscr{O}_X/\mathscr{I}$, assertion (i) is evident, and $Y^{(n)}$ is the closed subprescheme of X defined by the quasi-coherent ideal $\mathscr{I}^{n+1}$ of $\mathscr{O}_X$. Finally, to prove (iii), note that the data of λ_n makes the ideal $\mathscr{I}/\mathscr{I}^{n+1}$ of the augmentation ϕ_{0n} and its quotients $\mathscr{I}/\mathscr{I}^{k+1}$ $(1 \leqslant k \leqslant n)$ $\mathscr{O}_Y$-modules; it suffices to prove by induction on k that the $\mathscr{I}/\mathscr{I}^{k+1}$ are quasi-coherent $\mathscr{O}_Y$-modules and that the quotient $\mathscr{O}_Y$-module structure thereby induced on $\mathscr{I}^k/\mathscr{I}^{k+1}$ is the same as that defined in (16.1.2). The second assertion is immediate, $\mathscr{I}^k/\mathscr{I}^{k+1}$ being killed by $\mathscr{I}/\mathscr{I}^{n+1}$; the first follows by induction on k, since it is trivial for $k = 1$ and $\mathscr{I}/\mathscr{I}^{k+1}$ is an extension of $\mathscr{I}/\mathscr{I}^k$ by $\mathscr{I}^k/\mathscr{I}^{k+1}$ (1.4.17). ☐

Corollary (16.1.6). — *Under the general hypotheses of (16.1.5), if the immersion f is locally of finite presentation then the $\mathscr{G}r_n(f)$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

Proof. Indeed, with the notation from the proof of (16.1.5), $\mathscr{I}$ is an ideal of finite type of $\mathscr{O}_X$ (1.4.7), therefore the $\mathscr{I}^n/\mathscr{I}^{n+1}$ are $\mathscr{O}_Y$-modules of finite type, hence the conclusion. ☐

Corollary (16.1.7). — *Under the general hypotheses of (16.1.5), let $g : X \to Y$ be a morphism of preschemes, left inverse to f. Then, for every n, the composite morphism $(1, \lambda_n) : Y^{(n)} \xrightarrow{h_n} X \xrightarrow{g} Y$ defines a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$ right inverse to the augmentation ϕ_{0n}, making $\mathscr{O}_{Y^{(n)}}$ a quasi-coherent $\mathscr{O}_Y$-algebra; via these homomorphisms, the transition homomorphisms $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ $(n \leqslant m)$ are homomorphisms of $\mathscr{O}_Y$-algebras. Also, if g is locally of finite type, then the $\mathscr{O}_{Y^{(n)}}$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

Proof. The first assertion follows immediately from the definitions and (16.1.5). On the other hand, if g is locally of finite type, then f is locally of finite presentation (1.4.3, (v)); the $\mathscr{G}r_n(f)$ being then quasi-coherent $\mathscr{O}_Y$-modules of finite type by (16.1.6), the same goes for the $\mathscr{O}_Y$-modules $\mathscr{I}/\mathscr{I}^{n+1}$, being extensions of a finite number of the $\mathscr{G}r_k(f)$ (**III**, 1.4.17). ☐

Proposition (16.1.8). — *Let X be a locally Noetherian prescheme and let $j : Y \to X$ be an immersion. Then the $Y^{(n)}$ are locally Noetherian preschemes, the $\mathscr{G}r_n(j)$ are coherent $\mathscr{O}_Y$-modules, and $\mathscr{G}r_\bullet(j)$ is a coherent sheaf of rings on the space Y.*

Proof. Everything is local on X and Y, so we reduce to the case where X is affine and j is a closed immersion. All the assertions are then evident except the last; this follows from the fact that, if A is a Noetherian ring and $\mathfrak{I}$ is an ideal of A, then $\mathrm{gr}^{\bullet}_{\mathfrak{I}}(A)$ is a Noetherian ring, taking into account the exactness of the functor ψ^* and (**0**, 5.3.7). □

Proposition (16.1.9). — *Let X be a prescheme, let $j : Y \to X$ be an immersion locally of finite presentation, and let y be a point of Y. The following conditions are equivalent:* **IV-4** | 8

(a) There exists an open neighbourhood U of y in Y such that $j|U$ is a homeomorphism of U onto an open subset of X.

(b) There is an integer $n > 0$ such that the canonical homomorphism

$$(\phi_{n-1,n})_y : \mathscr{O}_{Y^{(n)},y} \longrightarrow \mathscr{O}_{Y^{(n-1)},y}$$

is bijective.

(c) There is an integer $n > 0$ such that $(\mathscr{G}r_n(j))_y = 0$.

In addition, if the integer n satisfies (b) *or* (c)*, then there is a neighbourhood V of y in Y such that $\mathscr{G}r_m(j)|V = 0$ for $m \geqslant n$ and that $\phi_{nm}|V : \mathscr{O}_{Y^{(m)}}|V \to \mathscr{O}_{Y^{(n)}}|V$ is bijective for $m \geqslant n$.*

Proof. Since the question is local on Y, we can restrict ourselves to the case where j is a closed immersion, with Y defined by a quasi-coherent ideal *of finite type* $\mathscr{I}$ of $\mathscr{O}_X$. The equivalence of (b) and (c), for a given n, is immediate; also, since $\mathscr{I}^n/\mathscr{I}^{n+1}$ is an $\mathscr{O}_X$-module of finite type, there is an open neighbourhood U of y in X such that $\mathscr{I}^n|U = \mathscr{I}^{n+1}|U$ (**0**, 5.2.2), and hence $\mathscr{I}^n|U = \mathscr{I}^m|U$ for $m \geqslant n$, proving the last assertions. To prove that (a) implies (b), we can restrict ourselves to the case where the underlying space of Y is equal to the underlying space of X and where $\mathscr{I}$ is generated by finitely many sections over X: since $\mathscr{I}$ is contained in the nilradical $\mathscr{N}$ of $\mathscr{O}_X$ (**I**, 5.1.2), it is nilpotent, which proves (b). Finally, to prove that (b) implies (a), we can likewise restrict ourselves to the case where $\mathscr{I}^n = \mathscr{I}^{n+1}$; therefore, for every $y \in Y$, since $\mathscr{I}_y \subset \mathfrak{m}_y$, maximal ideal of $\mathscr{O}_{X,y}$, we must have $\mathscr{I}_y^n = 0$ because of Nakayama's lemma, since $\mathscr{I}_y$ is an ideal of finite type. The set of $x \in X$ such that $\mathscr{I}_x^n = 0$ is therefore an open subset U of X containing Y (**0**, 5.2.2); since on the other hand $\mathscr{I}_x \neq 0$ for $x \notin Y$, we must have $U = Y$. □

Corollary (16.1.10). — *For the restriction of the immersion j to a neighbourhood of y in Y to be an open immersion (in other words, for j to be a* local isomorphism *at the point y), it is necessary and sufficient that $(\mathscr{G}r_1(j))_y = (\mathscr{N}_{Y/X})_y = 0$.*

Proof. The condition is clearly necessary, and the previous reasoning applied to $n = 1$ proves that it is sufficient. □

Remark (16.1.11). —

(i) Under the conditions of the definition (16.1.1), the projective limit of the projective system $(\mathscr{O}_{Y^{(n)}}, \phi_{nm})$ of sheaves of rings on Y is called the *normal invariant of infinite order* of f, and is sometimes denoted by $\mathscr{O}_{Y^{(\infty)}}$. When X is a locally Noetherian prescheme and $j : Y \to X$ is a closed immersion, so that Y is a closed subprescheme of X defined by a coherent ideal $\mathscr{I}$, the sheaf $\mathscr{O}_{Y^{(\infty)}}$ is precisely the *formal completion* of $\mathscr{O}_X$ along Y (**I**, 10.8.4), and $Y^{(\infty)} = (Y, \mathscr{O}_{Y^{(\infty)}})$ is the formal prescheme obtained by *completing* X along Y (**I**, 10.8.5). In all cases, one may say that $Y^{(\infty)}$ is the *formal neighbourhood* of Y in X (for the morphism f). In the particular case we have just considered, it is the formal prescheme that is the inductive limit of the infinitesimal neighbourhoods of order n.

(ii) Note that for a morphism of preschemes $f = (\psi, \theta) : Y \to X$, it can happen that the homomorphism $\theta^\# : \psi^*(\mathscr{O}_X) \to \mathscr{O}_Y$ is surjective without f being a local immersion and **IV-4** | 9 without f being injective. We obtain an example by taking Y to be a sum of preschemes Y_λ, all isomorphic to $\mathrm{Spec}(\mathscr{O}_x)$, where $x \in X$, and taking f to be the morphism equal to the canonical morphism on each Y_λ.

16.2. Functorial properties of the normal invariants of an immersion

(16.2.1). Let $f = (\psi, \theta) : Y \to X$ and $f' = (\psi', \theta') : Y' \to X'$ be two morphisms of ringed spaces such that $\theta^\#$ and $\theta'^\#$ are surjective; consider a commutative diagram of morphisms of ringed spaces

(16.2.1.1)
$$\begin{tikzcd} Y \arrow[r, "f"] & X \\ Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \end{tikzcd}$$

Let $u = (\rho, \lambda), v = (\sigma, \mu)$. We have $\rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X))$, and hence a commutative diagram of homomorphisms of sheaves of rings on Y':

$$\begin{tikzcd} \rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X)) \arrow[r, "\psi'^*(\mu^\#)"] \arrow[d, "\rho^*(\theta^\#)"'] & \psi'^*(\mathscr{O}_{X'}) \arrow[d, "\theta'^\#"] \\ \rho^*(\mathscr{O}_Y) \arrow[r, "\lambda^\#"'] & \mathscr{O}_{Y'} \end{tikzcd}$$

from which, if $\mathscr{I}$ and $\mathscr{I}'$ are the kernels of $\theta^\#$ and $\theta'^\#$, we conclude that $\psi'^*(\mu^\#)(\rho^*(\mathscr{I})) \subset \mathscr{I}'$, by the exactness of the functor ρ^*. It follows immediately that, for every integer n, $\psi'^*(\mu^\#)(\rho^*(\mathscr{I}^n)) \subset \mathscr{I}'^n$, which shows that $\psi'^*(\mu^\#)$ defines, on passing to the quotients, a homomorphism of sheaves of rings

(16.2.1.2)
$$\nu_n : \rho^*(\psi^*(\mathscr{O}_X)/\mathscr{I}^{n+1}) \longrightarrow \psi'^*(\mathscr{O}_{X'})/\mathscr{I}'^{n+1}$$

and therefore a morphism of ringed spaces $w_n = (\rho, \nu_n) : Y'^{(n)} \to Y^{(n)}$ (which, for $n = 0$, is none other than u). It follows immediately from this definition that the diagrams

$$\begin{tikzcd} Y^{(n)} \arrow[r, "h_{mn}"] & Y^{(m)} \arrow[r, "h_m"] & X \\ Y'^{(n)} \arrow[u, "w_n"] \arrow[r, "h'_{mn}"'] & Y'^{(m)} \arrow[u, "w_m"] \arrow[r, "h'_m"'] & X' \arrow[u, "v"'] \end{tikzcd} \qquad (n \leqslant m)$$

(where the horizontal arrows are the canonical morphisms (16.1.2)) are commutative.

By passage to the quotients via the morphisms (16.2.1.2), and taking into account the exactness of the functor ρ^*, we obtain a di-homomorphism of graded algebras (relative to the homomorphism $\lambda^\# : \rho^*(\mathscr{O}_Y) \to \mathscr{O}_{Y'}$) **IV-4** | 10

(16.2.1.3)
$$\mathrm{gr}(u) : \rho^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f')$$

(or, if one prefers, a ρ-morphism (**0**, 3.5.1) $\mathscr{G}r_\bullet(f) \to \mathscr{G}r_\bullet(f')$), and, in particular, a di-homomorphism of conormal sheaves

$$\mathrm{gr}_1(u) : \rho^*(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_1(f').$$

It is also immediate that these homomorphisms give rise to a commutative diagram

(16.2.1.4)
$$\begin{tikzcd} \rho^*(\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f))) \arrow[r] \arrow[d, "\mathbf{S}(\mathrm{gr}_1(u))"'] & \rho^*(\mathscr{G}r_\bullet(f)) \arrow[d, "\mathrm{gr}(u)"] \\ \mathbf{S}^\bullet_{\mathscr{O}_{Y'}}(\mathscr{G}r_1(f')) \arrow[r] & \mathscr{G}r_\bullet(f') \end{tikzcd}$$

where the horizontal arrows are the canonical homomorphisms (16.1.2.2).

Finally, if we have a commutative diagram of morphisms of ringed spaces

$$\begin{tikzcd} Y \arrow[r, "f"] & X \\ Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \\ Y'' \arrow[u, "u'"] \arrow[r, "f''"'] & X'' \arrow[u, "v'"'] \end{tikzcd}$$

where $f'' = (\psi'', \theta'')$ is such that $\theta''^{\#}$ is surjective, and if w'_n and w''_n are defined, respectively, from u', v' and from $u'' = u \circ u', v'' = v \circ v'$, then $w''_n = w_n \circ w'_n$, as follows immediately from the definitions and from (**0**, 3.5.5); we have also $\mathrm{gr}(u'') = \mathrm{gr}(u') \circ \rho'^*(\mathrm{gr}(u))$ if $u' = (\rho', \lambda')$. Therefore we can say that $Y^{(n)}$ and $\mathscr{G}r_\bullet(f)$ *depend functorially* on f.

Proposition (16.2.2). — *With the notation and hypotheses of (16.2.1), suppose also that f, f', u, and v are morphisms of preschemes. We have:*

(i) *The morphisms $w_n : Y'^{(n)} \to Y^{(n)}$ are morphisms of preschemes.*
(ii) *If $Y' = Y \times_X X'$, with u and f' the canonical projections, and if f is an immersion or v is flat, then $Y'^{(n)} = Y^{(n)} \times_X X'$.*
(iii) *If $Y' = Y \times_X X'$ and if v is flat (resp. if f is an immersion), the homomorphism*

$$\mathrm{Gr}(u) = \mathrm{gr}(u) \otimes 1 : \mathscr{G}r_\bullet(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_\bullet(f')$$

is bijective (resp. surjective).

Proof.

(i) The hypotheses immediately imply that, for every $y' \in Y'$, $\rho^*_{y'}(\theta^{\#}_{\psi'(y')})$ is a *local* homomorphism (**I**, 1.6.2), so w_n is a morphism of preschemes (**I**, 2.2.1). **IV-4** | 11

(ii) and (iii) If f is an immersion, we may restrict to the case where f is a closed immersion, with Y defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$ and $Y^{(n)}$ by the ideal $\mathscr{I}^{n+1}$; the assertions then follow from (**I**, 4.4.5).

Second, suppose that v is flat; we can restrict ourselves to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(A')$ are affines, A' being a flat A-module; so $Y' = \mathrm{Spec}(B')$ where $B' = B \otimes_A A'$; moreover, if $\mathfrak{I}$ is the kernel of the homomorphism $A \to B$, the kernel $\mathfrak{I}'$ of $A' \to B'$ is identified with $\mathfrak{I} \otimes_A A'$ by flatness, and

$$\mathfrak{I}'^n / \mathfrak{I}'^{n+1} = (\mathfrak{I}^n / \mathfrak{I}^{n+1}) \otimes_A A'.$$

It follows at once from (**0**, 4.3.3) that the $\mathscr{O}_{Y'}$-module $\mathscr{I}'^n / \mathscr{I}'^{n+1}$ is equal to

$$\begin{aligned} &\psi'^*(\sigma^*((\mathfrak{I}^n/\mathfrak{I}^{n+1})^\sim) \otimes_{\sigma^*(\mathscr{O}_X)} \mathscr{O}_{X'}) = \\ &\quad \psi'^*(\sigma^*((\mathfrak{I}^n/\mathfrak{I}^{n+1})^\sim)) \otimes_{\psi'^*(\sigma^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'}) = \rho^*(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^*(\psi^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'}) \end{aligned}$$

and in particular for $n = 0$, we have

$$\mathscr{O}_{Y'} = \rho^*(\mathscr{O}_Y) \otimes_{\rho^*(\psi^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'})$$

whence we obtain a canonical isomorphism from $\mathscr{I}'^n / \mathscr{I}'^{n+1}$ onto

$$\rho^*(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^*(\mathscr{O}_Y)} \mathscr{O}_{Y'} = (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$$

which proves (iii). Set now $C_n = \Gamma(Y, \mathscr{O}_{Y^{(n)}})$ and $C'_n = \Gamma(Y', \mathscr{O}_{Y'^{(n)}})$. Since $Y^{(n)}$ and $Y'^{(n)}$ are affine schemes (16.1.5), the kernel $\mathfrak{K}_n$ (resp. $\mathfrak{K}'_n$) of the homomorphism $C_n \to C_{n-1}$ (resp. $C'_n \to C'_{n-1}$) is $\Gamma(Y, \mathscr{I}^n/\mathscr{I}^{n+1})$ (resp. $\Gamma(Y', \mathscr{I}'^n/\mathscr{I}'^{n+1})$); therefore we can deduce from the above results that $\mathfrak{K}'_n = \mathfrak{K}_n \otimes_A A'$. Now, we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathfrak{K}_n \otimes_A A' \arrow[r] \arrow[d, "r"] & C_n \otimes_A A' \arrow[r] \arrow[d, "s_n"] & C_{n-1} \otimes_A A' \arrow[r] \arrow[d, "s_{n-1}"] & 0 \\
0 \arrow[r] & \mathfrak{K}'_n \arrow[r] & C'_n \arrow[r] & C'_{n-1} \arrow[r] & 0
\end{tikzcd}$$

where the left vertical arrow is bijective and both rows are exact (since A' is a flat A-module). It follows by induction that s_n is bijective for every n: this is true by hypothesis for $n = 0$, and the general case follows from the five lemma. This proves the second assertion of (ii).

□

Corollary (16.2.3). — *Let $g : X \to Y$ and $u : Y' \to Y$ be two morphisms of preschemes, let $X' = X \times_Y Y'$, and let $g' : X' \to Y'$ and $v : X' \to X$ be the canonical projections. Let $f : Y \to X$ be a Y-section of X (and hence an immersion), and let $f' = f_{(Y')} : Y' \to X'$ be the Y'-section of X' deduced from f by the base change u. We have:*

(i) *The morphism $w_n : {Y'_{f'}}^{(n)} \to Y_f^{(n)}$ corresponding to f, f', u, v (16.2.1) and the canonical morphism $h'_n : {Y'_{f'}}^{(n)} \to X'$ jointly identify ${Y'_{f'}}^{(n)}$ with the product $Y_f^{(n)} \times_X X'$.*

(ii) *If we endow $\mathscr{O}_{Y_f^{(n)}}$ (resp. $\mathscr{O}_{{Y'_{f'}}^{(n)}}$) with the structure of an $\mathscr{O}_Y$-algebra defined by g (resp. with the structure of an $\mathscr{O}_{Y'}$-algebra defined by g') (16.1.5, (iii)), then the homomorphism of $\mathscr{O}_{Y'}$-algebras*

(16.2.3.1)
$$\rho^*(\mathscr{O}_{Y_f^{(n)}}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{O}_{{Y'_{f'}}^{(n)}}$$

induced by the homomorphism ν_n (16.2.1.2) is bijective. Also, the homomorphism of $\mathscr{O}_{Y'}$-modules **IV-4** | 12

(16.2.3.2)
$$\mathrm{Gr}_1(u) : \mathscr{G}r_1(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_1(f')$$

is bijective.

Proof.

(i) First note that $f' : Y' \to X'$ and $u : Y' \to Y$ jointly identify Y' with the product $Y \times_X X'$ (for the structure morphisms $f : Y \to X$ and $v : X' \to X$) (14.5.12.1). The conclusion of (i) now follows from (16.2.2, (ii)), the morphism g being an immersion.

(ii) The commutative diagram

$$\begin{tikzcd}
Y_f^{(n)} \arrow[d, "h_n"] & {Y'_{f'}}^{(n)} \arrow[l, "w_n"'] \arrow[d, "h'_n"] \\
X \arrow[d, "g"] & X' \arrow[l, "v"'] \arrow[d, "g'"] \\
Y & Y' \arrow[l, "u"']
\end{tikzcd}$$

identifies ${Y'_{f'}}^{(n)}$ with the product $Y_f^{(n)} \times_X X'$ and identifies X' with the product $X \times_Y Y'$; hence, by (**I**, 3.3.9), the morphisms $g' \circ h'_n$ and w_n identify ${Y'_{f'}}^{(n)}$ with the product $Y_f^{(n)} \times_Y Y'$. Since $Y_f^{(n)}$ (resp. ${Y'_{f'}}^{(n)}$) is the affine prescheme over Y (resp. over Y') associated with the $\mathscr{O}_Y$-algebra $\mathscr{O}_{Y_f^{(n)}}$ (resp. the $\mathscr{O}_{Y'}$-algebra $\mathscr{O}_{{Y'_{f'}}^{(n)}}$), the bijectivity of the canonical homomorphism (16.2.3.1) follows from (**II**, 1.5.2). Finally, the canonical homomorphism (16.2.3.1) is compatible with the augmentations $\mathscr{O}_{Y_f^{(n)}} \to \mathscr{O}_Y$ and $\mathscr{O}_{{Y'_{f'}}^{(n)}} \to \mathscr{O}_{Y'}$; since $\mathscr{O}_{Y_f^{(n)}}$ is the direct sum (as an $\mathscr{O}_Y$-module) of $\mathscr{O}_Y$ and the augmentation ideal $\mathscr{I}/\mathscr{I}^{n+1}$, it follows that the canonical homomorphism (16.2.3.1), restricted to $(\mathscr{I}/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$, maps this module bijectively onto $\mathscr{I}'/\mathscr{I}'^{n+1}$. For $n = 1$, this shows that $\mathrm{Gr}_1(u)$ is bijective.

□

We note that, under the hypotheses of (16.2.3), the homomorphisms $\mathrm{Gr}_n(u)$ are *surjective* in view of the above, but are not bijective in general for $n \geqslant 2$. However:

Corollary (16.2.4). — *Under the hypotheses of (16.2.3), suppose that $u : Y' \to Y$ is a flat morphism (resp. that the $\mathscr{G}r_n(f)$ are flat $\mathscr{O}_Y$-modules for $n \leqslant m$). Then the homomorphism*

$$\mathrm{Gr}_n(u) : \mathscr{G}r_n(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_n(f')$$

is bijective for all n (resp. for $n \leqslant m$).

Proof. If u is flat, then $v : X' \to X$ is flat by base change, and in this case we already know that $\mathrm{Gr}(u)$ is bijective (16.2.2, (iii)). If the $\mathscr{G}r_n(f)$ are flat for $n \leqslant m$, then we first see by induction on n that the same holds for $\mathscr{I}/\mathscr{I}^{n+1}$ for $n \leqslant m$, because of the exact sequences

$$0 \longrightarrow \mathscr{I}^n/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^n \longrightarrow 0$$

(**0**, 6.1.2); in addition, we have the commutative diagram **IV-4** | 13

$$\begin{tikzcd}
0 \arrow[r] & (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^n) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathscr{I}'^n/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^n \arrow[r] & 0
\end{tikzcd}$$

in which the rows are exact (the first by flatness (**0**, 6.1.2)) and the last two vertical arrows are bijective by (16.2.2, (ii)); hence the conclusion. ☐

Remarks (16.2.5). —

(i) The reasoning of (16.2.2, (i)) still applies when the maps in (16.2.1.1) are morphisms of *locally ringed spaces* ($\mathbf{Err}_{\mathbf{III}}$, (**I**, 1.8.2)).

(ii) In (16.2.2, (ii)), the conclusion is no longer necessarily valid if we only suppose that v and f are morphisms of preschemes (f satisfying the condition of (16.1.1)). For example (with the notation of the proof of (16.2.2, (ii))), it can happen that $\mathfrak{I} = 0$ but the kernel $\mathfrak{I}'$ of $A' \to B' = B \otimes_A A'$ is not zero and that $B' \neq 0$, in which case we have $Y^{(n)} = Y$ for all n, but $Y'^{(n)} \neq Y'$. We have an example of this by taking $A = \mathbf{Z}$, $B = \mathbf{Q}$, $A' = \prod_{h=1}^{\infty}(\mathbf{Z}/m^h\mathbf{Z})$ where $m > 1$.

(16.2.6). Consider the special case of diagram (16.2.1.1) in which $X' = X$, v is the identity, X is a prescheme, Y is a subprescheme of X, Y' is a subprescheme of Y, and $f, u, f' = f \circ u$ are the canonical injections; by tensoring with $\mathscr{O}_{Y'}$ over $\rho^*(\mathscr{O}_Y)$, the di-homomorphism (16.2.1.3) gives a homomorphism of graded $\mathscr{O}_{Y'}$-algebras

(16.2.6.1) $$u^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f').$$

On the other hand, we identify $\mathscr{O}_Y$ with $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$ and $\mathscr{O}_{Y'}$ with $\rho^*(\mathscr{O}_Y)/\mathscr{I}_u$; since ρ^* is exact, we have $\rho^*(\mathscr{O}_Y) = \rho^*(\psi^*(\mathscr{O}_X))/\rho^*(\mathscr{I}_f) = \psi'^*(\mathscr{O}_X)/\rho^*(\mathscr{I}_f)$, and since $\mathscr{O}_{Y'}$ is also identified with $\psi'^*(\mathscr{O}_X)/\mathscr{I}_{f'}$, we see that $\mathscr{I}_u = \mathscr{I}_{f'}/\rho^*(\mathscr{I}_f)$. Thus, for every integer n, there is a canonical homomorphism $\mathscr{I}_{f'}^n/\mathscr{I}_{f'}^{n+1} \to \mathscr{I}_u^n/\mathscr{I}_u^{n+1}$, and hence a canonical homomorphism of graded $\mathscr{O}_{Y'}$-algebras

(16.2.6.2) $$\mathscr{G}r_\bullet(f') \longrightarrow \mathscr{G}r_\bullet(u).$$

Proposition (16.2.7). — *Let X be a prescheme, let Y be a subprescheme of X, let Y' be a subprescheme of Y, and let $j : Y' \to Y$ be the canonical injection. We then have an exact sequence of conormal sheaves ($\mathscr{O}_{Y'}$-modules)*

(16.2.7.1) $$j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0$$

where the arrows are the degree 1 components of the canonical homomorphisms (16.2.6.1) and (16.2.6.2).

Proof. Since the question is local, we may restrict to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(A/\mathfrak{I})$, and $Y' = \mathrm{Spec}(A/\mathfrak{K})$, where $\mathfrak{I}$ and $\mathfrak{K}$ are ideals of A such that $\mathfrak{I} \subset \mathfrak{K}$; everything reduces to seeing that the sequence of canonical morphisms $\mathfrak{I}/\mathfrak{K}\mathfrak{I} \to \mathfrak{K}/\mathfrak{K}^2 \to (\mathfrak{K}/\mathfrak{I})/(\mathfrak{K}/\mathfrak{I})^2 \to 0$ is exact, **IV-4** | 14
which is immediate given that the image of $\mathfrak{I}/\mathfrak{K}\mathfrak{I}$ in $\mathfrak{K}/\mathfrak{K}^2$ is $(\mathfrak{I} + \mathfrak{K}^2)/\mathfrak{K}^2$ and that $(\mathfrak{K}/\mathfrak{I})/(\mathfrak{K}/\mathfrak{I})^2$ is identified with $\mathfrak{K}/(\mathfrak{I} + \mathfrak{K}^2)$. ☐

It is easy to give examples where the sequence (16.2.7.1) extended on the left by 0 is not exact; with the above notation, it suffices to take $A = k[T]$, $\mathfrak{I} = AT^2$, $\mathfrak{K} = AT$, because then $(\mathfrak{I} + \mathfrak{K}^2)/\mathfrak{K}^2 = 0$ and $\mathfrak{I}/\mathfrak{K}\mathfrak{I} \neq 0$. See however (16.9.13) and (19.1.5) for some cases where the extended sequence is indeed exact.

16.3. Fundamental differential invariants of morphisms of preschemes

Definition (16.3.1). — Let $f : X \to S$ be a morphism of preschemes, and let $\Delta_f : X \to X \times_S X$ be the corresponding diagonal morphism, which is an immersion ((**I**, 5.3.9) and $\mathbf{Err_{III}}$, 10). We denote by $\mathscr{P}^n_f$ or $\mathscr{P}^n_{X/S}$, and call the *sheaf of principal parts of order n of the S-prescheme X*, the $\mathscr{O}_X$-augmented sheaf of rings that is the n-th normal invariant of Δ_f (16.1.2). We will also write $\mathscr{P}^\infty_f = \mathscr{P}^\infty_{X/S} = \varprojlim_n \mathscr{P}^n_{X/S}$, $\mathscr{G}r_n(\mathscr{P}_f) = \mathscr{G}r_n(\mathscr{P}_{X/S}) = \mathscr{G}r_n(\Delta_f)$ (16.1.2); the $\mathscr{O}_X$-module $\mathscr{G}r_1(\Delta_f)$, the augmentation ideal of $\mathscr{P}^1_{X/S}$, is also denoted by Ω^1_f or $\Omega^1_{X/S}$, and is called the $\mathscr{O}_X$-module of 1-*differentials* of f, or of X with respect to S, or of the S-prescheme X.

It follows from this definition that $\mathscr{P}^0_{X/S}$ is canonically identified with $\mathscr{O}_X$ (16.1.2).
We have (16.1.2.2) a canonical surjective homomorphism of graded $\mathscr{O}_X$-algebras

$$\mathbf{S}^\bullet_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}). \tag{16.3.1.1}$$

It also follows from Definition (16.3.1) that for every open U of X we have $\mathscr{P}^n_{f|U} = \mathscr{P}^n_f|U$, $\mathscr{P}^\infty_{f|U} = \mathscr{P}^\infty_f|U$, $\mathscr{G}r_n(\mathscr{P}_{f|U}) = \mathscr{G}r_n(\mathscr{P}_f)|U$, and $\Omega^1_{f|U} = \Omega^1_f|U$ (in other words, the notions introduced are *local* on X).

(16.3.2). Let p_1, p_2 be the two canonical projections of the product $X \times_S X$; since Δ_f is an X-section of $X \times_S X$ for both p_1 and p_2, *each* of these morphisms defines, for every n, a homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ that is right inverse to the augmentation $\mathscr{P}^n_{X/S} \to \mathscr{O}_X$ (16.1.7); equivalently, this defines on $\mathscr{P}^n_{X/S}$ *two structures of quasi-coherent augmented $\mathscr{O}_X$-algebra*; the corresponding $\mathscr{O}_X$-module structures on $\mathscr{G}r_n(\mathscr{P}_{X/S})$ are the same. We also have, by passing to the limit, two $\mathscr{O}_X$-algebra structures on $\mathscr{P}^\infty_{X/S}$.

(16.3.3). The morphism $s = (p_2, p_1)_S : X \times_S X \to X \times_S X$ is an *involutive automorphism* of $X \times_S X$, called the *canonical symmetry*, such that

$$p_1 \circ s = p_2, \qquad p_2 \circ s = p_1, \qquad s \circ \Delta_f = \Delta_f. \tag{16.3.3.1}$$

If we set $s = (\rho, \lambda)$, $p_i = (\pi_i, \mu_i)$ $(i = 1, 2)$, and $\Delta_f = (\delta, \nu)$, then $\lambda^\#$ is an isomorphism from $\rho^*(\pi_1^*(\mathscr{O}_X))$ onto $\pi_2^*(\mathscr{O}_X)$, and $\delta^*(\lambda^\#)$ leaves invariant $\delta^*(\mathscr{O}_{X\times_S X})$ and the kernel $\mathscr{I}$ of the homomorphism $\nu^\# : \delta^*(\mathscr{O}_{X\times_S X}) \to \mathscr{O}_X$. Therefore:

Proposition (16.3.4). — *The homomorphism $\sigma = \delta^*(\lambda^\#)$ induced from s (and also called the* canonical symmetry*) is an involutive automorphism of the projective system $(\mathscr{P}^n_{X/S})$ of $\mathscr{O}_X$-augmented sheaves of* **IV-4 | 15**
rings, and as a result also of the projective limit $\mathscr{P}^\infty_{X/S}$. This automorphism interchanges the two $\mathscr{O}_X$-algebra structures on $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$.

(16.3.5). In what follows, the two $\mathscr{O}_X$-algebra structures defined on the $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$ will play very different roles: *we will now agree, unless said otherwise, that when $\mathscr{P}^n_{X/S}$ or $\mathscr{P}^\infty_{X/S}$ is considered as an $\mathscr{O}_X$-algebra, it is the algebra structure induced by p_1.*

For every open U of X and every section $t \in \Gamma(U, \mathscr{O}_X)$, we will simply denote by $t.1$ or even t the image of t under the structure morphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^n_{X/S})$ (resp. $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^\infty_{X/S})$) (that is to say, the homomorphism corresponding to p_1).

Definition (16.3.6). — We denote by d^n_f, or $d^n_{X/S}$ (resp. d^∞_f, or $d^\infty_{X/S}$), or simply d^n (resp. d^∞), the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_f = \mathscr{P}^n_{X/S}$ (resp. $\mathscr{O}_X \to \mathscr{P}^\infty_f = \mathscr{P}^\infty_{X/S}$) induced by p_2. For every open U of X, and every $t \in \Gamma(U, \mathscr{O}_X)$, $d^n t$ (resp. $d^\infty t$) is called the *principal part of order n* (resp. *principal part of infinite order*) of t. We set $dt = d^1 t - t$, and we say that dt is the differential of t (an element of $\Gamma(U, \Omega^1_{X/S})$, also denoted $d_{X/S}(t)$).

It follows immediately [1] from this definition that we have

$$d(t_1 t_2) = t_1 dt_2 + t_2 dt_1 \tag{16.3.6.1}$$

for every t_1, t_2 in $\Gamma(U, \mathscr{O}_X)$; in other words, d is a *derivation* of the ring $\Gamma(U, \mathscr{O}_X)$ with values in the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

In all the notation introduced in (16.3.1) and (16.3.6), we will sometimes replace S by A when $S = \operatorname{Spec}(A)$.

(16.3.7). Suppose in particular that $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine schemes, B then being an A-algebra. Then Δ_f corresponds to the canonical surjective homomorphism $\pi : B \otimes_A B \to B$ such that $\pi(b \otimes b') = bb'$, with kernel $\mathfrak{J} = \mathfrak{J}_{B/A}$ (**0**, 20.4.1); $\mathscr{P}^n_f$ is the structure sheaf of the prescheme $\operatorname{Spec}(P^n_{B/A})$, where

$$P^n_{B/A} = (B \otimes_A B)/\mathfrak{J}^{n+1};$$

$\mathscr{G}r_\bullet(\mathscr{P}_f)$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the graded B-module

$$\operatorname{gr}^\bullet_{\mathfrak{J}}(B \otimes_A B) = \bigoplus_{n \geqslant 0} (\mathfrak{J}^n/\mathfrak{J}^{n+1});$$

in particular $\Omega^1_f = \Omega^1_{X/S}$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the B-module of 1-differentials of B over A, $\Omega^1_{B/A}$ (**0**, 20.4.3). The projection morphisms $p_1 : X \times_S X \to X$ and $p_2 : X \times_S X \to X$ correspond to the two homomorphisms of rings $j_1 : B \to B \otimes_A B$ and $j_2 : B \to B \otimes_A B$ such that $j_1(b) = b \otimes 1$ and $j_2(b) = 1 \otimes b$; thus (by the convention of (16.3.5)), $P^n_{B/A}$ is always considered as a B-algebra via the composite homomorphism $B \xrightarrow{j_1} B \otimes_A B \to P^n_{B/A}$; the ring homomorphism $B \xrightarrow{j_2} B \otimes_A B \to P^n_{B/A}$ is denoted by $d^n_{B/A}$ and corresponds to $d^n_{X/S}$ acting on $\Gamma(X, \mathscr{O}_X)$; for every $t \in B$, dt is equal to $d_{B/A}t$, defined in (**0**, 20.4.6) (cf. $\mathbf{Err_{IV}}$, 11).

If $\pi_n : B \otimes_A B \to P^n_{B/A}$ is the canonical homomorphism, then the preceding definitions give

$$\pi_n(b \otimes b') = b \cdot \pi_n(1 \otimes b') = b \cdot d^n_{B/A}(b') \quad \text{for } b \in B, b' \in B. \tag{16.3.7.1}$$

Proposition (16.3.8). — *The image of the canonical homomorphism $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ generates the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$.* **IV-4** | 16

Proof. We immediately reduce to the case where $X = \operatorname{Spec}(B)$ and $S = \operatorname{Spec}(A)$ are affine; the proposition then follows from (16.3.7.1), since π_n is surjective. We note that in general $d^n_{X/S}$ *is not surjective* (even for $n = 1$). □

Proposition (16.3.9). — *Suppose that $f : X \to S$ is a morphism locally of finite type. Then the $\mathscr{P}^n_f$ and the $\mathscr{G}r_n(\mathscr{P}_f)$ are quasi-coherent $\mathscr{O}_X$-modules of finite type.*

Proof. This follows from (16.1.6) and from the fact that Δ_f is locally of finite presentation (**I**, 4.3.1). □

[1] [Translator's note.] Locally, this is (**0**, 20.1.1).

16.4. Functorial properties of differential invariants

(16.4.1). Consider a commutative diagram of morphisms of preschemes

(16.4.1.1)
$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] \\ S & S' \arrow[l, "w"] \end{tikzcd}$$

We deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & X' \arrow[l, "u"'] \arrow[d, "\Delta_{f'}"] \\ X \times_S X & X' \times_{S'} X' \arrow[l, "v"] \end{tikzcd}$$

where v is the composite morphism (**I**, 5.3.5) and (**I**, 5.3.15):

(16.4.1.2)
$$X' \times_{S'} X' \xrightarrow{(p_1',p_2')_S} X' \times_S X' \xrightarrow{u \times_S u} X \times_S X.$$

Thus u and v give, as explained in (16.2.1), homomorphisms of augmented sheaves of rings

(16.4.1.3)
$$\nu_n : \rho^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{X'/S'}$$

(where we put $u = (\rho, \lambda)$); these homomorphisms form a projective system and therefore give, in the limit, a homomorphism of augmented sheaves of rings

(16.4.1.4)
$$\nu_\infty : \rho^*(\mathscr{P}^\infty_{X/S}) \longrightarrow \mathscr{P}^\infty_{X'/S'};$$

on the other hand, by passing to the quotients, the homomorphisms ν_n give rise to a di-homomorphism of graded algebras (relative to $\lambda^\#$):

(16.4.1.5)
$$\mathrm{gr}(u) : \rho^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'}).$$

(16.4.2). If we have a commutative diagram **IV-4** | 17

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] & X'' \arrow[l, "u'"'] \arrow[d, "f''"] \\ S & S' \arrow[l, "w"] & S'' \arrow[l, "w'"] \end{tikzcd}$$

we deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & X' \arrow[l, "u"'] \arrow[d, "\Delta_{f'}"] & X'' \arrow[l, "u'"'] \arrow[d, "\Delta_{f''}"] \\ X \times_S X & X' \times_{S'} X' \arrow[l, "v"] & X'' \times_{S''} X'' \arrow[l, "v'"] \end{tikzcd}$$

where v' is defined from u', w', f', f'' as v is from u, w, f, f'. We verify immediately that if $u'' = u \circ u'$ and $w'' = w \circ w'$, then the composite morphism $v \circ v'$ is equal to the morphism v'' deduced from u'', w'', f, f'' as v is deduced from u, w, f, f'. If we set $u' = (\rho', \lambda')$ and $u'' = (\rho'', \lambda'')$, it follows from (16.2.1) that the homomorphism $\nu''_n : \rho''^*(\mathscr{P}^n_{X/S}) \to \mathscr{P}^n_{X''/S''}$ is equal to the composite

$$\rho'^*(\rho^*(\mathscr{P}^n_{X/S})) \xrightarrow{\rho'^*(\nu_n^\#)} \rho'^*(\mathscr{P}^n_{X'/S'}) \xrightarrow{\nu'_n} \mathscr{P}^n_{X''/S''},$$

and there are analogous transitivity properties for the homomorphisms (16.4.1.4) and (16.4.1.5), allowing us to say that $\mathscr{P}^n_{X/S}$, $\mathscr{P}^\infty_{X/S}$, and $\mathscr{G}r_\bullet(\mathscr{P}_{X/S})$ *depend functorially on f*.

(16.4.3). We verify immediately (for example, by reducing to the affine case using (16.3.7)) that, with the notation of (16.4.1), the diagram

(16.4.3.1)
$$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d] & \mathscr{O}_{X'} \arrow[d] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

is commutative, where the vertical arrows define the algebra structures chosen in (16.3.5) (that is, those coming from the first projections); the same goes for the diagram

(16.4.3.2)
$$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d, "\rho^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

the vertical arrows here define the algebra structures coming from the second projections; besides, **IV-4** | 18
if σ and σ' are the canonical symmetries corresponding to f and f' (16.3.4), we have

$$\nu_n \circ \rho^*(\sigma) = \sigma' \circ \nu_n$$

which carries one of the preceding diagrams to the other. We therefore deduce from (16.4.3.1) a canonical homomorphism of *augmented $\mathscr{O}_{X'}$-algebras*

(16.4.3.3)
$$P^n(u) : u^*(\mathscr{P}^n_{X/S}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \mathscr{P}^n_{X'/S'}$$

and it follows from (16.4.3.2) that the diagram

(16.4.3.4)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "u^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ u^*(\mathscr{P}^n_{X/S}) \arrow[r, "P^n(u)"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

is commutative. We thus obtain a homomorphism of *graded $\mathscr{O}_{X'}$-algebras*

(16.4.3.5)
$$\mathrm{Gr}_\bullet(u) : u^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'})$$

and in particular a homomorphism of $\mathscr{O}_{X'}$-modules

(16.4.3.6)
$$\mathrm{Gr}_1(u) : \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/S'}$$

giving rise to a commutative diagram

(16.4.3.7)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "d_{X/S} \otimes 1"'] & \mathscr{O}_{X'} \arrow[d, "d_{X'/S'}"] \\ \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \arrow[r] & \Omega^1_{X'/S'} \end{tikzcd}$$

(16.4.4). When $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(A')$, $X = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(B')$ are affine, so that we have a commutative diagram of ring homomorphisms

$$\begin{tikzcd} B \arrow[r] & B' \\ A \arrow[r] \arrow[u] & A' \arrow[u] \end{tikzcd}$$

the image of $\mathfrak{I}_{B/A}$ in $B' \otimes_{A'} B'$ is contained in $\mathfrak{I}_{B'/A'}$, and the homomorphism ν_n corresponds to the homomorphism of rings $P^n_{B/A} \to P^n_{B'/A'}$ induced from the homomorphism $B \otimes_A B \to B' \otimes_{A'} B'$ by passing to quotients. The homomorphism (16.4.3.6) corresponds to the homomorphism defined in (**0**, 20.5.4.1), and the commutative diagram (16.4.3.7) to the diagram (**0**, 20.5.4.2).

Proposition (16.4.5). — *Suppose that $X' = X \times_S S'$, f' and u the canonical projections. Then the canonical homomorphisms $P^n(u)$ (16.4.3.3) and $\mathrm{Gr}_1(u)$ (16.4.3.6) are bijective.* **IV-4** | 19

Proof. We have $X' \times_{S'} X' = (X \times_S X) \times_S S'$, and it suffices to apply (16.2.3, (ii)), replacing g by the first projection $p_1 : X \times_S X \to X$ and f by the diagonal Δ_f. □

We note that under the hypotheses of (16.4.5) the homomorphism $\mathrm{Gr}_\bullet(u)$ (16.4.3.5) is *surjective*, but not bijective in general. However (16.2.4):

Corollary (16.4.6). — *Under the hypotheses of (16.4.5), suppose in addition that $w : S \to S'$ is flat (resp. that $\mathscr{G}r_n(\mathscr{P}_{X/S})$ are flat $\mathscr{O}_X$-modules for $n \leqslant m$); then the homomorphism*

$$\mathrm{Gr}_n(u) : u^*(\mathscr{G}r_n(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_n(\mathscr{P}_{X'/S'})$$

is bijective for each n (resp. for $n \leqslant m$).

Proof. Indeed, if w is flat, then so is $v : X' \times_{S'} X' \to X \times_S X$, so the conclusion follows from (16.2.4). □

(16.4.7). Let S be a prescheme, let $\mathscr{E}$ be a quasi-coherent $\mathscr{O}_S$-module, and set $X = \mathbf{V}(\mathscr{E})$ (**II**, 1.7.8), the vector bundle associated with $\mathscr{E}$, equal to $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$. Let $f : X \to S$ be the structure morphism. For every open U of S and every section $t \in \Gamma(U, \mathscr{E})$, t is identified with a section of $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ over U; let t' be its image in $\Gamma(f^{-1}(U), \mathscr{O}_X) = \Gamma(U, f_*(\mathscr{O}_X)) = \Gamma(U, \mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$, and set

$$\delta(t) = d^n_{X/S}(t') - t' \in \Gamma(f^{-1}(U), \mathscr{P}^n_{X/S}); \tag{16.4.7.1}$$

it is clear that δ is a di-homomorphism of modules (corresponding to the homomorphism of rings $\Gamma(U, \mathscr{O}_S) \to \Gamma(f^{-1}(U), \mathscr{O}_X)$) from $\Gamma(U, \mathscr{E})$ to $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$, whose image moreover belongs to the augmentation ideal of $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$. We deduce (by varying U) a canonical homomorphism of $\mathscr{O}_X$-algebras

$$f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{P}^n_{X/S} \tag{16.4.7.2}$$

and in view of the preceding remark, if $\mathscr{K}$ is the kernel ideal of the augmentation $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \to \mathscr{O}_S$, the image of $\mathscr{K}^{n+1}$ under (16.4.7.2) is zero; hence, after factoring by $\mathscr{K}^{n+1}$, we finally obtain a canonical homomorphism

$$\delta_n : f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})/\mathscr{K}^{n+1}) \longrightarrow \mathscr{P}^n_{X/S}. \tag{16.4.7.3}$$

Proposition (16.4.8). — *Under the conditions of (16.4.7), the homomorphisms δ_n are bijective and form a projective system of isomorphisms; we deduce an isomorphism of graded $\mathscr{O}_X$-algebras*

$$f^*(\mathbf{S}^\bullet_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}). \tag{16.4.8.1}$$

Proof. The fact that homomorphisms (16.4.7.3) form a projective system follows immediately from their definition. To prove they are isomorphisms, it suffices to prove that (16.4.8.1) is an isomorphism, since the filtrations on both sides of (16.4.7.3) are finite (Bourbaki, *Alg. comm.*, chap. III, §2, no 8, Corollary 3 to Theorem 1). To do this, consider the split exact sequence of $\mathscr{O}_S$-modules **IV-4** | 20

$$0 \longrightarrow \mathscr{E} \xrightarrow{u} \mathscr{E} \oplus \mathscr{E} \xrightarrow{v} \mathscr{E} \longrightarrow 0 \tag{16.4.8.2}$$

where, for every pair of sections s, t of $\mathscr{E}$ over an open U of S, we take $u(s) = (-s, s)$ and $v(s,t) = s + t$. We have

$$X \times_S X = \mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \otimes_{\mathscr{O}_S} \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) = \mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}))$$

((**II**, 1.4.6) and (**II**, 1.7.11)), and the diagonal morphism $X \to X \times_S X$ corresponds (**II**, 1.2.7) to the homomorphism of $\mathscr{O}_S$-algebras $\mathbf{S}(v) : \mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}) \to \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ (**II**, 1.7.4); thus, if $\mathscr{I}$ is the kernel of this homomorphism, then

$$\mathscr{P}^n_{X/S} = f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E})/\mathscr{I}^{n+1}).$$

The proposition now will be a consequence of the following lemma: □

Lemma (16.4.8.3). — *Let Y be a ringed space, $0 \to \mathscr{F}' \xrightarrow{u} \mathscr{F} \xrightarrow{v} \mathscr{F}'' \to 0$ an exact sequence of $\mathscr{O}_Y$-modules such that each point $y \in Y$ has an open neighbourhood V such that the sequence $0 \to \mathscr{F}'|V \to \mathscr{F}|V \to \mathscr{F}''|V \to 0$ is split. Let $\mathscr{I}$ be the kernel ideal of $\mathbf{S}(v)$:*

$$\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}) \longrightarrow \mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}''),$$

and let $\mathrm{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}))$ be the graded $\mathscr{O}_Y$-algebra associated to the $\mathscr{O}_Y$-algebra $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})$ endowed with the $\mathscr{I}$-preadic filtration. *Then the homomorphism of graded $\mathscr{O}_Y$-algebras*

$$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'') \longrightarrow \mathrm{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})) \tag{16.4.8.4}$$

(where the left-hand side is the graded tensor product of symmetric $\mathscr{O}_Y$-algebras endowed with their canonical gradings (**II**, 1.7.4) *and* (**II**, 2.1.2)*), induced by the canonical injection*

$$\mathscr{F}' \longrightarrow \mathscr{I} = \mathrm{gr}^1_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})),$$

is bijective.

Proof. The injection $\mathscr{F}' \to \mathscr{I}$ indeed canonically gives a homomorphism of graded $\mathscr{O}_Y$-algebras $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \to \mathrm{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}))$, and since the right-hand side is by definition a graded $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$-algebra, we obtain the canonical homomorphism (16.4.8.4) by tensoring the preceding one with $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$. To prove the lemma we can, being a local problem, restrict to the case where $\mathscr{F} = \mathscr{F}' \oplus \mathscr{F}''$, u and v the canonical homomorphisms. Then the graded algebra $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F})$ is canonically identified with the graded tensor product $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$ (**II**, 1.7.4), and it is immediate that $\mathscr{I}$ is therefore the ideal $\mathscr{I}' \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where $\mathscr{I}'$ is the augmentation ideal of $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}')$, that is to say the (direct) sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant 1$. We conclude that $\mathscr{I}^n = \mathscr{I}'^n \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where this time $\mathscr{I}'^n$ is the direct sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant n$; we therefore have $\mathscr{I}^n/\mathscr{I}^{n+1} = \mathbf{S}^n_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, which proves that (16.4.8.4) is bijective. □

Having proved the lemma, it remains to see that the homomorphism (16.4.8.1) is indeed the **IV-4** | 21
image under f^* of the homomorphism (16.4.8.4) corresponding to the exact sequence (16.4.8.2); this follows readily from the definitions of u (16.4.8.2) and δ (16.4.7.1), together with the definitions of the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ and of $d^n_{X/S}$ (16.3.5) and (16.4.3.6).

In particular:

Corollary (16.4.9). — *Under the conditions of (16.4.7), we have a canonical isomorphism*

$$\mathrm{gr}_1(\delta) : f^*(\mathscr{E}) \simeq \Omega^1_{X/S}. \tag{16.4.9.1}$$

Corollary (16.4.10). — *If $S = \mathrm{Spec}(A)$, $\mathscr{E} = \mathscr{O}_S^m$, so that*

$$X = \mathrm{Spec}(A[T_1, \dots, T_m]),$$

then $\mathscr{P}^n_{X/S}$ is canonically identified with the $\mathscr{O}_X$-algebra corresponding to the quotient $A[T_1, \dots, T_m]$-algebra
$A[T_1, \dots, T_m, U_1, \dots, U_m]/\mathfrak{K}^{n+1}$, where the U_i ($1 \leqslant i \leqslant m$) are m new indeterminates and $\mathfrak{K}$ is the ideal generated by $U_1, \dots, U_m$.

We thus recover in particular the structure of $\Omega^1_{X/S}$ in this case (**0**, 20.5.13).

Moreover, $d^n_{X/S}$ then sends a polynomial $F(T_1, \dots, T_m)$ to the class modulo $\mathfrak{K}^{n+1}$ of $F(T_1 + U_1, \dots, T_m + U_m)$, as follows from the definition (16.4.7.1).

Proposition (16.4.11). — *Let $f : X \to S$ be a morphism, let $g : S \to X$ be an S-section of X, and let $S^{(n)}$ be the n-th infinitesimal neighbourhood of S with respect to the immersion g (16.1.2). Then there exists a unique isomorphism of $\mathscr{O}_S$-algebras*

$$\varpi_n : g^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{O}_{S^{(n)}_g} \tag{16.4.11.1}$$

(via the $\mathscr{O}_S$-algebra structure on $\mathscr{O}_{S_g^{(n)}}$ defined by f (16.1.7)), making the diagram

(16.4.11.2)
$$\begin{tikzcd} \mathscr{O}_S = g^*(\mathscr{O}_X) \arrow[rr, "\lambda_n"] \arrow[dr, "g^*(d^n_{X/S})"'] & & \mathscr{O}_{S_g^{(n)}} \\ & g^*(\mathscr{P}^n_{X/S}) \arrow[ur, "\varpi_n"'] & \end{tikzcd}$$

commutative (where λ_n is the structure morphism).

Proof. In light of (**I**, 5.3.7), where we replace X, Y, S by S, X, S respectively and f by g, the diagrams

(16.4.11.3)
$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\ X \arrow[r, "(g\circ f, 1_X)_S"'] & X \times_S X \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\ X \arrow[r, "(1_X, g\circ f)_S"'] & X \times_S X \end{tikzcd}$$

identify S with the product of the $(X \times_S X)$-preschemes X and X via the morphisms Δ_f and $(g \circ f, 1_X)_S$ (resp. $(1_X, g \circ f)_S$). On the other hand, the diagrams **IV-4** | 22

(16.4.11.4)
$$\begin{tikzcd} X \arrow[r, "(g\circ f, 1_X)_S"] \arrow[d, "f"'] & X \arrow[d, "p_1"] \\ S \arrow[r, "g"'] & X \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "(1_X, g\circ f)_S"] \arrow[d, "f"'] & X \arrow[d, "p_2"] \\ S \arrow[r, "g"'] & X \end{tikzcd}$$

identify X with the product of the X-preschemes S and $X \times_S X$ via the morphisms g and p_1 (resp. p_2) (a special case of the associativity formula (**I**, 3.3.9.1)). We can say that Δ_f, considered as an X-section of $X \times_S X$ (relative to p_1 or p_2) plays the role of a *universal section* for the S-sections of X: each such section g is indeed obtained from it by the *base change* $(g \circ f, 1_X)_S : X \to X \times_S X$. The definition of the homomorphism ϖ_n and the fact that it is bijective therefore follow from these remarks and from (16.2.3, (ii)) applied to the first diagram (16.4.11.4). The commutativity of diagram (16.4.11.2) likewise follows from (16.2.3, (ii)), this time applied to the second diagram (16.4.11.4). To explain ϖ_n, we can restrict ourselves to the case where g is a closed immersion: Indeed, for every $s \in S$, there is an open neighbourhood W of s in S such that $g(W)$ is closed in an open set U of X, and it is clear that $g|W$ is a W-section of the morphism $U \cap f^{-1}(W) \to W$, the restriction of f. We can then suppose that S is a closed subprescheme of X defined by a quasi-coherent ideal $\mathscr{K}$. Then the preceding definitions show that if W is an open of S, t is a section of $\mathscr{O}_X$ over $f^{-1}(W)$, $\varpi_n(d^n t|W)$ is equal to the canonical image of t in $\Gamma(W, (\mathscr{O}_X/\mathscr{K}^{n+1})|W)$. The uniqueness of ϖ_n then follows since the image of $\mathscr{O}_X$ under $d^n_{X/S}$ generates the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$ (16.3.8). □

Corollary (16.4.12). — *Let k be a field, X a k-prescheme, x a point of X* rational over k. *Then $(\mathscr{P}^n_{X/k})_x \otimes_{\mathscr{O}_x} k(x)$ is canonically isomorphic (as an augmented $k(x)$-algebra) to $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$.*

Proof. It suffices to consider the unique k-section g of X such that $g(\mathrm{Spec}(k)) = \{x\}$. □

Corollary (16.4.13). — *Let $f : X \to S$ be a morphism, s a point of S, $X_s = X \times_S \mathrm{Spec}(k(s))$ the fibre of f in s. If $x \in X_s$ is* rational over $k(s)$, *$(\mathscr{P}^n_{X/S})_x \otimes_{\mathscr{O}_s} k(s)$ is canonically isomorphic to $\mathscr{O}_{X_s,x}/\mathfrak{m}_x'^{\,n+1}$, where $\mathfrak{m}'_x$ is the maximal ideal of $\mathscr{O}_{X_s,x}$; more precisely, this isomorphism sends $(d^n t)_x \otimes 1$* (where t is a section of $\mathscr{O}_X$ over an open neighbourhood of x in X) *to the class of $t_x \otimes 1$* modulo *$\mathfrak{m}_x'^{\,n+1}$.*

Proof. This follows from (16.4.5) and (16.4.12). □

The preceding corollaries justify the terminology "sheaf of principal parts of order n".

Proposition (16.4.14). — *Let $\rho : A \to B$ be a homomorphism of rings, and let S be a multiplicative subset of B. Then the canonical homomorphisms*

(16.4.14.1) $$S^{-1}P^n_{B/A} \longrightarrow P^n_{S^{-1}B/A}$$

deduced from the canonical homomorphisms $P^n_{B/A} \to P^n_{S^{-1}B/A}$ (16.4.4), form a projective system and are **IV-4** | 23
bijective.

Proof. It suffices to remark that $S^{-1}((B \otimes_A B)/\mathfrak{I}^{n+1}) = S^{-1}(B \otimes_A B)/(S^{-1}\mathfrak{I})^{n+1}$ by flatness, and that $S^{-1}(B \otimes_A B) = (S^{-1}B) \otimes_A (S^{-1}B)$ (**I**, 1.3.4). □

Corollary (16.4.15). — *The notation being that of (16.4.14), let R be a multiplicative subset of A such that $\rho(R) \subset S$. Then we have canonical isomorphisms*

(16.4.15.1) $$S^{-1}P^n_{B/A} \simeq P^n_{S^{-1}B/R^{-1}A}$$

forming a projective system.

Proof. It evidently suffices to define canonical isomorphisms

(16.4.15.2) $$P^n_{S^{-1}B/A} \simeq P^n_{S^{-1}B/R^{-1}A}$$

that is to say that we reduce to the case where $\rho(R)$ consists of *invertible* elements of B. But then the isomorphism (16.4.15.2) is simply induced by the canonical isomorphism $B \otimes_A B \to B \otimes_{R^{-1}A} B$ by passing to quotients (**I**, 1.5.3). □

Corollary (16.4.16). — *Let $f : X \to S$ be a morphism of preschemes, x a point of X, $s = f(x)$. Then we have canonical isomorphisms*

(16.4.16.1) $$(\mathscr{P}^n_{X/S})_x \simeq P^n_{\mathscr{O}_x/\mathscr{O}_s}$$

forming a projective system.

We deduce corresponding isomorphisms for the associated graded rings, and in particular a canonical isomorphism

(16.4.16.2) $$(\Omega^1_{X/S})_x \simeq \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}.$$

Corollary (16.4.17). — *Let k be a field, K the field of rational functions $k(T_1, \dots, T_r)$. Then, for every integer n, the homomorphism from $K[U_1, \dots, U_r]$ (U_i indeterminates) to $P^n_{K/k}$ which sends U_i to $d^n T_i - T_i.1$ is surjective and induces an isomorphism from the quotient $K[U_1, \dots, U_r]/\mathfrak{m}^{n+1}$ (where $\mathfrak{m}$ is the ideal generated by the U_i) to $P^n_{K/k}$.*

Proof. This follows from (16.4.8), (16.4.10) and (16.4.14), where we take $A = k$, $B = k[T_1, \dots, T_r]$ and $S = B - \{0\}$. □

We thus recover the fact that the dT_i form a basis of the K-vector space $\Omega^1_{K/k}$ (**0**, 20.5.10).

(16.4.18). Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of preschemes, and consider the canonical homomorphisms of augmented $\mathscr{O}_X$-algebras (16.4.3.3)

(16.4.18.1) $$g_{X/Y/Z} : \mathscr{P}^n_{X/Z} \longrightarrow \mathscr{P}^n_{X/Y}$$

(16.4.18.2) $$f_{X/Y/Z} : f^*(\mathscr{P}^n_{Y/Z}) \longrightarrow \mathscr{P}^n_{X/Z}.$$

Then $g_{X/Y/Z}$ is surjective, and its kernel is the sheaf of ideals generated by the image under $f_{X/Y/Z}$ of the augmentation ideal of $f^*(\mathscr{P}^n_{Y/Z})$.

Proof. First note that $g_{X/Y/Z}$ corresponds to the case in (16.4.3.3) where $X' = X$, $S' = Y$ and $S = Z$, **IV-4** | 24
$u = 1_X$, and $f_{X/Y/Z}$ to the case where we replace X', X, S, S' by X, Y, Z, Z respectively and u, f by f, g respectively.

We have a commutative diagram (**I**, 3.5)

(16.4.18.3)
$$\begin{tikzcd} X \arrow[r, "\Delta_f"] \arrow[dr, "f"'] & X \times_Y X \arrow[r, "j"] \arrow[d, "p"] & X \times_Z X \arrow[d, "f\times_Z f"] \\ & Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

where $j = (1_X, 1_X)_Z$ is an immersion, $j \circ \Delta_f = \Delta_{g\circ f}$, and p is the structure morphism. Since we can restrict ourselves to the case where X, Y and Z are affine, we can suppose that the immersions Δ_f, Δ_g and j are closed, so that $\mathscr{O}_X$ and $\mathscr{O}_{X\times_Y X}$ are identified respectively with $\mathscr{O}_{X\times_Z X}/\mathscr{I}$ and $\mathscr{O}_{X\times_Z X}/\mathscr{L}$, where $\mathscr{L} \supset \mathscr{I}$ are two quasi-coherent ideals corresponding respectively to the immersions $\Delta_{g\circ f}$ and j. The $\mathscr{O}_X$-algebra $\mathscr{P}^n_{X/Z}$ is identified with $\mathscr{O}_{X\times_Z X}/\mathscr{I}^{n+1}$, and $\mathscr{P}^n_{X/Y}$ is identified with $\mathscr{O}_{X\times_Y X}/(\mathscr{I}/\mathscr{L})^{n+1}$, which is to say with $\mathscr{O}_{X\times_Z X}/(\mathscr{I}^{n+1} + \mathscr{L})$, and therefore with the quotient of $\mathscr{P}^n_{X/Z}$ by $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$. But we know (*loc. cit*) that if p and j make $X \times_Y X$ the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$, so if $\mathscr{O}_Y$ is identified to $\mathscr{O}_{Y\times_Z Y}/\mathscr{K}$, where $\mathscr{K}$ is the ideal corresponding to Δ_g, $\mathscr{L}$ is equal to $(f \times_Z f)^*(\mathscr{K}).\mathscr{O}_{X\times_Z X}$ (**I**, 4.4.5). Since $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$ is the ideal of $\mathscr{P}^n_{X/Z}$ generated by the image of $\mathscr{L}$, we deduce the proposition. □

Corollary (16.4.19). — *With the notation of (16.4.18), we have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.19.1)
$$f^*(\Omega^1_{Y/Z}) \xrightarrow{f_{X/Y/Z}} \Omega^1_{X/Z} \xrightarrow{g_{X/Y/Z}} \Omega^1_{X/Y} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 5.7.1).

Proposition (16.4.20). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ a closed immersion, $\mathscr{K}$ the quasi-coherent sheaf of ideals of $\mathscr{O}_Y$ corresponding to j. It follows that $\mathscr{P}^n_{X/Y} = \mathscr{O}_X = \mathscr{O}_Y/\mathscr{K}$, the canonical homomorphism $j_{X/Y/Z} : j^*(\mathscr{P}^n_{Y/Z}) \to \mathscr{P}^n_{X/Z}$ is surjective, and its kernel is the ideal of $j^*(\mathscr{P}^n_{Y/Z})$ generated by $j^*(\mathscr{O}_Y \cdot d^n_{Y/Z}(\mathscr{K}))$ (it should be noted that $d^n_{Y/Z}(\mathscr{K})$ is a subsheaf of abelian groups of $\mathscr{P}^n_{Y/Z}$, but not an $\mathscr{O}_Y$-module in general).*

Proof. We know (**I**, 5.3.8) that the diagonal $\Delta_j : X \to X \times_Y X$ is an isomorphism, from which the first assertion follows. If ϖ_1 and ϖ_2 are the two canonical homomorphisms of algebras $\mathscr{O}_Y \to \mathscr{P}^n_{Y/Z}$ corresponding respectively to the two canonical projections p_1, p_2 of $Y \times_Z Y \to Y$, recall that by definition ((16.3.5) and (16.3.6)) ϖ_1 is the structure homomorphism of the $\mathscr{O}_Y$-algebra $\mathscr{P}^n_{Y/Z}$ and $\varpi_2 = d^n_{Y/Z}$. The $\mathscr{O}_X$-algebra $j^*(\mathscr{P}^n_{Y/Z})$ is therefore identified with $\mathscr{P}^n_{Y/Z}/\varpi_1(\mathscr{K})\mathscr{P}^n_{Y/Z}$ and its quotient by the ideal generated by $j^*(d^n_{Y/Z}(\mathscr{K}))$ to $\mathscr{P}^n_{Y/Z}/(\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K}))\mathscr{P}^n_{Y/Z}$. Now note that we have a commutative diagram **IV-4** | 25

$$\begin{tikzcd} Y \arrow[d, "\Delta_f"'] & X \arrow[l, "j"'] \arrow[d, "\Delta_{f\circ j}"] \\ Y \times_Z Y & X \times_Z X \arrow[l, "j\times_Z j"] \end{tikzcd}$$

identifying X with the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$ (**I**, 5.3.7). Since $j \times_Z j$ is an immersion, we therefore deduce from this remark and from (16.2.2) that if $\Delta^n_{Y/Z}$ and $\Delta^n_{X/Z}$ denote the infinitesimal neighbourhoods of order n of Y and X by the canonical immersions Δ_f and $\Delta_{f\circ j}$ respectively, then we have a diagram

$$\begin{tikzcd} \Delta^n_{Y/Z} \arrow[d] & \Delta^n_{X/Z} \arrow[l] \arrow[d] \\ Y \times_Z Y & X \times_Z X \arrow[l, "j\times_Z j"] \end{tikzcd}$$

making $\Delta^n_{X/Z}$ the product of the $(Y \times_Z Y)$-preschemes $\Delta^n_{Y/Z}$ and $X \times_Z X$. We can also say that $\mathscr{P}^n_{X/Z}$ is identified with the sheaf of rings $\mathscr{P}^n_{Y/Z} \otimes_{\mathscr{O}_{Y\times_Z Y}} \mathscr{O}_{X\times_Z X}$. But we see immediately (for example, by restricting to the affine case) that $\mathscr{O}_{X\times_Z X} = \mathscr{O}_{Y\times_Z Y}/(p_1^*(\mathscr{K}) + p_2^*(\mathscr{K}))\mathscr{O}_{Y\times_Z Y}$. Therefore $\mathscr{P}^n_{X/Z}$ is identified with the quotient of $\mathscr{P}^n_{Y/Z}$ by the ideal generated by the image in $\mathscr{P}^n_{Y/Z}$ of $p_1^*(\mathscr{K}) + p_2^*(\mathscr{K})$. But by definition this ideal is generated by $\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K})$. ☐

Corollary (16.4.21). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ an immersion. We have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.21.1) $$\mathscr{N}_{X/Y} \longrightarrow j^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 20.5.12.1).

Corollary (16.4.22). — *If $f : X \to S$ is a morphism locally of finite presentation, $\mathscr{P}^n_{X/S}$ and $\Omega^1_{X/S}$ are quasi-coherent $\mathscr{O}_X$-modules of finite presentation.*

Proof. We immediately reduce to the case where $S = \operatorname{Spec}(A)$ is affine, $X = \operatorname{Spec}(B)$, where $B = A[T_1, \dots, T_r]/\mathfrak{K}$, $\mathfrak{K}$ being an ideal of finite type of $C = A[T_1, \dots, T_r]$. We then apply (16.4.20) with $Z = S$, $Y = \operatorname{Spec}(C)$ and $\mathscr{K} = \widetilde{\mathfrak{K}}$. Then $j^*(\mathscr{P}^n_{Y/Z})$ is a free $\mathscr{O}_X$-module of finite rank (16.4.10) and the hypothesis on $\mathfrak{K}$ implies that $j^*(\mathscr{O}_Y.d^n_{Y/Z}(\mathscr{K}))$ generates a quasi-coherent $\mathscr{O}_X$-module of finite type; hence the conclusion. ☐

Proposition (16.4.23). — *Let X, Y be two S-preschemes, $Z = X \times_S Y$ their product, $p : X \times_S Y \to X$ and $q : X \times_S Y \to Y$ the canonical projections. Then the canonical homomorphism*

(16.4.23.1) $$p_{Z/X/S} \oplus q_{Z/Y/S} : p^*(\Omega^1_{X/S}) \oplus q^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{(X\times_S Y)/S}$$

is bijective.

Proof. The commutative diagram **IV-4** | 26

$$\begin{tikzcd} Y \arrow[d, "g"'] & X \times_S Y \arrow[l, "q"'] \arrow[d, "h"'] & X \times_S Y \arrow[l, "\mathrm{id}"'] \arrow[d, "p"] \\ S & S \arrow[l, "\mathrm{id}"] & X \arrow[l, "f"] \end{tikzcd}$$

gives us a factorization of the canonical *isomorphism* $P^n(p)$ (16.4.5)

$$p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y}$$

and similarly, switching X with Y, we have a factorization of the *isomorphism* $P^n(q)$

$$q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/X}.$$

This proves that the canonical homomorphism (16.4.18.1)

$$p_{Z/X/S} : p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \quad (\text{resp. } q_{Z/Y/S} : q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S})$$

is *injective*, and that the kernel of the canonical surjective homomorphism (16.4.18.2)

$$\mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y} \quad (\text{resp. } \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/X})$$

is complementary to the image of $p_{Z/X/S}$ (resp. $q_{Z/Y/S}$). On the other hand, by (16.4.18), this kernel is generated by the image under $q_{Z/Y/S}$ (resp. $p_{Z/X/S}$) of the augmentation ideal of $q^*(\mathscr{P}^n_{Y/S})$ (resp. $p^*(\mathscr{P}^n_{X/S})$). We conclude the proposition by considering the case $n = 1$. ☐

We immediately generalize (16.4.23) to the case of a product of any finite number of S-preschemes.

Remarks (16.4.24). —

(i) We will see (17.2.3) that when the morphism $f : X \to Y$ in (16.4.18) is *smooth*, the homomorphism $f_{X/Y/Z}$ in (16.4.19.1) is locally *left invertible* and in particular injective. Similarly, when the morphism $f \circ j : X \to Z$ of (16.4.20) is *smooth*, the homomorphism on the left in (16.4.21.1) is locally *left invertible* and *a fortiori* injective (17.2.5). In Chapter V, we will also give a variant, in the case of modules over a prescheme, of the "imperfection modules" studied in (**0**, 20.6), and the exact sequences where they occur.

(ii) Let X be a topological space, $\mathscr{A}$ a sheaf of rings over X and $\mathscr{B}$ an $\mathscr{A}$-algebra over X. Then it is clear that

$$U \longmapsto P^n_{\Gamma(U,\mathscr{B})/\Gamma(U,\mathscr{A})} \quad (U \text{ open in } X)$$

is a presheaf of augmented $\Gamma(U,\mathscr{B})$-algebras, and therefore the associated sheaf $\mathscr{P}^n_{\mathscr{B}/\mathscr{A}}$ is an augmented $\mathscr{B}$-algebra. In the particular case where X is a prescheme, $f = (\psi, \theta) : X \to S$ a morphism of preschemes, it follows easily from (16.4.16) and from the exactness of the functor $\varinjlim$ that $\mathscr{P}^n_{X/S}$ is canonically isomorphic to $\mathscr{P}^n_{\mathscr{O}_X/\psi^*(\mathscr{O}_S)}$. It follows that the formalism developed in the present paragraph could be considered as a particular case **IV-4** | 27
of a differential formalism for ringed spaces endowed with a sheaf of algebras over the structure sheaf. However, we did not start with this point of view, which is less intuitive and less convenient for applications. It also seems that, for various kinds of "varieties", the "global" constructions of the $\mathscr{P}^n$ analogous to those we have used here are also better suited for applications.

16.5. Relative tangent sheaves and bundles; derivations.

(16.5.1). Let $f = (\psi, \theta) : X \to S$ be a morphism of ringed spaces. For every $\mathscr{O}_X$-module $\mathscr{F}$, an *S-derivation* (or *(X/S)-derivation*, or *f-derivation*) *from $\mathscr{O}_X$ to $\mathscr{F}$* is any homomorphism of *sheaves of additive groups* $D : \mathscr{O}_X \to \mathscr{F}$ satisfying the following conditions:

(a) for every open V of X, and every pair of sections (t_1, t_2) of $\mathscr{O}_X$ over V, we have

$$D(t_1 t_2) = t_1 D(t_2) + D(t_1) t_2; \tag{16.5.1.1}$$

(b) for every open V of X, every section t of $\mathscr{O}_X$ over V, and every section s of $\mathscr{O}_S$ over an open U of S such that $V \subset f^{-1}(U)$, we have

$$D((s|V)t) = (s|V) D(t). \tag{16.5.1.2}$$

It is clear that this amounts to saying that, for all $x \in X$, the homomorphism of additive groups $D_x : \mathscr{O}_x \to \mathscr{F}_x$ is an *$\mathscr{O}_{f(x)}$-derivation*.

Another interpretation consists of considering the $\mathscr{O}_X$-algebra $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$ as equal to $\mathscr{O}_X \oplus \mathscr{F}$, the algebra structure being defined by the condition that for every open V of X, the product of two sections of $\mathscr{O}_X$ (resp. of a section of $\mathscr{O}_X$ and a section of $\mathscr{F}$) over V is defined by the ring structure of $\Gamma(V, \mathscr{O}_X)$ (resp. the $\Gamma(V, \mathscr{O}_X)$-module structure on $\Gamma(V, \mathscr{F})$), and the product of two sections of $\mathscr{F}$ over V is chosen to be zero; then $\mathscr{F}$ is an ideal of $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, the kernel of the canonical augmentation $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F}) \to \mathscr{O}_X$, and to say that D is an S-derivation from $\mathscr{O}_X$ to $\mathscr{F}$ means that $\iota_{\mathscr{O}_X} + D$ is an *$\mathscr{O}_S$-homomorphism of algebras* from $\mathscr{O}_X$ to $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, which, composed with the augmentation, gives $\iota_{\mathscr{O}_X}$.

The S-derivations from $\mathscr{O}_X$ to $\mathscr{F}$ clearly form a $\Gamma(X, \mathscr{O}_X)$-*module* $\operatorname{Der}_S(\mathscr{O}_X, \mathscr{F})$.

When $\mathscr{F} = \mathscr{O}_X$, an S-derivation of $\mathscr{O}_X$ to itself is simply called an *S-derivation of $\mathscr{O}_X$*.

Proposition (16.5.2). — *Let A be a ring, B an A-algebra, L a B-module; let $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, $\mathscr{F} = \widetilde{L}$. Then the map $D \mapsto \Gamma(D)$ which sends every S-derivation D from $\mathscr{O}_X$ to $\mathscr{F}$ to the map $\Gamma(D) : t \mapsto D(t)$ from B to L, is an isomorphism of B-modules from $\operatorname{Der}_S(\mathscr{O}_X, \mathscr{F})$ to $\operatorname{Der}_A(B, L)$ (cf. (**0**, 20.1.2)).*

Proof. This follows immediately from the given interpretation of S-derivations in terms of homo- **IV-4** | 28
morphisms of algebras, analogous to the interpretation given in (**0**, 20.1.6), and from the canonical correspondence between homomorphisms of $\mathscr{O}_X$-algebras and homomorphisms of B-algebras ((**I**, 1.3.13) and (**I**, 1.3.8)). □

Proposition (16.5.3). — *Let $f = (\psi, \theta) : X \to S$ be a morphism of preschemes.*

(i) *The differential $d_{X/S} : \mathscr{O}_X \to \Omega^1_{X/S}$ (16.3.6) is an S-derivation.*

(ii) *For every $\mathscr{O}_X$-module $\mathscr{F}$, the map $u \mapsto u \circ d_{X/S}$ is an isomorphism of $\Gamma(X, \mathscr{O}_X)$-modules*

$$\mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \simeq \mathrm{Der}_S(\mathscr{O}_X, \mathscr{F}). \tag{16.5.3.1}$$

Proof. Assertion (i) has already been noted in (16.3.6). On the other hand, it follows immediately (in light of (**0**, 20.4.8)) that $u \mapsto u \circ d_{X/S}$ is injective, by considering the restrictions of both sides to a stalk $\mathscr{O}_x$ and using (16.4.16.2). To see that the homomorphism (16.5.3.1) is surjective, consider an S-derivation $D : \mathscr{O}_X \to \mathscr{F}$; for every affine open $V = \mathrm{Spec}(B)$ of X, such that $f(V)$ is contained in an affine open $U = \mathrm{Spec}(A)$ of S, $D_V : B \to \Gamma(V, \mathscr{F})$ is an A-derivation, and therefore there exists a *unique* B-homomorphism $u_V : \Omega^1_{B/A} \to \Gamma(V, \mathscr{F})$ such that $D_V = u_V \circ d_{B/A}$ (**0**, 20.4.8); in addition, the uniqueness of u_V shows immediately that for an affine open $W \subset V$ we have $u_W = u_V|W$, and therefore the u_V define a homomorphism of $\mathscr{O}_X$-modules $u : \Omega^1_{X/S} \to \mathscr{F}$ answering the question. □

(16.5.4). With the notation of (16.5.1), for every open U of X, $\mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a $\Gamma(U, \mathscr{O}_X)$-module and it is clear that the map $U \mapsto \mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a presheaf; in fact, it is even a *sheaf* (and therefore an $\mathscr{O}_X$-module), in light of the pointwise characterization of S-derivations, seen in (16.5.1). This $\mathscr{O}_X$-module is denoted by $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ and is called the *sheaf of S-derivations from $\mathscr{O}_X$ to $\mathscr{F}$*, and what we have just seen is also expressed by the following corollary:

Corollary (16.5.5). — *For every $\mathscr{O}_X$-module $\mathscr{F}$, the homomorphism of $\mathscr{O}_X$-modules induced by $u \mapsto u \circ d_{X/S}$*

$$\mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \longrightarrow \mathscr{D}er_S(\mathscr{O}_X, \mathscr{F}) \tag{16.5.5.1}$$

is bijective.

Corollary (16.5.6). — (i) *If the morphism $f : X \to S$ is locally of finite presentation and if $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-module.*
(ii) *If in addition S is locally Noetherian and if $\mathscr{F}$ is coherent, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a coherent $\mathscr{O}_X$-module.*

Proof. The assertion (i) follows from the isomorphism (16.5.5.1), from (16.4.22), and (**I**, 1.3.12); the assertion (ii) follows from (**0**, 5.3.5). □

(16.5.7). We set

$$\mathfrak{G}_{X/S} = \mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \mathscr{D}er_S(\mathscr{O}_X, \mathscr{O}_X), \tag{16.5.7.1}$$

and say that it is the *sheaf of S-derivations of $\mathscr{O}_X$*, or even the *tangent sheaf of X relative to S*: it is therefore the *dual* of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. If f is locally of finite presentation, $\mathfrak{G}_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module; if in addition S is locally Noetherian, then $\mathfrak{G}_{X/S}$ is coherent (16.5.6). **IV-4** | 29

(16.5.8). Suppose in particular that $\Omega^1_{X/S}$ is a *locally free* $\mathscr{O}_X$-module (of finite rank) (which will be the case when f is *smooth* (17.2.3)); then $\mathfrak{G}_{X/S}$ is a locally free $\mathscr{O}_X$-module of the same rank as $\Omega^1_{X/S}$ at each point. More specifically, suppose that $\Omega^1_{X/S}$ is of rank n at a point x; then there are n sections s_i ($1 \leqslant i \leqslant n$) of $\mathscr{O}_X$ over an affine neighbourhood U of x such that the canonical images of the ds_i in $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$ form a basis of this $k(x)$-vector space; by virtue of Nakayama's lemma, the germs $(ds_i)_x = d(s_i)_x$ of the ds_i at the point x form a basis of the $\mathscr{O}_x$-module $(\Omega^1_{X/S})_x$, and therefore, by restricting U, we can suppose that the ds_i form a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$. So the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \mathfrak{G}_{X/S})$ is dual to the above; we denote by $(D_i)_{1 \leqslant i \leqslant n}$ or $\left(\frac{\partial}{\partial s_i}\right)_{1 \leqslant i \leqslant n}$ the dual basis of $(ds_i)_{1 \leqslant i \leqslant n}$, so that, by (16.5.3), we have

$$D_i s_j = \langle D_i, ds_j \rangle = \left\langle \frac{\partial}{\partial s_i}, ds_j \right\rangle = \delta_{ij} \quad \text{(Kronecker's symbol).} \tag{16.5.8.1}$$

Every $\Gamma(S, \mathscr{O}_S)$-derivation of the $\Gamma(S, \mathscr{O}_S)$-algebra $\Gamma(U, \mathscr{O}_X)$ is therefore written uniquely as

$$D = \sum_{i=1}^{n} a_i D_i = \sum_{i=1}^{n} a_i \frac{\partial}{\partial s_i},$$

where the a_i ($1 \leqslant i \leqslant n$) are sections of $\mathscr{O}_X$ over U. For every section $g \in \Gamma(U, \mathscr{O}_X)$, if we put $dg = \sum_{i=1}^n c_i ds_i$, then we have $c_i = \langle D_i, dg\rangle = D_i g$ by virtue of (16.5.8.1), in other words,

(16.5.8.2)
$$dg = \sum_{i=1}^{n} (D_i g) ds_i = \sum_{i=1}^{n} \frac{\partial g}{\partial s_i} ds_i.$$

(16.5.9). Let D_1, D_2 be two S-derivations of $\mathscr{O}_X$. For every open U of X, if D_1^U, D_2^U are the corresponding derivations of the ring $\Gamma(U, \mathscr{O}_X)$, the *bracket*

$$[D_1^U, D_2^U] = D_1^U \circ D_2^U - D_2^U \circ D_1^U$$

is also a derivation of this ring, and therefore the $\psi^*(\mathscr{O}_S)$-endomorphism of $\mathscr{O}_X$

(16.5.9.1)
$$[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1$$

is also an S-derivation; as we immediately check that this bracket satisfies the Jacobi identity, we have thus defined on $\mathrm{Der}_S(\mathscr{O}_X, \mathscr{O}_X)$ a $\Gamma(S, \mathscr{O}_S)$-*Lie algebra* structure. Since the definition of this structure commutes with the restriction to an open of X, we thus see that $\mathfrak{G}_{X/S}$ is canonically equipped with a $\psi^*(\mathscr{O}_S)$-*Lie algebra* structure. Note that the mapping $(D_1, D_2) \mapsto [D_1, D_2]$ is *not* $\Gamma(X, \mathscr{O}_X)$-*bilinear*.

(16.5.10). For every base change $g : S' \to S$, if we set $X' = X \times_S S'$, then we have seen in (16.4.5) that there is a canonical isomorphism

(16.5.10.1)
$$\Omega^1_{X/S} \otimes_S S' \simeq \Omega^1_{X'/S'},$$

from which we deduce, by (16.5.10.1), a canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3) **IV-4** | 30

(16.5.10.2)
$$\mathfrak{G}_{X/S} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} \longrightarrow \mathfrak{G}_{X'/S'},$$

which is neither injective nor surjective in general. However:

Proposition (16.5.11). — (i) *If $g : S' \to S$ is a flat morphism and if f is locally of finite type (resp. locally of finite presentation), then the homomorphism* (16.5.10.2) *is injective (resp. bijective).*

(ii) *If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module of finite type, then the homomorphism* (16.5.10.2) *is bijective.*

Proof. The assertion (ii) follows from Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3, prop. 7. The assertion (i) follows similarly from Bourbaki, *Alg. Comm.*, chap. I, §2, n°10, prop. 11 and from the fact that if f is locally of finite type (resp. locally of finite presentation), then $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of finite type (resp. of finite presentation) ((16.3.9) and (16.4.22)). □

(16.5.12). Since $\Omega^1_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module, we can consider the vector bundle over X defined by $\Omega^1_{X/S}$ (**II**, 1.7.8)

(16.5.12.1)
$$T_{X/S} = \mathbf{V}(\Omega^1_{X/S})$$

which is called the *tangent bundle of X relative to S*. We have therefore a canonical bijection (**II**, 1.7.9)

$$\Gamma(T_{X/S}/S) \simeq \mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \Gamma(X, \mathfrak{G}_{X/S})$$

by definition of $\mathfrak{G}_{X/S}$, and we can replace X by an open set U of X in this isomorphism; so we can say that the *tangent sheaf* of X relative to S is isomorphic to the *sheaf of germs of S-sections* of the tangent bundle of X relative to S. If $f : X \to Y$ is an S-morphism, we saw (16.4.19) that we have a canonical homomorphism $f_{X/Y/S} : f^*(\Omega^1_{Y/S}) \to \Omega^1_{X/S}$, which, having in mind that

$$\mathbf{V}(f^*(\Omega^1_{Y/S})) = \mathbf{V}(\Omega^1_{Y/S}) \times_Y X \quad (\mathbf{II}, 1.7.11)\,,$$

gives us an X-morphism $T_{X/S}(f) : T_{X/S} \to T_{Y/S} \times_Y X$. If $g : Y \to Z$ is a second S-morphism, we have $T_{X/S}(g \circ f) = (T_{Y/S}(g) \times 1_X) \circ T_{X/S}(f)$ (**0**, 20.5.4.1).

It follows from (16.5.10.1) and from (**II**, 1.7.11) that for every base change $g : S' \to S$ we have a canonical isomorphism

(16.5.12.2)
$$T_{X'/S'} \simeq T_{X/S} \times_S S' = T_{X/S} \times_X X'.$$

(16.5.13). For every point $x \in X$, we define the *tangent space of X at the point x* (relative to S) to be the set of points in the fibre $T_{X/S} \times_X \mathrm{Spec}(k(x))$ that are *rational over* $k(x)$, that is, the set

(16.5.13.1)
$$T_{X/S}(x) = \mathrm{Hom}_{k(x)}(\Omega^1_{X/S} \otimes_{\mathscr{O}_x} k(x), k(x)),$$

which is the *dual* of the $k(x)$-vector space $\Omega^1_{\mathscr{O}_x/\mathscr{O}_s}/\mathfrak{m}_x \cdot \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}$. When $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of *finite type*, then $T_{X/S}(x)$ is a vector space of finite rank over $k(x)$, and for every base change $g :$ **IV-4** | 31
$S' \to S$, and every point $x' \in X' = X \times_S S'$ over x, we have a canonical isomorphism

(16.5.13.2)
$$T_{X'/S'}(x') \simeq T_{X/S} \otimes_{k(x)} k(x').$$

If x is *rational over* $k(s)$, where $s = f(x)$ (so that $k(s) \to k(x)$ is an isomorphism), it follows from (16.4.13) that we have a canonical isomorphism

(16.5.13.3)
$$T_{X/S}(x) = T_{X_s/k(s)}(x) = \mathrm{Hom}_{k(s)}(\mathfrak{m}'_x/\mathfrak{m}'^2_x, k(x)),$$

where $\mathfrak{m}'_x$ is the maximal ideal of $\mathscr{O}_{X_s,x} = \mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$. In the case where S is the spectrum of a field k, we recover the definition of the Zariski tangent space of a point $x \in X$ *rational over* k, as the dual of $\mathfrak{m}_x/\mathfrak{m}_x^2$.

Let Y be a second S-prescheme and let $g : Y \to X$ be an S-morphism; then we have a canonical homomorphism of $\mathscr{O}_Y$-modules (16.4.19)

(16.5.13.4)
$$g_{Y/X/S} : g^*(\Omega^1_{X/S}) \longrightarrow \Omega^1_{Y/S}.$$

Now note that if $y \in Y$ and $x = g(y)$, then we have

$$g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y) = (\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)) \otimes_{k(x)} k(y)$$

and consequently, if $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module *of finite type*, then we can identify

$$\mathrm{Hom}_{k(y)}(g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y), k(y))$$

with $T_{X/S}(x) \otimes_{k(x)} k(y)$. We therefore deduce from the homomorphism (16.5.13.4) a homomorphism of *$k(y)$-vector spaces*

(16.5.13.5)
$$T_y(g) : T_{Y/S}(y) \longrightarrow T_{X/S}(x) \otimes_{k(x)} k(y)$$

called the *linear map tangent to g at the point y*. When y is *rational over* $k(s)$, we can identify $k(s)$, $k(y)$, and $k(x)$, and $T_y(g)$ is then a homomorphism of $k(s)$-vector spaces $T_{Y/S}(y) \to T_{X/S}(x)$; also note that in this case, $g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y)$ is identified with $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$, and the above homomorphism is therefore defined without any finiteness conditions on $\Omega^1_{X/S}$ and it is none other than the homomorphism $T_{Y/S}(g)$ (16.5.12) restricted to the fibre of $T_{Y/S}$ at the point y.

(16.5.14). The interpretation of derivations of an A-algebra B to a B-module L, given in (**0**, 20.1.1), translates to the language of preschemes in the following way.

Consider two morphisms of preschemes $f : X \to S$, $g : Y \to S$, and a closed subprescheme Y_0 of Y defined by a *square-zero* ideal $\mathscr{J}$ of $\mathscr{O}_Y$ (so that Y and Y_0 have the *same underlying topological space*). Suppose we are given an S-morphism $u_0 : Y_0 \to X$, so that we have a commutative diagram

(16.5.14.1)
$$\begin{tikzcd} X \arrow[d, "f"'] & Y_0 \arrow[l, "u_0"'] \arrow[d, "j"] \\ S & Y \arrow[l, "g"] \arrow[ul, dashed, "u"'] \end{tikzcd}$$

and we seek *S-morphisms* $u : Y \to X$ such that $u_0 = u \circ j$ (in other words, we ask whether the **IV-4** | 32
preceding diagram can be completed by the dotted arrow u while remaining *commutative*).

For that, consider an affine open $U = \mathrm{Spec}(C)$ of Y; its inverse image $j^{-1}(U)$ is the affine open $U_0 = \mathrm{Spec}(C/\mathfrak{L})$, where $\mathfrak{L} = \Gamma(U, \mathscr{J})$, a *square-zero* ideal in C; suppose that U is small enough so that $u_0(U_0)$ is contained in an affine open $V = \mathrm{Spec}(B)$ of X and that $g(U) = f(u_0(U_0))$ is contained in an affine open $W = \mathrm{Spec}(A)$ of S, so that B and C are A-algebras and $u_0|U_0$ corresponds to an A-homomorphism ψ from B to $C/\mathfrak{L}$; Let $P(U_0)$ be the set of restrictions $u|U$ of the sought morphisms; it corresponds canonically to the A-algebra homomorphisms $\phi : B \to C$ such that the *composite* $B \xrightarrow{\phi} C \to C/\mathfrak{L}$ *is equal to* ψ. So we know (**0**, 20.1.1) that the set of such homomorphisms is either empty or of the form $\phi_1 + \mathrm{Der}_A(B, \mathfrak{L})$; when $P(U_0)$ is not empty, the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ acts by addition on $P(U_0)$, which is therefore an *affine space* for the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ (or even a *principal homogeneous space* (or *torsor*) *under* $\mathrm{Der}_A(B, \mathfrak{L})$).

Now notice that, since $\mathfrak{L}$ is equipped with a B-module structure via ψ, we have an *isomorphism* $v \mapsto v \circ d_{B/A}$ of $\mathrm{Hom}_B(\Omega^1_{B/A}, \mathfrak{L})$ onto $\mathrm{Der}_A(B, \mathfrak{L})$ (**0**, 20.4.8). Besides, as $\mathfrak{L}$ is square-zero, therefore a $(C/\mathfrak{L})$-module, every B-homomorphism $v : \Omega^1_{B/A} \to \mathfrak{L}$ can be considered as a $(C/\mathfrak{L})$-homomorphism $\Omega^1_{B/A} \otimes_B (C/\mathfrak{L}) \to \mathfrak{L}$. As $\mathscr{J}$ is square-zero, it can be considered as a quasi-coherent $\mathscr{O}_{Y_0}$-module; let's introduce the $\mathscr{O}_{Y_0}$-module

$$\mathscr{G} = \mathscr{H}om(u_0^*(\Omega^1_{X/S}), \mathscr{J}); \tag{16.5.14.2}$$

it follows from the fact that $\Omega^1_{B/A} = \Gamma(V, \Omega^1_{X/S})$ (16.3.7) that we can write $\mathrm{Der}_A(B, \mathfrak{L}) = \Gamma(U_0, \mathscr{G})$.

As $P(U_0)$ is defined as a set of S-morphisms $U \to X$, it is clear that $U_0 \mapsto P(U_0)$ is a *sheaf of sets* $\mathscr{P}$ on Y_0. We can use this fact to prove that the map $h : \Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ defining the torsor structure on $P(U_0)$ is independent of choice of V and W and also that, if $U' \subset U$ is a second affine open of Y, U'_0 its inverse image in Y_0, then the diagram

$$\begin{tikzcd} \Gamma(U_0, \mathscr{G}) \times P(U_0) \arrow[r, "h"] \arrow[d] & P(U_0) \arrow[d] \\ \Gamma(U'_0, \mathscr{G}) \times P(U'_0) \arrow[r, "h'"'] & P(U'_0) \end{tikzcd} \tag{16.5.14.3}$$

is commutative (the vertical arrows being the restrictions). In light of the above remark, we reduce to proving the commutativity of the above diagram when h is defined as such from affine opens V, W and h' from affine opens $V' \subset V$ and $W' \subset W$. But because of the preceding description of h, **IV-4** | 33
this follows from the commutativity of the diagram (**0**, 20.5.4.2).

The mapping $\Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ therefore defines a homomorphism of *sheaves of sets* $m : \mathscr{G} \times \mathscr{P} \to \mathscr{P}$ such that, for all open sets U_0 for which $\Gamma(U_0, \mathscr{P}) \neq \varnothing$, $m_{U_0} : \Gamma(U_0, \mathscr{G}) \times \Gamma(U_0, \mathscr{P}) \to \Gamma(U_0, \mathscr{P})$ is an external law defining in $\Gamma(U_0, \mathscr{P})$ a torsor structure for the group $\Gamma(U_0, \mathscr{G})$.

(16.5.15). In general, when we are given a sheaf of sets $\mathscr{P}$ over a topological space Z, a sheaf of groups $\mathscr{G}$ (not necessarily commutative), and a homomorphism of sheaves of sets $m : \mathscr{G} \times \mathscr{P} \to \mathscr{P}$ such that, for every open $U \subset Z$ such that $\Gamma(U, \mathscr{P}) \neq \varnothing$, $m_U : \Gamma(U, \mathscr{G}) \times \Gamma(U, \mathscr{P}) \to \Gamma(U, \mathscr{P})$ makes $\Gamma(U, \mathscr{P})$ a *torsor* under the group $\Gamma(U, \mathscr{G})$, then we say that $\mathscr{P}$ is a *pseudo-torsor* (or *formally principal homogeneous sheaf*) under the sheaf of groups $\mathscr{G}$. We say that $\mathscr{P}$ is a *torsor* (or *principal homogeneous sheaf*) under $\mathscr{G}$ [2] if in addition $\Gamma(U, \mathscr{P}) \neq \varnothing$ for every open $U \neq \varnothing$ in a suitable basis for the topology of Z.

For the general theory of torsors, we refer to **[?]**; we will limit ourselves to recalling the canonical correspondence between isomorphism classes of torsors (for a *given* $\mathscr{G}$) and elements of the cohomology set $\mathrm{H}^1(Z, \mathscr{G})$. Consider a torsor $\mathscr{P}$ under $\mathscr{G}$ and an open cover (U_λ) of Z such that $\Gamma(U_\lambda, \mathscr{P}) \neq \varnothing$ for every λ; denote by p_λ an element of $\Gamma(U_\lambda, \mathscr{P})$. For every pair of indices λ, μ such that $U_\lambda \cap U_\mu \neq \varnothing$, there then exists a unique element $\gamma_{\lambda\mu}$ of $\Gamma(U_\lambda \cap U_\mu, \mathscr{G})$ such that $\gamma_{\lambda\mu} \cdot (p_\mu | U_\lambda \cap U_\mu) = p_\lambda | U_\lambda \cap U_\mu$; in addition, if λ, μ, ν are three indices such that $U_\lambda \cap U_\mu \cap U_\nu \neq \varnothing$, then the restrictions $\gamma'_{\lambda\mu}$, $\gamma'_{\mu\nu}$, $\gamma'_{\lambda\nu}$ of $\gamma_{\lambda\mu}$, $\gamma_{\mu\nu}$, $\gamma_{\lambda\nu}$ to $U_\lambda \cap U_\mu \cap U_\nu$ satisfy the condition $\gamma'_{\lambda\nu} = \gamma'_{\lambda\mu}\gamma'_{\mu\nu}$; in other words, $(\lambda, \mu) \mapsto \gamma_{\lambda\mu}$ is a 1-*cocycle* of the cover (U_λ) with values in $\mathscr{G}$. If, for every λ, p'_λ is a second element of $\Gamma(U_\lambda, \mathscr{P})$, then there exists a unique element $\beta_\lambda \in \Gamma(U_\lambda, \mathscr{G})$ such that $p'_\lambda = \beta_\lambda \cdot p_\lambda$, and the 1-cocycle $(\gamma'_{\lambda\mu})$ corresponding to the family (p'_λ) is given by $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$, that is, it is *cohomologous* to $\gamma_{\lambda\mu}$. Conversely, the data of a 1-cocycle $(\gamma_{\lambda\mu})$ defines, for every pair (λ, μ), an automorphism $\theta_{\lambda\mu}$ of the sheaf *of sets* $\mathscr{G} | U_\lambda \cap U_\mu$, namely the right translation by $\gamma_{\lambda\mu}$, and the fact that it is a cocycle shows that we can *glue* the sheaves of sets $\mathscr{G} | U_\lambda$ via the automorphisms $\theta_{\lambda\mu}$ (**0**, 3.3.1); we thus obtain a torsor under $\mathscr{G}$, denoted $\mathscr{P}$, and if we

[2] [Trans.] This is nowadays more commonly called a $\mathscr{G}$-torsor rather than a torsor *under* $\mathscr{G}$.

take for p_λ the unit section over U_λ, then the corresponding 1-cocycle is none other than the given 1-cocycle $(\gamma_{\lambda\mu})$; in addition, if we replace $(\gamma_{\lambda\mu})$ by a 1-cocycle $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$ cohomologous to it, then we check immediately that the torsor obtained is isomorphic to $\mathscr{P}$.

In particular, if $(\gamma_{\lambda\mu})$ is a 1-*coboundary*, in other words of the form $\gamma_{\lambda\mu} = \beta_\lambda \beta_\mu^{-1}$, then the torsor $\mathscr{P}$ obtained is *isomorphic to* $\mathscr{G}$ (considered as a torsor under itself by left translations); we say in this case that $\mathscr{P}$ is *trivial*, and the converse is evident.

In particular, it follows from (**III**, 1.3.1) that we have:

Proposition (16.5.16). — *Let Z be an affine scheme, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Z$-module; then every torsor under $\mathscr{G}$ is trivial.*

Returning to the problem considered in (16.5.14), we thus obtain: **IV-4** | 34

Proposition (16.5.17). — *Let X, Y be two S-preschemes, Y_0 a closed subprescheme of Y defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_Y$ such that $\mathscr{I}^2 = 0$, $j : Y_0 \to Y$ the canonical injection. Let $u_0 : Y_0 \to X$ be an S-morphism, and $\mathscr{P}$ the sheaf of sets on Y such that, for every open U of Y, $\Gamma(U, \mathscr{P})$ is the set of S-morphisms $u : U \to X$ such that $u_0|U_0 = u \circ (j|U_0)$, where $U_0 = j^{-1}(U)$. Then $\mathscr{P}$ has the structure of a pseudo-torsor under the $\mathscr{O}_{Y_0}$-module $\mathscr{G} = \mathscr{H}om_{\mathscr{O}_{Y_0}}(u_0^*(\Omega^1_{X/S}), \mathscr{I})$.*

In particular:

Corollary (16.5.18). — *With the notation of (16.5.17), suppose that Y is affine and $\Omega^1_{X/S}$ is of finite presentation; if there is an open cover (U_α) of Y, and, for every index α, an S-morphism $v_\alpha : U_\alpha \to X$ such that, if $U_\alpha^0 = j^{-1}(U_\alpha)$, we have $v_\alpha \circ (j|U_\alpha^0) = u_0|U_\alpha^0$, then there is an S-morphism $u : Y \to X$ such that $u \circ j = u_0$.*

Proof. Indeed, $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_{Y_0}$-module (**I**, 1.3.12); by (16.5.16) and the fact that Y_0 is then affine, the sheaf $\mathscr{P}$, which is by hypothesis a torsor under $\mathscr{G}$, and not only a pseudo-torsor, is *trivial*; if w is an isomorphism from $\mathscr{G}$ to $\mathscr{P}$ (as torsors under $\mathscr{G}$), the image under w of the zero section of $\mathscr{G}$ is the S-morphism sought. □

16.6. Sheaf of p-differentials and exterior differentials.

(16.6.1). Let $f : X \to S$ be a morphism of preschemes. We define the *sheaf of p-differentials of X relative to S* (p an integer) to be the *pth exterior power* (**0**, 4.1.5) of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$, denoted by

$$\Omega^p_{X/S} = \bigwedge^p(\Omega^1_{X/S}). \tag{16.6.1.1}$$

So we have $\Omega^0_{X/S} = \mathscr{O}_X$, and $\Omega^p_{X/S} = 0$ for $p < 0$; the $\Omega^p_{X/S}$ are the homogeneous components of the exterior algebra of $\Omega^1_{X/S}$

$$\Omega^\bullet_{X/S} = \bigwedge(\Omega^1_{X/S}) = \bigoplus_{p \in \mathbf{Z}} \bigwedge^p(\Omega^1_{X/S}), \tag{16.6.1.2}$$

which is therefore a quasi-coherent graded anti-commutative $\mathscr{O}_X$-algebra whose elements of degree 1 are square-zero. For every affine U of X, we have $\Gamma(U, \Omega^\bullet_{X/S}) = \bigwedge(\Gamma(U, \Omega^1_{X/S}))$, where $\Gamma(U, \Omega^1_{X/S})$ is considered as a $\Gamma(U, \mathscr{O}_X)$-module.

When $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, so that B is an A-algebra, we have (**0**, 4.1.5) $\Omega^p_{X/S} = (\Omega^p_{B/A})^\sim$, where $\Omega^p_{B/A} = \bigwedge^p \Omega^1_{B/A}$.

Theorem (16.6.2). — *There is one and only one endomorphism d of the sheaf of additive groups $\Omega^\bullet_{X/S}$ with the following properties:*

(i) *$d \circ d = 0$.*

(ii) *For every open set U of X and every section $f \in \Gamma(U, \mathscr{O}_X)$ we have $df = d_{X/S} f$.* **IV-4** | 35

(iii) *For every open set U of X, every pair of integers p, q and every pair of sections $\omega'_p \in \Gamma(U, \Omega^p_{X/S})$, $\omega''_q \in \Gamma(U, \Omega^q_{X/S})$, we have*

$$d(\omega'_p \wedge \omega''_q) = (d\omega'_p) \wedge \omega''_q + (-1)^p \omega'_p \wedge d\omega''_q. \tag{16.6.2.1}$$

Also, d is an endomorphism of graded $\psi^*(\mathscr{O}_X)$-modules of degree $+1$.

Proof. Suppose that we have proved the existence of an endomorphism d. For every affine open U of X, every section of $\Omega^p_{X/S}$ over U is (by (ii)) a linear combination of finitely many elements of the form $g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)$, where g and the f_i are sections of $\mathscr{O}_X$ over U (**0**, 20.4.7). The conditions (i) and (iii) then show, by induction on p, that we necessarily have

$$d(g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)) = dg \wedge df_1 \wedge df_2 \wedge \cdots \wedge df_p. \tag{16.6.2.2}$$

This therefore proves the *uniqueness* of d and the last claim of the theorem. By virtue of this uniqueness property, to show the existence of d, we can restrict ourselves to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affines. Now (Bourbaki, *Alg.*, chap. III, 3rd ed., §10), to define an A-antiderivation D of degree $+1$ on an exterior algebra $\bigwedge(M)$ (where M is a B-module and B an A-algebra), with values in a *graded anticommutative* A-algebra $C = \bigoplus_{n=0}^\infty C_n$ whose elements of degree 1 are square-zero, it suffices to choose arbitrarily an A-derivation D_0 from B to C_1 and an A-homomorphism D_1 from M to C_2; there then exists a unique A-antiderivation D from $\bigwedge(M)$ to C agreeing with D_0 on B and with D_1 on M.

In the present case, D_0 is necessarily equal to $d_{B/A}$ by (ii); we reduce to seeing, having (16.6.2.2) in mind, that there is an A-homomorphism u of $\Omega^1_{B/A}$ in $\Omega^2_{B/A}$ such that

$$u(g.df) = dg \wedge df \tag{16.6.2.3}$$

for all f, g in B; it suffices to show that there is an A-homomorphism $v : B \otimes_A \Omega^1_{B/A} \to \Omega^2_{B/A}$ such that

$$v(g.\omega) = dg \wedge \omega \tag{16.6.2.4}$$

for $g \in B$ and $\omega \in \Omega^1_{B/A}$. Finally, since $\Omega^1_{B/A} = \mathfrak{I}/\mathfrak{I}^2$ (where $\mathfrak{I} = \mathfrak{I}_{B/A}$ is the kernel of the canonical homomorphism $B \otimes_A B \to B$) and $\Omega^1_{B/A}$ is generated by elements of the form $g.df$, it is enough to define an A-homomorphism $w : B \otimes_A (B \otimes_A B) \to \Omega^2_{B/A}$ such that

$$w(g' \otimes g \otimes f) = dg' \wedge (g.df) \tag{16.6.2.5}$$

and such that w is zero on the image of $B \otimes_A \mathfrak{I}^2$. Since the right-hand side of (16.6.2.5) is A-trilinear in g', g, and f, the existence of w satisfying (16.6.2.5) is immediate. Since, on the other hand, $\mathfrak{I}$ is generated by elements of the form $1 \otimes x - x \otimes 1$ ($x \in B$), we reduce to checking that when $z = (1 \otimes x - x \otimes 1)(1 \otimes y - y \otimes 1)$ we have $w(g' \otimes z) = 0$. Now, since $z = 1 \otimes (xy) + (xy) \otimes 1 - x \otimes y - y \otimes x$, formula (16.6.2.4) shows that it is enough to see that we have $d(xy) - x.dy - y.dx = 0$, which is to **IV-4** | 36
say that d is a derivation.

It remains to be shown that d satisfies the condition (i). Now, the square of an antiderivation is a derivation (Bourbaki, *loc. cit.*), and since $\Omega^\bullet_{B/A}$ is generated by $\Omega^1_{B/A}$ as a B-algebra, it is enough to verify that $d(dz) = 0$ when $z \in B$ and when $z \in \Omega^1_{B/A}$; in the first case, this follows from the formula (16.6.2.3) when $g = 1$; for the second, we can restrict ourselves to the case where $z = g.df$ with f, g in B, and then we have, because of (16.6.2.1) and (16.6.2.3),

$$d(d(g.df)) = d(dg \wedge df) = (d(dg)) \wedge (df) - (dg) \wedge (d(df)) = 0.$$

□

Definition (16.6.3). — The antiderivation d defined in (16.6.2) (also denoted by $d_{X/S}$) is called the *exterior differential* on X (relative to S).

Proposition (16.6.4). — *For every base change $g : S' \to S$, if we put $X' = X \times_S S'$, the canonical morphism*

$$\Omega^\bullet_{X/S} \otimes_S S' \longrightarrow \Omega^\bullet_{X'/S'} \tag{16.6.4.1}$$

deduced from the isomorphism (16.5.10.1) is bijective. Also, if s is a section of $\Omega^\bullet_{X/S}$ over an open set U of X, $s \otimes 1$ its inverse image, section of $\Omega^\bullet_{X'/S'}$ over the inverse image U' of U in X', we have $d_{X'/S'}(s \otimes 1) = d_{X/S}(s) \otimes 1$.

Proof. The first claim is immediate, the formation of the exterior algebra of a module commutes with extending the scalar ring. To prove the second, we can, because of (16.6.2.2), restrict ourselves to the case where $s \in \Gamma(U, \mathscr{O}_X)$, and in this case the claim has already been proven (16.4.3.7). □

(16.6.5). Suppose that $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module of rank n at a point x, so that we have n sections $s_i \in \Gamma(U, \mathscr{O}_X)$ such that the ds_i form a basis for the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$ (16.5.8). Then, for every integer $p \geqslant 1$, the p-differentials $ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}$ (for $i_1 < i_2 < \cdots < i_p$ in $[1, n]$) form a basis of $\binom{n}{p}$ elements of $\Gamma(U, \Omega^p_{X/S})$ over $\Gamma(U, \mathscr{O}_X)$. Also the formula (16.6.2.2) shows that for every section $g \in \Gamma(U, \mathscr{O}_X)$, we have

$$d(g.ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}) = \sum_k (-1)^r \frac{\partial g}{\partial s_k} ds_{i_1} \wedge \cdots \wedge ds_{i_r} \wedge ds_k \wedge ds_{i_{r+1}} \wedge \cdots \wedge ds_{i_p} \tag{16.6.5.1}$$

where, on the right-hand side, k ranges over the $n - p$ indices distinct from the i_h, and i_r is the largest index $< k$.

We note that the relation $d(dg) = 0$ for every section $g \in \Gamma(U, \mathscr{O}_X)$ expresses itself in the form

$$D_i(D_j g) = D_j(D_i g) \quad \text{for } i \neq j;$$

in other words, the derivations D_i defined in (16.5.7) commute with each other.

16.7. The $\mathscr{P}^n_{X/S}(\mathscr{F})$.

(16.7.1). Let $f : X \to S$ be a morphism of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-module. We denote by $X^{(n)}_{\Delta_f}$ the *nth infinitesimal neighbourhood* of X for the diagonal morphism $\Delta_f : X \to X \times_S X$, by $h_n : X^{(n)}_{\Delta_f} \to X \times_S X$ **IV-4 | 37**
the canonical morphism (16.1.2), and consider the two composite morphisms

$$p_1^{(n)} : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_1} X, \quad p_2^{(n)} : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_2} X$$

so that, by definition, $p_1^{(n)}$ corresponds to the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ which we have chosen to define the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ (16.3.5), and $p_2^{(n)}$ to the homomorphism of sheaves of rings $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (16.3.6). Since $X^{(n)}_{\Delta_f}$ and X have the same underlying space, we can write

$$\mathscr{P}^n_{X/S} = (p_1^{(n)})_*((p_2^{(n)})^*(\mathscr{O}_X)). \tag{16.7.1.1}$$

More generally, we define

$$\mathscr{P}^n_{X/S}(\mathscr{F}) = (p_1^{(n)})_*((p_2^{(n)})^*(\mathscr{F})). \tag{16.7.1.2}$$

so that $\mathscr{P}^n_{X/S} = \mathscr{P}^n_{X/S}(\mathscr{O}_X)$; by definition, $\mathscr{P}^n_{X/S}(\mathscr{F})$ is an $\mathscr{O}_X$-module.

(16.7.2). Returning to the definition of the inverse image of modules on ringed spaces (**0**, 4.3.1), and bearing in mind that $X^{(n)}_{\Delta_f}$ and X have the same underlying space, we see that definition (16.7.1.2) can be written in the form

$$\mathscr{P}^n_{X/S}(\mathscr{F}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F}, \tag{16.7.2.1}$$

where it must be kept in mind that, in interpreting the symbol $\otimes$, $\mathscr{P}^n_{X/S}$ is endowed with the $\mathscr{O}_X$-module structure defined by *the homomorphism of sheaves of rings* $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$. It follows immediately from such formula (or directly from (16.7.1.2)) that $\mathscr{P}^n_{X/S}(\mathscr{F})$ is canonically equipped with a $\mathscr{P}^n_{X/S}$-module structure.

Proposition (16.7.3). — (i) *The functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$ from the category of $\mathscr{O}_X$-modules to the category of $\mathscr{P}^n_{X/S}$-modules is right exact, and commutes with arbitrary inductive limits; it is exact when $\mathscr{P}^n_{X/S}$ is flat.*

(ii) *If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module (resp. of finite type, resp. of finite presentation), then $\mathscr{P}^n_{X/S}(\mathscr{F})$ is quasi-coherent (resp. of finite type, resp. of finite presentation).*

Proof. The claims from (i) follow from (16.7.2.1) and the consideration of the symmetry of $\mathscr{P}^n_{X/S}$ (16.3.4). The claims from (ii) follow from the right exactness of the functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$. □

(16.7.4). The two $\mathscr{O}_X$-module structures on $\mathscr{P}^n_{X/S}$ define two commuting $\mathscr{O}_X$-module structures on $\mathscr{P}^n_{X/S}(\mathscr{F})$, and hence an $\mathscr{O}_X$-bimodule structure. It is convenient to denote the structure coming from the structure homomorphism $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (chosen in (16.3.5)) on the *left* and the one coming from the homomorphism $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ on the *right*. In other words, for every open U of X, and every triple $a \in \Gamma(U, \mathscr{O}_X)$, $b \in \Gamma(U, \mathscr{P}^n_{X/S})$, $t \in \Gamma(U, \mathscr{F})$, we have by definition

(16.7.4.1) $$a(b \otimes t) = (ab) \otimes t, \quad (b \otimes t)a = (b.d^n a) \otimes t = b \otimes (at) = (d^n a).(b \otimes t).$$

The $\mathscr{O}_X$-module structure coming from the definition (16.7.1.2) is therefore, under these conventions, the *left* $\mathscr{O}_X$-module structure. If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then the same is true for **IV-4** | 38 $\mathscr{P}^n_{X/S}(\mathscr{F})$ for any one of its $\mathscr{O}_X$-module structures. If also $\mathscr{F}$ is of finite type (resp. of finite presentation) and $f : X \to S$ is locally of finite type (resp. locally of finite presentation), $\mathscr{P}^n_{X/S}(\mathscr{F})$ is (for any one of its $\mathscr{O}_X$-module structures) of finite type (resp. of finite presentation), which is a consequence of (16.3.9) and (16.4.22).

(16.7.5). The definition (16.7.2.1) entails the existence of a homomorphism of sheaves of commutative groups

(16.7.5.1) $$d^n_{X/S,\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad (\text{also denoted } d^n_{X/S})$$

such that, in the notations of (16.7.4), we have

(16.7.5.2) $$d^n_{X/S,\mathscr{F}}(t) = 1 \otimes t$$

and consequently, because of (16.7.4.1)

(16.7.5.3) $$d^n_{X/S,\mathscr{F}}(at) = (1 \otimes t)a = (d^n_{X/S,\mathscr{F}}(t)).a$$

(16.7.5.4) $$d^n_{X/S,\mathscr{F}}(at) = (d^n_{X/S,\mathscr{F}}(a)).(1 \otimes t) = (d^n_{X/S,\mathscr{F}}(a))(d^n_{X/S,\mathscr{F}}(t)).$$

Therefore it is $\mathscr{O}_X$-linear for the *right* $\mathscr{O}_X$-module structure on $\mathscr{P}^n_{X/S}(\mathscr{F})$, and *semilinear* (relative to the automorphism σ (16.3.4)) for the *left* $\mathscr{O}_X$-module structure.

Proposition (16.7.6). — *The left $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}(\mathscr{F})$ is generated by the image of $\mathscr{F}$ under the homomorphism $d^n_{X/S,\mathscr{F}}$.*

Proof. This is an immediate consequence of (16.7.5.3) and of the particular case $\mathscr{F} = \mathscr{O}_X$ (16.3.8). □

(16.7.7). The canonical homomorphisms of sheaves of rings

$$\phi_{nm} : \mathscr{P}^m_{X/S} \longrightarrow \mathscr{P}^n_{X/S}$$

for $n \leqslant m$ (16.1.2) define, because of (16.7.2.1), canonical homomorphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad (n \leqslant m)$$

which are homomorphisms of $\mathscr{O}_X$-bimodules in light of (16.1.6) and (16.7.4.1); also we have commutative diagrams

$$\begin{tikzcd} \mathscr{P}^m_{X/S}(\mathscr{F}) \arrow[rr] & & \mathscr{P}^n_{X/S}(\mathscr{F}) \\ & \mathscr{F} \arrow[ul, "d^m_{X/S,\mathscr{F}}"] \arrow[ur, "d^n_{X/S,\mathscr{F}}"'] & \end{tikzcd}$$

We have therefore a projective system of $\mathscr{O}_X$-bimodules $(\mathscr{P}^n_{X/S}(\mathscr{F}))$, and we define

(16.7.7.1) $$\mathscr{P}^\infty_{X/S}(\mathscr{F}) = \varprojlim \mathscr{P}^n_{X/S}(\mathscr{F}).$$

Also, this shows that the homomorphisms (16.7.5.1) form a projective system of homomorphisms, and therefore define a canonical homomorphism

(16.7.7.2) $$d^\infty_{X/S,\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{P}^\infty_{X/S}(\mathscr{F}).$$

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(16.7.8). Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules; it follows immediately from the definition (16.7.2.1) that we have a canonical isomorphism of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}) \simeq \mathscr{P}^n_{X/S}(\mathscr{F}) \otimes_{\mathscr{P}^n_{X/S}} \mathscr{P}^n_{X/S}(\mathscr{G}) \tag{16.7.8.1}$$

(Bourbaki, *Alg.*, chap. II, 3rd ed., § 5, no. 1, prop. 3).

We conclude in particular (or see directly from definition (16.7.2.1)) that if $\mathscr{F}$ has an $\mathscr{O}_X$-algebra structure (not necessarily associative), $\mathscr{P}^n_{X/S}(\mathscr{F})$ has a canonical $\mathscr{P}^n_{X/S}$-algebra structure; the latter is associative (resp. commutative, resp. unital, resp. a Lie algebra) if $\mathscr{F}$ is so. Moreover, the canonical homomorphisms $\mathscr{P}^m_{X/S}(\mathscr{F}) \to \mathscr{P}^n_{X/S}(\mathscr{F})$ for $n \leqslant m$ (16.7.7) are then algebra dihomomorphisms; similarly, (16.7.5.1) is then an $\mathscr{O}_X$-algebra homomorphism when $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with the $\mathscr{O}_X$-algebra structure arising from its *right* $\mathscr{O}_X$-module structure.

With the same notation, we also have a canonical homomorphism of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})) \longrightarrow \mathscr{H}om_{\mathscr{P}^n_{X/S}}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{P}^n_{X/S}(\mathscr{G})) \tag{16.7.8.2}$$

(Bourbaki, *Alg.*, chap. II, 3rd ed., § 5, no. 3), which is bijective when $\mathscr{P}^n_{X/S}$ is an $\mathscr{O}_X$-module *locally free of finite type* (*loc. cit.*, prop. 7).

(16.7.9). Suppose we are in the situation described in (16.4.1); then from the canonical homomorphism $P^n(u)$ (16.4.3.3) we deduce immediately a canonical homomorphism of $\mathscr{O}_{X'}$-bimodules

$$u^*(\mathscr{P}^n_{X/S}(\mathscr{F})) \longrightarrow \mathscr{P}^n_{X'/S'}(u^*(\mathscr{F})). \tag{16.7.9.1}$$

We leave it to the reader to extend to this homomorphism the properties established in (16.4) for the case $\mathscr{F} = \mathscr{O}_X$.

Remark (16.7.10). — The definition of $\mathscr{P}^n_{X/S}(\mathscr{F})$ in the form (16.7.1.2) still makes sense when $\mathscr{F}$ is a sheaf of sets (the inverse image of a sheaf of sets by $p_2^{(n)}$ being defined in (**0**, 3.7.1)); a variant of this definition allows one to define the "jet scheme" (relative to S) of an arbitrary X-prescheme.

16.8. Differential operators. [3]

Definition (16.8.1). — Let $f = (\psi, \theta) : X \to S$ be a morphism of preschemes, $\mathscr{F}, \mathscr{G}$ two $\mathscr{O}_X$-modules, n an integer $\geqslant 0$. We say that a morphism $D : \mathscr{F} \to \mathscr{G}$ of sheaves of additive groups is a differential operator of order $\leqslant n$ (relative to S) if there is a homomorphism of $\mathscr{O}_X$-modules $u : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G}$ (where $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with its left $\mathscr{O}_X$-module structure (16.7.4)) such that $D = u \circ d^n_{X/S,\mathscr{F}}$.

It is clear, because of the existence of canonical morphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F})$$

for $n \leqslant m$ (16.7.7), that a differential operator of order $\leqslant n$ is a differential operator of order $\leqslant m$ **IV-4** | 40
for $n \leqslant m$. If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, then, for every open set U of X, $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$.

We say that a homomorphism $D : \mathscr{F} \to \mathscr{G}$ is a *differential operator* (relative to S) if, for every $x \in X$, there is an open neighbourhood U of x and an integer $n \geqslant 0$ such that $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$. The *order* of a differential operator D is the infimum of the integers n such that D is a differential operator of order $\leqslant n$ (and therefore $+\infty$ if there is no such integer); this order is always finite if X is *quasi-compact*. The differential operators of order 0 are exactly the homomorphisms of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G}$; the operators of order < 0 are *zero* by convention. For $n \geqslant 0$, a differential operator of order $\leqslant n$ is not in general a homomorphism of $\mathscr{O}_X$-modules, but is always a homomorphism of $\psi^*(\mathscr{O}_S)$-modules.

When $\mathscr{F} = \mathscr{O}_X$, a differential operator of order $\leqslant 1$ from $\mathscr{O}_X$ to $\mathscr{G}$ can be written uniquely in the form $v + D$, where $v : \mathscr{O}_X \to \mathscr{G}$ is an $\mathscr{O}_X$-homomorphism and D is an S-*derivation* (16.5.1) from $\mathscr{O}_X$ to $\mathscr{G}$; this follows from the structure of $P^1_{B/A}$ (**0**, 20.4.8).

[3] For a more general formalism, see Exposé VII of [**?**] (due to P. Gabriel).

(16.8.2). To describe in a more precise manner a differential operator of order $\leqslant n$, $D : \mathscr{F} \to \mathscr{G}$, it suffices, for every open set U of X whose image in S is contained in an affine open set V, to characterize the homomorphism $D = D_U : \Gamma(U, \mathscr{F}) \to \Gamma(U, \mathscr{G})$. If we put $\Gamma(V, \mathscr{O}_S) = A$, $\Gamma(U, \mathscr{O}_X) = B$, so that B is an A-algebra, we have $\Gamma(U, \mathscr{P}^n_{X/S}) = (B \otimes_A B)/\mathfrak{I}^{n+1}$, where, for brevity, $\mathfrak{I} = \mathfrak{I}_{B/A}$. Also put $M = \Gamma(U, \mathscr{F})$, $N = \Gamma(U, \mathscr{G})$; then the definition of D means that for every pair (U, V) satisfying the above, the A-homomorphism $D : M \to N$ factors through

$$M \longrightarrow ((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M \xrightarrow{v} N$$

where the first arrow is the canonical homomorphism $t \mapsto 1 \otimes t$, and v is a B-homomorphism, the B-module structure on its source coming from the first factor B (whereas, in forming the tensor product over B, the B-module structure on $(B \otimes_A B)/\mathfrak{I}^{n+1}$ comes from the second factor B). Note also that the B-module $((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M$ is isomorphic to $(B \otimes_A M)/\mathfrak{I}^{n+1}(B \otimes_A M)$, where $B \otimes_A M$ is regarded as a $(B \otimes_A B)$-module and its B-module structure comes from the homomorphism $b \mapsto b \otimes 1$ from B to $B \otimes_A B$. Let D' then be the B-homomorphism from $B \otimes_A M$ to N such that $D'(b \otimes t) = bD(t)$; the factorization condition on D is equivalently that D' vanish on the B-submodule $\mathfrak{I}^{n+1}(B \otimes_A M)$.

(16.8.3). It is clear that the set of differential operators of order $\leqslant n$ from $\mathscr{F}$ to $\mathscr{G}$ forms an additive group, denoted by $\mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G})$; when $\mathscr{F} = \mathscr{G} = \mathscr{O}_X$, we also write $\mathrm{Diff}^n_{X/S}$ instead of $\mathrm{Diff}^n_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$.

We saw in (16.8.1) that for two open sets $U \supset V$ of X, there is a canonical restriction homomorphism

$$\mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U) \longrightarrow \mathrm{Diff}^n_{V/S}(\mathscr{F}|V, \mathscr{G}|V)$$

from which we deduce that $U \mapsto \mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U)$ is a presheaf of additive groups; in fact, it is **IV-4** | 41
actually a *sheaf*, since for an open set U varying in X, the homomorphisms $u \mapsto u \circ d^n_{U/S, \mathscr{F}|U}$ are *isomorphisms* of sheaves of additive groups

$$\mathrm{Hom}_{\mathscr{O}_U}(\mathscr{P}^n_{U/S}(\mathscr{F}|U), \mathscr{G}|U) \simeq \mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U), \tag{16.8.3.1}$$

because of the fact that the image of $\mathscr{F}$ by $d^n_{X/S, \mathscr{F}}$ generates $\mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.6). We denote this sheaf by $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$, and we have:

Proposition (16.8.4). — *The isomorphisms* (16.8.3.1) *define an isomorphism of sheaves of additive groups*

$$\mathscr{H}\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}) \simeq \mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}). \tag{16.8.4.1}$$

When $\mathscr{F} = \mathscr{G} = \mathscr{O}_X$, we also write $\mathscr{D}\mathit{iff}^n_{X/S}$ instead of $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$; it follows from (16.8.4) that $\mathscr{D}\mathit{iff}^n_{X/S}$ is canonically identified with the *dual* of the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$; so we also write $\langle t, D\rangle$ instead of $u(t)$ if t is a section of $\mathscr{P}^n_{X/S}$ over an open set and if u is a homomorphism from $\mathscr{P}^n_{X/S}$ to $\mathscr{O}_X$ corresponding to D.

(16.8.5). Since $\mathscr{P}^n_{X/S}(\mathscr{F})$ has an $\mathscr{O}_X$-bimodule structure (16.7.4), we canonically obtain an $\mathscr{O}_X$-bimodule structure on $\mathscr{H}\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G})$, and hence on $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ by (16.8.4.1). More precisely, to the left $\mathscr{O}_X$-module structure on $\mathscr{P}^n_{X/S}(\mathscr{F})$ corresponds, because of (16.8.1), the left $\mathscr{O}_X$-module structure on $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ explained as follows: for every open set U of X, every section $a \in \Gamma(U, \mathscr{O}_X)$ and every differential operator $D : \mathscr{F}|U \to \mathscr{G}|U$, aD is the differential operator which, for every section $t \in \Gamma(U, \mathscr{F})$, makes correspond the section

$$(aD)(t) = a(D(t)) \tag{16.8.5.1}$$

of $\Gamma(U, \mathscr{G})$. Similarly, the right $\mathscr{O}_X$-module structure on $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ is made explicit as follows: under the same notations as above, Da is the operator which, to every $t \in \Gamma(U, \mathscr{F})$, makes correspond the section

$$(Da)(t) = D(at). \tag{16.8.5.2}$$

Proposition (16.8.6). — *If $f : X \to S$ is a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module of finite presentation and $\mathscr{G}$ a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ is a quasi-coherent $\mathscr{O}_X$-module for either of the structures defined in* (16.8.5).

Proof. The proposition follows from the fact that, under these hypotheses, $\mathscr{P}^n_{X/S}(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-module of finite presentation (16.7.4), together with (**I**, 1.3.12). ☐

(16.8.7). The set of differential operators (of unspecified order (16.8.1)) is denoted by $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$; we also see as in (16.8.3) that $U \mapsto \mathrm{Diff}_{U/S}(\mathscr{F}|U,\mathscr{G}|U)$ is a sheaf of additive groups, which we will denote by $\mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{F},\mathscr{G})$. It is immediate that $\mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{F},\mathscr{G})$ is the union of the increasing filtered family of its subsheaves $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G})$; if X is quasi-compact, $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$ is similarly the **IV-4** | 42
union of its subgroups $\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G})$ (16.8.1). The $\mathscr{O}_X$-bimodule structures on the $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G})$ therefore induce an $\mathscr{O}_X$-bimodule structure on $\mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{F},\mathscr{G})$, again described by (16.8.5.1) and (16.8.5.2).

Note that, for $n \leqslant m$, we have a commutative diagram

(16.8.7.1)
$$\begin{tikzcd}
\mathscr{H}\!\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}) \arrow[r, "\sim"] \arrow[d] & \mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G}) \arrow[d] \\
\mathscr{H}\!\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^m_{X/S}(\mathscr{F}),\mathscr{G}) \arrow[r, "\sim"] & \mathscr{D}\!\mathit{iff}^m_{X/S}(\mathscr{F},\mathscr{G})
\end{tikzcd}$$

where the horizontal arrows are the isomorphisms (16.8.4.1) and the left vertical arrow comes from the canonical homomorphism $\mathscr{P}^m_{X/S}(\mathscr{F}) \to \mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.7). For every open set U of X, we then equip $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F})) = \varprojlim \Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F}))$ with the projective-limit topology of the discrete topologies on the groups $\Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F}))$. This makes $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F}))$ a topological $\Gamma(U,\mathscr{O}_X)$-bimodule, so that $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ becomes a sheaf with values in the category of topological commutative groups (**0**, 3.2.6). Then, by [**?**, II, 1.11], the inductive limit of the system of sheaves of commutative groups $\big(\mathscr{H}\!\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G})\big)$ is precisely the sheaf of germs of *continuous* homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$ (the latter equipped with the discrete topology): indeed, the continuous homomorphisms from $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F}))$ to the discrete group $\Gamma(U,\mathscr{G})$ correspond bijectively to the inductive systems of group homomorphisms $\Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F})) \to \Gamma(U,\mathscr{G})$. We can furthermore express (16.8.4) by saying there is a canonical isomorphism

$$\mathscr{H}\!\mathit{om.cont}_{\mathscr{O}_X}(\mathscr{P}^\infty_{X/S}(\mathscr{F}),\mathscr{G}) \simeq \mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{F},\mathscr{G})$$

where the left-hand side denotes the sheaf of germs of continuous homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$.

Proposition (16.8.8). — *Let $\mathscr{F}$, $\mathscr{G}$ be two $\mathscr{O}_X$-modules, $D : \mathscr{F} \to \mathscr{G}$ a homomorphism of $\psi^*(\mathscr{O}_S)$-modules, n an integer $\geqslant 0$. The following conditions are equivalent:*

(a) D is a differential operator of order $\leqslant n$.
(b) For all sections a of $\mathscr{O}_X$ over an open set U, the homomorphism $D_a : \mathscr{F}|U \to \mathscr{G}|U$ such that, for every section t of $\mathscr{F}$ over an open set $V \subset U$, we have

(16.8.8.1)
$$D_a(t) = D(at) - aD(t)$$

is a differential operator of order $\leqslant n-1$.
(c) For every open set U of X, every family $(a_i)_{1\leqslant i\leqslant n+1}$ of $n+1$ sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, we have the identity

(16.8.8.2)
$$\sum_{H\subset I_{n+1}} (-1)^{\mathrm{Card}(H)}\Big(\prod_{i\in H} a_i\Big)D\Big(\Big(\prod_{i\notin H} a_i\Big)t\Big) = 0$$

(where I_{n+1} is the interval $1 \leqslant i \leqslant n+1$ of $\mathbb{N}$).

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Proof. Let us first prove the equivalence of (a) and (c). By definition, to prove that D is a differential operator of order $\leqslant n$, it suffices to prove this for the restriction $D|U : \mathscr{F}|U \to \mathscr{G}|U$ to every affine open subset U of X; on the other hand, property (c) holds for every open subset U of X if it holds for every affine open subset. We may therefore restrict ourselves to the case where $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine. By (16.8.2) (whose notation we retain), condition (a) then means that the A-homomorphism $D' : B \otimes_A M \to N$ defined by $D'(b \otimes t) = bD(t)$ vanishes on $\mathfrak{I}^{n+1}(B \otimes_A M)$; by (**0**, 20.4.4), this is equivalent to saying that D' vanishes on every element of the form

$$(\prod_{i=1}^{n+1}(a_i \otimes 1 - 1 \otimes a_i)).(1 \otimes t)$$

where $a_i \in B$ and $t \in M$. Now this element can be written as $\sum_{H \subset I_{n+1}}(\prod_{i \in H} a_i) \otimes ((\prod_{i \notin H} a_i)t)$, and its image under D' is the left-hand side of (16.8.8.2); this proves the equivalence of (a) and (c).

Let us now prove the equivalence of (b) and (c). We argue by induction on n, the assertion being trivial for $n = 0$. Writing a_{n+1} in place of a in condition (b), we see from the induction hypothesis that condition (b) means that, for every family $(a_i)_{1 \leqslant i \leqslant n}$ of n sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, one has

$$\sum_{H' \subset I_n} (-1)^{\mathrm{Card}(H')}(\prod_{i \in H'} a_i) D_{a_{n+1}}((\prod_{i \notin H'} a_i)t) = 0.$$

But if in this relation we replace $D_{a_{n+1}}$ by its definition (16.8.8.1), we see at once that, up to sign, we obtain the left-hand side of (16.8.8.2); hence the conclusion. □

Proposition (16.8.9). — *If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, and $D' : \mathscr{G} \to \mathscr{H}$ a differential operator of order $\leqslant n'$, then $D' \circ D : \mathscr{F} \to \mathscr{H}$ is a differential operator of order $\leqslant n + n'$.*

Proof. By hypothesis, we can write $D = u \circ d^n_{X/S,\mathscr{F}}$ and $D' = v \circ d^{n'}_{X/S,\mathscr{G}}$, where $u : \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \to \mathscr{G}$ and $v : \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G} \to \mathscr{H}$ are $\mathscr{O}_X$-homomorphisms. It is therefore enough to show that the composite homomorphism of sheaves of additive groups

$$\mathscr{F} \xrightarrow{d^n_{X/S,\mathscr{F}}} \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{d^{n'}_{X/S,\mathscr{G}}} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

factors as

$$\mathscr{F} \xrightarrow{d^{n+n'}_{X/S,\mathscr{F}}} \mathscr{P}^{n+n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{w} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

where w is an $\mathscr{O}_X$-homomorphism. It suffices to prove the following lemma.

Lemma (16.8.9.1). — *There is one and only one $\mathscr{O}_X$-homomorphism*

(16.8.9.2) $$\delta : \mathscr{P}^{n+n'}_{X/S} \longrightarrow \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}) = \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{P}^n_{X/S}$$

making the following diagram commute **IV-4** | 44

(16.8.9.3)
$$\begin{tikzcd}
\mathscr{O}_X \arrow[r, "d^{n+n'}_{X/S}"] \arrow[d, "d^n_{X/S}"'] & \mathscr{P}^{n+n'}_{X/S} \arrow[d, "\delta"] \\
\mathscr{P}^n_{X/S} \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S})
\end{tikzcd}$$

We will then have, indeed, a commutative diagram deduced from (16.8.9.3) by tensorization with $\mathscr{F}$

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "d^{n+n'}_{X/S,\mathscr{F}}"] \arrow[d, "d^n_{X/S,\mathscr{F}}"'] & \mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \arrow[d, "\delta\otimes 1"] \\
\mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F}))
\end{tikzcd}$$

and on the other hand, we verify immediately from the definition (16.7.5) that the diagram

$$\begin{tikzcd} \mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "u"] \arrow[d, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{G} \arrow[d, "d^{n'}_{X/S,\mathscr{G}}"] \\ \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F})) \arrow[r, "1\otimes u"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{G}) \end{tikzcd}$$

is commutative. We finish the proof by taking w to be the composite $\mathscr{O}_X$-homomorphism

$$\mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \xrightarrow{\delta\otimes 1} \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F})) \xrightarrow{1\otimes u} \mathscr{P}^{n'}_{X/S}(\mathscr{G}).$$

□

Proof of Lemma 16.8.9.1. It remains to prove Lemma (16.8.9.1). In view of (16.7.6), which proves the uniqueness of δ, we are reduced to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine. Putting $\mathfrak{J} = \mathfrak{J}_{B/A}$, it suffices to define a canonical homomorphism of B-modules

$$\phi : (B \otimes_A B)/\mathfrak{J}^{n+n'+1} \longrightarrow ((B \otimes_A B)/\mathfrak{J}^{n'+1}) \otimes_B ((B \otimes_A B)/\mathfrak{J}^{n+1})$$

the B-module structures on both sides being induced by the first factor B. Recall that in the tensor product on the right, $(B \otimes_A B)/\mathfrak{J}^{n'+1}$ must be regarded as a right B-module by its second B factor, **IV-4** | 45
and $(B \otimes_A B)/\mathfrak{J}^{n+1}$ as a left B-module by its first B factor (16.7.2). This is in turn equivalent to defining a homomorphism of B-modules

$$\phi_0 : B \otimes_A B \longrightarrow ((B \otimes_A B)/\mathfrak{J}^{n'+1}) \otimes_B ((B \otimes_A B)/\mathfrak{J}^{n+1})$$

and proving that it vanishes on $\mathfrak{J}^{n+n'+1}$. Such a homomorphism is defined at once by the condition

$$\phi_0(b \otimes b') = \pi_{n'}(b \otimes 1) \otimes \pi_n(1 \otimes b') \quad \text{for } b, b' \text{ in } B$$

with the notation of (16.3.7). Moreover, it is immediate that ϕ_0 is a homomorphism of *rings*. Now we may write

$$\phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes \pi_n(1\otimes 1) + \pi_{n'}(1\otimes b)\otimes \pi_n(1\otimes 1) - \pi_{n'}(1\otimes 1)\otimes \pi_n(1\otimes b)$$

and we have

$$\pi_{n'}(1\otimes b)\otimes \pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)b\otimes \pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes b\pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes \pi_n(b\otimes 1)$$

whence, finally,

$$\text{(16.8.9.4)} \quad \phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes \pi_n(1\otimes 1) + \pi_{n'}(1\otimes 1)\otimes \pi_n(b\otimes 1 - 1\otimes b).$$

A product of $n + n' + 1$ factors of the form (16.8.9.4) is therefore necessarily zero, since this is so for a product of $n+1$ factors of the form $\pi_n(b\otimes 1 - 1\otimes b)$ and for a product of $n'+1$ factors of the form $\pi_{n'}(b\otimes 1 - 1\otimes b)$. The conclusion therefore follows from (**0**, 20.4.4). □

Corollary (16.8.10). — *The sheaf $\mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$ (also denoted $\mathscr{D}\!\mathit{iff}_{X/S}$) is canonically endowed with a structure of sheaf of rings, and the $\mathscr{D}\!\mathit{iff}^n_{X/S}$ form an increasing filtration compatible with this structure.*

In particular, $\mathscr{D}\!\mathit{iff}^0_{X/S}$ is a sheaf of subrings of $\mathscr{D}\!\mathit{iff}_{X/S}$, which is canonically identified with $\mathscr{O}_X$ (16.8.1). The formulas (16.8.5.1) and (16.8.5.2) show that the structure of $\mathscr{O}_X$-bimodule of $\mathscr{D}\!\mathit{iff}_{X/S}$ comes from the multiplication on the left and on the right by sections of $\mathscr{O}_X$ considered as a sheaf of subrings of $\mathscr{D}\!\mathit{iff}_{X/S}$.

Remarks (16.8.11). — (i) Suppose that $\mathscr{F} = \oplus_{\lambda\in L}\mathscr{F}_\lambda$; then it is clear from (16.7.2.1) that $\mathscr{P}^n_{X/S}(\mathscr{F}) = \oplus_{\lambda\in L}\mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$. Since the functor $\mathscr{F} \mapsto \Gamma(U, \mathscr{F})$ commutes with the formation of arbitrary direct sums, $d^n_{X/S,\mathscr{F}}$ is the homomorphism whose restriction to each $\mathscr{F}_\lambda$ is $d^n_{X/S,\mathscr{F}_\lambda} : \mathscr{F}_\lambda \to \mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$; it follows at once that

$$\operatorname{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\lambda\in L} \operatorname{Diff}^n_{X/S}(\mathscr{F}_\lambda, \mathscr{G}),$$

and therefore also (**0**, 3.2.6)

$$\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\lambda\in L} \mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}_\lambda, \mathscr{G}).$$

Moreover, if $\mathscr{G} = \prod_{\mu\in M} \mathscr{G}_\mu$ (**0**, 3.2.6), we have

$$\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}) = \prod_{\mu\in M} \mathrm{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}_\mu),$$

Every homomorphism u from $\mathscr{P}^n_{X/S}(\mathscr{F})$ to $\mathscr{G}$ corresponds bijectively to the family of its **IV-4** | 46
composites $u_\mu : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G} \to \mathscr{G}_\mu$. Thus

$$\mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\mu\in M} \mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G}_\mu),$$

and consequently also

$$\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\mu\in M} \mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}_\mu).$$

(ii) So far, we have encountered differential operators $\mathscr{F} \to \mathscr{G}$ almost exclusively when $\mathscr{F}$ and $\mathscr{G}$ are locally free of finite rank; in that case, by (i), their structure is reduced locally to that of the sheaf $\mathscr{D}\!\mathit{iff}_{X/S}$. The latter will be studied below, in a particular case, in (16.11).

16.9. Regular and quasi-regular immersions

Definition (16.9.1). — Let X be a ringed space. We say that an ideal $\mathscr{I}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular) if, for every point $x \in \mathrm{Supp}(\mathscr{O}_X/\mathscr{I})$, there is an open neighbourhood U of x in X and a regular sequence (**0**, 15.2.2) (resp. quasi-regular sequence (**0**, 15.2.2)) of elements of $\Gamma(U, \mathscr{O}_X)$ which generates $\mathscr{I}|U$.

A regular (resp. quasi-regular) sequence of sections of $\mathscr{O}_X$ over U which generates $\mathscr{I}|U$ is called a *regular system* (resp. *quasi-regular system*) of generators of $\mathscr{I}|U$.

Definition (16.9.2). — Let $j : Y \to X$ be an immersion of preschemes, and let U be an open subset of X such that $j(Y) \subset U$ and j is a closed immersion of Y into U. We say that j is regular (resp. quasi-regular) if the closed subprescheme $j(Y)$ of U associated to j is defined by a regular (resp. quasi-regular) ideal of $\mathscr{O}_U$ (condition independent of the choice of U).

We say that a subprescheme Y of a prescheme X is *regularly immersed* (resp. *quasi-regularly immersed*) if the canonical injection $j : Y \to X$ is a regular immersion (resp. quasi-regular immersion). If Y is a closed subprescheme and $\mathscr{I}$ is the ideal of $\mathscr{O}_X$ that defines Y, it is the same as asking $\mathscr{I}$ to be *regular* (resp. *quasi-regular*).

For example, if A is an *integral* ring and f is a nonzero element of A, the closed subprescheme $V(f)$ of $\mathrm{Spec}(A)$ (isomorphic to $\mathrm{Spec}(A/(f))$) is *regularly immersed* in $\mathrm{Spec}(A)$.

Every regular ideal is quasi-regular (**0**, 15.2.2); every regular immersion is quasi-regular (cf. (16.9.11) for a converse).

Proposition (16.9.3). — *Let X be a ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$, $(f_i)_{1\leqslant i\leqslant m}$ a finite sequence of sections of $\mathscr{O}_X$ over X generating $\mathscr{I}$. For (f_i) to be a quasi-regular sequence* (**0**, 15.2.2), *it is necessary and sufficient that the following conditions be satisfied:*

(i) *The canonical images of f_i in $\mathscr{I}/\mathscr{I}^2$ form a basis of this $\mathscr{O}_X/\mathscr{I}$-module.*
(ii) *The canonical surjective homomorphism* (16.1.2.2)

$$\mathbf{S}^\bullet_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}\!\mathit{r}^\bullet_{\mathscr{I}}(\mathscr{O}_X)$$

is bijective.

Moreover, if this is so, every sequence $(f'_i)_{1\leqslant i\leqslant n}$ of n sections of $\mathscr{I}$ over X which generates $\mathscr{I}$ is quasi-regular.

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Proof. The two conditions in the statement merely express the definition given in (**0**, 15.2.2), taking into account the definition of the canonical homomorphisms (16.2.1.1). The last claim follows from the fact that, if a module M over a commutative ring A admits a basis of n elements, every system of n generators of M is a basis (Bourbaki, *Alg. comm.*, chap. II, §3, cor. 5 of th. 1). □

Corollary (16.9.4). — *Let X be a locally ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$. For $\mathscr{I}$ to be quasi-regular, it is necessary and sufficient that the following conditions hold:*

(i) *$\mathscr{I}$ is of finite type.*
(ii) *$\mathscr{I}/\mathscr{I}^2$ is a locally free $\mathscr{O}_X/\mathscr{I}$-module.*
(iii) *The canonical homomorphism*

(16.9.4.1) $$\mathbf{S}^{\bullet}_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}r^{\bullet}_{\mathscr{I}}(\mathscr{O}_X)$$

is bijective.

Proof. The necessity of the conditions follows immediately from (16.9.3). To see that they are sufficient, it is enough, by (16.9.3), to prove the following: if, at a point $x \in \operatorname{Supp}(\mathscr{O}_X/\mathscr{I})$, there are an open neighbourhood U of x in X and n sections f_i $(1 \leqslant i \leqslant n)$ of $\mathscr{I}$ over U whose canonical images in $\mathscr{I}/\mathscr{I}^2$ form a basis of $(\mathscr{I}/\mathscr{I}^2)|U$ over $(\mathscr{O}_X/\mathscr{I})|U$, then there is an open neighbourhood $V \subset U$ of x such that the $f_i|V$ generate $\mathscr{I}|V$. Now, by hypothesis, $\mathscr{I}_x \neq \mathscr{O}_x$, so $\mathscr{I}_x$ is contained in the maximal ideal of $\mathscr{O}_x$. Since $\mathscr{I}_x$ is an $\mathscr{O}_x$-module of finite type and the classes of $(f_i)_x$ in $\mathscr{I}_x/\mathscr{I}_x^2$ generate this $(\mathscr{O}_x/\mathscr{I}_x)$-module, Nakayama's lemma shows that the $(f_i)_x$ generate $\mathscr{I}_x$. Since $\mathscr{I}$ is of finite type, we conclude by (**0**, 5.2.2). ☐

Corollary (16.9.5). — *Let X be a locally ringed space, $\mathscr{I}$ a quasi-regular ideal of $\mathscr{O}_X$, $(f_i)_{1\leqslant i\leqslant n}$ a sequence of sections of $\mathscr{I}$ over X, x a point of* $\operatorname{Supp}(\mathscr{O}_X/\mathscr{I})$. *The following conditions are equivalent:*

(a) *There is an open neighbourhood U of x in X such that the $f_i|U$ form a quasi-regular sequence of elements of $\Gamma(U, \mathscr{O}_X)$ generating $\mathscr{I}|U$.*
(b) *The $(f_i)_x$ form a system of generators of $\mathscr{I}_x$ whose size is as small as possible.*
(b′) *The $(f_i)_x$ form a minimal set of generators of $\mathscr{I}_x$.*
(c) *If $\bar{f}_i$ is the canonical image of f_i in $\Gamma(X, \mathscr{I}/\mathscr{I}^2)$, the $(\bar{f}_i)_x$ form a basis for the $(\mathscr{O}_x/\mathscr{I}_x)$-module $\mathscr{I}_x/\mathscr{I}_x^2$.*

Proof. By hypothesis, $\mathscr{O}_x$ is a local ring, $\mathscr{I}_x$ an ideal of finite type of $\mathscr{O}_x$ contained in the maximal ideal of $\mathscr{O}_x$; the equivalence of (b), (b′) and (c) follows from Nakayama's lemma (Bourbaki, *Alg. comm.*, chap. II, §3, n°2, prop. 5). It is clear that (a) implies (c) because of (16.9.3); on the other hand, from (**0**, 5.2.2) it follows that if condition (c) is satisfied (and therefore so is (b)), there is a neighbourhood U of x in X such that $(\mathscr{I}/\mathscr{I}^2)|U$ has constant rank equal to n, and that the $f_i|U$ generate $\mathscr{I}|U$; it suffices now to apply the last assertion of (16.9.3) to U. ☐

Remarks (16.9.6). — (i) Under the general hypothesis of (16.9.5), for the sequence (f_i) to generate $\mathscr{I}$, it is not enough that the $(\bar{f}_i)_y$ form a basis of the $(\mathscr{O}_y/\mathscr{I}_y)$-module $(\mathscr{I}_y/\mathscr{I}_y^2)$ for all $y \in X$. An example is obtained by taking $X = \operatorname{Spec}(A)$, where A is a Dedekind ring, and **IV-4 | 48**
$\mathscr{I} = \widetilde{\mathfrak{I}}$, where $\mathfrak{I}$ is a *non-principal* prime ideal of A. Then $\mathscr{I}_y/\mathscr{I}_y^2 = 0$ at every point y other than the point $x \in X$ corresponding to $\mathfrak{I}$, while $\mathscr{I}_x/\mathscr{I}_x^2$ has rank 1 over the field $\mathscr{O}_x/\mathscr{I}_x$; moreover, $\mathscr{I}$ is plainly a regular ideal.

(ii) In (16.9.5), one cannot replace "quasi-regular" by "regular", even when X is a prescheme (cf. (16.9.12)). Indeed, let B denote the ring of germs at 0 in $\mathbb{R}$ of infinitely differentiable functions. It has a maximal ideal $\mathfrak{m}$ generated by the germ t of the identity map of $\mathbb{R}$ at 0, and the intersection $\mathfrak{n}$ of the $\mathfrak{m}^k$ for $k > 0$ is nonzero. Now let A be the quotient ring $B[T]/\mathfrak{n}TB[T]$, and let f_1, f_2 be the canonical images in A of the elements t and T of $B[T]$. The sequence (f_1, f_2) is *regular* in A: indeed, f_1 is not a zero divisor in A, because the relation $tP[T] \in \mathfrak{n}TB[T]$, for a polynomial $P \in B[T]$, implies that the products of t with the coefficients of P belong to $\mathfrak{n}$; it follows at once that the coefficients themselves belong to $\mathfrak{n}$, and hence $P[T] \in \mathfrak{n}TB[T]$. Since B/tB is isomorphic to $\mathbb{R}$, A/f_1A is isomorphic to the polynomial ring $\mathbb{R}[T]$ and is therefore integral; the image of f_2 in A/f_1A, being equal to T, is consequently not a zero divisor, proving the assertion. However, f_2 is a zero divisor in A, since for every nonzero element $x \in \mathfrak{n}$, the image of x in A is nonzero whereas the image of xT is zero. We conclude that the sequence (f_2, f_1) is *not regular* in A; on the other hand, the ideal $\mathfrak{I} = f_1A + f_2A$ is distinct from A, so the conditions (b), (b′) and (c) of (16.9.5) do not imply the condition (a) when we replace "quasi-regular" by "regular".

(16.9.7). If $X = \operatorname{Spec}(A)$ is an affine scheme, we say that an ideal $\mathfrak{I}$ of A is *regular* (resp. *quasi-regular*) if the ideal $\mathscr{J} = \widetilde{\mathfrak{I}}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular). We note that this notion is *local* and does not imply the existence of a *system of generators* of $\mathfrak{I}$ forming a regular (resp. quasi-regular) sequence in A, as Example (16.9.6, (i)) shows; it does, however, imply this when A is *local* (16.9.5).

The proposition (16.9.4) can be translated in terms of quasi-regular immersions in the following manner:

Proposition (16.9.8). — *Let $j : Y \to X$ be a morphism of preschemes; for j to be a quasi-regular immersion, it is necessary and sufficient that j satisfies the following conditions:*

(i) j is an immersion locally of finite presentation.
(ii) The conormal sheaf $\mathscr{G}r^1(j) = \mathscr{N}_{Y/X}$ (16.1.2) is a locally free $\mathscr{O}_Y$-module.
(iii) The canonical homomorphism

$$\mathbf{S}^{\bullet}_{\mathscr{O}_Y}(\mathscr{G}r^1(j)) \longrightarrow \mathscr{G}r^{\bullet}(j)$$

(16.1.2.2) is bijective.

Proof. Since the question is local on Y, we may restrict ourselves to the case where j is the canonical injection of a closed subprescheme Y of X; the translation of (16.9.4) into (16.9.8) then follows from the description of $\mathscr{G}r^1(j)$ and $\mathscr{G}r^{\bullet}(j)$ in terms of the ideal $\mathscr{I}$ of $\mathscr{O}_X$ defining Y (16.1.3, (ii)). ☐

Corollary (16.9.9). — *Let Y be a prescheme, X a Y-prescheme, and $j : Y \to X$ a Y-section of X, so that the n-th normal invariant $\mathscr{A}^{(n)}$ of j (16.1.2) is an augmented $\mathscr{O}_Y$-algebra (16.1.7); take $\mathscr{A}^{(\infty)} = \varprojlim \mathscr{A}^{(n)}$. For j to be a quasi-regular immersion, it is necessary and sufficient that j be locally of finite presentation and that every $y \in Y$ admit an affine open neighbourhood U with coordinate ring C such that $\mathscr{A}^{(\infty)}|U$ is isomorphic, as an augmented topological $\mathscr{O}_U$-algebra, to $\mathscr{O}_U[[T_1, \ldots, T_n]]$.*

Proof. We may reduce to the case where j is a closed immersion by restricting to a sufficiently small neighbourhood of y (see the argument of (16.4.11)); then $\mathscr{O}_Y$ is identified with the quotient algebra $\mathscr{O}_X/\mathscr{I}$, and the canonical surjective homomorphism $\mathscr{O}_X \to \mathscr{O}_Y$ admits a right inverse (16.1.7). We may therefore suppose that $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$ are affine, with B an augmented A-algebra whose augmentation ideal $\mathfrak{I}$ is of finite type. Since $\mathscr{A}^{(n)}$ is then identified with $(B/\mathfrak{I}^{n+1})^{\sim}$, the corollary follows from the equivalence of (b) and (c) in (**0**, 19.5.4), since $B/\mathfrak{I} = A$. ☐ **IV-4** | 49

We note that, in the affine case under consideration, the fact that j is a quasi-regular immersion is also equivalent, by (**0**, 19.5.4), to saying that B is a *formally smooth* A-algebra for the $\mathfrak{I}$-preadic topology.

We also note that the condition that j is locally of finite presentation is always satisfied when $X \to Y$ is locally of finite type (1.4.3, (v)).

Proposition (16.9.10). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, $j : Y \to X$ the canonical injection, y a point of Y.*

(i) *For there to exist an open neighbourhood U of y in X such that the restriction $Y \cap U \to U$ of j be a regular immersion, it is necessary and sufficient that the kernel $\mathscr{I}_y$ of the surjective homomorphism $\mathscr{O}_{X,y} \to \mathscr{O}_{Y,y}$ is generated by a regular sequence of elements of $\mathscr{O}_{X,y}$.*
(ii) *For the immersion j to be regular, it is necessary and sufficient that it is quasi-regular.*

Proof. (i) We can reduce to the case where Y is a subprescheme defined by a *coherent* ideal $\mathscr{I}$ of $\mathscr{O}_X$. The condition is clearly necessary. Conversely, if $\mathscr{I}_y$ is generated by a regular sequence $(s_i)_y$, where the s_i are sections of $\mathscr{I}$ over an open neighbourhood U of y in X, we may suppose that the s_i generate $\mathscr{I}|U$ (**0**, 5.5.2) and form a regular sequence (**0**, 15.2.4), whence the assertion.

(ii) The fact that a quasi-regular immersion is regular follows from (i) and the identification of quasi-regular and regular sequences in $\mathscr{O}_{X,y}$ formed by elements of the maximal ideal (**0**, 15.1.11).

☐

When, without any Noetherian hypothesis on X, the kernel $\mathscr{I}_y$ of $\mathscr{O}_{X,y} \to \mathscr{O}_{Y,y}$ is generated by a regular sequence of elements of $\mathscr{O}_{X,y}$, we say that the immersion j is *regular at the point y*.

Corollary (16.9.11). — *Let X be a locally Noetherian prescheme; then every quasi-regular ideal of $\mathscr{O}_X$ is regular.*

Remarks (16.9.12). — (i) We note that a regular immersion is not in general a flat morphism, and hence *a fortiori* is not a regular morphism in the sense of (6.8.1).

(ii) Let A be a local Noetherian ring; it follows immediately from (16.9.4) and from (**0**, 17.1.1) that for A to be *regular*, it is necessary and sufficient that its maximal ideal $\mathfrak{m}$ is *quasi-regular* (or regular, which amounts to the same thing given that A is Noetherian). For an affine Noetherian scheme X to be *regular*, it is necessary and sufficient that for every closed point $x \in X$, the canonical injection $\operatorname{Spec}(k(x)) \to X$ be a *regular* immersion.

Proposition (16.9.13). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, Y' a subprescheme of Y, such that the canonical injection $j : Y' \to Y$ is regular. Then the sequence of $\mathscr{O}_{Y'}$-modules*

(16.9.13.1) $$0 \longrightarrow j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0$$

is exact; furthermore, for every $x \in X$, there is an open neighbourhood U of x such that the restrictions to U of the homomorphisms of (16.9.13.1) form a split exact sequence. **IV-4** | 50

Let us first prove the following lemma:

Lemma (16.9.13.2). — *Let A be a ring, $\mathfrak{I}$ an ideal of A, $A' = A/\mathfrak{I}$, $(f_i)_{1 \leqslant i \leqslant r}$ a sequence of elements of A which is A'-regular, $\mathfrak{K} = \sum_i f_i A$, $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, $\mathfrak{K}' = \sum_i f_i A'$, so that $C = A/\mathfrak{L}$ is isomorphic to $A'/\mathfrak{K}'$. For every integer $n > 0$, and every integer $N \geqslant n$, we have the relation*

(16.9.13.3) $$\mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I}\mathfrak{K}^n + (\mathfrak{I} \cap \mathfrak{K}^N).$$

Proof. It is plainly enough to prove that every element of the left-hand side belongs to the right-hand side, and an induction reduces us to the case $N = n + 1$. An element of the left-hand side of (16.9.13.3), being in $\mathfrak{K}^n$, can be written as $P(f_1, \ldots, f_r)$, where $P \in A[T_1, \ldots, T_r]$ is homogeneous of degree n. If f'_i is the canonical image of f_i in A', the hypothesis $P(f_1, \ldots, f_r) \in \mathfrak{I}$ means that $P(f'_1, \ldots, f'_r) = 0$. But $P(f'_1, \ldots, f'_r) \in \mathfrak{K}'^n$, so the canonical image of $P(f'_1, \ldots, f'_r)$ in $\mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is zero. Now the hypothesis that (f_i) is A'-regular implies that the canonical homomorphism $\mathbf{S}^n_C(\mathfrak{K}'/\mathfrak{K}'^2) \to \mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is bijective (**0**, 15.1.9); we conclude that the coefficients of P are in $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$. It follows at once that $P(f_1, \ldots, f_r) \in \mathfrak{I}\mathfrak{K}^n + \mathfrak{K}^{n+1}$, and since $P(f_1, \ldots, f_r) \in \mathfrak{I}$, we finally have $P(f_1, \ldots, f_r) \in \mathfrak{I}\mathfrak{K}^n + (\mathfrak{I} \cap \mathfrak{K}^{n+1})$, which proves the lemma. □

Taking the quotients of both sides of (16.9.13.3) by $\mathfrak{I}\mathfrak{K}^n$, we see that the relations (16.9.13.3), for $N \geqslant n$, imply

(16.9.13.4) $$(\mathfrak{I} \cap \mathfrak{K}^n)/\mathfrak{I}\mathfrak{K}^n \subset \bigcap_{N \geqslant n} \mathfrak{K}^N \cdot (A/(\mathfrak{I}\mathfrak{K}^n)).$$

We deduce the following corollary.

Corollary (16.9.13.5). — *Suppose that the hypotheses of (16.9.13.2) are satisfied and also that the ring A is Noetherian and $\mathfrak{K}$ is contained in the radical of A. Then for every integer $n > 0$,*

(16.9.13.6) $$\mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I}\mathfrak{K}^n.$$

Proof. Indeed, the second member of (16.9.13.4) is then zero, given that $A/\mathfrak{I}\mathfrak{K}^n$ is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. III, §3, n°3, prop. 6). □

Taking in particular $n = 2$ in (16.9.13.6), and observing that $\mathfrak{L}^2 = \mathfrak{I}^2 + \mathfrak{I}\mathfrak{K} + \mathfrak{K}^2 = \mathfrak{I}\mathfrak{L} + \mathfrak{K}^2$, we have $\mathfrak{I}\mathfrak{L} \subset \mathfrak{I} \cap \mathfrak{L}^2$ and therefore

$$\mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I}\mathfrak{L} + (\mathfrak{I} \cap \mathfrak{K}^2) = \mathfrak{I}\mathfrak{L} + \mathfrak{I}\mathfrak{K}^2 = \mathfrak{I}\mathfrak{L},$$

in other words,

(16.9.13.7) $$\mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I}\mathfrak{L},$$

which may also be expressed by saying that the canonical homomorphism

$$\mathfrak{I}/\mathfrak{I}\mathfrak{L} \longrightarrow (\mathfrak{I} + \mathfrak{L}^2)/\mathfrak{L}^2$$

is bijective.

Proof (16.9.13). Having demonstrated the lemmas, let us prove the first claim of (16.9.13): It is clearly enough to prove that, at a point $x \in Y'$, the sequence of stalks of the sheaves appearing **IV-4** | 51 in (16.9.13.1) is exact. Put $A = \mathscr{O}_{X,x}$. We may write $\mathscr{O}_{Y,x} = A' = A/\mathfrak{I}$, where $\mathfrak{I}$ is an ideal contained in the maximal ideal of A, and then $\mathscr{O}_{Y',x} = A'/\mathfrak{K}'$, where $\mathfrak{K}'$ is generated by an A'-*regular* sequence of elements of A' which are the images of elements of an A-regular sequence in A belonging to the maximal ideal of A. If $\mathfrak{K}$ is the ideal generated by the latter elements and $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, then $\mathscr{O}_{Y',x} = A/\mathfrak{L}$; since we are in the situation of (16.9.13.5), the canonical homomorphism $\mathfrak{I}/\mathfrak{I}\mathfrak{L} \to (\mathfrak{I} + \mathfrak{L}^2)/\mathfrak{L}^2$ is bijective. But this shows that the sequence

$$0 \longrightarrow \mathfrak{I}/\mathfrak{I}\mathfrak{L} \longrightarrow \mathfrak{L}/\mathfrak{L}^2 \longrightarrow (\mathfrak{L}/\mathfrak{I})/(\mathfrak{L}/\mathfrak{I})^2 \longrightarrow 0$$

is exact (see the proof of (16.2.7)), and the modules appearing in this sequence are precisely the stalks at x of the sheaves in (16.9.13.1). The second claim follows from the fact that $\mathscr{N}_{Y'/Y}$ is a locally free $\mathscr{O}_{Y'}$-module (16.9.8) and Bourbaki, *Alg.*, chap. II, 3rd ed., §1, n°11, prop. 21. □

16.10. Differentially smooth morphisms.

Definition (16.10.1). — We say that a morphism of preschemes $f : X \to S$ is differentially smooth (or that X is differentially smooth over S) if it satisfies the following conditions:

(i) $\Omega^1_{X/S}$ is a locally projective $\mathscr{O}_X$-module, which is to say that every point $x \in X$ admits an affine open neighbourhood U such that $\Gamma(U, \Omega^1_{X/S})$ is a projective $\Gamma(U, \mathscr{O}_X)$-module (not necessarily of finite type).

(ii) The canonical morphism (16.3.1.1)

$$\mathbf{S}^{\bullet}_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}\!r_{\bullet}(\mathscr{P}_{X/S})$$

is bijective.

In particular, if $\Omega^1_{X/S}$ is locally free of finite rank, the $\mathscr{P}^n_{X/S}$ are locally free of finite rank $\mathscr{O}_X$-modules (being extensions of such modules).

We say that f is *differentially smooth at a point* $x \in X$ if there is an open neighbourhood U of x in X such that $f|U$ is differentially smooth.

We will see later (17.12.4) that a smooth morphism is differentially smooth, which justifies the terminology; but the converse is not true; indeed, a *monomorphism* $f : X \to S$ is differentially smooth, since $\Omega^1_{X/S} = 0$ because of (**I**, 5.3.8), and consequently the surjective homomorphism (16.3.1.1) is clearly bijective; however a monomorphism is not even necessarily flat, neither *a fortiori* smooth. Let us limit ourselves to proving the following proposition:

Proposition (16.10.2). — *Let A be a ring, B a formally smooth A-algebra for the discrete topology* (**0**, 19.3.1)*. Then* $\operatorname{Spec}(B)$ *is differentially smooth over* $\operatorname{Spec}(A)$*.*

Proof. Indeed, $B \otimes_A B$ is then (with the discrete topology) a formally smooth B-algebra (by one or the other canonical homomorphisms $b \mapsto b \otimes 1$, $b \mapsto 1 \otimes b$ of B to $B \otimes_A B$) (**0**, 19.3.5, (iii)); therefore **IV-4** | 52 $B \otimes_A B$ is a formally smooth A-algebra with the discrete topology (**0**, 19.3.5, (ii)). Taking $\mathfrak{I} = \mathfrak{I}_{B/A}$, it follows that $B \otimes_A B$ is a formally smooth A-algebra for the $\mathfrak{I}$-preadic topology (**0**, 19.3.8); since by hypothesis $B = (B \otimes_A B)/\mathfrak{I}$ is a formally smooth A-algebra for the discrete topology, the proposition follows from the equivalence of (a) and (b) in (**0**, 19.5.4). □

Proposition (16.10.3). — *For a morphism $f : X \to S$ to be differentially smooth, it is necessary and sufficient that for every $x \in X$, there is an affine open neighbourhood of x, of ring A, such that $\Gamma(U, \mathscr{P}^{\infty}_{X/S})$ is an augmented topological A-algebra isomorphic to the completion $\widehat{B}$, where $B = \mathbf{S}_A(V)$, V being a projective A-module and B being endowed with the B^+-preadic topology (where B^+ is the augmentation ideal). If $\Omega^1_{X/S}$ is locally free of finite rank, we can replace $\widehat{B}$ with the ring of formal series $A[[T_1, \ldots, T_n]]$.*

Proof. The notion of a differentially smooth morphism is clearly local on X, so we reduce to the case where $S = \operatorname{Spec}(B)$, $X = \operatorname{Spec}(C)$. Consider $C \otimes_B C$ as a C-algebra (by the first factor); put $\mathfrak{I} = \mathfrak{I}_{C/B}$ and endow $C \otimes_B C$ with the $\mathfrak{I}$-preadic topology; we can apply to the topological C-algebra $C \otimes_B C$ and to the ideal $\mathfrak{I}$ of $C \otimes_B C$ the equivalence of (b) and (c) of (**0**, 19.5.4), given that $(C \otimes_B C)/\mathfrak{I} = C$ is clearly a formally smooth C-algebra for the discrete topologies. The topology of $\Gamma(U, \mathscr{P}^{\infty}_{X/S})$ is clearly the projective limit topology of this ring (16.1.11). □

We note that the integer n of the proposition (16.10.3) is the *rank* of $\Omega^1_{X/S}$ in the point x. We shall see (17.13.5) that when f is differentially smooth and locally of finite type, n is equal to the dimension of the fibre $f^{-1}(f(x))$.

Proposition (16.10.4). — *Let $f : X \to S$, $g : S' \to S$ be two morphisms, and take $X' = X \times_S S'$, $f' = f_{(S')} : X' \to S'$.*

(i) *If f is differentially smooth, the same is true for f'.*

(ii) *Conversely, if g is faithfully flat and quasi-compact, and if f' is differentially smooth and $\Omega^1_{X'/S'}$ is a finite type $\mathscr{O}_{X'}$-module, f is differentially smooth and $\Omega^1_{X/S}$ is a finite type $\mathscr{O}_X$-module.*

Proof. Indeed, if f is differentially smooth, the $\mathscr{G}r_n(\mathscr{P}_{X/S})$ are *flat* $\mathscr{O}_X$-modules; therefore, by (16.4.6), the homomorphisms $\mathscr{G}r_n(\mathscr{P}_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}_{X'/S'})$ are bijective for every n. In view of the commutativity of diagram (16.2.1.3), it follows from Definition (16.10.1) that f' is differentially smooth. On the other hand, if g is faithfully flat and quasi-compact, it also follows from (16.4.6) that $\mathscr{G}r_n(\mathscr{P}_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}_{X'/S'})$ is bijective for every n. Suppose also that f' is differentially smooth and $\Omega^1_{X'/S'}$ is of finite rank. Since the canonical projection $X' \to X$ is a faithfully flat and quasi-compact morphism, it follows first from (2.5.2) that $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module locally free of finite rank, then from (2.2.7) that the canonical homomorphism (16.3.1.1) is bijective, and therefore f is differentially smooth. □

Proposition (16.10.5). — *For a morphism locally of finite type $f : X \to S$ to be differentially smooth, it is necessary and sufficient that the diagonal immersion $\Delta_f : X \to X \times_S X$ be quasi-regular.*

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Proof. Being a local problem, we can reduce to the case where S and X are affines, and therefore the diagonal subprescheme of $X \times_S X$ is closed. The hypothesis that f is locally of finite type implies that Δ_f is locally of finite presentation (**I**, 4.3.1), therefore the diagonal prescheme of $X \times_S X$ is defined by an ideal $\mathscr{I}$ of finite type, and $\Omega^1_{X/S} = \mathscr{I}/\mathscr{I}^2$ is an $\mathscr{O}_X$-module of finite type. The proposition is now immediate from the comparison of the conditions of (16.10.1) and (16.9.4). □

Remark (16.10.6). — Let $f : X \to S$ be a morphism such that the $\mathscr{O}_X$-module $\Omega^1_{X/S}$ is locally free of finite rank. It follows from (**0**, 20.4.7) that every $x \in X$ has an open neighbourhood such that there is a finite family $(z_\lambda)_{\lambda\in L}$ of sections of $\mathscr{O}_X$ over U for which $(dz_\lambda)_{\lambda\in L}$ forms a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

16.11. Differential operators on a differentially smooth S-prescheme

(16.11.1). Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda\in L}$ a family of sections of $\mathscr{O}_X$ over U such that the dz_λ form a system of generators of $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. Let m be an integer or the symbol ∞, and take, for every λ

(16.11.1.1) $$\zeta_\lambda = \delta z_\lambda = d^m z_\lambda - z_\lambda \in \Gamma(U, \mathscr{P}^m_{X/S}).$$

We will also use the usual notations from analysis; for every $\mathbf{p} = (p_\lambda) \in \mathbb{N}^{(L)}$ (so that $p_\lambda = 0$ except for a finite number of indices), we put

(16.11.1.2) $$|\mathbf{p}| = \sum_\lambda p_\lambda, \quad \mathbf{p}! = \prod_\lambda (p_\lambda!)$$

(16.11.1.3) $$\binom{\mathbf{p}}{\mathbf{q}} = \mathbf{p}!/(\mathbf{q}!(\mathbf{p}-\mathbf{q})!), \quad \text{for } \mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}, \mathbf{q} \leqslant \mathbf{p}$$

and we adopt the convention that $\binom{\mathbf{p}}{\mathbf{q}} = 0$ when $\mathbf{q} \not\leqslant \mathbf{p}$,

(16.11.1.4) $$\mathbf{z}^{\mathbf{p}} = \prod_\lambda (z_\lambda)^{p_\lambda}, \quad \boldsymbol{\zeta}^{\mathbf{p}} = \prod_\lambda (\zeta_\lambda)^{p_\lambda}.$$

We therefore have, with this notation

(16.11.1.5) $$d^m(\mathbf{z}^{\mathbf{p}}) = (d^m(\mathbf{z}))^{\mathbf{p}} = (\boldsymbol{\zeta} + \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q}\leqslant\mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}} \mathbf{z}^{\mathbf{p}-\mathbf{q}} \boldsymbol{\zeta}^{\mathbf{q}}$$

$$\boldsymbol{\zeta}^{\mathbf{p}} = (d^m \mathbf{z} - \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q} \leqslant \mathbf{p}} (-1)^{|\mathbf{p}-\mathbf{q}|} \binom{\mathbf{p}}{\mathbf{q}} \mathbf{z}^{\mathbf{p}-\mathbf{q}} d^m(\mathbf{z}^{\mathbf{q}}). \tag{16.11.1.6}$$

Since the dz_λ generate $\Omega^1_{X/S}$, are the images of the δz_λ, and the canonical morphism (16.3.1.1) is surjective, we conclude that, for finite m, the δz_λ generate the $\mathscr{O}_U$-algebra $\mathscr{P}^m_{U/S}$ (Bourbaki, *Alg. comm.*, chap. III, §2, n°8, cor. 2 to th. 1). Therefore the $\boldsymbol{\zeta}^{\mathbf{p}}$ (for $|\mathbf{p}| \leqslant m$) generate the $\mathscr{O}_U$-module $\mathscr{P}^m_{U/S}$. A differential operator $D \in \mathrm{Diff}^m_{U/S}$ is consequently entirely determined by the values $\langle \boldsymbol{\zeta}^{\mathbf{p}}, D\rangle$ for $|\mathbf{p}| \leqslant m$, or, equivalently by (16.11.1.5) and (16.11.1.6), by the values of $\langle d^m(\mathbf{z}^{\mathbf{p}}), D\rangle =$ **IV-4** | 54
$D(\mathbf{z}^{\mathbf{p}})$ for $|\mathbf{p}| \leqslant m$; more precisely, it follows from (16.11.1.5) that

$$D(\mathbf{z}^{\mathbf{p}}) = \langle d^m(\mathbf{z}^{\mathbf{p}}), D\rangle = \sum_{\mathbf{q} \leqslant \mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}} \langle \boldsymbol{\zeta}^{\mathbf{q}}, D\rangle \mathbf{z}^{\mathbf{p}-\mathbf{q}}. \tag{16.11.1.7}$$

Theorem (16.11.2). — *Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda \in L}$ a family of sections of $\mathscr{O}_X$ over U such that the family $(dz_\lambda)_{\lambda\in L}$ generates $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. The following conditions are equivalent:*

(a) *$f|U$ is differentially smooth and (dz_λ) is a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$.*
(b) *There is a family $(D_{\mathbf{p}})_{\mathbf{p}\in\mathbb{N}^{(L)}}$ of differential operators of $\mathscr{O}_U$ to itself verifying the conditions*

$$D_{\mathbf{p}}(\mathbf{z}^{\mathbf{q}}) = \binom{\mathbf{q}}{\mathbf{p}} \mathbf{z}^{\mathbf{q}-\mathbf{p}}, \quad (\mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}). \tag{16.11.2.1}$$

Also, when these conditions are satisfied, the family $(D_{\mathbf{p}})$ is uniquely determined by the conditions (16.11.2.1) and satisfies the relations

$$D_{\mathbf{q}} \circ D_{\mathbf{p}} = D_{\mathbf{p}} \circ D_{\mathbf{q}} = \frac{(\mathbf{p}+\mathbf{q})!}{\mathbf{p}!\mathbf{q}!} D_{\mathbf{p}+\mathbf{q}} \quad (\mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}). \tag{16.11.2.2}$$

Finally, if L is finite, for every integer m, the $D_{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ form a basis of the $\mathscr{O}_U$-module $\mathscr{D}\mathit{iff}^m_{U/S}$, in other words, every differential operator of order $\leqslant m$ on U can be written uniquely as

$$D = \sum_{|\mathbf{p}| \leqslant m} a_{\mathbf{p}} D_{\mathbf{p}}$$

where the $a_{\mathbf{p}}$ are sections of $\mathscr{O}_X$ over U.

Proof. Note first that because of (16.11.1.6) and (16.11.1.5), we verify immediately that the conditions (16.11.2.1) are equivalent to

$$\langle \boldsymbol{\zeta}^{\mathbf{p}}, D_{\mathbf{q}}\rangle = \delta_{\mathbf{pq}} \quad \text{(Kronecker's symbol)}. \tag{16.11.2.3}$$

The existence of the family $(D_{\mathbf{p}})$ verifying these conditions implies first (by taking $|\mathbf{p}| = 1$) that the dz_λ are linearly independent, and therefore form a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$. Then, for every integer $m \geqslant 1$, we deduce similarly from (16.11.2.3) that the $\boldsymbol{\zeta}^{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ are linearly independent; it follows that the canonical homomorphism (16.3.1.1) is injective, and therefore bijective, which proves that (b) implies (a). The converse follows immediately from Definition (16.10.1): the fact that the $\boldsymbol{\zeta}^{\mathbf{p}}$ form a basis of $\mathscr{P}^m_{U/S}$ for $|\mathbf{p}| \leqslant m$ implies the existence and uniqueness of a family of homomorphisms $u_{\mathbf{q},m} : \mathscr{P}^m_{U/S} \to \mathscr{O}_U$ ($|\mathbf{q}| \leqslant m$) such that $\langle \boldsymbol{\zeta}^{\mathbf{p}}, u_{\mathbf{q},m}\rangle = \delta_{\mathbf{pq}}$ for $|\mathbf{p}| \leqslant m$, $|\mathbf{q}| \leqslant m$. For a given value of $\mathbf{q}$, the differential operators corresponding to $u_{\mathbf{q},m}$ for $m \geqslant |\mathbf{q}|$ are identified with the same operator $D_{\mathbf{q}}$. This proves that (a) implies (b), and also that the family $(D_{\mathbf{q}})$ is uniquely determined and that, if L is finite, the $D_{\mathbf{p}}$ with $|\mathbf{p}| \leqslant m$ form a basis of the $\mathscr{O}_U$-module $\mathscr{D}\mathit{iff}^m_{U/S}$, the dual of $\mathscr{P}^m_{U/S}$. Finally, the relations (16.11.2.2) follow immediately by evaluating the three operators in question on the $\mathbf{z}^{\mathbf{r}}$ and using the fact that the $\boldsymbol{\zeta}^{\mathbf{r}}$ for $|\mathbf{r}| \leqslant m$ generate $\mathscr{P}^m_{U/S}$. □

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Remarks (16.11.3). — (i) The fact that, by (16.11.2.2), the $D_{\mathbf{p}}$ commute pairwise naturally does not imply that the $\mathscr{O}_U$-algebra $\mathscr{D}\mathit{iff}_{U/S}$ is commutative: the $D_{\mathbf{p}}$ do not commute with multiplication by sections of $\mathscr{O}_U$ unless $\mathbf{p} = 0$.

(ii) The indices $\mathbf{p}$ such that $|\mathbf{p}| = 1$ are the $\boldsymbol{\epsilon}_\lambda = (\epsilon_{\lambda\mu})_{\mu\in L}$ where $\epsilon_{\lambda\mu} = 0$ if $\mu \neq \lambda$ and $\epsilon_{\lambda\lambda} = 1$; when L is finite, the operators $D_{\boldsymbol{\epsilon}_\lambda}$ are exactly the S-derivations D_λ introduced in (16.5.7). We note that in general (contrary to what happens in classical analysis), it is not true that a differential operator of any order can be written as a linear combination of powers of D_i (cf. (16.12)).

(iii) For every integer $r \geqslant 1$, we can define the notion of *differentially smooth up to order* r by replacing in (16.10.1) the condition (ii) by the condition that the homomorphisms

$$\mathbf{S}^m_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_m(\mathscr{P}_{X/S})$$

are bijective *for every* $m \leqslant r$. The argument of (16.11.2) proves also that if, in the condition (a), we replace "differentially smooth" by "differentially smooth up to order r", this condition is equivalent to (b) by restricting ourselves to $\mathbf{p} \in \mathbb{N}^{(L)}$, $\mathbf{q} \in \mathbb{N}^{(L)}$ such that $|\mathbf{p}| \leqslant r$, $|\mathbf{q}| \leqslant r$.

16.12. Case of characteristic zero: Jacobian criterion for differentially smooth morphisms.

(16.12.1). We say that a prescheme X is *of characteristic* p (p equal to zero or a prime number) if, for every affine open set U of X, the ring $\Gamma(U, \mathscr{O}_X)$ is of characteristic p (**0**, 21.1.1). It follows from (**0**, 21.1.3) that for X to be of characteristic 0, it is necessary and sufficient that for every *closed* point x of X, the residue field $k(x)$ is of characteristic 0, or even that X can be given a structure of $\mathbb{Q}$-prescheme (necessarily unique).

Theorem (16.12.2). — *Let X be a prescheme of characteristic* 0*, $f : X \to S$ a morphism. If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module (not necessarily of finite type), f is differentially smooth.*

Proof. The problem being local on X, we can suppose there is a family (z_λ) of sections of $\mathscr{O}_X$ over X such that the (dz_λ) is a basis for the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. Applying the criterion (16.11.2), it is enough for the operators

$$D_{\mathbf{p}} = (\mathbf{p}!)^{-1} \prod_\lambda D_\lambda^{p_\lambda}$$

(where the D_λ are the coordinate forms corresponding to the basis (dz_λ)) to verify the relations (16.11.2.1), which is a consequence of the fact that the D_λ are derivations. □

(16.12.3). The theorem above is not true if we discard the hypothesis that X is of characteristic 0. For example, if $S = \mathrm{Spec}(k)$, where k is a field of characteristic $p > 0$, $X = \mathrm{Spec}(K)$ where $K = k(\alpha)$ where $\alpha \notin k$, $\alpha^p \in k$, we verify immediately that $\Omega^1_{X/S}$ has rank 1, and that the morphism $X \to S$ **IV-4** | 56
is differentially smooth up to order $p - 1$ (16.11.3, (iii)), but not up to order p. However, the proof of (16.12.2) proves that if $\Omega^1_{X/S}$ is locally free, and if $n!1_{\mathscr{O}_X}$ is invertible in $\Gamma(X, \mathscr{O}_X)$, then X is differentially smooth over S up to order n.

§17. Smooth morphisms, unramified (or net) morphisms, and étale morphisms.

In this section, we return to the notions studied in (**0**, 19), expressed in the geometric language of schemes and from the global point of view, for preschemes locally of finite presentation over a given base prescheme.

Most of the results (except those of (17.7), (17.8), (17.9), (17.13), and (17.16)) reduce, moreover, to variants of properties already encountered in (**0**, 19).

For more specific results on étale morphisms, the reader should consult (18).

17.1. Formally smooth morphisms, formally unramified morphisms, formally étale morphisms.

Definition (17.1.1). — Let $f : X \to Y$ be a morphism of preschemes. We say that f is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) if, for all affine schemes Y', all closed subschemes Y'_0 of Y' defined by a nilpotent ideal $\mathscr{J}$ of $\mathscr{O}_{Y'}$, and every morphism $Y' \to Y$, the map

(17.1.1.1) $$\mathrm{Hom}_Y(Y', X) \longrightarrow \mathrm{Hom}_Y(Y'_0, X)$$

induced by the canonical map $Y'_0 \to Y'$, is *surjective* (resp. *injective*, resp. *bijective*).

One also says that X is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) over Y.

It is clear that for f to be formally étale, it is necessary and sufficient for f to be formally smooth and formally unramified.

Remark (17.1.2). —

(i) Suppose that $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, so that f comes from a homomorphism of rings $\phi : A \to B$. According to (**0**, 19.3.1) and (**0**, 19.10.1), saying that f is formally smooth (resp. formally unramified, resp. formally étale) means that, via ϕ, B is a *formally smooth* (resp. *formally unramified*, resp. *formally étale*) A-algebra, for the *discrete* topologies on A and B.

(ii) To verify that f is formally smooth (resp. formally unramified, resp. formally étale), we can, in Definition (17.1.1), restrict to the case where $\mathscr{J}^2 = 0$. To see this, if f satisfies the corresponding condition of Definition (17.1.1) in the particular case $\mathscr{J}^2 = 0$, and if we have $\mathscr{J}^n = 0$, then we consider the closed subscheme Y'_j of Y' defined by the sheaf of ideals $\mathscr{J}^{j+1}$ for $0 \leqslant j \leqslant n-1$, so that Y'_j is a closed subscheme of Y'_{j+1} defined by a square-zero sheaf of ideals; the hypotheses imply that each of the maps

$$\mathrm{Hom}_Y(Y'_{j+1}, X) \longrightarrow \mathrm{Hom}_Y(Y'_j, X) \quad (0 \leqslant j \leqslant n-1)$$

is surjective (resp. injective, resp. bijective); by composition, we conclude that the same holds for (17.1.1.1). **IV** | 57

(iii) Note that the properties of the morphism f defined in (17.1.1) are properties of the *representable functor* ($\mathbf{0}_{\mathbf{III}}$, 8.1.8)

$$Y' \longmapsto \mathrm{Hom}_Y(Y', X)$$

from the category of Y-preschemes to the category of sets; they keep a meaning for *any* contravariant functor with the same domain and codomain, representable or not.

(iv) Assume that the morphism f is formally unramified (resp. formally étale); consider an *arbitrary* Y-prescheme Z and a closed subprescheme Z_0 of Z defined by a *locally nilpotent* sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_Z$. Then the map

(17.1.2.1) $$\mathrm{Hom}_Y(Z, X) \longrightarrow \mathrm{Hom}_Y(Z_0, X)$$

induced by the canonical injection $Z_0 \to Z$, is still injective (resp. bijective). To see this, let (U_α) be an affine open covering of Z such that the sheaves of ideals $\mathscr{J}|U_\alpha$ are nilpotent, and for each α, let U^0_α be the inverse image of U_α in Z_0, which is the closed subprescheme of U_α defined by $\mathscr{J}|U_\alpha$. Let $f_0 : Z_0 \to X$ be a Y-morphism; by hypothesis, for each α, there is at most one (resp. one and only one) Y-morphism $f_\alpha : U_\alpha \to X$ whose restriction to Z_0 coincides with $f_0|U_\alpha$. We immediately conclude that if f_α and f_β are defined, then, for each affine open $V \subset U_\alpha \cap U_\beta$, we have $f_\alpha|V = f_\beta|V$, as the restrictions of these morphisms to the inverse image V_0 of V in Z_0 coincide. There is therefore at most one (resp. one and only one) Y-morphism $f : Z \to X$ whose restriction to Z_0 coincides with f_0.

Proposition (17.1.3). —

(i) *A monomorphism of preschemes is formally unramified; an open immersion is formally étale.*

(ii) *The composition of two formally smooth (resp. formally unramified, resp. formally étale) morphisms is formally smooth (resp. formally unramified, resp. formally étale).*

(iii) *If $f : X \to Y$ is a formally smooth (resp. formally unramified, resp. formally étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two formally smooth (resp. formally unramified, resp. formally étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.*
(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if $g \circ f$ is formally unramified, then so is f.*
(vi) *If $f : X \to Y$ is a formally unramified morphism, then so is $f_{\mathrm{red}} : X_{\mathrm{red}} \to Y_{\mathrm{red}}$.*

Proof. According to (**I**, 5.5.12), it suffices to prove (i), (ii), and (iii). The assertions in (i) are both trivial. To prove (ii), consider two morphisms $f : X \to Y$, $g : Y \to Z$, an affine scheme Z', a closed subscheme Z'_0 of Z' defined by a nilpotent ideal, and a morphism $Z' \to Z$. Suppose that f and g are formally smooth, and consider a Z-morphism $u_0 : Z'_0 \to X$; the hypothesis on g implies that **IV | 58**
there exists a Z-morphism $v : Z' \to Y$ such that $f \circ u_0 = v \circ j$ (where $j : Z'_0 \to Z'$ is the canonical injection); the hypothesis on f then implies that there exists a morphism $u : Z' \to X$ such that $f \circ u = v$ and $u \circ j = u_0$, therefore $(g \circ f) \circ u$ is equal to the given morphism $Z' \to Z$ and $u \circ j = u_0$, which proves that $g \circ f$ is formally smooth; we argue the same way when we suppose that f and g are formally unramified.

Finally, to prove (iii), let $X' = X_{S'}$, $Y' = Y_{S'}$, $f' = f_{S'}$; consider an affine scheme Y'', a closed subscheme Y''_0 defined by a nilpotent sheaf of ideals, and a morphism $g : Y'' \to Y'$ making Y'' a Y'-prescheme; we then know by (**I**, 3.3.8) that $\mathrm{Hom}_{Y'}(Y'', X')$ is canonically identified with $\mathrm{Hom}_Y(Y'', X)$, and $\mathrm{Hom}_{Y'}(Y''_0, X')$ with $\mathrm{Hom}_Y(Y''_0, X)$, and the conclusion follows immediately from Definition (17.1.1). ☐

We note that a *closed immersion* is not necessarily formally smooth.

Proposition (17.1.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is formally unramified. Then, if $g \circ f$ is formally smooth (resp. formally étale), so is f.*

Proof. Let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent sheaf of ideals, $h : Y' \to Y$ a morphism, $j : Y'_0 \to Y'$ the canonical injection, and $u_0 : Y'_0 \to X$ a Y-morphism, so that $f \circ u_0 = h \circ j$. Suppose that $g \circ f$ is formally smooth; then there exists a morphism $u : Y' \to X$ such that $u \circ j = u_0$ and $(g \circ f) \circ u = g \circ h$. But these two relations imply that $f \circ u$ and h are Z-morphisms from Y' to Y such that $(f \circ u) \circ j = h \circ j$; by virtue of the hypothesis that g is formally unramified, we get that $f \circ u = h$, in other words that u is a Y-morphism; thus f is formally smooth. Taking into account (17.1.3, (v)), this proves the proposition. ☐

Corollary (17.1.5). — *Suppose that g is formally étale; then, for $g \circ f$ to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that f is.*

Proof. This follows from (17.1.4) and (17.1.3, (ii) and (v)). ☐

Proposition (17.1.6). — *Let $f : X \to Y$ be a morphism of preschemes.*

(i) *Let (U_α) be an open covering of X and, for each α, let $i_\alpha : U_\alpha \to X$ be the canonical injection. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each $f \circ i_\alpha$ is.*
(ii) *Let (V_λ) be an open covering of Y. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each of the restrictions $f^{-1}(V_\lambda) \to V_\lambda$ of f is.*

Proof. First note that (ii) is a consequence of (i): if $j_\lambda : V_\lambda \to Y$ and $i_\lambda : f^{-1}(V_\lambda) \to X$ are the canonical injections, then the restriction $f_\lambda : f^{-1}(V_\lambda) \to V_\lambda$ of f is such that $j_\lambda \circ f_\lambda = f \circ i_\lambda$; if f is formally smooth (resp. formally unramified), then so is $f \circ i_\lambda$ since i_λ is formally étale (17.1.3); but since j_λ is formally étale, this means that f_λ is formally smooth (resp. formally unramified), by virtue of (17.1.5). Conversely, if all the f_λ are formally smooth (resp. formally unramified), the same applies to $j_\lambda \circ f_\lambda$ (17.1.3), so also to f in virtue of (i).

If we take into account that the i_α are formally étale, everything comes down to proving that if **IV | 59**
the $f \circ i_\alpha$ are formally smooth (resp. formally unramified), then the same applies to f.

Therefore let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent ideal $\mathscr{J}$, which we may assume to satisfy $\mathscr{J}^2 = 0$ (17.1.2, (ii)), and finally let $g : Y' \to Y$ be a morphism. Suppose we are given a Y-morphism $u_0 : Y'_0 \to X$; denote by W_α (resp. W^0_α) the prescheme induced by Y' (resp. Y'_0) on the open subset $u_0^{-1}(U_\alpha)$ (we recall that Y' and Y'_0 share the *same underlying topological space*). Let us first suppose that the $f \circ i_\alpha$ are *formally unramified*, and show that, if u' and

u'' are two Y-morphisms from Y' to X whose restrictions to Y'_0 coincide, then we have $u' = u''$. Indeed, taking into account (17.1.2, (iv)), the hypothesis that the $f \circ i_\alpha$ are formally unramified implies that for all α, we have $u'|W_\alpha = u''|W_\alpha$, since the restrictions of both Y-morphisms to W^0_α coincide. Hence the conclusion follows.

Now suppose that the $f \circ i_\alpha$ are *formally smooth* and prove the existence of a Y-morphism $u : Y' \to X$ whose restriction to Y'_0 is u_0. Now, since Y' is an *affine scheme*, we can apply (16.5.17), the hypotheses of which are satisfied, and the conclusion of which precisely proves the existence of u. □

We can therefore say that the notions introduced in (17.1.1) are *local* on X and Y, which always allows, in virtue of (17.1.2, (i)), to be reduced to the study of formally smooth (resp. formally unramified, resp. formally étale) *algebras*.

17.2. General properties of differentials

Proposition (17.2.1). — *For a morphism $f : X \to Y$ to be formally unramified, it is necessary and sufficient that $\Omega^1_f = 0$ (what we still write $\Omega^1_{X/Y} = 0$ (16.3.1)).*

Proof. Taking into account (17.1.6), we reduce to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and the conclusion then follows from (**0**, 20.7.4) and the interpretation of $\Omega^1_{X/Y}$ in this case (16.3.7). □

Corollary (17.2.2). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms. For f to be formally unramified, it is necessary and sufficient that the canonical morphism (16.4.19)*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is surjective.

Proof. This is an immediate consequence of (17.2.1) and the exact sequence (16.4.19.1). □

Proposition (17.2.3). — *Let $f : X \to Y$ be a formally smooth morphism.*

(i) *The $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is locally projective (16.10.1). If f is locally of finite type, then $\Omega^1_{X/Y}$ is locally free and of finite type.*

(ii) *For all morphisms $g : Y \to Z$, the sequence (16.4.19) of $\mathscr{O}_X$-modules*

(17.2.3.1) $$0 \longrightarrow f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.3.1) form a split *exact sequence.*

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Proof.

(i) We know (16.3.9) that if f is locally of finite type, then Ω^1_f is an $\mathscr{O}_X$-module of finite type. To prove that, in all cases, it is locally projective, we can reduce, by virtue of (17.1.6), to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and the result follows from the hypothesis on f and from (**0**, 20.4.9) and (**0**, 19.2.1).

(ii) Again, we can restrict to the case where X, Y, and Z are affine (17.1.6), and the conclusion in this case follows from the interpretation of the sheaves of modules in the sequence (17.2.3.1) and from (**0**, 20.5.7).

□

Corollary (17.2.4). — *If $f : X \to Y$ is formally étale, then, for all morphisms $g : Y \to Z$, the canonical homomorphism of $\mathscr{O}_X$-modules*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is bijective.

Proof. This follows from the exactness of the sequence (17.2.3.1) and from the fact that we then have $\Omega^1_{X/Y} = 0$ (17.2.1). □

Proposition (17.2.5). — *Let $f : X \to Y$ be a morphism, X' a subprescheme of X such that the composite morphism $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) is formally smooth. Then the sequence of $\mathscr{O}_{X'}$-modules (16.4.21)*

$$0 \longrightarrow \mathscr{N}_{X'/X} \longrightarrow \Omega^1_{X/Y} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/Y} \longrightarrow 0 \tag{17.2.5.1}$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.5.1) form a split *exact sequence.*

Proof. By virtue of (17.1.6), we reduce to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, and $X' = \mathrm{Spec}(B/\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of B. The conormal sheaf $\mathscr{N}_{X'/X}$ then corresponds to the B-module $\mathfrak{J}/\mathfrak{J}^2$ (16.1.3), and the conclusion follows from (**0**, 20.5.14). ☐

Proposition (17.2.6). — *Let X and Y be two preschemes, $f : X \to Y$ a morphism locally of finite type. The following conditions are equivalent:*

(a) *f is a monomorphism.*
(b) *f is radicial and formally unramified.*
(c) *For each $y \in Y$, the fibre $f^{-1}(y)$ is empty or $k(y)$-isomorphic to* $\mathrm{Spec}(k(y))$ *(in other words, it is reduced to a single point z such that $k(y) \to \mathscr{O}_z/\mathfrak{m}_y\mathscr{O}_z$ is an isomorphism).*

Proof. The fact that (a) implies (c) follows from (8.11.5.1). It is clear that (c) implies that f is radicial; let us prove that it also follows from (c) that $\Omega^1_{X/Y} = 0$, which will prove that (c) implies (b) (17.2.1). Note that the $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is quasi-coherent of finite type (16.3.9). It follows from (**I**, 9.1.13.1) that, for $(\Omega^1_{X/Y})_x = 0$, it is necessary and sufficient that if we set $Y_1 = \mathrm{Spec}(k(y))$, $X_1 = f^{-1}(y) = X \times_Y Y_1$, then we have $(\Omega^1_{X_1/Y_1})_x = 0$; but as the morphism $f_1 : X_1 \to Y_1$ induced by f is formally unramified by virtue of the hypothesis (c) (17.1.3), the conclusion follows from (17.2.1). Finally, let us prove that (b) implies (a); for this, consider the diagonal morphism $g = \Delta_f : X \to X \times_Y X$; since f is radicial, g is surjective (1.8.7.1); on the other hand, $\Omega^1_{X/Y}$ is by definition the conormal sheaf $\mathscr{G}r_1(g)$ of the immersion g (16.3.1), and to say that f is formally unramified therefore means that $\mathscr{G}r_1(g) = 0$ (17.2.1). In addition, g is locally of finite presentation (1.4.3.1); therefore the hypothesis IV | 61
$\mathscr{G}r_1(g) = 0$ implies that g is an open immersion (16.1.10); being surjective, this immersion is an isomorphism, hence f is a monomorphism (**I**, 5.3.8). ☐

17.3. Smooth morphisms, unramified morphisms, étale morphisms

Definition (17.3.1). — We say that a morphism $f : X \to Y$ is *smooth* (resp. *unramified*, or *net* [4] resp. *étale*) if it is locally of finite presentation and formally smooth (resp. formally unramified, resp. formally étale).

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over* Y.

We will see later (17.5.2) that this definition of a smooth morphism coincides with the definition already given in (6.8.1); until then, we will exclusively use definition (17.3.1).

It is clear that saying that f is étale means that it is *both* smooth and unramified.

Remarks (17.3.2). —

(i) Note that definition (17.3.1) can be phrased using only the functor
$$Y' \longmapsto \mathrm{Hom}_Y(Y', X)$$
considered in (17.1.2, (iii)) because to say that f is locally of finite presentation is equivalent to saying that the preceding functor *commutes with projective limits of affine schemes* (8.14.2).

(ii) Let A be a ring and B an A-algebra. We say that B is a *smooth* (resp. *unramified*, resp. *étale*) A-algebra if the corresponding morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is smooth (resp. unramified, resp. étale). It is equivalent to saying that B is an A-algebra *of finite presentation* (1.4.6) that is furthermore formally smooth (resp. formally unramified, resp. formally étale) for the discrete topologies.

[4] The words "net" and "formally net" seem preferable to the established terminology "unramified" (resp. "formally unramified") and will be used almost exclusively beginning with Chapter V. In this chapter, we have retained the old terminology so as not to conflict with (**0**, 19.10).

(iii) It follows from (17.1.6) and the definition of a morphism locally of finite presentation (1.4.2) that the notion of a smooth (resp. unramified, resp. étale) morphism is *local on X and on Y*.

Proposition (17.3.3). —

(i) *An open immersion is étale. For an immersion to be unramified, it is necessary and sufficient for it to be locally of finite presentation.*

(ii) *The composition of two smooth (resp. unramified, resp. étale) morphisms is smooth (resp. unramified, resp. étale).*

(iii) *If $f : X \to Y$ is a smooth (resp. unramified, resp. étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are smooth (resp. unramified, resp. étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.* IV | 62

(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if g is locally of finite type and if $g \circ f$ is unramified, then f is unramified.*

Proof. This follows from (1.4.3) and (17.1.3). □

Proposition (17.3.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is unramified. Then, if $g \circ f$ is smooth (resp. unramified, resp. étale), so is f.*

Proof. As g and $g \circ f$ are locally of finite presentation, so is f (1.4.3, (v)); the conclusion thus follows from (17.1.4) and (17.1.3, (v)). □

Corollary (17.3.5). — *Suppose that g is étale; then, for f to be smooth (resp. unramified, resp. étale) it is necessary and sufficient that $g \circ f$ is.*

Proof. This follows from (17.3.4) and (17.3.3, (ii)). □

Proposition (17.3.6). — *Let $g : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation. For an S-morphism $f : X \to Y$ to be unramified, it is necessary and sufficient that the canonical homomorphism* (16.4.19)

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is surjective.

Proof. As f is locally of finite presentation (1.4.3, (v)), the proposition follows from (17.2.2). □

Definition (17.3.7). — Let $f : X \to Y$ be a morphism. We say that f is *smooth* (resp. *unramified*, resp. *étale*) at a point $x \in X$, if there exists an open neighbourhood U of x in X such that the restriction $f|U$ is a smooth (resp. unramified, resp. étale) morphism from U to Y.

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over Y at the point x*.

Taking Remark (17.3.2, (iii)) into account, saying that f is smooth (resp. unramified, resp. étale) is equivalent to saying that it is smooth (resp. unramified, resp. étale) at every point of X.

It is clear that the set of points of X at which the morphism $f : X \to Y$ is smooth (resp. unramified, resp. étale) is *open* in X.

Proposition (17.3.8). — *For all preschemes Y and all locally free $\mathscr{O}_Y$-modules $\mathscr{E}$ of finite type, the vector bundle prescheme $\mathbf{V}(\mathscr{E})$* (**II**, 1.7.8) *associated to $\mathscr{E}$ is a smooth Y-prescheme.*

Proof. By (17.3.2, (iii)), we may restrict to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{V}(\mathscr{E}) = \operatorname{Spec}(A[T_1, \ldots, T_r])$; since $A[T_1, \ldots, T_r]$ is a formally smooth A-algebra for the discrete topologies (**0**, 19.3.2) and is of finite presentation, this proves the proposition (17.3.2, (ii)). □

Corollary (17.3.9). — *Under the hypotheses of* (17.3.8), *the projective prescheme $\mathbf{P}(\mathscr{E})$* (**II**, 4.1.1) *is a smooth Y-prescheme.*

Proof. We can still restrict to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{P}(\mathscr{E}) = \mathbf{P}^r_Y$. We then know (**II**, 2.3.14) that we have a finite open cover of $\mathbf{P}^r_A$ by the $D_+(T_i)$ $(0 \leqslant i \leqslant r)$ respectively equal to the spectrum of the ring $S_{(f)}$, where we wrote S for $A[T_1, \ldots, T_r]$ and f for T_i; but it follows immediately from the definition of $S_{(f)}$ (**II**, 2.2.1), that this ring, in this case, is isomorphic to $A[T_0, \ldots, T_{i-1}, T_{i+1}, \ldots, T_r]$; hence the corollary follows by (17.3.8). □

17.4. Characterizations of unramified morphisms. IV | 63

Theorem (17.4.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X. The following properties are equivalent:*

(a) f is unramified at the point x.
(b) The diagonal morphism $\Delta_f : X \to X \times_Y X$ is a local isomorphism at the point x.
b') If one sets $\Delta_f = (\psi, \theta)$, $Z = X \times_Y X$ and $z = \psi(x)$, the homomorphism $\theta^\#_z : \mathscr{O}_{Z,z} \to \mathscr{O}_{X,x}$ is bijective.
b") For every morphism $g : Y' \to Y$, every point $y' \in Y'$ above $y = f(x)$, and every Y'-section s' of $X' = X \times_Y Y'$ such that $x' = s'(y')$ lies above x, the section s' is a local isomorphism at y'.
(c) $(\Omega^1_{X/Y})_x = 0$.
(d) The $k(y)$-prescheme $f^{-1}(y)$ is unramified over $k(y)$ at the point x.
*d') The point x is isolated in $X_y = f^{-1}(y)$ (equivalently (**Err$_{\mathbf{III}}$**, 20), the morphism f is quasi-finite at x), and the ring $\mathscr{O}_{X_y,x}$ is a field that is a finite separable extension of $k(y)$.*
d") The ring $\mathscr{O}_{X_y,x} = \mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is a field that is a finite separable extension of $k(y)$.
(e) The ring $\mathscr{O}_{X,x}$ is a formally unramified $\mathscr{O}_{Y,y}$-algebra for the discrete topologies.

Proof. Since f is locally of finite type, the $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is of finite type (16.3.9); hence $(\Omega^1_{X/Y})_x = 0$ is equivalent to the existence of an open neighbourhood U of x such that $\Omega^1_{X/Y}|U = 0$. This proves the equivalence of a) and c). On the other hand, if $A = \mathscr{O}_{Y,y}$ and $B = \mathscr{O}_{X,x}$, then $(\Omega^1_{X/Y})_x = \Omega^1_{B/A}$ (16.4.15), and the equivalence of c) and e) follows from (**0**, 20.7.4).

Since d') involves only properties of the morphism $X_y \to \operatorname{Spec}(k(y))$, the equivalence of a) and d') will imply, *ipso facto*, the equivalence of d) and d'). Moreover, d') and d") are equivalent, since, X_y being a $k(y)$-prescheme locally of finite type, saying that $\mathscr{O}_{X_y,x}$ is a *finite* $k(y)$-algebra is equivalent to saying that x is an isolated point of X_y (**I**, 6.4.4).

We now prove the equivalence of b) and b'). We may restrict to the case where $Y = \operatorname{Spec}(R)$ and $X = \operatorname{Spec}(S)$ are affine and f is of finite presentation. Then $Z = \operatorname{Spec}(S \otimes_R S)$, and Δ_f corresponds to the canonical surjective homomorphism $S \otimes_R S \to S$, whose kernel $\mathfrak{J}$ is an ideal of finite type (**0**, 20.4.4). If $\mathscr{J} = \widetilde{\mathfrak{J}}$, the $\mathscr{O}_Z$-module $\psi_*(\mathscr{O}_X) = \mathscr{O}_Z/\mathscr{J}$ is therefore of finite presentation; the hypothesis that $\theta^\#_z : \mathscr{O}_{Z,z} \to (\psi_*(\mathscr{O}_X))_z = \mathscr{O}_{X,x}$ is bijective then implies that, after replacing X if necessary by a suitable open neighbourhood of x, the homomorphism $\theta : \mathscr{O}_Z \to \psi_*(\mathscr{O}_X)$ is itself bijective (**0**, 5.2.7). This shows that b') implies b); the converse is obvious.

On the other hand, the equivalence of b) and b") follows from (**I**, 5.3.7), without any finiteness hypothesis on f: giving a Y'-section $s' : Y' \to X'$ is equivalent to giving a Y-morphism $h = g' \circ s' :$ IV | 64
$Y' \to X$ (where $g' : X' \to X$ is the canonical projection), so that $s' = (1_{Y'}, h)_X$, and then the diagram

(17.4.1.1)
$$\begin{tikzcd} Y' \arrow[r, "s'"] \arrow[d, "h"'] & X' = Y' \times_Y X \arrow[d, "h \times 1_X"] \\ X \arrow[r, "\Delta_f"'] & X \times_Y X \end{tikzcd}$$

identifies Y' with the $(X \times_Y X)$-prescheme product of X and X'. Hence ((**I**, 4.3.2)) if Δ_f is an isomorphism local at the point x, s' is an isomorphism local at the point y' (since $x = h(y')$), which proves that b) implies b''). The converse is obtained by applying b'') to the case $Y' = X$, $y' = x$, $g = f$, and $s' = \Delta_f$.

To finish the proof of (17.4.1), it suffices to prove the implications d'') $\Rightarrow$ c) $\Rightarrow$ b) $\Rightarrow$ d'').

d'') $\Rightarrow$ c): Since $\Omega^1_{X/Y}$ is an $\mathscr{O}_X$-module of finite type, Nakayama's lemma shows that condition c) is equivalent to $(\Omega^1_{X/Y})_x/\mathfrak{m}_y(\Omega^1_{X/Y})_x = 0$, that is, by (16.4.5), to $(\Omega^1_{X_y/\operatorname{Spec}(k(y))})_x = 0$. We are thus reduced to the case where Y is the spectrum of a field k and X is an algebraic k-prescheme. The hypothesis that $\mathscr{O}_{X,x}$ is a field k', finite extension of k, implies first that x is a *closed* point in X ((**I**, 6.4.2)), then that x is a *maximal* point, hence is an isolated point of X. Replacing X by the open $\{x\}$ of X, one may suppose $X = \operatorname{Spec}(k')$; but then the hypothesis that k' is a finite separable extension of k implies $\Omega^1_{k'/k} = 0$ (**0**, 20.6.20), which proves c).

c) $\Rightarrow$ *b*): We saw above that one then has $\Omega^1_{X/Y}|U = 0$ for a neighbourhood open U of x in X; the assertion *b*) follows then from the definition of $\Omega^1_{X/Y}$ (16.3.1) and (16.1.9).

b) $\Rightarrow$ *d''*): Replacing X by an open neighbourhood of x, we may suppose that Δ_f is an open immersion. If $f_y : X_y \to \mathrm{Spec}(k(y))$ denotes the morphism obtained from f by base change, then Δ_{f_y} is also an open immersion (**I**, 5.3.4). Since condition d'') concerns only X_y, we may reduce to the case where Y is the spectrum of a field k and X is the spectrum of a k-algebra A of finite type. What must be proved is that A is a *finite separable* k-algebra, that is, a finite product of finite separable extensions of k. If K is an algebraically closed extension of k, saying this is equivalent to saying that $A \otimes_k K$ is a finite separable K-algebra (4.6.1); we may therefore suppose that k is algebraically closed. We first show that A is finite over k. It is enough to show that every closed point x of X is isolated: the set of such points will then be open and discrete in X, and hence finite because it is quasi-compact (X being Noetherian), which proves the assertion by (**I**, 6.4.4). But then $k(x) = k$, since k is algebraically closed (**I**, 6.4.3); consequently there is a Y-section s of X such that $s(Y) = \{x\}$. By (17.4.1.1), $\{x\}$ is the inverse image of the diagonal $\Delta_X(X)$ under a morphism $X \to X \times_Y X$, and therefore $\{x\}$ is open in X by hypothesis b). Thus we have shown that A is a *finite* k-algebra. Since a finite k-algebra is a finite product of finite local k-algebras, to express that Δ_f is an open immersion we may reduce to the case where A is a finite *local* k-algebra. The residue field of A, being a finite extension of k, is necessarily equal to k; hence (**I**, 3.4.9) the underlying space of $X \times_k X$ consists of a single point, and Δ_f is necessarily an *isomorphism*. But since A is a k-algebra, the canonical homomorphism $A \otimes_k A \to A$ can only be bijective if $A = k$. ☐ **IV** | 65

Remark (17.4.2). — Suppose only that f is *locally of finite type*. Then $\Omega^1_{X/Y}$ is still an $\mathscr{O}_X$-module of finite type (16.3.9), and Δ_f is an immersion locally of finite presentation (1.4.3.1). The entire proof of (17.4.1) remains valid, provided a) is replaced by: *the restriction of f to a suitable neighbourhood of x is a formally unramified morphism*. One sees moreover that in this case the restriction of f to a suitable neighbourhood of x is a morphism *locally quasi-finite*.

Corollary (17.4.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation. The following properties are equivalent:*

- *(a) f is unramified.*
- *(b) The diagonal morphism $\Delta_f : X \to X \times_Y X$ is an open immersion.*
- *b') For every morphism $Y' \to Y$, every Y'-section of $X' = X \times_Y Y'$ is an open immersion.*
- *(c) $\Omega^1_{X/Y} = 0$.*
- *(d) For every $y \in Y$, the $k(y)$-prescheme $f^{-1}(y)$ is unramified over $k(y)$.*
- *d') For every $y \in Y$, the $k(y)$-prescheme $f^{-1}(y)$ is isomorphic to a prescheme of the form $\coprod_{\lambda \in L} \mathrm{Spec}(K_\lambda)$, where each K_λ is a finite separable extension of $k(y)$.*
- *(e) For every $x \in X$, the ring $\mathscr{O}_{X,x}$ is a formally unramified $\mathscr{O}_{Y,f(x)}$-algebra for the discrete topologies.*

Corollary (17.4.3). — *If $f : X \to Y$ is unramified, then f is locally quasi-finite (**Err$_{\text{III}}$**, 20).*

Proof. Indeed, by ((17.4.1), *d'*)), for every $y \in Y$, the fibre $f^{-1}(y)$ is isomorphic to a prescheme of the form $\coprod_{\lambda \in L} \mathrm{Spec}(K_\lambda)$, where the K_λ are finite separable extensions of $k(y)$; hence $f^{-1}(y)$ is a discrete topological space. ☐

Proposition (17.4.4). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are local Noetherian rings, and let k be the residue field of A. Then the conditions equivalent a) to e) of theorem (17.4.1) are also equivalent to each of the following:*

- *f) $\widehat{B} \otimes_{\widehat{A}} k$ is a field, finite separable extension of k (which implies that $\widehat{B}$ is an $\widehat{A}$-algebra finite).*
- *f') $\widehat{B}$ is a formally unramified $\widehat{A}$-algebra for the adic topologies.*

If moreover $k(x) = k(y)$, or if k is separably closed, these conditions are also equivalent to:

- *f'') The homomorphism $\widehat{A} \to \widehat{B}$ is surjective.*

Proof. First note, by the same argument as in (**0**, 19.3.6), that saying that B is a formally unramified A-algebra for the preadic topologies is equivalent to saying that $\widehat{B}$ is a formally unramified $\widehat{A}$-algebra for the adic topologies. On the other hand, the hypothesis that f is locally of finite type implies that $\Omega^1_{B/A}$ is a B-module of finite type (16.3.9), and hence is separated for the $\mathfrak{n}$-preadic topology (where $\mathfrak{n}$ is the maximal ideal of B) (**0**, 7.3.5); consequently, $\Omega^1_{B/A} = 0$ is equivalent to $\widehat{\Omega^1_{B/A}} = 0$. By (**0**, 20.7.4), it is therefore equivalent to say that B is a formally unramified A-algebra for the discrete topologies or for the preadic topologies. This proves the equivalence of e) and f′).

If $\mathfrak{m}$ is the maximal ideal of A, then **IV** | 66

$$k = A/\mathfrak{m} = \widehat{A}/\mathfrak{m}\widehat{A}, \qquad \widehat{B} \otimes_{\widehat{A}} k = \widehat{B}/\mathfrak{m}\widehat{B} = \widehat{B} \otimes_B (B/\mathfrak{m}B).$$

It follows from (**0**, 7.3.5) that $\widehat{B}/\mathfrak{m}\widehat{B}$ is the completion of $B/\mathfrak{m}B = B \otimes_A k$ for the $\mathfrak{n}$-preadic topology; this proves the equivalence of d″) and f). Finally, when $k(x) = k(y)$, or when k is separably closed, condition f) implies that the homomorphism $\widehat{A}/\mathfrak{m}\widehat{A} \to \widehat{B}/\mathfrak{m}\widehat{B}$ is bijective. Condition f) also implies that $\widehat{B}$ is a quasi-finite $\widehat{A}$-algebra (**0**, 7.4.4), hence a finite one, since $\widehat{A}$ is complete and $\widehat{B}$ is separated for the $\mathfrak{m}$-preadic topology, $\mathfrak{m}\widehat{B}$ being an ideal of definition of $\widehat{B}$ (**0**, 7.4.1). Nakayama's lemma therefore shows that $\widehat{A} \to \widehat{B}$ is surjective. Thus f) implies f″), and the converse is immediate. □

(17.4.5). Given an S-prescheme Y and two S-morphisms $f : X \to Y$, $g : X \to Y$, one canonically deduces an S-morphism $(f,g)_S : X \to Y \times_S Y$. We call the *prescheme of coincidences* of f and g the inverse image under $(f,g)_S$ of the diagonal $\Delta_{Y|S}$; it is therefore a subprescheme of X, and is a *closed* subprescheme when Y is an S-scheme (**I**, 5.4.1).

Proposition (17.4.6). — *Let $h : Y \to S$ be an unramified morphism, and let $f : X \to Y$, $g : X \to Y$ be two S-morphisms. Then the prescheme C of coincidences of f and g is a sub-prescheme induced on an open of X; if moreover Y is an S-scheme ((**I**, 5.4.1)), C is a closed sub-prescheme of X.*

Proof. Indeed, since $\Delta_h : Y \to Y \times_S Y$ is an open immersion (17.4.2), the inverse image by $(f,g)_S$ of $\Delta_{Y|S}$ is a sub-prescheme induced on an open of X ((**I**, 4.4.1)). The last assertion follows from (17.4.5). □

Corollary (17.4.7). — *Under the hypotheses of (17.4.6), let x be a point of X such that the two composed morphisms* $\mathrm{Spec}(k(x)) \to X \xrightarrow{f} Y$ *and* $\mathrm{Spec}(k(x)) \to X \xrightarrow{g} Y$ *are equal. Then there exists an open neighbourhood U of x such that $f|U = g|U$. If moreover Y is an S-scheme, there exists a closed open neighbourhood X' of x in X such that $f|X' = g|X'$. If finally X is connected, then $f = g$.*

This follows from (17.4.6) and ((**I**, 5.3.17)).

Corollary (17.4.8). — *Under the hypotheses of (17.4.6), suppose that the structural morphism $\varphi = h \circ f = h \circ g$ of X in S is closed. Let s be a point of S; let X_s be the $k(s)$-prescheme $\varphi^{-1}(s)$ and suppose that the two morphisms composed $X_s \to X \xrightarrow{f} Y$ and $X_s \to X \xrightarrow{g} Y$ are equal. Then there exists a neighbourhood open V of s in S such that $f|\varphi^{-1}(V) = g|\varphi^{-1}(V)$. If moreover Y is an S-scheme and φ is open, one can take V open and closed. If finally S is connected, $f = g$.*

Proof. It follows from (17.4.7) that the prescheme C of coincidences of f and g is induced on an open subset of X and contains X_s. Since φ is closed, there exists an open neighbourhood V of s such that $\varphi^{-1}(V) \subset C$. If moreover Y is an S-scheme, then C is closed; hence, when φ is also open, $\varphi(X - C)$ is both open and closed in S, and its complement V in S is therefore an open-and-closed neighbourhood of s such that $\varphi^{-1}(V) \subset C$. If finally S is connected, then $V = S$, and consequently $f = g$. □

Proposition (17.4.9). — *Let Y be a connected prescheme, $f : X \to Y$ an unramified and separated morphism. Then every Y-section g of X is an isomorphism of Y onto a connected component open of X, and the application $g \mapsto g(Y)$ is a bijection of $\Gamma(X/Y)$ onto the set of connected components Z of X (necessarily* open *in X) such that the restriction of f to Z is an isomorphism. In particular, if g' and g'' are two Y-sections of X such that $g'(y) = g''(y)$ for some $y \in Y$, then $g' = g''$.*

Proof. It follows from (17.4.1, b'')) that a Y-section s of X is also an open immersion, and since X is a Y-scheme, s is also a closed immersion ((**I**, 5.4.7)); it follows that s is an isomorphism of Y onto a sub-prescheme of X that is both open and closed, hence a connected component open of X, and since $s(Y)$ is connected, it is necessarily a connected component. The rest of the proposition is immediate. □

Remark (17.4.10). — In view of Remark (17.4.2), in statements (17.4.6) through (17.4.9) the word "unramified" may everywhere be replaced by "formally unramified and locally of finite type".

17.5. Characterizations of smooth morphisms. **IV** | 67

Theorem (17.5.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$. The following conditions are equivalent:*

(a) f is smooth at the point x.
(b) f is flat at the point x and the $k(y)$-prescheme $f^{-1}(y)$ is smooth on $k(y)$ at the point x (6.8.1).
b') f is regular at the point x (6.8.1).
(c) $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-algebra formally smooth for the discrete topologies.

Proof. One may reduce to the case where $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(C)$, where $C = B/\mathfrak{J}$, $B = A[T_1, \dots, T_n]$ being a polynomial algebra and $\mathfrak{J}$ an ideal *of finite type*. The equivalence of a) and c) follows from (**0**, 22.6.4). On the other hand, applying this result to the morphism locally of finite type $f^{-1}(y) \to \operatorname{Spec}(k(y))$, one sees that the equivalence of b) and b') follows from the equivalence of a) and b) in (6.8.6). It remains to prove the equivalence of a) and b).

Let us first show that a) implies b). Denote by $\mathfrak{p}$ the prime ideal $\mathfrak{i}_x$ of C, and by $\mathfrak{r}$ the prime ideal $\mathfrak{i}_y$ of A; then $\mathfrak{p} = \mathfrak{q}/\mathfrak{J}$, where $\mathfrak{q}$ is a prime ideal of B, and $\mathfrak{r}$ is the inverse image of $\mathfrak{q}$ in A. Hypothesis a) first implies, by (17.3.3), that $f^{-1}(y)$ is smooth over $k(y)$ at x, and it remains to prove that $C_{\mathfrak{p}} = \mathscr{O}_{X,x}$ is a *flat* $A_{\mathfrak{r}}$-module. Since $C_{\mathfrak{p}}$ is a formally smooth $A_{\mathfrak{r}}$-algebra and B is a formally smooth A-algebra (for the discrete topologies), the Jacobian criterion (**0**, 22.6.4), together with (**0**, 19.1.12), implies that there exist in $\mathfrak{J}$ a system of r polynomials u_i $(1 \leqslant i \leqslant r)$ and r indices j_i $(1 \leqslant i \leqslant r)$ such that the images of the u_i in $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}^2$ generate this $B_{\mathfrak{q}}$-module and

(17.5.1.1) $$\det\left(\frac{\partial u_i}{\partial T_{j_k}}\right) \notin \mathfrak{q}.$$

Now let $g : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ be the structural morphism. The fibre $g^{-1}(y)$ is the spectrum of the regular ring $k(y)[T_1, \dots, T_n]$ (**0**, 17.3.7); hence the Noetherian local ring $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ at a point of this fibre is regular. Condition (17.5.1.1) implies that the canonical images v_i of the u_i in the maximal ideal $\mathfrak{m}$ of $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ are *linearly independent* modulo $\mathfrak{m}^2$. Indeed, otherwise there would exist polynomials $w_i \in B$ $(1 \leqslant i \leqslant r)$, not all belonging to $\mathfrak{q}$, such that $\sum_{i=1}^{r} w_i u_i \in \mathfrak{q}^2$. Differentiating with respect to the T_{j_k} would then give $\sum_{i=1}^{r} w_i(\partial u_i/\partial T_{j_k}) \in \mathfrak{q}$ for $1 \leqslant k \leqslant r$, contradicting (17.5.1.1), since $\mathfrak{q}$ is prime. It follows from (**0**, 17.1.7) that (v_i) is a *regular sequence* in $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$. But since g is locally of finite presentation and B is a flat A-module, (11.3.8) shows that the canonical images u'_i of the u_i in $B_{\mathfrak{q}}$ also form a regular sequence and that $B_{\mathfrak{q}}/(\sum_i u'_i B_{\mathfrak{q}})$ is a flat $A_{\mathfrak{r}}$-module. Since the images of the u'_i in $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}^2$ generate this $B_{\mathfrak{q}}$-module, Nakayama's lemma gives $\sum_i u'_i B_{\mathfrak{q}} = \mathfrak{J}_{\mathfrak{q}}$; hence $C_{\mathfrak{p}} = B_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}$ is indeed a flat $A_{\mathfrak{r}}$-module. **IV** | 68

Finally, let us prove that b) implies a). With the same notation, the hypothesis that $B_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}} = C_{\mathfrak{p}}$ is a flat $A_{\mathfrak{r}}$-module implies that the canonical homomorphism $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}} \to B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ is injective ($\mathbf{0_I}$, 6.1.2), so that $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}}$ is identified with an ideal of $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$. Since $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ is a formally smooth $A_{\mathfrak{r}}$-algebra for the discrete topologies, the Jacobian criterion (**0**, 22.6.4), applied to

$$C_{\mathfrak{p}} = (B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}})/(\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}}),$$

together with (**0**, 19.1.12), proves, by hypothesis b), the existence of r polynomials $v_i \in k(y)[T_1, \dots, T_n]$ and r indices j_i $(1 \leqslant i \leqslant r)$ such that their images in

$$(\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}})/(\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}})^2$$

generate this $(B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}})$-module and

(17.5.1.2) $$\det\left(\frac{\partial v_i}{\partial T_{j_k}}\right) \notin \mathfrak{q}B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}.$$

For each i, choose an element $u_i \in \mathfrak{J}$ whose canonical image is v_i. It follows from (17.5.1.2) that the u_i satisfy (17.5.1.1); on the other hand, Nakayama's lemma shows that the images u'_i of the u_i in $\mathfrak{J}_{\mathfrak{q}}$ generate this $B_{\mathfrak{q}}$-module. The Jacobian criterion (**0**, 22.6.4), together with (**0**, 19.1.12), then proves that $C_{\mathfrak{p}} = B_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}$ is a formally smooth $A_{\mathfrak{r}}$-algebra for the discrete topologies. Q.E.D. ☐

Corollary (17.5.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be smooth (in the sense of (17.3.1)), it is necessary and sufficient that f be regular (6.8.1), i.e. that f be flat and that for every $y \in Y$, $f^{-1}(y)$ be a $k(y)$-prescheme geometrically regular (6.7.6).*

One has thus established the equivalence of the two definitions of "smooth morphism" given in (6.8.1) and (17.3.1).

Proposition (17.5.3). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are local Noetherian rings. Then the conditions equivalent a) to c) of (17.5.1) are also equivalent to each of the following:*

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d) B is an A-algebra formally smooth for the preadic topologies.
d') $\widehat{B}$ is an $\widehat{A}$-algebra formally smooth for the adic topologies.

If moreover $k(x) = k(y)$, these conditions are also equivalent to:

d'') $\widehat{B}$ is an $\widehat{A}$-algebra isomorphic to a formal power series algebra $\widehat{A}[[T_1, \dots, T_n]]$.

Proof. The equivalence of condition c) of (17.5.1) and condition d) follows from the equivalence of a) and d) in the Jacobian criterion (**0**, 22.6.4), and the equivalence of d) and d') follows from (**0**, 19.3.6). Moreover, d'') implies d') without any hypothesis on the residue fields (**0**, 19.3.4). Finally, if $\mathfrak{m}$ denotes the maximal ideal of $\widehat{A}$, hypothesis d') implies that $\widehat{B}/\mathfrak{m}\widehat{B}$ is a complete Noetherian local $k(y)$-algebra, formally smooth for its adic topology (**0**, 19.3.5); the hypothesis $k(y) = k(x)$ therefore implies that $\widehat{B}/\mathfrak{m}\widehat{B}$ is $k(y)$-isomorphic to a formal power series algebra $k(y)[[T_1, \dots, T_n]]$ (**0**, 19.6.4). On the other hand, $\widehat{A}[[T_1, \dots, T_n]]$ is a flat $\widehat{A}$-module and a complete Noetherian local $\widehat{A}$-algebra; it follows from (**0**, 19.7.1.5) that this algebra is isomorphic to $\widehat{B}$. Thus d') implies d'') under the additional hypothesis $k(x) = k(y)$. □

Remark (17.5.4). — Suppose that Y is a locally Noetherian prescheme, and $f : X \to Y$ a morphism locally of finite type. The criterion (17.5.3, d)), together with (**0**, 22.1.4) shows that for proving f is smooth, one can apply the definition (17.1.1), *restricting* to the case where the affine scheme Y' is the spectrum of a local Artinian ring.

Proposition (17.5.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that Y is reduced at the point y. Then, for f to be smooth at the point x, it is necessary and sufficient that f be universally open in a neighbourhood of x in $f^{-1}(y)$ and that $f^{-1}(y)$ be a $k(y)$-prescheme geometrically regular at the point x.*

Proof. By (17.5.1), everything reduces to seeing that, if $f^{-1}(y)$ is a $k(y)$-prescheme geometrically regular at the point x, it is equivalent to say that f is flat or universally open in a neighbourhood of x in $f^{-1}(y)$. But if f is flat at the point x, it is in a neighbourhood of x in X (11.1.1) and hence universally open in this neighbourhood (2.4.6). Reciprocally, the hypothesis that f is universally open in a neighbourhood of x in $f^{-1}(y)$ and that $f^{-1}(y)$ is a $k(y)$-prescheme geometrically regular at the point x implies that f is flat at the point x, since $\mathscr{O}_{Y,y}$ is reduced (15.2.2). □

Corollary (17.5.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that Y is reduced and geometrically unibranch (6.15.1) at the point y. Then, for f to be smooth at the point x, it is necessary and sufficient that f be equidimensional at the point x and that $f^{-1}(y)$ be a $k(y)$-prescheme geometrically regular at the point x.*

Observing that the set of points at which f is equidimensional is open (13.3.2), one sees that the corollary follows from (17.5.5) and the Chevalley criterion (14.4.4).

Proposition (17.5.7). — *Let $f : X \to Y$ be a morphism locally of finite presentation, smooth at a point $x \in X$; set $y = f(x)$. Then, for $\mathscr{O}_{X,x}$ to be reduced (resp. integrally closed, resp. geometrically unibranch), it is necessary and sufficient that $\mathscr{O}_{Y,y}$ be so.*

This has been proved in (11.3.13) and (11.3.14) completed by $\mathbf{Err_{IV}}$, 30.

Proposition (17.5.8). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, smooth at a point $x \in X$; set $y = f(x)$. Then:*

(i) $\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{Y,y}) + \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_{Y,y}} k(y))$.
(ii) $\operatorname{coprof}(\mathscr{O}_{X,x}) = \operatorname{coprof}(\mathscr{O}_{Y,y})$.
(iii) *For $\mathscr{O}_{X,x}$ to have the property (S_n) (5.7.2) (resp. (R_n) (5.8.2)), it is necessary and sufficient that $\mathscr{O}_{Y,y}$ have it. In particular, for $\mathscr{O}_{X,x}$ to be regular, it is necessary and sufficient that $\mathscr{O}_{Y,y}$ be so.*

These are special cases of (6.1.2), (6.3.2), (6.4.1) and (6.5.3).

17.6. Characterizations of étale morphisms. **IV** | 70

Theorem (17.6.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$. The following conditions are equivalent:*

a) *f is étale at the point x.*
a′) *f is smooth at the point x and non ramified at the point x.*
b) *f is smooth at the point x and locally quasi-finite (**Err$_{\mathbf{III}}$**, 20).*
c) *f is flat at the point x and non ramified at the point x.*
c′) *f is flat at the point x and the ring $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is a field, extension finite separable of $k(y)$.*
d) *$\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-algebra formally étale for the discrete topologies.*

Proof. The equivalence of a) and a') follows immediately from the definitions; that of a) and d) follows from the equivalence of a) and c) in (17.4.1) and the equivalence of a) and d) in (17.5.1). The equivalence of c) and c') follows from the equivalence of a) and d') in (17.4.1). The fact that a') implies c') follows from (17.5.1); conversely, if c') holds, then f is regular (hence smooth by (17.5.1)) at x, because if K is a finite separable extension of a field k, then, for every extension k' of k, $\operatorname{Spec}(K \otimes_k k')$ is regular, being a finite sum of spectra of fields. The fact that a') implies b) follows from (17.4.1, d')) and (17.5.1, b)). It remains to show that b) implies c); since one already knows from (17.5.1) that f is flat at x, it is enough to prove that $f^{-1}(y)$ is a $k(y)$-prescheme unramified over $k(y)$. In other words, one is reduced to proving that b) implies c) when $Y = \operatorname{Spec}(k)$ is the spectrum of a field k and $X = \operatorname{Spec}(A)$, where A is a finite local k-algebra ($\mathbf{0_I}$, 7.4.1). By hypothesis b), A is a formally smooth k-algebra for the discrete topologies, which here coincide with the preadic topologies; hence ($\mathbf{0}$, 19.6.5) A is a regular local ring, and therefore a field since it is Artinian. It then follows from ($\mathbf{0}$, 19.6.5.1) that A is a finite separable extension of k. □

Corollary (17.6.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation. The following conditions are equivalent:*

(a) *f is étale.*
a′) *f is smooth and non ramified.*
(b) *f is smooth and locally quasi-finite (**Err$_{\mathbf{III}}$**, 20).*
(c) *f is flat and non ramified.*
c′) *f is flat, and every fibre $f^{-1}(y)$ is a $k(y)$-scheme sum of spectra of fields, extensions finite and separable of $k(y)$.*
c″) *f is flat, and for every $y \in Y$ and every algebraically closed extension k' of $k(y)$, the "geometric fibre" $f^{-1}(y) \otimes_{k(y)} k'$ is a sum of spectra of fields isomorphic to k'.*

Proposition (17.6.3). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are local Noetherian rings, and let k be the residue field of A. Then the conditions equivalent a) to d) of (17.6.1) are also equivalent to each of the following:*

e) *$\widehat{B}$ is an $\widehat{A}$-algebra formally étale for the adic topologies.*
e′) *$\widehat{B}$ is an $\widehat{A}$-module free and $\widehat{B} \otimes_{\widehat{A}} k$ is a field, extension finite and separable of k (which implies that $\widehat{B}$ is an $\widehat{A}$-algebra finite).*

If moreover $k(x) = k(y)$, or if k is separably closed, these conditions are also equivalent to:

e″) *The canonical homomorphism $\widehat{A} \to \widehat{B}$ is bijective.*

Proposition (17.6.4). — *Under the hypotheses of (17.6.3), if f is étale at the point x, then* $\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{Y,y})$.

This is a particular case of (17.5.8, (i)) since x is isolated in its fibre $f^{-1}(y)$.

17.7. Properties of descent, passage to the limit, and constructibility. **IV** | 72

Proposition (17.7.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, $g : Y' \to Y$ a morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$ and $g' : X' \to X$ the canonical projections. Let x' be a point of X' and set $x = g'(x')$, $y' = f'(x')$.*

(i) If f' is non ramified at the point x', then f is non ramified at the point x.

(ii) Suppose moreover that g is flat at the point y'. Then, if f' is smooth (resp. étale) at the point x', f is smooth (resp. étale) at the point x.

Proof. Set $y = f(x) = g(y')$, so that $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$; note that f' is locally of finite presentation.

(i) Since the property for a morphism (locally of finite presentation) of being non ramified at a point only involves the fibre of the morphism at that point (17.4.1, d)), one may reduce to the case where Y and Y' are spectra of fields. But then (17.4.1, d')) it suffices to say that if (f') is non ramified at x (resp. x'), or that it is étale at x (resp. x').

(ii) Since f' is flat at x' (17.5.1), the hypothesis that g is flat at y' implies f is flat at x (by transitivity, 2.2.11, (iv)). The fact that f is smooth (resp. étale) at x then follows from (17.5.1) (resp. (17.6.1)) since the fibre $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ and being smooth (resp. étale) at x' implies $f^{-1}(y)$ is geometrically regular (resp. a sum of spectra of fields) at x (6.7.8) and this implies that X is a $k(y)$-prescheme geometrically regular at x, hence f is smooth at x by (17.5.1). □

Corollary (17.7.2). — *(i) With the notation of (17.7.1), let U (resp. U') be the set of points of X (resp. X') where f (resp. f') is non ramified; then $U' = g'^{-1}(U)$.*

(ii) Suppose moreover that g is flat, and let V (resp. V') be the set of points of X (resp. X') where f (resp. f') is smooth; then $V' = g'^{-1}(V)$.

Proof. (i). Since the property of being non ramified only involves the fibre ((17.4.1), d)), and the fibre of f' at x' is $f^{-1}(y) \otimes_{k(y)} k(y')$, the result follows from the fact that non-ramification is stable under base change of fields ((6.7.8)).

(ii). By (i), one has $U' \cap V' = g'^{-1}(U \cap V)$. Since f' is smooth at x' means f' is flat at x' and f' is non ramified at x' on the fibre, the result follows from the fact that flatness descends by flat morphisms ((2.2.11), (iv)) and from ((17.7.1), (ii)). □

Corollary (17.7.3). — *(i) Suppose g surjective; then, for f to be non ramified, it is necessary and sufficient that f' be so.*

(ii) Let $f : X \to Y$ be an S-morphism, $g : S' \to S$ a morphism faithfully flat, or that g be quasi-compact. Suppose that f is locally of finite presentation, or that g is locally of finite presentation. Then, for f to be smooth (resp. non ramified, resp. étale), it is necessary and sufficient that f' be so.

Proof. (i). The sufficiency follows from base change ((17.3.3), (ii)). For the necessity, by ((17.7.2), (i)), $U' = g'^{-1}(U)$; if f' is non ramified, then $U' = X'$, and since g is surjective, g' is surjective, hence $U = X$.

(ii). The sufficiency is again ((17.3.3), (ii)). For the necessity, the base change of a smooth (resp. non ramified, resp. étale) morphism is smooth (resp. non ramified, resp. étale) ((17.3.3), (ii)); for the converse, use ((17.7.1)) and the fact that a faithfully flat quasi-compact morphism is surjective ((2.2.13)). □

Proposition (17.7.4). — *Let $f : X \to Y$ be a morphism, $g : Y' \to Y$ a morphism flat and locally of finite presentation, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$ and $g' : X' \to X$ the canonical projections. Let V (resp. V') be the set of $x \in X$ (resp. $x' \in X'$) where f (resp. f') has one of the following properties:*

(i) locally of finite type;

(ii) locally of finite presentation;

(iii) flat;

(iv) non ramified;

(v) smooth;

(vi) étale.

Then $V' = g'^{-1}(V)$ (in other words, for f' to have the property considered at a point x', it is necessary and sufficient that f have this property at x).

Lemma (17.7.5). — *Let $f : X \to Y$ be a faithfully flat morphism and locally of finite presentation, $g : Y \to Z$ a morphism such that $g \circ f : X \to Z$ has one of the following properties: being*

(i) locally of finite type;
(ii) locally of finite presentation;
(iii) of finite type.

Then g has the same property. If moreover f is quasi-compact or g quasi-separated, the same conclusion holds for the property:

(iv) of finite presentation.

Remark (17.7.6). — Regarding the assertion relative to property (iv), one cannot suppress the hypothesis that f is quasi-compact; without this, it would imply that every morphism $g : Y \to Z$ quasi-compact and locally of finite presentation would be of finite presentation, a conclusion known to be erroneous (**I**, 6.4). Indeed, one can only restrict to the case where Z is affine, hence Y quasi-compact; there is then a finite covering of Y by affine opens U_i such that the restrictions $g|U_i$ are of finite presentation; it would then suffice to take for X the prescheme *sum* of the U_i, for $f : X \to Y$ the canonical morphism, which is evidently faithfully flat and locally of finite presentation (**I**, 6.5); $g \circ f$ would be of finite presentation by virtue of the choice of the U_i and of (**I**, 6.5), whence our assertion.

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Proposition (17.7.7). — *Let $f : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$. Suppose that g is flat at the point x. Then, if h is lisse (resp. non ramified, resp. étale) at the point x, f is lisse (resp. non ramified, resp. étale) at the point y.*

Proof. Set $s = h(x) = f(y)$. To say that h is non ramified at the point x (resp. that f is non ramified at the point y) means that $h^{-1}(s)$ is étale on $k(s)$ at the point x (resp. that $f^{-1}(s)$ is étale on $k(s)$ at the point y) (17.4.1) and (17.6.1); as the morphism $g_s : h^{-1}(s) \to f^{-1}(s)$ deduced from g is flat at the point x, one can reduce to proving the proposition when h is lisse or étale. Moreover, as h is then flat at x (17.5.1), f is flat at the point y (by transitivity, (2.2.11), (iv)). It therefore suffices to say that $f^{-1}(s)$ is lisse (resp. étale) on $k(s)$ at y, and one is thus reduced to the case where $S = \mathrm{Spec}(k)$ is the spectrum of a field.

(i) Case of smooth morphisms. As g is a morphism locally of finite presentation (**I**, 4.3), (v)), there is a neighbourhood U of x in X in which g is flat (11.3.1) and h lisse. Moreover $g(U)$ is an open neighbourhood of y in Y (2.4.6); replacing X by U and Y by $g(U)$, one can suppose that g is *faithfully flat and lisse*, and one is reduced to proving that f is then lisse. If k' is an algebraically closed extension of k, $X \otimes_k k'$ is then lisse on k' (17.7.3), (ii)); it suffices to prove that $Y \otimes_k k'$ is lisse on k' (since $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is faithfully flat); one can therefore restrict to the case where k is *algebraically closed*. The set of points of Y rational on k is then very dense in Y (10.4.8) and the set of points of Y where Y is lisse on k is open, so it suffices to prove that Y is lisse on k at each rational point y; but at such a point, saying that Y is lisse on k is equivalent to saying that Y is *regular* at y (17.5.1) and (6.7.8)). One has $y = g(x)$ for some $x \in X$ and by hypothesis (17.5.1) X is regular at the point x; as X and Y are then locally Noetherian and g is flat, Y is indeed regular at the point y (6.5.1), (i)).

(ii) Case of étale morphisms. By (i) one already knows that f is étale at the point y; by virtue of (17.6.1), it suffices to show that f is *quasi-finite* at the point y. Or, as $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-module faithfully flat ($\mathbf{0}_I$, (6.6.2)), $\mathscr{O}_{Y,y}$ is identified with a sub-k-module of $\mathscr{O}_{X,x}$ and as $\mathscr{O}_{X,x}$ is by hypothesis a k-algebra finite, so is $\mathscr{O}_{Y,y}$. □ **IV** | 75

Proposition (17.7.8). — *With the notations being those of* (8.8.1), *suppose X_α and Y_α locally of finite presentation on S_α. Let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism, $f : X \to Y$ the corresponding S-morphism.*

(i) Let x be a point of X, x_λ its canonical projection in X_λ. For f to be lisse (resp. non ramified, resp. étale) at the point x, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ is lisse (resp. non ramified, resp. étale) at the point x_λ.

(ii) Suppose moreover that X_α is quasi-compact. For f to be lisse (resp. non ramified, resp. étale), it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ is lisse (resp. non ramified, resp. étale).

Proof. (i) By hypothesis, for every λ, $f^{-1}(y) = f_\lambda^{-1}(y_\lambda) \otimes_{k(y_\lambda)} k(y)$; the part of the statement concerning non ramified morphisms is thus clear, and it suffices to consider the case of smooth morphisms. As f and f_λ are locally of finite presentation, it reduces to the same by (17.5.1) to saying that $f^{-1}(y)$ is geometrically regular at the point x or that $f_\lambda^{-1}(y_\lambda)$ is geometrically regular at the point x_λ (6.7.8). The proposition then results from (17.5.1) and (11.2.6).

(ii) For every λ, let U_λ be the open set of $x_\lambda \in X_\lambda$ where f_λ is lisse (resp. non ramified, resp. étale); let V_λ be its inverse image in X. By hypothesis, for every $x \in X$, there exists a λ such that f_λ is lisse (resp. non ramified, resp. étale) at the point x_λ. As, by (17.3.3), for $\lambda \leqslant \mu$ we have $V_\lambda \subset V_\mu$, and as X is quasi-compact, there exists an index μ such that $X = V_\mu$. Furthermore (17.7.3), (ii)), one concludes that for some $\nu \geqslant \mu$, $U_\nu = X_\nu$, which means f_ν is lisse (resp. non ramified, resp. étale) by (17.3.3). □

Corollary (17.7.9). — *Let $S = \mathrm{Spec}(A)$ be an affine scheme, $f : X \to S$ a morphism. The following conditions are equivalent:*

- *a) f is a morphism of finite presentation and lisse (resp. non ramified, resp. étale).*
- *b) There exists an affine Noetherian scheme $S_0 = \mathrm{Spec}(A_0)$, a morphism of finite type $f_0 : X_0 \to S_0$ and a morphism $S \to S_0$ such that the S-prescheme $X_0 \otimes_{S_0} S$ is S-isomorphic to X and that f_0 is lisse (resp. non ramified, resp. étale).*
- *c) The conditions of b) are satisfied, and moreover A_0 is a sub-$\mathbf{Z}$-algebra of finite type of A, and the morphism $S \to S_0$ corresponds to the canonical injection $A_0 \to A$.*

Proof. The proof from (17.7.8) is the same as that of (11.2.7) from (11.2.6). □

Proposition (17.7.10). — *Let $f : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$. Suppose that h is flat at the point x and that f is non ramified at the point y. Then f is étale at the point y and g is flat at the point x.* **IV | 76**

Proof. (One will see further on (18.4.9) that one can in fact dispense with the hypothesis that h is locally of finite presentation.)

The question being local on S, X and Y, one can suppose that S, X and Y are affine, f, g, h are morphisms of finite presentation, h flat and f non ramified; taking into account (11.2.7) and (17.7.9), one can furthermore suppose that S, X and Y are *Noetherian*; finally one can reduce to the case where $S = \mathrm{Spec}(\mathscr{O}_{S,s})$ (where $s = f(y) = h(x)$). The ring $A = \mathscr{O}_{S,s}$ being a local Noetherian ring, there exists a complete local Noetherian ring B, of residue field algebraically closed, and a local homomorphism $A \to B$ making B an A-module faithfully flat ($\mathbf{0_{III}}$, (10.3.1)). Replacing X and Y by $X \times_S \mathrm{Spec}(B)$ and $Y \times_S \mathrm{Spec}(B)$, one concludes by (2.5.1) and (17.7.1) that one can reduce to the case where A is complete and with residue field algebraically closed. The hypothesis that f is non ramified at the point y implies (17.4.4) that $\mathscr{O}_{Y,y}$ is an $\mathscr{O}_{S,s}$-algebra finite, and as the residue field of $\mathscr{O}_{S,s}$ is algebraically closed, the homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y}$ is *surjective* (17.4.4). But on the other hand, the hypothesis that h is flat at the point x implies that the composed homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective ($\mathbf{0}_I$, (6.5.1)), whence $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y}$ is bijective; this shows that f is étale at the point y (17.6.3); furthermore, $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-module flat, and g is flat at the point x. □

Proposition (17.7.11). — *Let S be a prescheme, X, Y two S-preschemes locally of finite presentation on S. For every $s \in S$, denote X_s, Y_s, f_s the preschemes and morphism deduced from X, Y, f by the change of base $\mathrm{Spec}(k(s)) \to S$. Then:*

- *(i) The set of $x \in X$ such that, if s is the image of x in S, $f_s : X_s \to Y_s$ is lisse (resp. non ramified, resp. étale, resp. differentially lisse) at the point x, is locally constructible.*
- *(ii) Suppose moreover that f is of finite presentation. The set of $y \in Y$ such that, if s is the image of y in S, f_s is lisse (resp. non ramified, resp. étale, resp. differentially lisse) at all the points of $f_s^{-1}(y)$, is locally constructible.*
- *(iii) Suppose X and Y of finite presentation on S. The set of $s \in S$ such that f_s is lisse (resp. non ramified, resp. étale, resp. differentially lisse) is locally constructible.*

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Proof. Let E be the set of $x \in X$ satisfying the property considered in (i). Denoting by $g : X \to S$ and $h : Y \to S$ the structural morphisms, one can first replace S by the reduced sub-prescheme S'

of S having $\overline{\{g(x)\}}$ as underlying space, X by $g^{-1}(S')$, Y by $h^{-1}(S')$; the fibres of X and X' (resp. Y and Y') at the points of S' being the same. Now let us restrict to the case where S is *integral* and where $\eta = g(x) = h(y)$ (where $y = f(x)$) is its generic point.

1º Suppose first that $x \in E$. The local rings $\mathscr{O}_{X_\eta,x}$ and $\mathscr{O}_{Y_\eta,y}$ are respectively equal to $\mathscr{O}_{X,x}$ and $\mathscr{O}_{Y,y}$; as the property of smoothness of a morphism of finite presentation at a point depends only on the local ring of the morphism at that point and on the local ring of its image (17.5.1), one sees that the hypothesis $x \in E$ means the morphism f is lisse at the point x; it then has this property at the points of an open neighbourhood of x in X, and it suffices to apply (17.3.3), (iii)) to obtain the conclusion.

2º Suppose second that $x \in X - E$, and that the morphism f_η is *not flat* at the point x. The conclusion results from the following lemma which specifies (11.2.8):

Lemma (17.7.11.1). — *Let $g : X \to S$, $h : Y \to S$ be two morphisms locally of finite presentation, $f : X \to Y$ an S-morphism, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent. Then the set E of $x \in X$ such that $\mathscr{F}_{g(x)}$ is $f_{g(x)}$-flat at the point x is locally constructible.*

3º Suppose now that $x \in X - E$, that the morphism f_η is *flat* at the point x, but that f_η does not lisse at the point x. Note that to say f_η is flat at the point x means that f itself is flat at the point x and replacing X by a neighbourhood of x, (11.1.1); one concludes that it is also true of f_s for every $s \in S$, and since for every $y \in Y$, $f^{-1}(y) = f_s^{-1}(y)$, it comes back to the same to say that $f_{g(x)}$ is lisse at the point x' or to say that f is lisse at the point x'. But the set of $x' \in X$ where f is lisse is open (17.5.1), and since it does not contain x by hypothesis, it also does not contain $\overline{\{x\}}$, which achieves the proof for the first property considered in (i).

We prove secondly in (i) when the property in question is that of being *étale*. Note that this property for f_s at the point x means that f_s is at the same time *lisse* and *quasi-finite* at the point x (17.6.1). Or, it comes back to the same to say that $f_{g(x)}$ is quasi-finite, or that f is itself quasi-finite at this point; it results from (13.1.4) that the set of points x such that $f_{g(x)}$ is quasi-finite at the point x is open in X, and *a fortiori* locally constructible; the conclusion thus results from the fact that the set of x such that $f_{g(x)}$ is lisse is also locally constructible. **IV** | 78

Let us pass to the proof of (i) when the property in question is that of being *differentially lisse*. Let $p : X \times_Y X \to X$ be the second canonical projection; the second projection $X_s \times_{Y_s} X_s \to X_s$ is just p_s for every $s \in S$, and it results from (17.12.5) that for $f_{g(x)}$ to be differentially lisse at the point x, it is necessary and sufficient that $p_{g(x)}$ be lisse at $\Delta_f(x)$. As p is locally of finite presentation, the set of points $z \in X \times_Y X$ such that $p_{g(p(z))}$ is lisse at the point z is locally constructible, and the intersection of this set with the locally closed set $\Delta_f(X)$ is also locally constructible; the restriction of p to $\Delta_f(X)$ being an isomorphism on X, one sees that the set of $x \in X$ such that $f_{g(x)}$ is differentially lisse at the point x is locally constructible.

Finally, consider the property of being non ramified; note that the diagonal morphism $\Delta_f : X \to X \times_Y X$ is an immersion locally of finite presentation (**I**, 4.3.1) and for every $s \in S$, the diagonal morphism $\Delta_{f_s} : X_s \to X_s \times_{Y_s} X_s$ is just $(\Delta_f)_s$; saying that $f_{g(x)}$ is non ramified at the point x means that $(\Delta_f)_{g(x)}$ is an isomorphism local at the point x (17.4.1), and since this is an immersion locally of finite presentation, it suffices from (17.9.1) to say that $(\Delta_f)_{g(x)}$ is étale at the point x; it suffices therefore to apply what has been seen above for the property of being étale.

The assertions (ii) and (iii) of (17.7.11) follow from (i), the theorem of Chevalley on images of constructible sets (1.8.4), and (when h is the structural morphism) similar arguments. □

17.8. Criteria for smoothness and non-ramification by fibres. **IV** | 79

Proposition (17.8.1). — *Let $g : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation. For an S-morphism $f : X \to Y$ to be non ramified, it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \to g^{-1}(s)$ deduced from f by the change of base* $\operatorname{Spec}(k(s)) \to S$*, be non ramified.*

Proof. This follows immediately from the fact that, for a morphism locally of finite presentation, being non ramified is a property of the *fibres* of this morphism (17.4.1), *d*)), and from the fact that for every $y \in Y$, one has $f^{-1}(y) = f_s^{-1}(y)$ where $s = g(y)$. □

Proposition (17.8.2). — *Let $g : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation; suppose moreover that h is flat. For an S-morphism $f : X \to Y$ to be lisse (resp. étale), it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \to g^{-1}(s)$ be lisse (resp. étale). When this is the case, the morphism g is étale and flat at all points of $f(X)$.*

Proof. One knows in fact (11.3.10) that for every $s \in S$, f_s is flat at all points of $f(X)$, and it suffices that for every $s \in S$, f_s be lisse at all points of $f(X)$; but for a morphism locally of finite presentation plat, the fact of being lisse is a property of the *fibres* of this morphism (17.5.1), *b*)), and for every $y \in Y$, $f^{-1}(y) = f_s^{-1}(y)$ where $s = g(y)$. □

Remark (17.8.3). — The preceding proofs show (taking into account (11.3.10)) that if the hypotheses on g and h are the same as above, then, for f to be non ramified (resp. lisse, resp. étale) *at a point* $x \in X$, it suffices that, if one sets $s = h(x)$, f_s be non ramified (resp. lisse, resp. étale) at the point x.

17.9. Étale morphisms and open immersions.

Theorem (17.9.1). — *Let $f : X \to Y$ be a morphism. The following properties are equivalent:*

- *a) f is an open immersion.*
- *b) f is a flat monomorphism locally of finite presentation.*
- *c) f is étale and radiciel.*

Proof. It results from (**I**, 4.3), (i)) that *a*) implies *b*). The condition *b*) implies that for every $y \in Y$, the fibre $f^{-1}(y)$ is either empty or isomorphic to $\operatorname{Spec}(k(y))$ (8.11.5.1), hence *b*) implies *c*), by virtue of (17.6.2), *c*)). It remains to show that *c*) implies *a*). The question being local on X and on Y (since f is injective), one can reduce to the case where Y is affine and f of finite presentation. As f is flat, this is an *open* morphism (2.4.6), hence, replacing Y by $f(X)$, one can suppose that f is *surjective*. For every morphism $Y' \to Y$, $f' = f_{(Y')} : X_{(Y')} \to Y'$ is again étale, radiciel, surjective and of finite presentation; in other words f is a *universal homeomorphism*, that is to say f is of finite type and is separated (**I**, 8.7.1), (ii)). The hypothesis that f is radiciel and of finite type implies that f is quasi-finite; hence (8.11.1) f is a finite morphism. To prove that f is an isomorphism, one can reduce to the case where $Y = \operatorname{Spec}(A)$, A being a local ring. As f is of finite presentation, on has $X = \operatorname{Spec}(B)$, where B is an A-module flat of finite presentation (**I**, 4.7), hence *free* (Bourbaki, *Alg. comm.*, chap. II, § 5, nº 2, cor. 2 of th. 1). Moreover, if $\mathfrak{m}$ is the maximal ideal of A and k its residue field, $B/\mathfrak{m}B$ is by hypothesis a field, which is both radiciel extension and étale extension of k (17.6.1); hence $B/\mathfrak{m}B$ is isomorphic to k, and as B is a free A-module, B is isomorphic to A. C.Q.F.D. □

Corollary (17.9.2). — *Let X be a connected prescheme; if $f : X \to Y$ is a closed étale immersion, then f is an isomorphism of X onto a connected component of Y.*

Proof. Indeed f is étale and radiciel, hence an isomorphism from X onto a sub-prescheme of Y; but by hypothesis $f(X)$ is closed in Y, hence since open and closed, and since $f(X)$ is connected, it is a connected component of Y. □

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Corollary (17.9.3). — *Let $f : X \to Y$ be a morphism étale (resp. étale and separated). Then every Y-section $g : Y \to X$ of X is an open immersion (resp. open and closed); moreover the application $g \mapsto g(Y)$ is a bijection of the set $\Gamma(X/Y)$ of Y-sections of X to the set of open (resp. open and closed) parts Z of X such that the restriction of f to Z is a surjective morphism and radiciel of Z onto Y.*

Proof. Indeed, the fact that g is an open immersion follows already from the fact that f is non ramified (17.4.1), $b'')$), and the restriction of f to the open $g(Y)$ of X is an isomorphism. Inversely, if Z is an open of X such that $f|Z$ is a surjective morphism and radiciel, since f is étale, $f|Z$ is an isomorphism by virtue of (17.9.1), which proves the corollary. □

Corollary (17.9.4). — *Let Y be a connected prescheme and separated, $f : X \to Y$ a morphism étale. Then every Y-section g of X is an isomorphism of Y onto a connected component of X, and the application $g \mapsto g(Y)$ is a bijection of $\Gamma(X/Y)$ to the set of connected components of X such that the restriction of f to Z is a surjective morphism and radiciel of Z onto Y.*

Corollary (17.9.5). — *Let $g : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation; suppose moreover that h is flat. For an S-morphism $f : X \to Y$ to be an open immersion (resp. isomorphism), it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \to g^{-1}(s)$ deduced from f by the change of base* $\operatorname{Spec}(k(s)) \to S$, *be an open immersion (resp. isomorphism).*

Proof. Indeed, if f_s is an open immersion for every $s \in S$, it results from (17.8.3) that f is étale, and as for every $y \in Y$, $f^{-1}(y) = f_s^{-1}(y)$ with $s = g(y)$, f is radiciel; hence f is an open immersion by virtue of (17.9.1). If moreover f_s is surjective for every $s \in S$, f is surjective, hence an isomorphism. □

The following proposition specifies (10.4.11):

Proposition (17.9.6). — *Let S be a prescheme, X an S-prescheme of finite presentation. Every S-endomorphism of X which is a monomorphism is an automorphism of X.*

Proof. Let $f : X \to S$ be the structural morphism, g the S-endomorphism considered. The question being local on S, one can suppose S affine. Using (8.9.1) and (8.10.5), (i *bis*)), one is reduced to the case where $S = \operatorname{Spec}(A)$, where A is a $\mathbf{Z}$-algebra of finite type, and hence X is a $\mathbf{Z}$-prescheme of finite type. It follows already from (10.4.11) that g is a bijective morphism, since a monomorphism is radiciel ((8.11.5.1)); it will thus suffice to show that g is an open immersion, and since g is radiciel, it will suffice, by virtue of (17.9.1), to prove that g is étale. Moreover, as the set of points where g is étale is open and X is a Jacobson prescheme ((10.4.7)), it suffices to show that g is étale at every *closed* point z of X ((10.3.1)). Set $z' = g(z)$; it follows from (10.4.11), (i) that z' is also a closed point of X, and since g is a monomorphism, the map $k(z') \to k(z)$ deduced from g is an isomorphism. For g to be étale at the point z, it is thus necessary and sufficient that the canonical homomorphism $\widehat{\mathscr{O}}_{X,z'} \to \widehat{\mathscr{O}}_{X,z}$ be bijective ((17.6.3), $e'')$). We will show for this purpose that for every integer n, the canonical homomorphism $\mathscr{O}_{X,z'}/\mathfrak{m}_{z'}^{n+1} \to \mathscr{O}_{X,z}/\mathfrak{m}_z^{n+1}$ is bijective, from which the conclusion will immediately follow. One can suppose that z belongs to the finite set $T_{p,d}$ of closed points t of X such that $k(t)$ is an extension of $\mathbf{F}_p$ whose degree divides d; in which case the same holds for z'. Denote by $\mathscr{J}$ the coherent Ideal of $\mathscr{O}_X$ such that $\mathscr{J}_x = \mathscr{O}_x$ for $x \notin T_{p,d}$, and $\mathscr{J}_x = \mathfrak{m}_x^{n+1}$ for $x \in T_{p,d}$. Let $T_{p,d,n}$ be the closed sub-prescheme of X defined by $\mathscr{J}$, $j : T_{p,d,n} \to X$ the canonical injection; the composed morphism $g \circ j : T_{p,d,n} \to X$ maps the set $T_{p,d}$ into itself, and for every $x \in T_{p,d,n}$, the homomorphism $\mathscr{O}_{X,g(x)} \to \mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1}$ factors as

$$\mathscr{O}_{X,g(x)} \longrightarrow \mathscr{O}_{X,g(x)}/\mathfrak{m}_{g(x)}^{n+1} \longrightarrow \mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1}.$$

By ((**I**, 4.9)), there therefore exists a unique endomorphism g' of $T_{p,d,n}$ such that $j \circ g' = g \circ j$. As j is a monomorphism, and g is also one by hypothesis, one deduces that g' is also a monomorphism, and the proposition will be proven if one shows that g' is an *automorphism* of $T_{p,d,n}$. But, for every $x \in T_{p,d,n}$, we have seen that $k(x)$ is a finite field, and as $\mathscr{O}_{X,x}$ is Noetherian, each of the $\mathfrak{m}_x^h/\mathfrak{m}_x^{n+1}$ is a $k(x)$-vector space of finite rank; from which one concludes immediately that the local ring $\mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1}$ at the point x has a finite number of elements, and *a fortiori* is a $\mathbf{Z}$-module of finite type. As $T_{p,d,n}$ is the sum of the preschemes $\operatorname{Spec}(\mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1})$ in finite number, it is a finite $\mathbf{Z}$-prescheme; the endomorphism g' is therefore also finite (**II**, 6.1.5, (v)), hence proper. But a proper monomorphism is a closed immersion ((8.11.5)); if $T_{p,d,n} = \operatorname{Spec}(B)$, g' corresponds to a surjective endomorphism $\varphi : B \to B$ of the ring B. But the set B is finite, hence φ is necessarily bijective. □

17.10. Relative dimension of a smooth prescheme over another. **IV | 81**

Definition (17.10.1). — Let $f : X \to Y$ be a morphism locally of finite type. The *relative dimension* of f at a point $x \in X$ (or the *dimension of X relative to Y at the point x*) is the positive integer $\dim_x f = \dim_x(f^{-1}(f(x)))$.

To say that f is *quasi-finite* at the point x ($\mathbf{Err_{III}}$, 20) is equivalent to saying that $\dim_x f = 0$. One has seen (13.1.3) that the function $x \mapsto \dim_x f$ is upper semi-continuous. One notes that, even when the morphism f has the property (S_1) (that is to say (6.8.1) is flat and such that its fibres have no embedded associated prime cycle), the function $x \mapsto \dim_x f$ is not necessarily continuous, as shown by the example where $Y = \mathrm{Spec}(k)$, where k is a field, and $X = \mathrm{Spec}(k[\mathrm{U}, \mathrm{V}, \mathrm{W}]/\mathfrak{p}\mathfrak{q})$, where $\mathfrak{p} = (\mathrm{W})$ and $\mathfrak{q} = (\mathrm{U}) + (\mathrm{V} - \mathrm{W})$, the prime ideals of $k[\mathrm{U}, \mathrm{V}, \mathrm{W}]$ (X being thus the reunion, in the affine space of 3 dimensions, of a plane and of a line non parallel to this plane).

To say that a morphism locally of finite presentation $f : X \to Y$ is étale at a point $x \in X$ means that f is lisse at the point x and that $\dim_x f = 0$ (17.6.1).

Proposition (17.10.2). — *Let $f : X \to Y$ be a smooth morphism. For every $x \in X$, the $\mathscr{O}_{X,x}$-Module locally free Ω^1_f (17.2.3) has rank equal to $\dim_x f$ in x (which implies that $x \mapsto \dim_x f$ is a* continuous *function on X).*

Proof. Indeed, if $y = f(x)$, $X_y = f^{-1}(y)$ is lisse over $k(y)$, and if $f_y : X_y \to \mathrm{Spec}(k(y))$ is the structural morphism, one has $\Omega^1_{f_y} = \Omega^1_f \otimes_{\mathscr{O}_x} k(y)$ (16.4.5); one can thus reduce to the case where $Y = \mathrm{Spec}(k)$ is the spectrum of a field. But (16.4.12) $(\Omega^1_{X/k})_x \otimes_{\mathscr{O}_x} k(x)$ is then k-isomorphic to $\mathfrak{m}_x/\mathfrak{m}_x^2$, and since $\mathscr{O}_x$ is a regular local ring (17.5.1), $\mathrm{rg}_k(\mathfrak{m}_x/\mathfrak{m}_x^2) = \dim(\mathscr{O}_x)$ ($\mathbf{0}$, (17.1.1)); hence $\mathrm{rg}_{\mathscr{O}_x}(\Omega^1_{X/k})_x = \dim(\mathscr{O}_x)$. But since $k(x) = k$, one has $\dim_x f = \dim(\mathscr{O}_x)$ (5.2.3). C.Q.F.D. □

We will prove a reciprocal of this result further on (17.15.5).

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Corollary (17.10.3). — *Let $f : X \to Y$, $g : Y \to Z$ be two smooth morphisms. Then for every $x \in X$, one has*

$$\dim_x(g \circ f) = \dim_x f + \dim_{f(x)} g.$$

Proof. Indeed, $g \circ f$ is smooth (17.3.3), hence the three $\mathscr{O}_X$-Modules $\Omega^1_{X/Y}$, $\Omega^1_{X/Z}$ and $f^*(\Omega^1_{Y/Z})$ are locally free ((17.2.3) and ($\mathbf{0_I}$, 5.4.5)); moreover the rank at x of $f^*(\Omega^1_{Y/Z})$ is equal to the rank at y of $\Omega^1_{Y/Z}$. The equality (17.10.3) is thus a consequence of (17.10.2) and of the exactness of the suite (17.2.3). □

Corollary (17.10.4). — *Let $f : X \to Y$ be a smooth morphism, X' a sub-prescheme of X such that the composed morphism $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) be lisse. Then the conormal sheaf $\mathscr{N}_{X'/X}$ is an $\mathscr{O}_{X'}$-Module locally free, and for every $x \in X'$, one has*

(17.10.4.1) $$\dim_x f = \dim_x(f \circ j) + \mathrm{rg}_{\mathscr{O}_{X',x}}(\mathscr{N}_{X'/X})_x.$$

Proof. Indeed, $\Omega^1_{X/Y} \otimes_{\mathscr{O}_x} \mathscr{O}_{X'}$ and $\Omega^1_{X'/Y}$ are then locally free and the exact sequence (17.2.5.1) is locally split in a neighbourhood of each point of X', hence $\mathscr{N}_{X'/X}$ is also locally free, and the relation (17.10.4.1) results from the exactness of the suite (17.2.5.1). □

17.11. Smooth morphisms of smooth preschemes.

Theorem (17.11.1). — *Let $f : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X; set $y = g(x)$, $s = f(y) = h(x)$. The following conditions are equivalent:*

a) f is lisse at the point y and g is lisse at the point x.
b) g and h are lisse at the point x.
c) h is lisse at the point x, and the canonical homomorphism (16.4.18)

$$(g^*(\Omega^1_{Y/S}))_x \longrightarrow (\Omega^1_{X/S})_x \tag{17.11.1.1}$$

is left invertible (in other words is an isomorphism onto a direct factor of $(\Omega^1_{X/S})_x$).

c′) h is lisse at the point x, and the canonical homomorphism

$$((\Omega^1_{Y/S})_y \otimes_{\mathscr{O}_{Y,y}} k(y)) \otimes_{k(y)} k(x) \longrightarrow (\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x) \tag{17.11.1.2}$$

is injective.

Suppose moreover that the homomorphism $k(y) \to k(x)$ is bijective. Then the preceding conditions are also equivalent to the following:

d) h is lisse at the point x, and the canonical application $\mathrm{T}_{X/S}(x) \to \mathrm{T}_{Y/S}(y)$ (16.5.12) from the tangent vector space at x to the tangent vector space at y to Y is surjective.

Proof. The fact that a) implies b) results trivially from (17.3.3), (ii)); b) implies c) by application of (17.2.3), (ii)). To see that c) is equivalent to c'), note that by virtue of (17.2.3), (i)), $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-Module locally free of finite type; it suffices to apply (**0**, (19.1.12)) to the ring $\mathscr{O}_{X,x}$ and the homomorphism (17.11.1.1) of $\mathscr{O}_{X,x}$-modules of finite type. When $k(x) = k(y)$, the tangent linear application $\mathrm{T}_{X/S}(x) \to \mathrm{T}_{Y/S}(y)$ is the transpose of (17.11.1.2), by (16.5.12), hence the equivalence of c') and d) in this case.

It remains to prove that c) implies a). One can restrict to the case where $S = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $X = \mathrm{Spec}(C)$ are affine. The hypothesis c) implies ((17.2.3), (i)) that $(\Omega^1_{X/S})_x = \Omega^1_{C_\mathfrak{m}/A}$ is a $C_\mathfrak{m}$-module locally free: indeed (16.10.6) there exist elements t_i $(1 \leqslant i \leqslant r)$ of $B_\mathfrak{p} = \mathscr{O}_{Y,y}$ such that the differentials $d_{B/A}(t_i)$ engender the $B_\mathfrak{p}$-module $\Omega^1_{B/A_\mathfrak{p}}$ and their images in $\Omega^1_{C_\mathfrak{m}/A}$ are part of a base of this $C_\mathfrak{m}$-module. As $\Omega^1_{Y/S}$ (resp. $\Omega^1_{X/S}$) is an $\mathscr{O}_Y$-Module (resp. $\mathscr{O}_X$-Module) of finite presentation (16.4.22), one can, replacing when needed X and Y by suitable open affine neighbourhoods of x and y respectively, suppose that $\Omega^1_{C/A}$ is a C-module libre and that the t_i are images of elements s_i $(1 \leqslant i \leqslant r)$ of B such that the $d_{B/A}(s_i)$ are part of a base of the C-module $\Omega^1_{C/A}$. Let φ be the A-homomorphism from $B' = A[\mathrm{T}_1, \dots, \mathrm{T}_r]$ to B such that $\varphi(\mathrm{T}_i) = s_i$ for every i; the di-homomorphism corresponding $\Omega^1_{B'/A} \to \Omega^1_{B/A}$ (**0**, (20.5.2)) transforms the $d_{B'/A}(\mathrm{T}_i)$, which form a base of $\Omega^1_{B'/A}$ (**0**, (20.4.13)), into the $d_{B/A}(s_i)$ and is hence *surjective*; if $Y' = \mathrm{Spec}(B')$ and if $u : Y \to Y'$ is the S-morphism corresponding to φ, one concludes that u is *non ramified* (17.2.2). If one proves that the composed morphism $u \circ g : X \to Y'$ is *étale* at the point y, then from (17.3.5) that g is lisse at the point x; finally, as the structural morphism $f' : Y' \to S$ is lisse (17.3.8), $f = f' \circ u$ will be lisse at the point y by (17.3.5). That for the proof that c) implies a), one sees thus that one can restrict to the case where $Y' = Y$, and hence suppose that f is a smooth morphism *lisse*. IV | 83

By virtue of (17.8.2), one can reduce to the case where $S = \mathrm{Spec}(k)$ is the spectrum of a field; moreover, by (16.4.5) and (17.7.1), (ii)), one can suppose that k is algebraically closed, and one can suppose that f and h are both lisse, and that the canonical homomorphism

$$g^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is left invertible (**0**, (19.1.12)). Then the set of points of X rational on k is very dense in X (10.4.8), hence, to prove that g is lisse, it suffices to prove that g is lisse at every point $x \in X$ *rational on* k, or equivalently that at such a point, $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-module flat and $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ a regular ring (17.5.1). Or, $y = g(x)$ is the image of x, B= $\mathscr{O}_{Y,y}$, the residue field of B is equal to k, it is the same as that of A, hence x (resp. y) is a closed point of X (resp. Y) (**I**, (6.4.2)), and one has $\dim(A) = \dim(B) = n$. Since f is non ramified (17.7.1), the homomorphism $\widehat{A} \to \widehat{B}$ is surjective (17.4.4), f'')); but as $\dim(\widehat{A}) = \dim(\widehat{B}) = n$, and $\widehat{A}$ being a regular ring, integral, the homomorphism $\widehat{A} \to \widehat{B}$ is also *injective* (**0**, (16.3.10)). This homomorphism is thus bijective, which achieves the proof that f is étale at the point x (17.6.3), e'')). ☐

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Corollary (17.11.2). — *Under the general hypotheses of (17.11.1), the following conditions are equivalent:*

a) f is lisse at the point y and g is étale at the point x.
b) h is lisse at the point x and g is étale at the point x.
c) h is lisse at the point x, and the canonical homomorphism

$$(g^*(\Omega^1_{X/S}))_x \longrightarrow (\Omega^1_{X/S})_x$$

is bijective.

c′) h is lisse at the point x, and the canonical homomorphism

$$((\Omega^1_{Y/S})_y \otimes_{\mathscr{O}_{Y,y}} k(y)) \otimes_{k(y)} k(x) \longrightarrow (\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x)$$

is bijective.

Suppose moreover that the homomorphism $k(y) \to k(x)$ is bijective. Then the preceding conditions are also equivalent to the following:

d) h is lisse at the point x, and the canonical application $\mathrm{T}_{X/S}(x) \to \mathrm{T}_{Y/S}(y)$ (16.5.12) is bijective.

Proof. Each of conditions a), b), c) of (17.11.2) is equivalent to the conjunction of the corresponding condition of (17.11.1) and of the fact that g is non ramified at the point x, taking into account (17.2.2) in what concerns condition c); whence the equivalence of a), b) and c). The equivalence of c) and c') results from the equivalence of the corresponding conditions of (17.11.1) and from Nakayama's lemma; the equivalence of c') and d) when $k(x) = k(y)$ is immediate by transposition (16.5.12). ☐

Corollary (17.11.3). — *Let $h : X \to S$ be a smooth morphism, s_i ($1 \leqslant i \leqslant r$) sections of $\mathscr{O}_X$ above X (which are also sections of $h_*(\mathscr{O}_X)$ above S), $g : X \to S[\mathrm{T}_1, \ldots, \mathrm{T}_r] = \mathbf{V}^r_S$ the S-morphism corresponding to the homomorphism of $\mathscr{O}_S$-Modules $\mathscr{O}^r_S \to h_*(\mathscr{O}_X)$ defined by these sections (**II**, (1.2.7)). For g to be lisse (resp. étale) at a point $x \in X$, it is necessary and sufficient that the $(d_{X/S}(s_i))_x$ form part of a base (resp. constitute a base) of the $\mathscr{O}_{X,x}$-module $(\Omega^1_{X/S})_x$.*

Proof. It suffices to apply (17.11.1) (resp. (17.11.2)) taking for f the structural morphism $S[\mathrm{T}_1, \ldots, \mathrm{T}_r] \to S$. ☐

Corollary (17.11.4). — *For a morphism $h : X \to S$ to be lisse at a point $x \in X$, it is necessary and sufficient that there exist an open neighbourhood U of x, an integer r, and an S-morphism étale $U \to S[\mathrm{T}_1, \ldots, \mathrm{T}_n]$.*

Proof. The condition is evidently sufficient since the structural morphism $S[\mathrm{T}_1, \ldots, \mathrm{T}_n] \to S$ is lisse (17.3.8). To show that it is necessary, note that since h is lisse at the point x, there is an open neighbourhood U of x such that $\Omega^1_{X/S}|U$ is locally free (16.10.6) for obtaining (restricting to U) sections s_i of $\mathscr{O}_X$ above U such that the $d_{X/S}(s_i)$ form a base of $\Omega^1_{X/S}|U$; one concludes by applying (17.11.3). ☐

Proposition (17.11.5). — *Let $f : Y \to S$, $h : X \to S$ be two smooth morphisms. For an S-morphism $g : X \to Y$ to be an open immersion, it is necessary and sufficient that g be a monomorphism of preschemes and that for every $x \in X$, if one sets $y = g(x)$, one has $\dim_y(f) = \dim_x(h)$ (for a generalisation of this proposition, see (18.10.5)).*

Proof. The condition is evidently necessary; let us prove that it is sufficient. By virtue of (17.9.5), one is immediately reduced to the case where $S = \mathrm{Spec}(k)$ is the spectrum of a field, and by virtue of (2.7.1), (x)), one can suppose k algebraically closed. By virtue of (17.9.1), it suffices to prove that g is étale at the *closed* points (or equivalently, rational on k) of X (10.4.8). So let x be such a point, and set $y = g(x)$, which is also rational on k and hence closed. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are regular local rings, and let k be the residue field of A. Let $d = \dim(A)$ and $(t_i)_{1 \leqslant i \leqslant d}$ a regular system of parameters for A. Set $B = \mathscr{O}_{X,x}$, $C = B/\mathfrak{m}_y B$. As g is a monomorphism, so is the morphism $\mathrm{Spec}(C) \to \mathrm{Spec}(k(y)) = \mathrm{Spec}(k)$ deduced from g by base change (**I**, (3.3.12)); but this means the corresponding homomorphism $u : k \to C$ is *surjective* (hence bijective), since u admits a left inverse $v : C \to k$, and $u \circ v$ and the identity of $k \to C$ give the same morphism, which, composed with

u, gives the identity of $k \to C$. As by hypothesis B is a regular ring of dimension d, the images of the t_i in B form a regular system of parameters for B ($\mathbf{0}$, (17.1.7)); the condition (17.6.3), e'')) is thus verified by g at the point x, which achieves the proof. □

17.12. Smooth subpreschemes of a smooth prescheme. Smooth morphisms and differentially smooth morphisms. IV | 85

Theorem (17.12.1). — *Let $f : X \to S$, $h : Y \to S$ be two morphisms locally of finite presentation, $j : Y \to X$ an immersion, y a point of Y, $x = j(y)$. The following conditions are equivalent:*

a) h is lisse at the point y and f is lisse at the point x.
b) f is lisse at the point x, and the canonical homomorphism (16.4.21)

$$(\mathscr{N}_{Y/X})_y \longrightarrow (j^*(\Omega^1_{X/S}))_y$$

is left invertible.
c) h is lisse at the point y and there exists an open neighbourhood U of y in Y such that $j|U : U \to X$ is a regular immersion (16.9.2).

Corollary (17.12.2). — *With the notations of (17.12.1), suppose that f is lisse at the point x, and let Y be a closed sub-prescheme of X defined by an Ideal $\mathscr{J}$ of $\mathscr{O}_X$, and S-lisse at the point x. Let $(g_i)_{1\leqslant i\leqslant r}$ be sections of $\mathscr{J}$ above X. The following conditions are equivalent:*

a) The $(g_i)_x$ form a system of generators of the $\mathscr{O}_{X,x}$-module $\mathscr{J}_x$ whose number of elements is the smallest possible.
b) If g'_i is the canonical image of g_i in $\Gamma(X, \mathscr{J}/\mathscr{J}^2)$, the $(g'_i)_x$ form a base of the $(\mathscr{O}_{X,x}/\mathscr{J}_x)$-module $\mathscr{J}_x/\mathscr{J}_x^2$.
c) The images of the $d_{X/S}g_i$ in $(\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x)$ are linearly independent over $k(x)$, and the $(g_i)_x$ engender $\mathscr{J}_x$.
d) There exists an open neighbourhood U of x in X and sections g_j $(r+1 \leqslant j \leqslant n)$ of $\mathscr{O}_X$ above U such that the S-morphism $u : U \to S[\mathrm{T}_1, \dots, \mathrm{T}_n]$ corresponding to the homomorphism $\mathscr{O}_S^n \to f_*(\mathscr{O}_U)$ defined by the sections $g_i|U$ $(1 \leqslant i \leqslant r)$ and g_j $(r+1 \leqslant j \leqslant n)$ is an étale morphism, and such that the reciprocal image of the sub-prescheme $Y' = S[\mathrm{T}_{r+1}, \dots, \mathrm{T}_n]$ is the open $U \cap j(Y)$.

Proof. As $(g'_i)_x$ is the canonical image of $(g_i)_x$ and b) results from (17.10.4) (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, prop. 5). By virtue of (17.12.1), b)), $\mathscr{J}_x/\mathscr{J}_x^2$ is a free $(\mathscr{O}_{X,x}/\mathscr{J}_x)$-module libre of rank r; and the equivalence of b) and c) results from Bourbaki, *loc. cit.* Moreover, if a) is verified, $(g'_i)_x$ is the image of $(g_i)_x$ in $\mathscr{O}_{X,x}/\mathscr{J}_x$; the equivalence of a) and b) follows from the fact that $\mathscr{J}_x$ is of finite type and $\mathscr{J}_x/\mathscr{J}_x^2$ free, by virtue of the Nakayama lemma.

Moreover, if a) is verified, the images of the $d_{X/k}g_i$ engender $(\Omega^1_{X/k})_x \otimes 1$ for $1 \leqslant i \leqslant r$; the d_i for $r+1 \leqslant i \leqslant n$ already engender $\Omega^1_{k(x)/k}$. As $(\Omega^1_{X/S})_x$ is free of rank n, it results from Bourbaki, *loc. cit.*, that replacing X by an open neighbourhood of x, one can suppose that there exist $n-r$ sections g_i $(r+1 \leqslant i \leqslant n)$ of $\mathscr{O}_X$ above X such that the $d_{X/S}g_i$ for $1 \leqslant i \leqslant n$ form a base of $(\Omega^1_{X/S})_x$; the fact that the corresponding étale morphism u is étale at the point x results from (17.11.3); it is then étale in a neighbourhood of x; replacing X by this neighbourhood, u can be supposed étale. Moreover, it is immediate that Y (identified, by restriction of $j(Y)$ to X) is a closed sub-prescheme of $u^{-1}(Y')$, and by virtue of (17.2.5) and (17.11.2), the restriction to Y of u can be supposed étale. One deduces that for every $y' \in Y'$, $Y \cap u^{-1}(y')$ is an open immersion (17.9.6)), whence $Y \to u^{-1}(Y')$ is an open immersion, and we conclude by (17.11.3) that this proves that a) implies d). The reciprocal $d) \Rightarrow a)$ follows from (17.11.3). □ IV | 86

Corollary (17.12.3). — *Let $f : X \to S$ be a morphism locally of finite presentation, $u : S \to X$ an S-section of X. For f to be lisse at a point $x \in X$, it is necessary and sufficient that u be a quasi-regular immersion in a neighbourhood of $s = f(x)$.*

Proof. The conclusion results from (17.12.1), since $f \circ u = 1_S$ is lisse. □

Proposition (17.12.4). — *Every smooth morphism $f : X \to S$ is differentially smooth (in other words (16.10.5) $\Delta_f : X \to X \times_S X$ is a quasi-regular immersion). In particular, the $\mathscr{P}^n_f = \mathscr{P}^n_{X/S}$ and the $\mathrm{gr}_n(\mathscr{P}_{X/S})$ are $\mathscr{O}_X$-Modules locally free of finite type.*

Proof. It suffices to note that the structural morphism $p_1 : X \times_S X \to X$ is lisse (17.3.3), (iii)) and that Δ_f is an X-section for p_1; the conclusion results from (17.12.3). □

Proposition (17.12.5). — *Let $f : X \to Y$ be a morphism locally of finite presentation. The following conditions are equivalent:*

a) f is differentially smooth.
b) For every morphism $g : Y' \to Y$, if one sets $X' = X \times_Y Y'$ and $f' = f_{(Y')} : X' \to Y'$, for every Y'-section s' of X', f' is lisse at all points of $s'(Y')$.
c) The second projection $p_2 : X \times_Y X \to X$ is lisse at all the points of the diagonal $\Delta_f(X)$.

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Proof. The condition c) is a particular case of b): it suffices to take $Y' = X$ in b), f' is then the second projection p_2, and Δ_f is an X-section for p_2. In second place, c) implies a), since p_2 is a morphism locally of finite presentation, and if p_2 is lisse at the points of $\Delta_f(X)$, Δ_f is a quasi-regular immersion (17.12.3), hence f is differentially smooth. It remains to show that a) implies b). One can suppose that $Y' = Y$, and $f' = f$, and hence show that if f is differentially smooth, then for every Y-section s of X, f is lisse at the points of $s(Y)$. Or, if $m : G \times_S G \to G$ is the morphism defining the group prescheme structure, $m \circ (s \times 1_G) : S \times_S G \to G \times_S G \to G$ is an S-isomorphism of G, "left translation" τ_s, transforming the identity section of G into the points of $s(S)$; hence one concludes that G is lisse on S at the points of $s(S)$. □

Corollary (17.12.6). — *With the notations being those of* (8.8.1)*, suppose X_α quasi-compact, and $f_\alpha : X_\alpha \to S_\alpha$ locally of finite presentation. For $f : X \to S$ to be differentially lisse, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $f_\lambda : X_\lambda \to S_\lambda$ be differentially lisse (in which case $f_\mu : X_\mu \to S_\mu$ is differentially lisse for $\mu \geqslant \lambda$).*

Proof. The sufficiency of the condition and the last assertion result from (16.10.4). To prove that the condition is necessary, note that $\Delta_f : X \to X \times_S X$ and $p_2 : X \times_S X \to X$ are deduced by change of base from $\Delta_{f_\lambda} : X_\lambda \to X_\lambda \times_{S_\lambda} X_\lambda$ and $p_{2,\lambda} : X_\lambda \times_{S_\lambda} X_\lambda \to X_\lambda$. By virtue of (17.12.5), for every $z \in X \times_S X$ in $\Delta_f(X)$, there exists an affine open $U(z)$ of z in $X \times_S X$ such that p_2 is lisse in $U(z)$; by virtue of (8.2.11) and (17.7.8), there exists an index $\lambda(z)$ and a neighbourhood $U_{\lambda(z)}$ of z in $X_{\lambda(z)} \times_{S_{\lambda(z)}} X_{\lambda(z)}$ such that $p_{2,\lambda(z)}$ is lisse in $U_{\lambda(z)}$. As X is quasi-compact, one can cover $\Delta_f(X)$ by a finite number of such neighbourhoods; by virtue of (8.3.4), there exists a λ superior to all the $\lambda(z_i)$ such that the inverse images of the $U_{\lambda(z_i)}$ in $X_\lambda \times_{S_\lambda} X_\lambda$ form a covering; it results then from (17.12.5) that this index λ answers the question. □

Example (17.12.7). — Let S be a prescheme, G an S-*prescheme in groups*, locally of finite presentation on S. For G to be differentially smooth on S, it is necessary and sufficient that G be lisse on S at all the points of the "identity section" e of G. The condition is indeed necessary by (17.12.5). Inversely, suppose it is fulfilled; for every morphism $S' \to S$, $G' = G \times_S S'$ is then an S'-prescheme in groups locally of finite presentation on S'; on can therefore, by virtue of (17.12.5), restrict to proving that for every S-section s of G, f is lisse at the points of $s(S)$. But if $m : G \times_S G \to G$ is the morphism that defines the group prescheme structure, $m \circ (s \times 1_G) : S \times_S G \to G \times_S G \to G$ is an S-isomorphism of G, the "left translation" τ_s, transforming the identity section of G into the points of $s(S)$; whence one concludes that G is lisse on S at the points of $s(S)$.

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Remark (17.12.8). — An S-prescheme in groups G, of finite type (and even finite) on a locally Noetherian prescheme S, can be differentially smooth on S without being lisse (or even *flat*) on S. Consider for example for S the spectrum of the dual number ring $D = k[\mathrm{T}]/\mathrm{T}^2k[\mathrm{T}]$ over a field k, for G the spectrum of the D-algebra direct composition $E = D \oplus k$. One defines on G a structure of S-scheme in groups by defining a "diagonal application" δ, homomorphism of E in $E \otimes_D E$: if e', e'' are the idempotents, canonical images of 1 of D in the factors D and k of E, one easily verifies that such an application is obtained by taking $\delta(e') = e' \otimes e' + e'' \otimes e''$ and $\delta(e'') = e' \otimes e'' + e'' \otimes e'$. The identity section of the group scheme G corresponds to the homomorphism $E \to D$ which is the identity in D and 0 in k; it is an isomorphism from S onto a connected component of G, and *a fortiori* G is differentially lisse on S (17.12.6), but it is clear that G is not S-flat.

17.13. Transversal morphisms.

(17.13.1). Let S be a prescheme, X, Y, X' three S-preschemes, $i : Y \to X$ an S-immersion, $f : X' \to X$ an S-morphism. Set $Y' = Y \times_X X'$, and let $g : Y' \to Y$, $j : Y' \to X'$ be the canonical projections, so that one has the commutative diagram

(17.13.1.1)
$$\begin{tikzcd} Y \arrow[d, "i"'] & Y' \arrow[l, "g"'] \arrow[d, "j"] \\ X & X' \arrow[l, "f"] \end{tikzcd}$$

and that j is an S-immersion. One has then a commutative diagram of $\mathscr{O}_{Y'}$-Modules quasi-coherent

(17.13.1.2)
$$\begin{tikzcd} g^*(\mathscr{N}_{Y/X}) \arrow[r] & f^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \arrow[r] & g^*(\Omega^1_{Y/S}) \arrow[r] & 0 \\ \mathscr{N}_{Y'/X'} \arrow[r] \arrow[u, "\mathrm{gr}_1(g)"] & \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \arrow[r] \arrow[u] & \Omega^1_{Y'/S} \arrow[r] \arrow[u] & 0 \end{tikzcd}$$

where the lower line is the exact suite (16.4.21) applied to j, the upper line comes from the same exact suite for i, by application of the exact functor to the right g^* (which thus leaves it exact); $\mathrm{gr}_1(g)$ is defined in (16.2.1), and the commutativity of the two squares results from (**0**, (20.5.7.3)) and (**0**, (20.5.11.3)). We have seen moreover ((16.2.2), (iii)) that $\mathrm{gr}_1(g)$ is here surjective, from which one deduces from (17.13.1.2) the exact sequence

(17.13.1.3)
$$g^*(\mathscr{N}_{Y/X}) \xrightarrow{\alpha} \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \longrightarrow \Omega^1_{Y'/S} \longrightarrow 0.$$

Proposition (17.13.2). — *With the notations of (17.13.1), let x' be a point of Y', $x = g(x')$ its image in Y; suppose X and Y lisse on S at the point x, X' lisse on S at the point x'. Let m, $m - c$ be the relative dimensions of X and Y on S at the point x (17.10.1), n that of X' on S at the point x'. The following conditions are equivalent:*

a) Y' is lisse on S at the point x' and of relative dimension $n - c$.

b) The homomorphism $\alpha \otimes 1 : g^*(\mathscr{N}_{Y/X}) \otimes_{\mathscr{O}_{Y'}} k(x') \to \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} k(x')$ deduced from the homomorphism α of (17.13.1.3) is injective.

Let s be the image of x in S, and suppose that x' is rational on $k(s)$ (s being the image of x' in S). Then the conditions a) and b) are also equivalent to the following:

b') The transposed homomorphism of $\alpha \otimes 1$

$$\mathrm{T}_{X'/S}(x') \longrightarrow (\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} k(x'))^\vee = \mathrm{T}_{X/S}(x)/\mathrm{T}_{Y/S}(x)$$

(cf. (16.5.12)) is surjective.

Definition (17.13.3). — With the notations of (17.13.1), we say that the morphism f is *transversal* to Y at the point x', relatively to S, if X and Y are lisse on S at the point x, X' lisse on S at the point x', and the equivalent conditions a), b) of (17.13.2) are satisfied. When there is no confusion to fear, we suppress the mention of the prescheme S and simply say that f is transversal to Y at the point x'.

Remarks (17.13.4). — (i) Suppose that X, Y and X' are flat and locally of finite presentation on S. For every $s \in S$, denote X_s, Y_s, X'_s, Y'_s the fibres of X, Y, X', Y' at the point s, $f_s : X'_s \to X_s$ the morphism deduced from f by change of base. It results then from (17.5.1) that for f to be a transversal morphism to Y at the point x', it is necessary and sufficient that, if s is the image of x in S, f_s be transversal to Y_s at the point x', relatively to $k(s)$.[5]

(ii) One has seen in (17.13.2) that the set of points $x' \in Y'$ where f is transversal is open in Y'; if Y' is moreover *proper* on S, one deduces that the set of $s \in S$ such that f_s is transversal to Y at *all* the points of Y'_s, relatively to S, is *open* in S.

[5][Trans.] The original reference (17.5.9) is an erratum in the original; there is no item 17.5.9. The correct reference is (17.5.1).

(iii) The property of being lisse in a point, as well as the notion of relative dimension on S in a point, being stable by every change of base $S' \to S$ ((17.3.3) and (4.2.7)), the same holds for the property of a morphism $f : X' \to X$ of being transversal to a sub-prescheme of X at a point, relatively to S.

(iv) The condition b) of (17.13.2) also expresses that the homomorphism $\alpha : g^*(\mathscr{N}_{Y/X}) \to \Omega^1_{X/S} \otimes_{\mathscr{O}_x} \mathscr{O}_{Y'}$ is *universally injective* with respect to Y' (11.9.18).

(17.13.5). Consider now a prescheme S, three S-preschemes X, Y, Z, two S-morphisms $f : Y \to X$, **IV | 92**
$g : Z \to X$; set $T = Y \times_X Z$; one then knows (**I**, 5.3.5) that there is a commutative diagram

(17.13.5.1)
$$\begin{tikzcd} X & T = Y \times_X Z \arrow[l] \\ X \times_S X \arrow[u, "\Delta"] & Y \times_S Z \arrow[l, "u"] \arrow[u, "j"'] \end{tikzcd}$$

making T the product of the $(X \times_S X)$-preschemes X and $Y \times_S Z$, where $u = f \times_S g$. As Δ is an S-immersion (**I**, 5.3.9), one is in the situation of the diagram (17.13.1.1); what corresponds to $\Omega^1_{X/S}$ in (17.13.1) is then $(\Omega^1_{Y/S} \otimes_{\mathscr{O}_y} \mathscr{O}_{Y \times_S Z}) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_z} \mathscr{O}_{Y \times_S Z})$ by (16.4.23). On the other hand, what corresponds to the $\mathscr{O}_Y$-Module $\mathscr{N}_{Y/X}$ in (17.13.1) is here by definition $\Omega^1_{X/S}$ (16.3.1); there therefore corresponds to (17.13.1.3) an exact sequence

(17.13.5.2)
$$\Omega^1_{X/S} \otimes_{\mathscr{O}_x} \mathscr{O}_T \xrightarrow{\rho} (\Omega^1_{Y/S} \otimes_{\mathscr{O}_y} \mathscr{O}_T) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_z} \mathscr{O}_T) \xrightarrow{\sigma} \Omega^1_{T/S} \longrightarrow 0$$

where it remains to make the homomorphisms ρ and σ precise. Taking into account first (16.4.23) and (**0**, 20.5.2), one sees that if $p : T \to Y$, $q : T \to Z$ are the canonical projections, one has, with the notations of (16.4.19),

(17.13.5.3)
$$\sigma = p_{T/Y/S} + q_{T/Z/S}.$$

On the other hand, to evaluate ρ, one uses the commutativity of the left square in (17.13.1.2), which in the present case reduces first to explicating the canonical homomorphism $\rho' : \Omega^1_{X/S} \to \Omega^1_{(X \times_S X)/S} \otimes_{\mathscr{O}_{X \times_S X}} \mathscr{O}_X$ defined in (16.4.21) applied to the immersion Δ. One can reduce to the case where $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ are affine, and then ρ' corresponds to the homomorphism δ of (**0**, 20.5.11.2); one sees that ρ' is the difference of the two homomorphisms $\pi^{(1)}$ and $\pi^{(2)}$ of $\Omega^1_{X/S}$ into $\Omega^1_{(X \times_S X)/S} \otimes_{\mathscr{O}_{X \times_S X}} \mathscr{O}_X$, corresponding respectively to the first and second projections of $X \times_S X$ into X, by (16.4.3.3). To obtain ρ, one must then consider the homomorphism $\rho'' : \Omega^1_{(X \times_S X)/S} \otimes_{\mathscr{O}_{X \times_S X}} \mathscr{O}_{(Y \times_S Z)} \to \Omega^1_{(Y \times_S Z)/S}$ corresponding by (16.4.3.3) to the morphism u of (17.13.5.1), then, after tensorisation by $\mathscr{O}_T$, form the composition $(\rho'' \otimes 1) \circ (\rho' \otimes 1)$; it follows from what precedes that one has, with the notations of (16.4.18),

(17.13.5.4)
$$\rho = (f_{Y(X/S)} \otimes 1_{\mathscr{O}_T}) \oplus (-g_{Z(X/S)} \otimes 1_{\mathscr{O}_T}).$$

Applying (17.13.2) to the situation of the diagram (17.13.5.1) (taking into account ((17.3.3), (iv)), **IV | 93**
which implies that $X \times_S X$ is smooth over S at x if X is so) gives the

Corollary (17.13.6). — *Let S be a prescheme, X, Y, Z three S-preschemes, $f : Y \to X$, $g : Z \to X$ two S-morphisms; set $T = Y \times_X Z$, and let $p : T \to Y$, $q : T \to Z$ be the canonical projections. Let t be a point of T, $y = p(t)$, $z = q(t)$, $x = f(y) = g(z)$. Suppose that X is smooth over S at the point x, of relative dimension m, Y (resp. Z) smooth over S at the point y (resp. z), of relative dimension $m + a$ (resp. $m + b$), a and b being positive or negative integers. Then the following conditions are equivalent:*

a) T is smooth over S at the point t, of relative dimension $m + a + b$.

b) The homomorphism

$$\rho \otimes 1 : \Omega^1_{X/S} \otimes_{\mathscr{O}_x} k(t) \longrightarrow (\Omega^1_{Y/S} \otimes_{\mathscr{O}_y} k(t)) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_z} k(t))$$

where ρ is given by (17.13.5.4), is injective.

c) The morphism $T = Y \times_S Z \to X \times_S X$ is transversal to the diagonal of $X \times_S X$ at the point t.

If t is rational over the residue field of its image s in S, these conditions are also equivalent to the following:

b′) The homomorphism

$$\mathrm{T}_y(f) - \mathrm{T}_s(g) : \mathrm{T}_{Y/S}(y) \oplus \mathrm{T}_{Z/S}(z) \longrightarrow \mathrm{T}_{X/S}(x) \tag{17.13.6.1}$$

(cf. (16.5.12.5)*) is surjective.*

Moreover, when conditions a) *and* b) *are verified at t, they are so in a neighbourhood of t in T, and restricting T to such a neighbourhood, the sequence*

$$0 \longrightarrow \Omega^1_{X/S} \otimes_{\mathscr{O}_x} \mathscr{O}_T \xrightarrow{\rho} (\Omega^1_{Y/S} \otimes_{\mathscr{O}_y} \mathscr{O}_T) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_z} \mathscr{O}_T) \xrightarrow{\sigma} \Omega^1_{T/S} \longrightarrow 0 \tag{17.13.6.2}$$

(where σ is given by (17.13.5.3)) is exact.

The only point remaining to prove in (17.13.6) concerns b′), and results from the fact that, under the hypothesis that t is rational over $k(s)$, the homomorphism $\rho \otimes 1$ is the transpose *of $\mathrm{T}_y(f) - \mathrm{T}_s(g)$, by (17.13.5.4).*

When the equivalent conditions of (17.13.6) are satisfied, one says that f and g form a couple of transversal morphisms at the point t, relatively to S*; here again, one often suppresses the mention of S.*

(17.13.7). Consider in particular the case where Y and Z are *sub-preschemes* of X, f and g being the canonical injections; T is then the sub-prescheme "intersection" $\inf(Y, Z)$ of X (**I**, 4.4.3), and one has $t = y = z = x$. Instead of saying that the couple (f, g) is transversal at the point x, one says then that Y and Z *cut transversally at the point x* (relatively to S). Noting X_s, Y_s, Z_s, T_s the fibres of X, **IV | 94**
Y, Z, T at the point s, image of x in S, and taking into account ((5.2.3)), ((5.1.9)) and (**0**, 16.5.12), one sees that for this (when X, Y, Z are smooth over S at the point x) it suffices that T be smooth over S at the point x, and that one has the relation

$$\operatorname{codim}_x(\mathrm{T}_s, \mathrm{X}_s) = \operatorname{codim}_x(\mathrm{Y}_s, \mathrm{X}_s) + \operatorname{codim}_x(\mathrm{Z}_s, \mathrm{X}_s). \tag{17.13.7.1}$$

Proposition (17.13.8). — *Let S be a prescheme, X an S-prescheme locally of finite presentation over S, Y, Z two sub-preschemes of X, $T = \inf(Y, Z)$ the sub-prescheme "intersection" of Y and Z, x a point of T. The following conditions are equivalent:*

- *a) The canonical injection $i : Y \to X$ is a transversal morphism to Z at the point x, relatively to S (17.13.3).*
- *a′) The canonical injection $j : Z \to X$ is a transversal morphism to Y at the point x, relatively to S.*
- *a″) Y and Z cut transversally at the point x, relatively to S.*
- *b) The homomorphism*

$$\rho \otimes 1 : \Omega^1_{X/S} \otimes_{\mathscr{O}_x} k(x) \longrightarrow (\Omega^1_{Y/S} \otimes_{\mathscr{O}_y} k(x)) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_z} k(x))$$

is injective.

When x is rational over the residue field $k(s)$ (s image of x in S), these conditions are also equivalent to the following:

b′) The homomorphism

$$\mathrm{T}_x(i) - \mathrm{T}_x(j) : \mathrm{T}_{Y/S}(x) \oplus \mathrm{T}_{Z/S}(x) \longrightarrow \mathrm{T}_{X/S}(x)$$

is surjective.

Moreover, when conditions a) *and* b) *are verified at x, they are so in a neighbourhood of x in T, and restricting X to a neighbourhood of x, the sequence (17.13.6.2) is exact, and one has a canonical isomorphism*

$$\mathscr{N}_{T/X} \xrightarrow{\sim} (\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_y} \mathscr{O}_T) \oplus (\mathscr{N}_{Z/X} \otimes_{\mathscr{O}_z} \mathscr{O}_T). \tag{17.13.8.1}$$

Proof. The conditions a), a'), a'') all imply that X, Y, Z are smooth over S at the point x. Let more- **IV | 95**
over m, $m - a$, $m - b$ be the relative dimensions of X, Y, Z over S at the point x. It results then from (17.13.2) applied replacing X' by Z and Y' by $T = Y \times_X Z$, that the conditions a), a') are both equivalent to saying that T is smooth over S at the point x and of relative dimension $m - a - b$ at this point; but by (17.13.7), this means precisely that condition a'') is verified, whence the equivalence of a), a') and a''). The equivalence of a'') and b) (or b') when x is rational over $k(s)$) has been proved in (17.13.6), as well as the fact that if these conditions are satisfied at the point x, they are so in a neighbourhood of x, and the exactness of the sequence (17.13.6.2) in such a neighbourhood. It remains to define the canonical isomorphism (17.13.8.1). Let us denote α and $-\beta$ the homomorphisms appearing in the second member of (17.13.5.4), γ and δ those appearing in the

second member of (17.13.5.2). Saying that the sequence (17.13.6.2) is exact signifies that, in the category of $\mathscr{O}_T$-Modules, $\Omega^1_{X/S} \otimes_{\mathscr{O}_x} \mathscr{O}_T$ identifies canonically with the *fibre product* of $\Omega^1_{Y/S} \otimes_{\mathscr{O}_y} \mathscr{O}_T$ and $\Omega^1_{Z/S} \otimes_{\mathscr{O}_z} \mathscr{O}_T$ over $\Omega^1_{T/S}$, for the homomorphisms γ and δ. The same reasoning as in (**0**, 18.1.2 and 18.1.3), where one replaces rings and ideals respectively by Modules and sub-Modules, furnishes a commutative diagram where the 3rd and 4th rows and columns are exact by the smoothness hypotheses. The fact that the composed homomorphism $\varepsilon = \gamma \circ \alpha = \delta \circ \beta$ is the homomorphism corresponding to the canonical injection $T \to X$ results from (**0**, 20.5.2.7). As the diagonal of the preceding diagram is exact, $\mathrm{Ker}(\varepsilon)$ identifies canonically with $\mathscr{N}_{T/X}$, taking into account that T is smooth over S (17.2.5); hence one has a canonical isomorphism (17.13.8.1) which is the reciprocal of ι. □

(17.13.9). One can generalise the results of (17.13.5) and (17.13.6) to a finite number of S-morphisms $f_i : Y_i \to X$ ($i \in I$, I a finite set). For this, let X^I be the product of the S-preschemes Y_i (for the composed morphisms $Y_i \to X \to S$), T the product of the X-preschemes Y_i (for the morphisms f_i). Let then Y be the product of the X-preschemes Y_i (for the composed morphisms $Y_i \xrightarrow{f_i} X \to S$), T the product of the X-preschemes Y_i (for the morphisms f_i). One proves as in (**I**, 5.3.5) that one has a commutative diagram

(17.13.9.1)
$$\begin{tikzcd} X & T \arrow[l, "v"'] \\ X^I \arrow[u, "\Delta"] & Y \arrow[l, "u"] \arrow[u, "j"'] \end{tikzcd}$$

(where u is the product (over S) of the f_i), making T the product of the X^I-preschemes X and Y. By recurrence on the number of elements of I, it results from (16.4.23) that $\Omega^1_{X^I/S}$ identifies canonically to the direct sum of the $p_i^*(\Omega^1_{X/S})$ (resp. of the $q_i^*(\Omega^1_{Y_i/S})$), where $q_i : Y \to Y_i$ are the canonical projections. Suppose now that X is smooth over S at a point x (hence in a neighbourhood of this point ((17.3.3), (iv))), and restricting X to a neighbourhood of x, one has ((17.2.5)) an exact sequence of $\mathscr{O}_X$-Modules locally free $0 \to \mathscr{J} \to (\Omega^1_{X/S})^I \to \Omega^1_{X/S} \to 0$. Setting $\mathscr{J} = \mathscr{J}_1(\Delta)$; one deduces from the preceding sequence an exact sequence of $\mathscr{O}_T$-Modules locally free IV | 96

$$0 \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_x} \mathscr{O}_T \longrightarrow (\Omega^1_{X/S})^I \otimes_{\mathscr{O}_x} \mathscr{O}_T \xrightarrow{\sigma} \Omega^1_{X/S} \otimes_{\mathscr{O}_x} \mathscr{O}_T \longrightarrow 0$$

where σ is none other than the canonical homomorphism which, to each family of sections $(t'_i)_{i\in I}$ of $\Omega^1_{X/S}$ above an open of T, sends their *sum*. On the other hand, what corresponds here to the second vertical arrow of the diagram (17.13.1.2) is the homomorphism

(17.13.9.2)
$$\tau : (\Omega^1_{X/S})^I \otimes_{\mathscr{O}_x} \mathscr{O}_T \longrightarrow \bigoplus_{i\in I} (\Omega^1_{Y_i/S} \otimes_{\mathscr{O}_{y_i}} \mathscr{O}_T)$$

which, to every family $(t'_i \otimes 1)_{i\in I}$, associates the sum of the $((f_i)_{Y_i(X/S)}(t'_i)) \otimes 1$, with the notation of (16.4.18). The homomorphism ρ which corresponds to α of (17.13.1.3) is then the *restriction* to $\mathscr{J} \otimes_{\mathscr{O}_x} \mathscr{O}_T$ of the preceding homomorphism τ. These hypotheses and notations being specified, one can apply to the situation of (17.13.9.1) the proposition (17.13.2), which gives the

Corollary (17.13.10). — *Under the general hypotheses of (17.13.9), let t be a point of T, y_i ($i \in I$) its projections in the Y_i, x the point of X equal to each of the $f_i(y_i)$. Suppose that X is smooth over S at the point x and of relative dimension d, and that each of the Y_i is smooth over S at the point y_i; let $d + c_i$ (c_i a positive or negative integer) be the relative dimension of Y_i at the point y_i. Then the following conditions are equivalent:*

a) T is smooth over S at the point t, of relative dimension $d + \sum_{i\in I} c_i$.
b) The homomorphism

$$\rho \otimes 1 : \mathscr{J} \otimes_{\mathscr{O}_x} k(t) \longrightarrow \bigoplus_{i\in I} (\Omega^1_{Y_i/S} \otimes_{\mathscr{O}_{y_i}} k(t))$$

is injective.

When t is rational over the residue field $k(s)$ (s the image of t in S), these conditions are also equivalent to the following: IV | 97 *b′) The homomorphism transpose of* $\rho \otimes 1$

$$\bigoplus_{i\in I} \mathrm{T}_{Y_i/X}(y_i) \longrightarrow (\mathrm{T}_{X/S}(x))^I/\delta(\mathrm{T}_{X/S}(x))$$

(where δ *is the diagonal application) is surjective.*

It suffices to note that X^I *is smooth over* S *at the point* x *and of relative dimension* $d\cdot\mathrm{Card}(I)$*, and* Y *smooth over* S *at the point* t *and of relative dimension* $d\cdot\mathrm{Card}(I)+\sum_{i\in I} c_i$*.*

When the equivalent conditions of (17.13.10) are verified, one says further that the f_i *form a* family of transversal morphisms at the point t, relatively to S. *One sees further that the set of points* $t\in T$ *where this holds is* open *in* T*. The remark ((17.13.4), (ii)) shows moreover that if* X *and the* Y_i *are flat and locally of finite presentation over* S*, and if moreover* T *is* proper *over* S*, the set of* $s\in S$ *such that the* f_i *form a family of transversal morphisms at* all *the points of* T_s*, relatively to* S*, is* open *in* S*.*

(17.13.11). Consider in particular the case, generalising (17.13.7), where the Y_i ($i\in I$) are *sub-preschemes* of X, the f_i being the canonical injections, so that $T=\inf_{i\in I}(Y_i)$ is also the sub-prescheme "intersection" of the Y_i, and $t=y_i=x$ for every i; instead of saying that the f_i form a family of transversal morphisms at the point x, one says further that the Y_i *cut transversally at the point* x (relatively to S). The condition a) of (17.13.10) is expressed further by the relation that generalises (17.13.7.1):

(17.13.11.1)
$$\mathrm{codim}_x(\mathrm{T}_s,\mathrm{X}_s)=\sum_{i\in I}\mathrm{codim}_x((\mathrm{Y}_i)_s,\mathrm{X}_s).$$

Moreover, one has the following property which extends (17.13.8.1), and of which we give another demonstration when I has 2 elements:

Corollary (17.13.12). — *When the* Y_i *cut transversally at the point* x*, one has a canonical isomorphism*

(17.13.12.1)
$$\mathscr{N}_{T/X}\xrightarrow{\sim}\bigoplus_{i\in I}(\mathscr{N}_{Y_i/X}\otimes_{\mathscr{O}_{y_i}}\mathscr{O}_T).$$

Proof. One can reduce to the case where the Y_i are closed sub-preschemes of X, defined by quasi-coherent Ideals $\mathscr{J}_i$, so that T is defined by the Ideal $\mathscr{J}=\sum_i\mathscr{J}_i$. By definition of the conormal sheaf of an immersion (16.1.3), the canonical homomorphism $\bigoplus_i\mathscr{J}_i\to\mathscr{J}$ gives, by passage to quotients, a *surjective* homomorphism

(17.13.12.2)
$$\bigoplus_{i\in I}(\mathscr{N}_{Y_i/X}\otimes_{\mathscr{O}_{y_i}}\mathscr{O}_T)\longrightarrow\mathscr{N}_{T/X}.$$

But here the $\mathscr{O}_T$-Modules on both sides of (17.13.12.2) are locally free and of the same rank (since c_i is the rank of $\mathscr{N}_{Y_i/X}$), by (17.2.5) and the condition a) of (17.13.10); one concludes then from Bourbaki, *Alg. comm.*, chap. II, §3, nº 2, cor. de la prop. 6, that (17.13.12.2) is bijective, and (17.13.12.1) is the reciprocal isomorphism. □ IV | 98

17.14. Local and infinitesimal characterisations of smooth morphisms, of non ramified morphisms and of étale morphisms. IV | 98

Proposition (17.14.1). — *Let* $f:X\to Y$ *be a morphism locally of finite presentation,* x *a point of* X*,* $y=f(x)$*. For* f *to be lisse (resp. non ramified, resp. étale) at the point* x*, it is necessary and sufficient that the following condition be satisfied:*

For every local scheme $Y'=\mathrm{Spec}(A')$ *of closed point* y'*, every morphism* $h:Y'\to Y$ *such that* $h(y')=y$*, every closed sub-scheme* $Y'_0=\mathrm{Spec}(A'/\mathfrak{J})$ *of* Y'*, where* $\mathfrak{J}$ *is an Ideal of square zero, and every* Y*-morphism* $g_0:Y'_0\to X$ *such that* $g_0(y')=x$*, there exists* at least one *(resp.* at most one*, resp.* a unique*)* Y*-morphism* $g:Y'\to X$ *whose restriction to* Y'_0 *is* g_0*.*

Proof. Taking into account the definitions (17.1.1) and (17.3.1), it suffices to show that the condition is sufficient in the cases of smooth morphisms and non ramified morphisms respectively.

(i) Case of smooth morphisms. One can restrict to the case where $Y=\mathrm{Spec}(A)$ and $X=\mathrm{Spec}(C)$ are affine, with $B=A[\mathrm{T}_1,\ldots,\mathrm{T}_n]$ an A-algebra of polynomials and C a quotient $B/\mathfrak{R}$, $\mathfrak{R}$ being a finitely generated ideal of B. Using (17.7.2), (ii)) and (17.7.9), one can further suppose S, X and Y are *Noetherian*; finally by virtue of ($\mathbf{0_{III}}$, (9.2.3)), one is reduced to the Noetherian case. It suffices to establish that $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-algebra formally smooth for the *pre-adic topologies*. But as $\mathfrak{m}'$ is

nilpotent, one sees that the condition of the statement is satisfied when one replaces the condition $\mathfrak{m}'\mathfrak{J} = 0$ by $\mathfrak{J} \subset \mathfrak{m}'$; and considering the rings $A'/\mathfrak{m}'^i\mathfrak{J}$ and reasoning by recurrence on j, this identity results from the hypothesis applied, by recurrence on n, to $A' = C/\mathfrak{r}^n$ and $\mathfrak{J}' = (r^n + (\mathfrak{J}/\mathfrak{J}^2))/\mathfrak{r}^n$.

(ii) Case of non ramified morphisms. We take up the notations of the proof of (17.14.1), (ii)); to prove that $\mathfrak{J}/\mathfrak{J}^2$ of C is null, note that C is then a local Noetherian ring, being the local ring of an affine open sub-prescheme of $X \times_Y X$, which is locally of finite type on X, hence on Y. The hypothesis, applied to $A' = C$ and $\mathfrak{J}' = \mathfrak{J}/\mathfrak{J}^2$, shows that the two applications $B \xrightarrow{p_1} C \to C/\mathfrak{r}^n$ and $B \xrightarrow{p_2} C \to C/\mathfrak{r}^n$ are identical; but as $C/\mathfrak{r}^n$ is artinian, this identity results from the hypothesis applied to $A' = C/\mathfrak{r}^n$, hence $\Omega^1_{B/A} = 0$ by (**0**, (20.4.7)). ☐

Proposition (17.14.2). — *Let Y be a prescheme locally Noetherian, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. For f to be lisse (resp. non ramified, resp. étale) at the point x, it is necessary and sufficient that it satisfies the condition of* (17.14.1) *where one supposes that the local ring A' is artinian, that $\mathfrak{m}'\mathfrak{J} = 0$, where $\mathfrak{m}'$ is the maximal ideal of A', and that the residue field $A'/\mathfrak{m}'$ of A' is equal to $k(x)$.*

Remark (17.14.3). — One notes that, under the hypotheses of (17.14.2), if moreover the residue field $k(x)$ is a *finite* extension of $k(y)$, then the A'-modules $\mathfrak{m}'^j/\mathfrak{m}'^{j+1}$ are $k(y)$-vector spaces of finite rank, hence *a fortiori* A' is an $\mathscr{O}_{Y,y}$-algebra finite.

17.15. The case of preschemes over a base field. We recall first (6.7.7), (6.7.8) and (6.8.1) the **IV** | 99

Proposition (17.15.1). — *Let k be a field, X a prescheme locally of finite type on k, x a point of X. For X to be smooth at a point x, it is necessary and sufficient that for every finite radiciel extension k' of k (or only for every finite radiciel extension k' of k), the local ring $\mathscr{O}_{X,x} \otimes_k k'$ be regular. If $k(x)$ is a separable extension of k (in particular if k is perfect), this amounts to saying that $\mathscr{O}_{X,x}$ is regular.*

Corollary (17.15.2). — *Under the conditions of* (17.15.1), *for X to be lisse on k, it is necessary and sufficient that X be geometrically regular on k* (6.7.6), *which implies that X is reduced. If k is perfect, X is lisse on k if and only if X is regular.*

Proposition (17.15.3). — *Let k be a field, X a prescheme locally of finite type on k, x a point of X, $n = \dim_x(X)$. Let $(g_i)_{1 \leqslant i \leqslant n}$ be a family of n sections of $\mathscr{O}_X$ above X, and let $f : X \to Y = \operatorname{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n])$ be the morphism corresponding to the k-homomorphism $k[\mathrm{T}_1, \dots, \mathrm{T}_n] \to \Gamma(X, \mathscr{O}_X)$ sending T_i to g_i. The following conditions are equivalent (and imply that X is lisse on k at the point x and hence regular at the point x):*

a) f is étale at the point x.
b) The images $d_{X/k}g_i$ form a base of the $\mathscr{O}_{X,x}$-module $(\Omega^1_{X/k})_x$.
c) The images $d_{X/k}g_i$ engender the $\mathscr{O}_{X,x}$-module $(\Omega^1_{X/k})_x$.

If f is étale at the point x, X is lisse on k at the point x since $Y = \operatorname{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n])$ is lisse on k (17.3.8); *the fact that a) implies b) is thus a particular case of* (17.11.3). *As b) trivially implies c), it remains to see that c) implies a).*

Proof. Note first that hypothesis c) entails that the homomorphism $(f^*(\Omega^1_{Y/k}))_x \to (\Omega^1_{X/k})_x$ is surjective, hence, replacing X by an open neighbourhood of x, one can suppose that the homomorphism $f^*(\Omega^1_{Y/k}) \to \Omega^1_{X/k}$ is surjective (Bourbaki, *Alg. comm.*, chap. II, § 5, nº 1, prop. 2); hence f is non ramified ((17.2.2)). Let us first show that one can reduce to the case where x is rational over k. Indeed, if one sets $k' = k(x)$ and $X' = X \otimes_k k'$, $Y' = Y \otimes_k k' = \operatorname{Spec}(k'[\mathrm{T}_1, \dots, \mathrm{T}_n])$, there exists a point $x' \in X'$ above x such that $k(x') = k'$. To prove that f is étale at the point x, it suffices to show that $f' = f_{(k')} : X' \to Y'$ is étale at the point x' ((17.7.1), (ii)); moreover f' is non ramified ((17.3.3), (iii)) and one has $\dim_{x'}(X') = n$ ((4.2.7)). In the same way, replacing k' by an algebraically closed extension of k', one can suppose that k is algebraically closed. Set then $y = f(x)$, $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$; as the residue field of B is equal to k, the same is true of that of A, hence x (resp. y) is a closed point of X (resp. Y) ((**I**, 6.4.2)) and one has therefore $\dim(A) = \dim(B) = n$. Since f is non ramified, the homomorphism $\widehat{A} \to \widehat{B}$ is surjective ((17.4.4), f'')); but as $\dim(\widehat{A}) = \dim(\widehat{B}) = n$, and $\widehat{A}$, being a regular ring, is integral, the homomorphism $\widehat{A} \to \widehat{B}$ is also injective (**0**, 16.3.10).

This homomorphism is therefore bijective, which completes the proof that f is étale at the point x ((17.6.3), e'')). □

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Corollary (17.15.4). — *Under the general hypotheses of (17.15.3), suppose moreover that $k(x)$ is a finite and separable extension of k and that the germs $(g_i)_x$ belong to $\mathfrak{m}_x$. Then the conditions a), b), c) of (17.15.3) are also equivalent to each of the following:*

d) The n germs $(g_i)_x$ engender the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_{X,x}$.

*d′) The ring $\mathscr{O}_{X,x}$ is regular and the $(g_i)_x$ form a regular system of parameters for this ring (**0**, (17.1.6)).*

Proof. Indeed, d') trivially implies d). On the other hand, as $k(x)$ is a finite extension and separable of k, one has $\Omega^1_{k(x)/k} = 0$ (**0**, (20.6.20)) and $k(x) = \mathscr{O}_{X,x}/\mathfrak{m}_x$ is a k-algebra formally lisse for the discrete topologies (**0**, (19.6.1)), hence the exact suite (**0**, (20.5.14.1)) applies with $A = k$, $B = \mathscr{O}_{X,x}$, $\mathfrak{R} = \mathfrak{m}_x$, and furnishes an isomorphism canonical

$$\delta : \mathfrak{m}_x/\mathfrak{m}_x^2 \xrightarrow{\sim} (\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x).$$

From this one first deduces the equivalence of conditions c) and d), by virtue of the Nakayama lemma. On the other hand, if f is étale at the point x, the ring $\mathscr{O}_{X,x}$ is regular and of dimension n (**I**, (6.4.2)), and n elements that engender this ideal maximal form necessarily a regular system of parameters for $\mathscr{O}_{X,x}$ (**0**, (17.1.6)), which proves that a) implies d'). □

Proposition (17.15.5). — *Let k be a field, X a prescheme locally of finite type on k, x a point of X, $n = \dim_x(X)$. The following conditions are equivalent:*

a) X is lisse on k at the point x.

b) X is differentially smooth on k at the point x.

c) $(\Omega^1_{X/k})_x$ is an $\mathscr{O}_{X,x}$-module libre of rank n.

d) $(\Omega^1_{X/k})_x$ is an $\mathscr{O}_{X,x}$-module admitting a system of n generators.

Proof. We have already proved that a) implies c) (17.10.2); c) implies d) trivially; c) also implies b) (17.12.4). The fact that d) implies a) results from (**0**, (20.4.7) and (17.15.3), c)). The fact that b) implies a) is shown in the proof of (17.15.3). □

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Corollary (17.15.6). — *Let X be a prescheme of finite type on a field k. For X to be lisse on k, it is necessary and sufficient that the $\mathscr{O}_X$-Module $\Omega^1_{X/k}$ be locally free, and that the local rings of X at maximal points be fields, separable extensions of k (the latter condition being automatically satisfied if k is a perfect field and X a* reduced *prescheme).*

Proof. The conditions are necessary, for if X is lisse on k, it results from (17.10.2) that $\Omega^1_{X/k}$ is locally free; on the other hand, X is regular, hence *a fortiori* reduced, hence at every maximal point x of X, $\mathscr{O}_{X,x}$ is a field, which must be a k-algebra formally smooth for the discrete topologies ((17.5.1)), hence a separable extension of k (**0**, 19.6.1).

The conditions are sufficient. One can indeed reduce to the case where X is connected, hence $\Omega^1_{X/k}$ is locally free of constant rank n. For every maximal point x of X, one then has $\mathrm{rg}_{k(x)}\, \Omega^1_{k(x)/k} = n$ since $\mathscr{O}_{X,x} = k(x)$; as by hypothesis $k(x)$ is a separable extension of k, one has $\Upsilon_{k(x)/k} = 0$ (**0**, 20.6.3), hence $\deg.\mathrm{tr}_k\, k(x) = n$ by Cartier's equality (**0**, 21.7.1). All the irreducible components of X therefore have the same dimension n ((5.2.1)), and one concludes that X is lisse on k at every point by virtue of the fact that c) implies a) in ((17.15.5)). □

Corollary (17.15.7). — *If k is of characteristic 0, then, for X to be lisse on k at a point x, it is necessary and sufficient that $(\Omega^1_{X/k})_x$ be an $\mathscr{O}_{X,x}$-module libre.*

Proof. This results from (16.12.2) and the equivalence of b) and a) in (17.15.5). □

Proposition (17.15.8). — *Let k be a field, X a prescheme locally of finite type on k, x a point of X; set $n = \dim_x(X)$, $r = \dim(\mathscr{O}_{X,x})$, so that $n - r = \deg.\mathrm{tr}_k\, k(x)$ (5.2.3). Let $(g_i)_{1\leqslant i\leqslant n}$ be a family of n sections of $\mathscr{O}_X$ above X, such that $(g_i)_x \in \mathfrak{m}_x$ for $1 \leqslant i \leqslant r$; let $f : X \to Y = \mathrm{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n])$*

be the morphism corresponding to the k-homomorphism $k[\mathrm{T}_1, \ldots, \mathrm{T}_n] \to \Gamma(X, \mathscr{O}_X)$ sending T_i to g_i. The following conditions are equivalent (and imply that X is lisse on k at the point x and that $k(x)$ is a separable extension of k):

a) f is étale at the point x.

b) The $(g_i)_x$ $(1 \leqslant i \leqslant r)$ engender $\mathfrak{m}_x$ and the images in $\Omega^1_{k(x)/k}$ of the $(d_{X/k}g_i)_x$ for $r+1 \leqslant i \leqslant n$ engender $\Omega^1_{k(x)/k}$.

Corollary (17.15.9). — *Let X be a prescheme locally of finite type on a field k, x a point of X; set $n = \dim_x(X)$, $r = \dim(\mathscr{O}_{X,x})$. The following conditions are equivalent:*

a) The ring $\mathscr{O}_{X,x}$ is regular and $k(x)$ is a separable extension of k.

b) X is lisse on k at the point x, and the canonical homomorphism

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x) \longrightarrow \Omega^1_{k(x)/k} \longrightarrow 0$$

is exact.

c) There exist n sections g_i of $\mathscr{O}_X$ above an open neighbourhood U of x, such that $(g_i)_x \in \mathfrak{m}_x$ for $1 \leqslant i \leqslant r$ and the morphism $U \to \mathrm{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n])$ corresponding to the g_i (cf. (17.15.8)) is étale at the point x.

d) There exist n sections g_i $(1 \leqslant i \leqslant n)$ of $\mathscr{O}_X$ above an open neighbourhood U of x, such that $(g_i)_x$ $(1 \leqslant i \leqslant r)$ engender $\mathfrak{m}_x$ and that the images in $\Omega^1_{k(x)/k}$ of the $(d_{U/k}g_i)_x$ for $r+1 \leqslant j \leqslant n$ engender $\Omega^1_{k(x)/k}$.

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Remarks (17.15.10). — (i) For a prescheme X of finite type over k to be lisse over k, it does not suffice that $\Omega^1_{X/k}$ be locally free, as shown by the example where $X = \mathrm{Spec}(K)$, K being a finite non separable extension of k.

(ii) When k is not perfect, it may happen that X is lisse over k without $k(x)$ being separable over k. One has an example of this by taking $X = \mathrm{Spec}(k[\mathrm{T}])$ and for x the point corresponding to the principal prime ideal $(\mathrm{T}^p - \lambda)$ ($p > 0$ characteristic of k, $\lambda \in k - k^p$).

(iii) However, if X is a prescheme lisse over k, the set of closed points of X such that $k(x)$ is separable (and finite) over k is *dense* in X. Indeed, let $f : X \to \mathrm{Spec}(k)$ be the structural morphism; for every $x_0 \in X$, there is an open neighbourhood U of x_0 in X and a factorisation of $f|U : U \xrightarrow{g} \mathrm{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n]) \xrightarrow{h} \mathrm{Spec}(k)$ where g is étale (17.11.4). Since one can restrict to the case where k is not perfect, hence infinite, the set of points of the open set $g(U)$ which are *rational* over k is non empty; if y is such a point and $x \in U$ a point above y, x is closed in X and $k(x)$ is separable over $k(y) = k$ (17.6.2).

Proposition (17.15.11). — *Let X be a prescheme of finite type on a field k. The following conditions are equivalent:*

a) X is étale on k.

b) X is non ramified on k.

c) X is isomorphic to $\mathrm{Spec}(A)$, where A is a finite and separable k-algebra.

Proof. This results from (17.6.2) and (17.4.2), taking into account that the condition b) implies here that X is finite on k. □

Proposition (17.15.12). — *Let k be a field, X a prescheme locally of finite type over k. For there to exist an everywhere dense open subset U of X which is lisse over k, it is necessary and sufficient that for every maximal point x of X, X be reduced at this point and that $k(x)$ be a separable extension of k. For there to exist an everywhere dense open subset U of X such that U_{red} be lisse over k, it is necessary and sufficient that for every maximal point x of X, $k(x)$ be a separable extension of k.*

Proof. The second assertion results evidently from the first. To say that there exists an everywhere dense open U lisse over k signifies that X is lisse over k at each of its maximal points. It is necessary for this that $\mathscr{O}_{X,x}$ be regular (17.15.1) and *a fortiori* reduced, hence a *field*, equal to $k(x)$; moreover (17.15.1), $k(x) \otimes_k k'$ must be regular for every radiciel extension k' of k, hence $k(x)$ must be a separable extension of k (4.3.5); the reciprocal is immediate, by virtue of (17.15.1). □

Corollary (17.15.13). — *Let X be an algebraic prescheme over a field k. Then there exist an everywhere dense open subset U in X and a finite radiciel extension k' of k such that $(U_{(k')})_{\mathrm{red}}$ is lisse over k'.*

Proof. Indeed, there is such an extension k' such that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k' (4.6.6), which is equivalent to saying (4.6.1) that for every maximal point x' of $X_{(k')}$, $k(x')$ is a separable extension of k'. One concludes therefore from (17.15.12) that there exists an everywhere dense open subset U' of $X_{(k')}$ such that U'_{red} is lisse over k'. But the morphism $X_{(k')} \to X$ is a homeomorphism (2.4.5), hence U' is of the form $U_{(k')}$, where U is an everywhere dense open subset in X. □ **IV | 104**

Proposition (17.15.14). — *Let X be an algebraic prescheme over a field k, of dimension $\leqslant 1$. Then there exists a finite radiciel extension k' of k such that the normalised $(X_{(k')})_{\mathrm{red}}$ is lisse on k'.*

Proposition (17.15.15). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be lisse at a point $x \in X$, it is necessary and sufficient that f be flat at the point x and that Ω^1_f be a $\mathscr{O}_X$-Module locally free in a neighbourhood of x, of rank at x equal to* $\dim_x f$.

Proof. The necessity of the conditions results from (17.5.1) and (17.10.2). Inversely, if these conditions are satisfied, and if $y = f(x)$, it suffices to show (17.5.1) that $f^{-1}(y)$ is lisse on $k(y)$ at the point x; but this follows from the definition of $\dim_x f$ (17.10.1), from (16.4.5), and from (17.15.5). □

17.16. Quasi-sections of flat or smooth morphisms. The statements of this number complement those of (14.5), by means of hypotheses of flatness. **IV | 105**

Proposition (17.16.1). — *Let $f : X \to S$ be a morphism flat locally of finite presentation. Let $s \in S$, x a closed point of X_s such that $\mathscr{O}_{X_s,x}$ is a Cohen-Macaulay ring; then there exists an open neighbourhood U of s in S and a sub-prescheme $X' \subset f^{-1}(U)$ of X such that $x \in X'$ and the morphism $X' \to U$, restriction of f, is flat, quasi-finite and of finite presentation.*

Corollary (17.16.2). — *Let $f : X \to S$ be a morphism faithfully flat and locally of finite presentation. Then there exists a morphism $g : S' \to S$, faithfully flat, locally of finite presentation and locally quasi-finite, such that there exists an S-morphism $S' \to X$ (in other words, such that there exists an S'-section of $X' = X \times_S S'$ (**I**, (3.3.14))). If S is quasi-compact (resp. quasi-compact and quasi-separated), one can suppose S' affine (resp. the morphism g of finite presentation and quasi-finite).*

Corollary (17.16.3). — *(i) Let $f : X \to S$ be a smooth morphism lisse. Let $s \in S$, x a closed point of X_s such that the residue field $k(x)$ is separable on $k(s)$; then, in the conclusion of (17.16.1), one can take the morphism $X' \to U$, restriction of f, to be étale.*

(ii) Let $f : X \to S$ be a smooth morphism lisse surjective. Then, in the conclusions of (17.16.2), one can further suppose that $g : S' \to S$ be étale.

Proposition (17.16.4). — *Let S be a prescheme quasi-compact and quasi-separated, $f : X \to S$ a morphism of finite presentation such that, for every $s \in S$, $X_s = f^{-1}(s)$ is a prescheme proper on $k(s)$. Then there exists a finite family $(S_i)_{i \in I}$ of sub-preschemes affines of S, of finite presentation on S, pairwise disjoint, of reunion S, and such that, for every $i \in I$, there exist two morphisms of finite presentation $S''_i \xrightarrow{g_i} S'_i \xrightarrow{h_i} S_i$, where h_i is étale and g_i flat and radiciel, and an S-morphism $S''_i \to X$ (in other words, such that there exists an S''_i-section of $X''_i = X \times_S S''_i \to S_i$).*

Corollary (17.16.5). — *Let $f : X \to S$ be a surjective morphism locally of finite presentation. Then there exists a morphism $g : S' \to S$ surjective, locally of finite presentation and locally quasi-finite, such that there exists an S-morphism $S' \to X$ (in other words, such that there exists an S'-section of $X' = X \times_S S'$). If S is quasi-compact (resp. quasi-compact and quasi-separated), one can suppose S' affine (resp. S' affine and the morphism g of finite presentation and quasi-finite).*

Proposition (17.16.6). — *Let S be a prescheme quasi-compact and quasi-separated, $f : X \to S$ a morphism of finite presentation such that, for every $s \in S$, $X_s = f^{-1}(s)$ is a prescheme proper on $k(s)$. Then there exists a finite family $(S_i)_{i \in I}$ of sub-preschemes affines of S, of finite presentation on S, pairwise disjoint, of reunion S, and such that, for every i, the morphism $X_i = X \times_S S_i \to S_i$ is proper and flat. If moreover, for every $s \in S$, X_s is finite on $k(s)$, one can take the S_i such that each of the morphisms $X_i \to S_i$ factors as*

$$X_i \xrightarrow{u_i} X'_i \xrightarrow{h_i} S_i$$

where u_i and h_i are finite and locally free, u_i is étale and surjective, and h_i is flat.

§18. Supplement on étale morphisms. Henselian local rings and strictly local rings

18.1. A remarkable equivalence of categories.

Theorem (18.1.1). — *Let $\phi : A \to B$ be a local homomorphism of local rings, $\mathfrak{m}$ the maximal ideal of A, $B_0 = B/\mathfrak{m}B$. The following conditions are equivalent:*

a) *ϕ is a formally étale homomorphism (for the preadic topologies defined by $\mathfrak{m}$ and $\mathfrak{m}B$ respectively).*
b) *B is a* Noetherian *ring, B is $\mathfrak{m}B$-adically* separated and complete, *and B_0 is a formally étale $A/\mathfrak{m}$-algebra (for the discrete topologies).*

Proof. We have already seen ((**0**, 19.7.1)) that the conditions are equivalent when A is a *Noetherian* **IV-4** | 109
complete local ring (with B also Noetherian complete local). Now let us handle the general case.

We first prove that *a*) implies *b*). By (**0**, 19.3.5, ii) and (**0**, 19.6.4), we see that B is Noetherian and $\mathfrak{m}B$-adically separated and complete, and it remains to show that B_0 is a formally étale $(A/\mathfrak{m})$-algebra. But if C_0 is an $(A/\mathfrak{m})$-algebra and $\mathfrak{J}_0$ a nilpotent ideal of C_0, the ring C_0 can be considered as an A-algebra, and $\mathfrak{J}_0$ as an ideal of this A-algebra; moreover, C_0 is separated and complete for the $\mathfrak{m}$-preadic topology since $\mathfrak{m}C_0 = 0$. The hypothesis on ϕ then implies that the canonical map $\mathrm{Hom}_{A\text{-alg}}(B, C_0) \to \mathrm{Hom}_{A\text{-alg}}(B, C_0/\mathfrak{J}_0)$ is bijective; but since $\mathfrak{m}C_0 = 0$, both sides identify respectively with $\mathrm{Hom}_{(A/\mathfrak{m})\text{-alg}}(B_0, C_0)$ and $\mathrm{Hom}_{(A/\mathfrak{m})\text{-alg}}(B_0, C_0/\mathfrak{J}_0)$, hence B_0 is formally étale over $A/\mathfrak{m}$.

Conversely, suppose *b*) is satisfied, and let C be an A-algebra, separated and complete for the $\mathfrak{m}$-preadic topology, $\mathfrak{J}$ an ideal of C such that $\mathfrak{J}^2 = 0$ (which is sufficient by (**0**, 19.3.6, ii)). We must show that the canonical map $\mathrm{Hom}(B, C) \to \mathrm{Hom}(B, C/\mathfrak{J})$ is bijective (the Hom being taken in the category of continuous A-algebra homomorphisms for the $\mathfrak{m}$-preadic topologies on B, C, and $C/\mathfrak{J}$). Let us pose $A_n = A/\mathfrak{m}^{n+1}$, $B_n = B/\mathfrak{m}^{n+1}B$, $C_n = C/\mathfrak{m}^{n+1}C$, $\mathfrak{J}_n$ the image of $\mathfrak{J}$ in C_n; since B and C are separated and complete, we have $B = \varprojlim B_n$ and $C = \varprojlim C_n$, and since the B_n are Artinian, and $C_n/\mathfrak{J}_n = (C/\mathfrak{J})_n$, we are reduced to proving that $\mathrm{Hom}_{A_n\text{-alg}}(B_n, C_n) \to \mathrm{Hom}_{A_n\text{-alg}}(B_n, C_n/\mathfrak{J}_n)$ is bijective for all n, i.e. that B_n is a formally étale A_n-algebra (for the discrete topologies). Now, by the already established special case ((**0**, 19.7.1)), B is formally étale over the completion $\widehat{A}$ of A; in particular, B_n is a formally étale A_n-algebra, since $A_n = \widehat{A}/\mathfrak{m}^{n+1}\widehat{A}$, which concludes the proof. □

(18.1.2). Under the hypotheses and with the notation of (18.1.1), the formally étale homomorphism $\phi : A \to B$ is also *faithfully flat* ((**0**, 19.7.1, i)). In particular, the image in B of every non-zero-divisor of A is a non-zero-divisor of B; the image in B of every element of A not belonging to $\mathfrak{m}$ is *invertible* in B; therefore ϕ extends to a homomorphism $\phi' : A_{\mathfrak{m}} \to B$, where $A_{\mathfrak{m}}$ is the localisation of A at $\mathfrak{m}$ (which coincides with A when A is already local). It is immediate that ϕ' is also a formally étale homomorphism. On the other hand, if $\mathfrak{n}$ is a maximal ideal of B, then $\mathfrak{n} \supset \mathfrak{m}B$ since $\mathfrak{m}B$ is contained in the radical of B ((**0**, 7.1.10)); the quotient $B/\mathfrak{n}$ is therefore a quotient of B_0, and conversely, any maximal ideal of B_0 is the image of a maximal ideal of B; in other words, the maximal ideals of B correspond bijectively to those of B_0. Since B_0 is a formally étale $\kappa(\mathfrak{m})$-algebra (where $\kappa(\mathfrak{m}) = A/\mathfrak{m}$), B_0 is a *finite* product of *finite separable extensions* of $\kappa(\mathfrak{m})$ ((**0**, 20.7.3)).

(18.1.3). We now specialise to the case where B is a *local* ring. Then B_0 is a finite separable extension of $\kappa(\mathfrak{m})$, therefore there exists a unique (by (**0**, 20.5.9)) polynomial $F \in A[\mathrm{T}]$ *monic* and such that
B_0 is isomorphic to $\kappa(\mathfrak{m})[\mathrm{T}]/(F_0)$, where F_0 is the canonical image of F in $\kappa(\mathfrak{m})[\mathrm{T}]$; moreover, F_0 is **IV-4** | 110
a product of *distinct* irreducible factors in $\kappa(\mathfrak{m})[\mathrm{T}]$, the image $\bar{\xi}$ of T in B_0 is a root of F_0, and B_0 is isomorphic to $\kappa(\mathfrak{m})[\mathrm{T}]/(g_0)$ where g_0 is the minimal polynomial of $\bar{\xi}$ over $\kappa(\mathfrak{m})$, so g_0 is *separable* and divides F_0. By a well-known property of Henselian rings ((**0**, I.8.3)), B is isomorphic to $\widehat{A}[\mathrm{T}]/(g)$ where $g \in \widehat{A}[\mathrm{T}]$ is a monic polynomial lifting g_0, $\widehat{A}$ being the completion of A.

(18.1.4). We return to the general case. To say that $B_0 = B/\mathfrak{m}B$ is a formally étale $\kappa(\mathfrak{m})$-algebra is equivalent to saying (in the notation of (18.1.2)) that the set $\mathrm{Hom}_{\kappa(\mathfrak{m})\text{-alg}}(B_0, \Omega)$ is *finite* for every algebraically closed extension Ω of $\kappa(\mathfrak{m})$, and that $\Omega^1_{B_0/\kappa(\mathfrak{m})} = 0$ ((**0**, 20.7.3)). Furthermore, the formally étale homomorphism $\phi : A \to B$ is a formally smooth homomorphism ((**0**, 19.3.1)), hence a flat homomorphism ((**0**, 19.7.1, i)).

Theorem (18.1.2). — *Let S be a prescheme, S_0 a closed sub-prescheme of S whose underlying space is identical to that of S. Then the functor*

$$X \longmapsto X \times_S S_0$$

from the category of S-preschemes étale over S, to the category of S_0-preschemes étale over S_0, is an equivalence of categories.

Proof. Let us first show that this functor is *fully faithful*. Let X, Y be two S-preschemes étale over S, and set $X_0 = X \times_S S_0$, $Y_0 = Y \times_S S_0$. If $Z = X \times_S Y$, the set $\mathrm{Hom}_S(X, Y)$ is in canonical bijective correspondence with the set of X-sections $\Gamma(Z/X)$, and likewise $\mathrm{Hom}_{S_0}(X_0, Y_0)$ is in canonical bijective correspondence with $\Gamma(Z_0/X_0)$, where $Z_0 = Z \times_S S_0 = X_0 \times_{S_0} Y_0$. Now Z is étale over X, Z_0 étale over X_0, and X_0 (resp. Z_0) is a closed sub-prescheme of X (resp. Z) having the same underlying space. The open subsets of Z are therefore the *same* as the open subsets of Z_0 having the corresponding properties, and our assertion follows from (17.9.3).

To complete the proof, it suffices to show that for every S_0-prescheme X_0 étale over S_0, there exists an S-prescheme X étale over S and an S_0-isomorphism $X_0 \xrightarrow{\sim} X \times_S S_0$. By virtue of Proposition (18.1.1), there exists an open covering (U_α) of X_0 and for each α, an S-prescheme V_α which is étale over S, and finally an S_0-isomorphism $\theta_\alpha : U_\alpha \xrightarrow{\sim} V_\alpha \times_S S_0$. Furthermore, by virtue of the first part of the proof, there exists a unique S_0-isomorphism $\varphi_{\alpha\beta}$ from $\theta_\alpha(U_\alpha \cap U_\beta)$ to $\theta_\beta(U_\alpha \cap U_\beta)$, corresponding to the identity automorphism of $U_\alpha \cap U_\beta$, and it is immediate, for the same reason, that these isomorphisms satisfy the glueing condition ((**0**, 4.1.7)). Hence there exists an S-prescheme X such that the V_α identify canonically with sub-preschemes induced on the open subsets of X, the θ_α identify with S_0-isomorphisms which coincide on the intersections $U_\alpha \cap U_\beta$, and define an S_0-isomorphism $X_0 \xrightarrow{\sim} X \times_S S_0$. It is clear that X is étale over S (17.3.2), which completes the proof. □

Corollary (18.1.3). — *Let S be a prescheme, X an S-prescheme étale over S, S' an S-prescheme, S'_0 a closed sub-prescheme of S' having the same underlying space. Then the canonical map $X(S')_S \to X(S'_0)_S$* (**I**, 3.4.3) *is bijective.*

Proof. Indeed, set $X' = X \times_S S'$, $X'_0 = X \times_S S'_0$, so that $X(S')_S = \Gamma(X'/S')$ and $X(S'_0)_S = \Gamma(X'_0/S'_0)$; the corollary follows from the fact that the functor defined in (18.1.2.thm) (with S and S_0 replaced by S' and S'_0 respectively) is fully faithful (or directly from (17.9.3)). □

18.2. Coverings.

(18.2.1). Given a ring A and a commutative A-algebra B which is *finite* and a *free* A-module, recall (Bourbaki, *Alg.*, ch. VIII, § 12, no. 2) that one defines on B an *A-linear form* $\mathrm{Tr}_{B/A}$, the "trace form"; one deduces from it the definition of a *symmetric A-bilinear form* (also called "trace form")

$$(x, y) \longmapsto \mathrm{Tr}_{B/A}(xy) \tag{18.2.1.1}$$

whose datum is equivalent to that of the associated A-linear map $\mathrm{astr}_{B/A} : B \to \check{B}$, where $\check{B}$ denotes the *dual* of the A-module B, equal to its transpose. When A is a *field*, it is equivalent to say that this bilinear form is *non-degenerate* or that B is a *separable* A-algebra (Bourbaki, *Alg.*, ch. IX, § 2, prop. 5). **IV-4** | 111

Let $f : A \to A'$ be a ring homomorphism; set $B' = B \otimes_A A'$, and let $g : B \to B'$ be the canonical homomorphism; the image by g of a base of the A-module B is then a base of the A'-module B', and it follows from the definitions that, for all $x \in B$,

$$\mathrm{Tr}_{B'/A'}(g(x)) = f(\mathrm{Tr}_{B/A}(x)). \tag{18.2.1.2}$$

(18.2.2). Consider now a *ringed space* $(X, \mathscr{O}_X)$ and let $\mathscr{B}$ be an $\mathscr{O}_X$-Algebra which, as an $\mathscr{O}_X$-Module, is *locally free of finite rank*; then, for every open subset $U \subset X$ such that $\mathscr{B}|U$ is (as an $\mathscr{O}_U$-Module) isomorphic to $\mathscr{O}_U^n$ (for some n depending on U), $\Gamma(U, \mathscr{B})$ is a $\Gamma(U, \mathscr{O}_X)$-algebra which, as a $\Gamma(U, \mathscr{O}_X)$-module, is free of finite rank, and thus defines a $\Gamma(U, \mathscr{O}_X)$-linear form $\mathrm{Tr}_{\Gamma(U,\mathscr{B})/\Gamma(U,\mathscr{O}_X)}$, which we also denote by $\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X,U}$; from this we deduce an associated linear map

$$\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X,U} : \Gamma(U, \mathscr{B}) \longrightarrow \Gamma(U, \mathscr{B})^\vee = \Gamma(U, \check{\mathscr{B}}).$$

Furthermore, it follows from (18.2.1.2) that these linear maps are compatible with the restriction operations from U to a smaller open subset, and thus define an $\mathscr{O}_X$-*Module homomorphism*, also called the *trace homomorphism*:

$$\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X} : \mathscr{B} \longrightarrow \mathscr{O}_X \tag{18.2.2.1}$$

and also an $\mathscr{O}_X$-*Module homomorphism*

$$\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X} : \mathscr{B} \longrightarrow \check{\mathscr{B}} \tag{18.2.2.2}$$

called the *homomorphism associated to the trace*, and equal to its transpose. It also follows from (18.2.1.2) that for every $x \in X$, we have

$$(\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X})_x = \mathrm{Tr}_{\mathscr{B}_x|\mathscr{O}_{X,x}} \tag{18.2.2.3}$$

$$(\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X})_x = \mathrm{astr}_{\mathscr{B}_x|\mathscr{O}_{X,x}}\,. \tag{18.2.2.4}$$

Finally, under the conditions of (18.2.1), if we set $X = \mathrm{Spec}(A)$ and $\mathscr{B} = \widetilde{B}$, the form $\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X}$ (resp. the homomorphism $\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X}$) corresponds to the form $\mathrm{Tr}_{B/A}$ (resp. the A-module homomorphism $\mathrm{astr}_{B/A}$), as also follows from (18.2.1.2).

Proposition (18.2.3). — *Let $f : X \to Y$ be a finite morphism of preschemes and set $\mathscr{B} = f_*(\mathscr{O}_X)$. The following conditions are equivalent:*

a) *f is étale.*

a′) *f is a flat morphism of finite presentation, and for every $x \in X$, if we set $y = f(x)$, $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is a finite separable extension of $\kappa(y)$.*

b) *$\mathscr{B}$ is a locally free $\mathscr{O}_Y$-Module and for every $y \in Y$, $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a finite separable $\kappa(y)$-algebra (hence the direct composite of a finite number of fields, finite separable extensions of $\kappa(y)$).*

c) *$\mathscr{B}$ is a locally free $\mathscr{O}_Y$-Module and the homomorphism $\mathrm{astr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \check{\mathscr{B}}$ (18.2.2) is bijective.*

Proof. Since f is quasi-compact, the equivalence of a) and a') has already been shown (17.6.2). To prove the rest of the proposition, we may restrict to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, B being a finite A-algebra and $\mathscr{B} = \widetilde{B}$. To say that f is a morphism of finite presentation is equivalent to saying that B is an A-module of finite presentation (1.4.7). If in addition f is flat, then B is a flat A-module, so B is an A-module projective, known (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 2, cor. 2 du th. 1) to be equivalent to saying that B is an A-Module *projective*; thus $\mathscr{B}$ is a locally free $\mathscr{O}_Y$-Module (*loc. cit.*, no. 2, th. 1), and the converse is immediate. On the other hand, $f^{-1}(y)$ is nothing but the spectrum of the $\kappa(y)$-algebra $\mathscr{B}(y) = \mathscr{B}_y/\mathfrak{m}_y\mathscr{B}_y$, which completes the proof of the equivalence of a') and b). To see that b) is equivalent to c), let us note that the second assertion of b) is equivalent to the fact that $\mathrm{astr}_{\mathscr{B}(y)|\kappa(y)} : \mathscr{B}(y) \to \mathscr{B}(y)^\vee$ is bijective; since $\mathscr{B}(y) = \mathscr{B}_y/\mathfrak{m}_y\mathscr{B}_y$ and $\check{\mathscr{B}}_y$ are free $\mathscr{O}_{Y,y}$-modules, it follows from (18.2.2.4) and Bourbaki, *Alg. comm.*, ch. II, § 3, no. 3, cor. of prop. 6, that the homomorphism $\mathrm{astr}_{\mathscr{B}_y|\mathscr{O}_{Y,y}} : \mathscr{B}_y \to \check{\mathscr{B}}_y$ is itself bijective; the converse being evident, this completes the proof. **IV-4** | 112

When an $\mathscr{O}_X$-Algebra $\mathscr{B}$ satisfies the equivalent conditions b) and c) of (18.2.3), we say that $\mathscr{B}$ is a *finite étale $\mathscr{O}_X$-Algebra*. When $X = \mathrm{Spec}(A)$ is affine and $\mathscr{B} = \widetilde{B}$, this amounts, by virtue of (18.2.3), to saying that B is a finite étale A-algebra, or that B is a finite A-algebra that is étale (in the sense of (17.3.2)). □

Corollary (18.2.4). — *Let $f : X \to Y$ be a morphism of finite type and of finite presentation, and set $\mathscr{B} = f_*(\mathscr{O}_X)$. Let y be a point of Y. The following conditions are equivalent:*

a) *There exists an open neighbourhood U of y in Y such that the restriction $f^{-1}(U) \to U$ of f is an étale morphism.*

b) *$\mathscr{B}_y$ is a locally free $\mathscr{O}_{Y,y}$-module of finite type and $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a separable $\kappa(y)$-algebra.*

Proof. It is clear that a) implies b) by virtue of (18.2.3). On the other hand, $\mathscr{B}$ is an $\mathscr{O}_Y$-Module of finite presentation (1.6.3) and (1.4.7), hence, if $\mathscr{B}_y$ is a locally free $\mathscr{O}_{Y,y}$-module, there exists an open neighbourhood U of y in Y such that $\mathscr{B}|U$ is a locally free $\mathscr{O}_U$-Module ((**0**, 5.2.7)); moreover, by hypothesis, the homomorphism $\mathrm{astr}_{\mathscr{B}_y|\mathscr{O}_{Y,y}} : \mathscr{B}_y \to \check{\mathscr{B}}_y$ being bijectif, one can also suppose by

(**0**, 5.2.7) that one can choose U so that the homomorphism $\mathrm{astr}_{\mathscr{B}|U,\mathscr{O}_U}$ is bijective. The fact that $b)$ implies $a)$ then follows from (18.2.3). □ **IV-4** | 113

Corollary (18.2.5). — *Let Y be a quasi-compact or locally Noetherian prescheme, $f : X \to Y$ a morphism of finite type and of finite presentation; set $\mathscr{B} = f_*(\mathscr{O}_X)$. Suppose that for every* closed *point y of Y, $\mathscr{B}_y$ is a locally free $\mathscr{O}_{Y,y}$-module and $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a separable $\kappa(y)$-algebra. Then f is étale.*

Proof. Indeed, it follows from (18.2.4) that every closed point y of Y has an open neighbourhood U such that the restriction $f^{-1}(U) \to U$ of f is étale; the conclusion follows from the fact that in the two cases considered, every non-empty closed subset of Y contains a closed point ((5.1.11) and (**0**, 2.1.3)). □

Corollary (18.2.6). — *If $f : X \to Y$ is a finite étale morphism, and if we set $\mathscr{B} = f_*(\mathscr{O}_X)$, then the $\mathscr{O}_Y$-homomorphism $\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \mathscr{O}_Y$ (18.2.2) (also denoted Tr_f) is surjective.*

Proof. The question being local, we may, by virtue of (18.2.3), suppose that $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, with B a free A-module; as by virtue of (18.2.3) the bilinear form (18.2.1.1) is non-degenerate, this implies in particular that the linear form $\mathrm{Tr}_{B/A}$ is surjective. □

Remarks (18.2.7). — (i) When $f : X \to Y$ is a finite morphism such that $f_*(\mathscr{O}_X)$ is a locally free $\mathscr{O}_Y$-Module (resp. locally free of rank n), we say further that f is a *finite locally free* morphism (resp. *locally free of rank n*). This condition, by virtue of (18.2.3), is satisfied if f is a finite étale morphism, but does not imply by itself that f is étale, as shown by the example where $X = \mathrm{Spec}(K)$ and $Y = \mathrm{Spec}(k)$ are spectra of fields, K being a finite non-separable extension of k. When $f : X \to Y$ is a finite étale morphism, we say further that X is an *étale covering* of Y. One notes that in this case, f is universally open and universally closed, and in particular $f(X)$ is a subset that is both open and closed of Y.

We say that an étale covering X of Y is *trivial* if X is a *sum* of a finite number of preschemes *isomorphic* to Y. We say that an étale covering X of Y is *locally trivial* if the morphism $f : X \to Y$ is such that for every point $y \in Y$ there is an open neighbourhood U for which the covering $f^{-1}(U)$ of U is trivial.

(ii) Let $f : X \to Y$ be a finite morphism, locally free of rank n; set $\mathscr{B} = f_*(\mathscr{O}_X)$ and let $u = \mathrm{astr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \check{\mathscr{B}}$; we deduce from this an n-th exterior power homomorphism $\bigwedge^n u : \bigwedge^n \mathscr{B} \to \bigwedge^n \check{\mathscr{B}} = (\bigwedge^n \mathscr{B})^\vee$ of *invertible* $\mathscr{O}_Y$-Modules, and by (**0**, 5.4.2) an element

(18.2.7.1) $$d_{X/Y} \in \Gamma(Y, (\bigwedge^n \check{\mathscr{B}}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \check{\mathscr{B}}))$$

which we call the *discriminant* of X over Y. Furthermore, since $(\bigwedge^n \check{\mathscr{B}}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \check{\mathscr{B}})$ is the dual of $(\bigwedge^n \mathscr{B}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \mathscr{B})$, $d_{X/Y}$ can also be identified with a homomorphism

(18.2.7.2) $$(\bigwedge^n \mathscr{B}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \mathscr{B}) \longrightarrow \mathscr{O}_Y$$

and we denote by $\mathscr{D}_{X/Y}$ the quasi-coherent ideal of $\mathscr{O}_Y$ of finite type, *image* of the homomorphism (18.2.7.2), also called the *discriminant ideal* of X over Y.

For the homomorphism u to be bijective, it is necessary and sufficient that $\bigwedge^n u$ be bijective, or again that the section $d_{X/Y}$ have an *invertible* germ at every point $y \in Y$, which is also written $d_{X/Y}(y) \neq 0$ for every y ((**0**, 5.5.2)). It amounts to the same to say that the discriminant ideal $\mathscr{D}_{X/Y}$ is *equal* to $\mathscr{O}_Y$. **IV-4** | 114

The terminology "covering" introduced in (18.2.7), (i), is justified by the following proposition:

Proposition (18.2.8). — *Let $f : X \to Y$ be a morphism étale, separated and of finite type, and for every $y \in Y$, let $n(y)$ be the geometric number of points of $f^{-1}(y)$. Then the function $y \mapsto n(y)$ is lower semi-continuous on Y. For it to be continuous at a point y (and hence constant in a neighbourhood of y), it is necessary and sufficient that there exists an open neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is a* finite *(étale) morphism.*

Proof. Since f is quasi-finite (17.6.1) and locally of finite presentation, it amounts to saying that f is finite or that f is proper (8.11.1): moreover, each fibre $f^{-1}(y)$ is geometrically reduced over $\kappa(y)$. The conclusions therefore follow from (15.5.9), (i) and (ii) and from the fact that f is flat. □

Corollary (18.2.9). — *Let Y be a connected prescheme, $f : X \to Y$ a morphism étale, separated and of finite type. For f to be finite (in other words, for X to be an étale covering of Y), it is necessary and sufficient that all the fibres of f have the same geometric number of points.*

Remarks (18.2.10). — (i) The example of the "affine line with a doubled point" ((**I**, 5.5.11)) shows that a morphism étale of finite type of Noetherian preschemes may not be separated; for this example, the first assertion of (18.2.8) is no longer exact.

(ii) For a separated morphism of finite type and étale $f : X \to Y$ to make X a *locally trivial* covering, it is necessary and sufficient that for every $x \in X$ there exist an open neighbourhood U of $y = f(x)$ and a U-section g of $f^{-1}(U)$ such that $g(y) = x$. Indeed, the condition is obviously necessary; the fact that it is also sufficient follows from the fact that every fibre $f^{-1}(y)$ is finite ((17.6.1)) and (**I**, 6.2.2), from the characterisation of sections of a Y-scheme étale (17.9.3) and from Proposition (18.2.8).

18.3. Finite algebras and étale algebras.

Proposition (18.3.1). — *Let A be a ring, B an A-algebra of finite presentation.*

(i) For B to be an unramified A-algebra, it is necessary and sufficient that B be an A-module of finite presentation and that $(B \otimes_A B)$ be a projective *$(B \otimes_A B)$-module.*

(ii) Suppose moreover that B is a finite A-algebra. For B to be an étale A-algebra, it is necessary and sufficient that B be an A-module projective and a $(B \otimes_A B)$-module projective.

Proof. The $(B \otimes_A B)$-module structure on B is that coming from the canonical ring homomorphism **IV-4** | 115
$B \otimes_A B \to B$, which is surjective and has kernel $\mathfrak{J}_{B/A}$ ((**0**, 20.4.1)).

(i) To say that the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is locally of finite presentation (1.4.6). To say that B is an unramified A-algebra is equivalent to saying that it is an A-algebra of finite presentation (1.4.7). To say that B is unramified is equivalent to saying that $(B \otimes_A B)/\mathfrak{J}_{B/A}$ is an open and closed subset of $\mathrm{Spec}(B \otimes_A B)$ and we know that for this, it is necessary and sufficient that $\mathfrak{J}_{B/A}$ be a *direct factor ideal* of $B \otimes_A B$ (Bourbaki, *Alg. comm.*, ch. II, § 4, no. 3, prop. 15); but this amounts to saying that the $(B \otimes_A B)$-module quotient $(B \otimes_A B)/\mathfrak{J}_{B/A}$ is projective (Bourbaki, *Alg.*, ch. II, 3^e^ éd., § 2, no. 2, prop. 4).

(ii) Recall that a flat A-module of finite presentation is projective, and conversely (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 2, cor. 2 du th. 1), so the assertion (ii) follows from that of (i) and from (17.6.2). □

Proposition (18.3.2). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$-preadic topology, A is separated and complete; set $A_0 = A/\mathfrak{J}$. Then the functor*

$$B \longmapsto B \otimes_A A_0$$

is an equivalence of the category of finite étale A-algebras, to the category of finite étale A_0-algebras.

Proof. We will first prove the following lemma:

Lemma (18.3.2.1). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$-preadic topology, A is separated and complete.*

(i) Every projective A-module of finite type M is separated and complete for the $\mathfrak{J}$-preadic topology, hence a projective limit of the $(A/\mathfrak{J}^{n+1})$-modules projective $M/\mathfrak{J}^{n+1}M$.

(ii) Conversely, set $A_n = A/\mathfrak{J}^{n+1}$, and let (M_n) be a projective system of A_n-modules, such that, for every n, the homomorphism $M_{n+1} \otimes_{A_{n+1}} A_n \to M_n$ deduced from the di-homomorphism of transition $M_{n+1} \to M_n$ is bijective. Suppose moreover that the M_n are projective and M_0 of finite type. Then $M = \varprojlim M_n$ is a projective A-module of finite type such that the canonical homomorphism $M \otimes_A A_0 \to M_0$ is bijective.

Proof. (i) There is a free A-module of finite type L such that M is isomorphic to a direct factor of L; as L is separated for the $\mathfrak{J}$-preadic topology, so is every sub-module N of L since $\mathfrak{J}^{n+1}N \subset \mathfrak{J}^{n+1}L$; in particular M is separated for this topology, and since the surjective homomorphism $f : L \to M$ is continuous for the $\mathfrak{J}$-preadic topology, its kernel N is closed for the topology induced by that of L; since L is complete and f a strict morphism, we conclude that M is complete (Bourbaki, *Top. gén.*, ch. IX, 2^e^ éd., § 3, no. 1, prop. 4).

(ii) It follows from Nakayama's lemma that M_0 is generated by a finite family (x_{i0}) of r elements and if for all n, x_{in} is an element of M_n whose image $x_{i,n-1}$ is the image in M_{n-1}, then (x_{in}) $(1 \leqslant i \leqslant r)$ is a system of generators of M_n (Bourbaki, *Alg. comm.*, ch. II, § 3, cor. 2 de la prop. 4). Setting, for all n, $L_n = A_n^r$; if $(e_{in})_{1\leqslant i\leqslant r}$ is the canonical base of L_n, let $u_n : L_n \to M_n$ be the A-linear map such that $u_n(e_{in}) = x_{in}$ for all i. By hypothesis, we have a split exact sequence

$$0 \longrightarrow N_n \xrightarrow{v_n} L_n \xrightarrow{u_n} M_n \longrightarrow 0$$

and since $L_n = L_{n+1}/\mathfrak{J}^{n+1}L_{n+1}$ and $M_n = M_{n+1}/\mathfrak{J}^{n+1}M_{n+1}$, the vertical arrows in the commutative diagram **IV-4** | 116

$$\begin{tikzcd} 0 \arrow[r] & N_{n+1} \arrow[r, "v_{n+1}"] \arrow[d] & L_{n+1} \arrow[r, "u_{n+1}"] \arrow[d] & M_{n+1} \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & N_n \arrow[r, "v_n"] & L_n \arrow[r, "u_n"] & M_n \arrow[r] & 0 \end{tikzcd}$$

are all three surjective. Now we have $M = \varprojlim M_n$ and $L = A^r = \varprojlim L_n$; setting $N = \varprojlim N_n$, $u = \varprojlim u_n$, $v = \varprojlim v_n$, by (**0**, 13.2.2) the exact sequence

(18.3.2.2) $$0 \longrightarrow N \xrightarrow{v} L \xrightarrow{u} M \longrightarrow 0$$

is exact. Moreover, since for every n, v_n is left-invertible and M_n is a projective A_n-module, by virtue of ((**0**, 19.1.8)) the exact sequence (18.3.2.2) is split, which proves the lemma. □

This being established, let us first show that the functor of (18.3.2) is *fully faithful*. Set, as in the lemma, $A_n = A/\mathfrak{J}^{n+1}$; let B, C be two finite étale A-algebras, and set, for every n, $B_n = B \otimes_A A_n$, $C_n = C \otimes_A A_n$; by virtue of (18.3.1) and (18.3.2.1), B and C are separated and complete for the $\mathfrak{J}$-preadic topology, and we have $B = \varprojlim B_n$, $C = \varprojlim C_n$; moreover, every A-algebra homomorphism $u : B \to C$ is continuous for the $\mathfrak{J}$-preadic topologies, and hence gives a projective system of A_n-algebra homomorphisms $u_n = u \otimes 1 : B_n \to C_n$, of which it is the projective limit; the converse being evident, we have a canonical bijection

$$\mathrm{Hom}_{A\text{-alg.}}(B, C) \xrightarrow{\sim} \varprojlim \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n).$$

But since B and C are finite étale A-algebras, it follows also from (18.1.2.thm) that the canonical map

$$\mathrm{Hom}_{A_{n+1}\text{-alg.}}(B_{n+1}, C_{n+1}) \longrightarrow \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n)$$

is *bijective* for $n \geqslant 0$, which proves that the canonical map $\mathrm{Hom}_{A\text{-alg.}}(B, C) \to \mathrm{Hom}_{A_0\text{-alg.}}(B_0, C_0)$ is bijective.

To complete the proof of (18.3.2), it suffices to show that for every finite étale A_0-algebra B_0, there exists a finite étale A-algebra B and an A_0-isomorphism $B_0 \xrightarrow{\sim} B \otimes_A A_0$. Now, by (18.1.2.thm), there is a projective system (B_n) such that B_n is an étale A_n-algebra and that the homomorphisms $B_{n+1} \otimes_{A_{n+1}} A_n \to B_n$ are bijective. It follows from (18.3.1) and (18.3.2.1) that the A-algebra $B = \varprojlim B_n$ is a projective A-module of finite type and that B_0 is isomorphic to $B \otimes_A A_0$. To prove that B is a finite étale A-algebra, it suffices, by (18.2.5), to show that for every maximal ideal $\mathfrak{m}$ of A, $B_\mathfrak{m}/\mathfrak{m}B_\mathfrak{m}$ is a separable $(A/\mathfrak{m})$-algebra. Now, since $\mathfrak{J}$ is contained in the radical of A ((**0**, 7.1.10)), we have $\mathfrak{J} \subset \mathfrak{m}$, so $\mathfrak{m}_0 = \mathfrak{m}/\mathfrak{J}$, we have $A_0/\mathfrak{m}_0 = A/\mathfrak{m}$ and $B_\mathfrak{m}/\mathfrak{m}B_\mathfrak{m} = (B_0)_{\mathfrak{m}_0}/\mathfrak{m}_0(B_0)_{\mathfrak{m}_0}$; the conclusion therefore follows from the fact that B_0 is a finite étale A_0-algebra (18.2.5). □ **IV-4** | 117

(18.3.3). *Example.* — Proposition (18.3.2) applies in particular when A is a *local* ring, separated and complete, $\mathfrak{J}$ being the maximal ideal of A, so that A_0 is a field and the category of finite étale A_0-algebras is identical to that of finite separable A_0-algebras, hence isomorphic to direct composites of fields, finite separable extensions of A_0. In particular, if the field A_0 is *separably closed*, these extensions are all identical to A_0, and hence every étale covering of $\mathrm{Spec}(A)$ is *trivial* (18.2.7) by virtue of (18.3.2).

Theorem (18.3.4). — *Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is separated and complete for the $\mathfrak{J}$-preadic topology, $A_0 = A/\mathfrak{J}$. Let $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A_0)$. Let X be a proper S-scheme, and set $X_0 = X \times_S S_0$. Then the functor*

$$Z \longmapsto Z \times_X X_0$$

from the category of finite étale X-schemes over X, to the category of finite étale X_0-schemes over X_0, is an equivalence of categories.

Proof. Let us first show that this functor is *fully faithful*. Let Z' and Z'' be two finite étale X-schemes over X. Set $S_n = \operatorname{Spec}(A/\mathfrak{J}^{n+1})$, $X_n = X \times_S S_n$, $Z'_n = Z' \times_S S_n$, $Z''_n = Z'' \times_S S_n$. It follows from ((**III**, 5.4.1)) that we have a canonical bijection $\operatorname{Hom}_X(Z', Z'') \xrightarrow{\sim} \varprojlim \operatorname{Hom}_{X_n}(Z'_n, Z''_n)$. Now, by virtue of (18.1.2.thm), the canonical map $\operatorname{Hom}_{X_{n+1}}(Z'_{n+1}, Z''_{n+1}) \to \operatorname{Hom}_{X_n}(Z'_n, Z''_n)$ is bijective, which proves our assertion.

It remains to show that if $\mathscr{B}_0$ is a finite étale $\mathscr{O}_{X_0}$-Algebra, there exists an $\mathscr{O}_X$-Algebra $\mathscr{B}$, finite and étale, and an isomorphism $\mathscr{B}_0 \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_0}$. It follows from (18.1.2.thm) that there is a projective system $(\mathscr{B}_n)$, where $\mathscr{B}_n$ is an $\mathscr{O}_{X_n}$-Algebra finite and étale, and the second comparison theorem ((**III**, 5.1.4)) proves the existence of a coherent $\mathscr{O}_X$-Module $\mathscr{B}$ and of a projective system of isomorphisms $\mathscr{B}_n \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_n}$. The data of an Algebra structure on $\mathscr{B}$ is equivalent to that of a homomorphism $\mathscr{F} \otimes \mathscr{F} \to \mathscr{F}$ rendering the tensor powers commutative diagrams where only tensor powers of $\mathscr{F}$ appear; by (**III**, 5.1.3), (**I**, 10.11.4) and (**I**, 10.11.7), $\mathscr{B}$ is endowed in a natural way with a structure of $\mathscr{O}_X$-Algebra for which the isomorphism $\mathscr{B}_0 \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_0}$ is an isomorphism of Algebras. Moreover, $\mathscr{B}$ is an $\mathscr{O}_X$-Module *locally free*; this also follows from (**III**, 5.1.3), (**I**, 10.11.4), (**I**, 10.11.7) and from the fact that, in the category of $\mathscr{O}_X$-Modules, the locally free $\mathscr{O}_X$-Modules $\mathscr{F}$ can be defined as those for which the functors $\mathscr{G} \mapsto \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ are *exact*. Finally, to see that $\mathscr{B}$ is a finite étale $\mathscr{O}_X$-Algebra, it suffices (18.2.5) to show that for every *closed* point $x \in X$, $\mathscr{B}_x \otimes_{\mathscr{O}_{X,x}} \kappa(x)$ is a separable $\kappa(x)$-algebra. But since the structural morphism $f : X \to S$ is *proper*, $f(x)$ is a closed point of S, so it belongs to S_0, since $\mathfrak{J}$ is contained in the radical of A ((**0**, 7.1.10)); the conclusion therefore follows from the fact that $X_0 = f^{-1}(S_0)$ and that $\mathscr{B}_0$ is a finite étale $\mathscr{O}_{X_0}$-algebra. □

18.4. Local structure of unramified morphisms and of étale morphisms. **IV-4** | 118

Lemma (18.4.1). — *Let A be a ring, B a finite monogenic A-algebra, u a generator of the A-algebra B, $F \in A[\mathrm{T}]$ a polynomial such that $F(u) = 0$, F' the derivative polynomial; set $u' = F'(u)$. Then the ideal of B, annihilator of $\Omega^1_{B/A}$, contains $u'B$; it is equal to $u'B$ if the ideal $\mathfrak{J}$ of $A[\mathrm{T}]$ of polynomials G such that $G(u) = 0$, is generated by F, in other words if the surjective canonical homomorphism $\varphi : A[\mathrm{T}]/F.A[\mathrm{T}] \to B$ transforming the image of T into u, is bijective.*

Proof. Set $C = A[\mathrm{T}]$, so that $B = C/\mathfrak{J}$. We have the exact sequence ((**0**, 20.5.12.1))

$$\mathfrak{J}/\mathfrak{J}^2 \longrightarrow \Omega^1_{C/A} \otimes_C B \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

and $\Omega^1_{B/A}$ therefore identifies with the quotient $B/\mathfrak{J}'$, where $\mathfrak{J}'$ is the ideal generated by the elements $G'(u)$, where G ranges over a system of generators of the ideal $\mathfrak{J}$ ((**0**, 20.5.13)); whence the lemma immediately follows. □

Proposition (18.4.2). — *Let the notation be those of (18.4.1), let $\mathfrak{q}$ be a prime ideal of B. Then:*

- *(i) If $\mathfrak{q}$ does not contain u', $B_\mathfrak{q}$ is an $A_\mathfrak{p}$-algebra formally unramified ($\mathfrak{p}$ being the inverse image of $\mathfrak{q}$ in A); in other terms, $\operatorname{Spec}(B_u)$ is formally unramified over $\operatorname{Spec}(A)$.*
- *(ii) Suppose moreover that F is unitaire and generates $\mathfrak{J}$. Then, for $\operatorname{Spec}(B)$ to be étale over $\operatorname{Spec}(A)$ at the point $\mathfrak{q}$, it is necessary and sufficient that $u' \notin \mathfrak{q}$.*

Proof. The hypothesis that $u' \notin \mathfrak{q}$ implies that $\Omega^1_{B_\mathfrak{q}/A_\mathfrak{p}} = 0$ ((**0**, 20.5.9)), hence (i) follows from (17.2.1). Moreover, under the hypotheses of (ii), B is a free A-module by virtue of the Euclidean division; as the annihilator $\mathfrak{J}'$ of $\Omega^1_{B/A}$ is then equal to $u'B$ by virtue of (18.4.1) and $\Omega^1_{B/A}$ is a B-module of finite presentation (16.4.22), the annihilator of $\Omega^1_{B_\mathfrak{q}/A_\mathfrak{p}}$ is equal to $u'B_\mathfrak{q}$ (Bourbaki, *Alg. comm.*, ch. II, § 2, no. 4, formule (g)), and (ii) follows from (i) and from the implication $c) \Rightarrow a)$ in (17.6.1). □

Corollary (18.4.3). — *Let the notation be those of (18.4.2), suppose that F is unitaire and generates $\mathfrak{J}$. Then, for B to be an étale A-algebra, it is necessary and sufficient that u' be invertible in B (or, what amounts to the same, that the ideal of $A[\mathrm{T}]$ generated by F and F' be equal to $A[\mathrm{T}]$).*

Proof. In view of (18.4.2), (ii), saying that $\operatorname{Spec}(B)$ is étale over $\operatorname{Spec}(A)$ means that u' does not belong to any prime ideal of B, i.e. that it is invertible in B. □

We say that a monic polynomial $F \in A[\mathrm{T}]$ such that the ideal of $A[\mathrm{T}]$ generated by F and F' is equal to $A[\mathrm{T}]$ is *separable*; it is immediate that this definition coincides with the usual definition (Bourbaki, *Alg.*, ch. V, § 7, no. 6) when A is a field.

Lemma (18.4.4). — *Let A be a local ring, B a finite monogenic A-algebra, u a generator of the A-algebra B. Let $\mathfrak{n}_i$ ($1 \leqslant i \leqslant r$) be the maximal ideals of B such that* $\operatorname{Spec}(B)$ *is formally unramified over* $\operatorname{Spec}(A)$ *at the points $\mathfrak{n}_i$. Then there exists a monic unitaire polynomial $F \in A[\mathrm{T}]$ such that $F(u) = 0$; such a polynomial F satisfies the conditions $F'(u) \notin \mathfrak{n}_i$ for all i ($1 \leqslant i \leqslant r$). Moreover, if k is the residue field of A, $f \in k[\mathrm{T}]$ the minimal polynomial of the image of u in $B \otimes_A k$, there exists an $F \in A[\mathrm{T}]$ such that $F(u) = 0$; such an F satisfies the conditions $F'(u) \notin \mathfrak{n}_i$ for $1 \leqslant i \leqslant r$.* **IV-4** | 119

Proposition (18.4.5). — *Let A be a local ring, k its residue field, B an A-algebra. Suppose moreover that the residue field k is* infinite, *or else that B is a local ring. Let n be the rank of $L = B \otimes_A k$ over k. For B to be a formally unramified A-algebra (resp. étale), it is necessary and sufficient that there exist a monic separable polynomial $F \in A[\mathrm{T}]$ such that B is isomorphic to a quotient of $A[\mathrm{T}]/F.A[\mathrm{T}]$ (resp. isomorphic to $A[\mathrm{T}]/F.A[\mathrm{T}]$). Moreover, one can (and necessarily must) suppose that F is of degree n (resp. of degree n).*

Proof. The conditions are sufficient by virtue of (18.4.2), without assuming either that k is infinite or that B is local. To see that the conditions are necessary, note that if B is a formally unramified A-algebra, L is a finite algebra and separable over k, hence a direct composite of a finite number of fields, finite separable extensions k_j of k ($1 \leqslant j \leqslant r$), k_j being generated by an element $\bar{\bar{\xi}}_j$, of minimal polynomial f_j ($1 \leqslant j \leqslant r$) (Bourbaki, *Alg.*, ch. V, § 11, no. 4, prop. 4). Let us show that by virtue of the hypotheses made on k or B, there exists an element $\bar{\bar{\xi}}$ of L generating the k-algebra L. This is immediate if B is local, since then $r = 1$. Otherwise, k being assumed infinite, one can suppose that the irreducible polynomials $f_j \in k[\mathrm{T}]$ are all *distinct* (by replacing each $\bar{\bar{\xi}}_j$ by $\bar{\bar{\xi}}_j + a_j$, for a suitable element $a_j \in k$); if we set $f = f_1 f_2 \ldots f_r$, it is clear that L is isomorphic to $k[\mathrm{T}]/f.k[\mathrm{T}]$ in both cases, hence generated by an element $\bar{\bar{\xi}}$ of minimal polynomial $f \in k[\mathrm{T}]$ of degree n. Note now that $\mathfrak{m}B$ is contained in the radical of B; as $1, \bar{\bar{\xi}}, \ldots, \bar{\bar{\xi}}^{n-1}$ form a base of L over k, it follows by Nakayama's lemma that $1, u, \ldots, u^{n-1}$ generate the A-module B, and hence there exists a monic unitaire polynomial $F \in A[\mathrm{T}]$ of degree n such that $F(u) = 0$; moreover, since $\bar{\bar{\xi}}$ is a root of the canonical image of F in $k[\mathrm{T}]$, this image is necessarily equal to f. But then $F'(\bar{\bar{\xi}})$ of F in L is $f'(\bar{\bar{\xi}})$, and as $f'(\bar{\bar{\xi}}) \neq \mathfrak{n}_i$ by (18.4.4), this proves that $F'(u)$ is invertible in B, hence F is a separable polynomial. Finally, if B is an étale A-algebra, B being an A-module flat of finite presentation (1.4.7), and $1, u, \ldots, u^{n-1}$ form a base of the A-module B (Bourbaki, *Alg. comm.*, ch. II, § 3, no. 2, prop. 5), in other words the A-algebra B is isomorphic to $A[\mathrm{T}]/F.A[\mathrm{T}]$. ☐

Theorem (18.4.6). — *(Chevalley). —*

- *(i) Let $f : X \to Y$ be a morphism locally of finite type, x a point of X, $y = f(x)$, and set $A = \mathscr{O}_{Y,y}$. For $\mathscr{O}_{X,x}$ to be an A-algebra formally unramified (resp. formally étale), it is necessary and sufficient that there exist a monic unitaire polynomial $F \in A[\mathrm{T}]$ and a maximal ideal $\mathfrak{n}$ such that $B = A[\mathrm{T}]/F.A[\mathrm{T}]$ (resp. an algebra quotient B of $A[\mathrm{T}]/F.A[\mathrm{T}]$) such that $\mathscr{O}_{X,x}$ is A-isomorphic to $B_{\mathfrak{n}}$ and that, if u is the image of T in B, one has $F'(u) \notin \mathfrak{n}$.*
- *(ii) Suppose in addition that f is locally of finite presentation. Then, for f to be étale at the point x, it is necessary and sufficient that we can moreover take $B = A[\mathrm{T}]/F.A[\mathrm{T}]$.*

Proof. The conditions are sufficient by virtue of (18.4.2). To see that they are necessary, and taking **IV-4** | 120
into account the remark (17.1.2), when f is formally unramified and quasi-finite, since f is affine, it follows from (8.12.8) that there exists a finite A-algebra C and a maximal ideal $\mathfrak{r}$ such that $\mathscr{O}_{X,x}$ is A-isomorphic to $C_{\mathfrak{r}}$. Moreover, by (17.4.1.2) the residue field $C_{\mathfrak{r}}/\mathfrak{r}C_{\mathfrak{r}}$ is a finite separable extension of $k = A/\mathfrak{m}$. Let $\mathfrak{r}_i$ ($1 \leqslant i \leqslant h$) be the maximal ideals of the semi-local ring C *other than* $\mathfrak{r}$; there exists an element $u \in C$ belonging to all the $\mathfrak{r}_i$ and such that its image in $C/\mathfrak{r}$ equals v (Bourbaki, *Alg. comm.*, ch. II, § 1, no. 2, prop. 5). We shall now show that the sub-A-algebra $B = A[u]$ of C is finite on B and $\mathfrak{n} = \mathfrak{r} \cap B$ answers the question.

To handle the case where $\mathscr{O}_{X,x}$ is a formally ramified A-algebra, it suffices to prove that $B_{\mathfrak{n}}$ is **IV-4** | 121
isomorphic to $C_{\mathfrak{r}}$; indeed, $B_{\mathfrak{n}}$ will then be formally unramified over A and the existence of the polynomial $F \in A[\mathrm{T}]$ having the properties of the statement will follow from (18.4.4). Note now that, since f is formally unramified, $B_{\mathfrak{n}}$ is a $B'_{\mathfrak{n}}$-algebra formally étale; with the preceding notations,

since B' does not belong to $\mathfrak{n}'$ by hypothesis, the morphism $\mathrm{Spec}(B') \to \mathrm{Spec}(A)$ is étale at the point $\mathfrak{n}'$ by (18.4.2), (ii). As by hypothesis $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is étale at the point $\mathfrak{n}$, one concludes by (17.3.4) that $\mathrm{Spec}(B) \to \mathrm{Spec}(B')$ is étale at the point $\mathfrak{n}$; but since this morphism is an immersion, it cannot be étale at a point unless it is a local isomorphism at that point (17.9.1), hence $B_{\mathfrak{n}}$ and $B'_{\mathfrak{n}'}$ are isomorphic.

Finally, suppose that $\mathscr{O}_{X,x}$ is a formally étale A-algebra; with the preceding notations, by (17.1.5) that $B_{\mathfrak{n}}$ is a $B'_{\mathfrak{n}}$-algebra formally étale; but since the homomorphism $B'_{\mathfrak{n}'} \to B_{\mathfrak{n}}$ is surjective, this can only happen if this homomorphism is *bijective* ((**0**, 19.10.3), (i)). This completes the proof of (18.4.6). ☐

Corollary (18.4.7). — *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X. For f to be formally unramified at the point x, it is necessary and sufficient that there exist an open neighbourhood U of x such that $f|U$ factorises as $U \xrightarrow{j} X' \xrightarrow{h} Y$, where h is an étale morphism and j is a closed immersion.*

Proof. We can obviously reduce to the case where $Y = \mathrm{Spec}(R)$ is affine and f is of finite type. If $A = \mathscr{O}_{Y,y}$, with $y = f(x)$, the condition for f to be formally unramified at the point x is equivalent, by virtue of (17.4.1.2), to saying that $\mathscr{O}_{X,x}$ is a formally unramified A-algebra. If it is so, we can apply (18.4.6), (i); replacing Y as needed by an affine neighbourhood of y, we may suppose (with the notation of (18.4.6)) that the polynomial F is the image in $A[\mathrm{T}]$ of a monic unitaire polynomial $G \in R[\mathrm{T}]$. We then set $X' = \mathrm{Spec}(R[\mathrm{T}]/G.R[\mathrm{T}])$; let x' be the image of the point $\mathfrak{n}$ of $\mathrm{Spec}(B)$ by the morphism corresponding to the composite homomorphism $R[\mathrm{T}]/G.R[\mathrm{T}] \to A[\mathrm{T}]/F.A[\mathrm{T}] \to B$. It follows from (18.4.2) that the canonical morphism $h : X' \to Y$ is étale at the point x, hence, restricting as needed U and X', we can suppose h étale. On the other hand, by virtue of (**I**, 6.5.1), (ii)) and (1.7.2), it follows from the fact that we have a local homomorphism $\varphi : \mathscr{O}_{X',x'} \to \mathscr{O}_{X,x}$ that this homomorphism corresponds to a morphism $j : U \to X'$ from an open neighbourhood U of x in X'; in applying (**I**, 6.5.1), (i)), suppose that $h \circ j = f|U$, hence j is of finite type ((**I**, 6.3.4), (v)); finally, the homomorphism φ is surjective by (18.4.6), (i), hence it follows from ((**I**, 6.5.4), (i)) that we can, restricting further U and X', suppose that j is a closed immersion. This proves the necessity of the condition stated; the sufficiency is immediate (17.1.3), (i) and (ii). ☐ **IV-4** | 122

Corollary (18.4.8). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be non ramified (resp. étale), it is necessary and sufficient that there exist a family of flat morphisms $g_\alpha : Y'_\alpha \to Y$ and, for each α, an open subset U_α in $X'_\alpha = X \times_Y Y'_\alpha$, such that, if $g'_\alpha : X'_\alpha \to X$ and $f'_\alpha : X'_\alpha \to Y'_\alpha$ are the canonical projections, the $g'_\alpha(U_\alpha)$ form a covering of X, and that each of the composed morphisms $U_\alpha \to X'_\alpha \xrightarrow{f'_\alpha} Y'_\alpha$ is a closed immersion (resp. open). One can moreover take the g_α to be étale.*

Proof. The necessity of the condition for étale morphisms is trivial, taking a single Y'_α equal to X, the morphism g_α corresponding to the diagonal, and the open subset $U \subset X \times_Y X$ being the diagonal. When f is non ramified, the necessity of the condition follows from (18.4.7); on the other hand we need to find an open covering (V_α) of X such that for each α, $f|V_\alpha$ factorises as $V_\alpha \xrightarrow{j_\alpha} X'_\alpha \xrightarrow{g_\alpha} Y$, where j_α is a closed immersion and g_α an étale morphism. Then $j_\alpha : V_\alpha \to Y'_\alpha$ is a V_α-section of X'_α, and since $f|V_\alpha$ is étale, s_α is a V_α-section of X'_α, and since g_α is étale, the morphism s_α is an immersion ouverte (17.4.1), and it suffices to take $U_\alpha = s_\alpha(V_\alpha)$ to answer the question.

The sufficiency of the conditions follows from (17.7.1); one concludes at each point x of $g'(U_\alpha)$, hence in all of X since the $g'_\alpha(U_\alpha)$ cover X. ☐

Proposition (18.4.9). — *Let S be a prescheme, $f : X \to S$ a morphism, $h : Y \to S$ an S-morphism, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$. The following conditions are equivalent:*

- a) *h is étale at the point y and g is flat at the point x.*
- b) *h is non ramified at the point y and f is flat at the point x.*

Proof. Since $f = h \circ g$, $a)$ implies that f is flat at the point x (2.1.6), and it is obvious that h being non ramified at the point y, hence $a)$ implies $b)$.

To prove that $b)$ implies $a)$, we can first suppose that h is non ramified (replacing Y by an open neighbourhood of y); then, replacing S by an open neighbourhood of $s = h(y) = f(x)$, one can suppose that there exists an étale morphism $u : S' \to S$, a point y' of $Y' = Y_{(S')}$ above y in Y' and

an open neighbourhood V of y' in Y' such that, if the restriction of h' to V is a closed immersion (18.4.8). If one proves that h' is étale at the point y', it will follow that h is étale at the point y (17.7.1), (ii); moreover, $f' = f_{(S')} : X_{(S')} \to S'$ is flat at every point x' above x. As the projections $v : Y' \to Y$, $w : X' \to X$ are étale morphisms (hence flat), if one proves that $g' = g_{(S')} : X_{(S')} \to Y_{(S')}$ is flat at the point x', it will follow from (2.2.11), (iv) that g is flat at the point x. We can hence reduce to the case where h is a *closed immersion* of finite presentation, f being supposed flat at the point x **IV-4** | 123
and f is of finite type. Let $\mathscr{J}$ be the quasi-coherent Ideal of $\mathscr{O}_S$ of finite type defining the closed sub-prescheme Y of S. The hypothesis that f is flat at the point x implies that the homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{X,x}$ is injective (($\mathbf{0}$, 6.5.1)); *a fortiori* the homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y}$ is injective, which means that $\mathscr{J}_s = 0$, since it is the kernel of the preceding homomorphism. Since $\mathscr{J}$ is of finite type, there exists an open neighbourhood U of s in S such that $\mathscr{J}|U = 0$ (($\mathbf{0}$, 5.2.2)). We can hence suppose that h is an open immersion, and it is clear that g is flat at the point x. □

Proposition (18.4.10). — *Let us denote by $\mathbf{P}(f, x)$ a property satisfying the following conditions:*

1° *For every morphism $f : X \to Y$ and every local isomorphism $h : Y \to Z$, $\mathbf{P}(f, x)$ is equivalent to $\mathbf{P}(h \circ f, x)$ for $x \in X$.*

2° *For every morphism $f : X \to Y$, every étale morphism $g : Y' \to Y$, every point $x \in X$, if we set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$ and if $x' \in X'$ is above x, the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(f', x')$ are equivalent ("invariance by étale base change").*

Let then S be a prescheme, $f : X \to S$ and $h : Y \to S$ two morphisms, $g : X \to Y$ an S-morphism, x a point of X, and suppose h étale at the point y. Then the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(g, x)$ are equivalent.

Proof. With the same notations as in the proof of (18.4.9) and replacing S (resp. Y) by an open neighbourhood of $s = h(y)$ (resp. y), we may find, by (18.4.8), an étale morphism $u : S' \to S$, a point y' of $Y' = Y_{(S')}$ and an open neighbourhood V of y' in Y' such that h' restricted to V is an open immersion; if $x' \in X'$ is any point above x, $\mathbf{P}(f', x')$ (resp. $\mathbf{P}(g', x')$) is then equivalent to $\mathbf{P}(f, x)$ (resp. $\mathbf{P}(g, x)$). The hypothesis implies that $\mathbf{P}(g, x)$ is then equivalent to $\mathbf{P}(h \circ g, x)$, and we deduce the conclusion. □

Let then S be a prescheme, $f : X \to S$ and $h : Y \to S$ two morphisms, $g : X \to Y$ an S-morphism, x a point of X, and suppose h étale at the point y. Then the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(g, x)$ are equivalent.

(18.4.11). *Examples.* — One can take for the property $\mathbf{P}(f, x)$ any one of the following, taking into account (17.7.4), (ii):

(i) f is flat at the point x (2.2.11), (iv);
(ii) f is locally of finite presentation and of coprofondeur $\leqslant n$ at the point x ((6.8.1) and (6.7.8));
(iii) f is locally of finite presentation and of Cohen-Macaulay at the point x ((6.8.1) and (6.7.8));
(iv) f is locally of finite presentation and possesses the property (S_n) at the point x ((6.8.1) and (6.7.8));
(v) f is locally of finite presentation and possesses the property (R_n) at the point x ((6.8.1) and (6.7.8));
(vi) f is locally of finite presentation and normal at the point x ((6.8.1) and (6.7.8));
(vii) f is locally of finite presentation and reduced at the point x ((6.8.1) and (6.7.8));
(viii) f is non ramified at the point x (17.7.4);
(ix) f is smooth at the point x (17.7.4);
(x) f is étale at the point x (17.7.4).

Corollary (18.4.12). — *(i) Let $f : X \to Y$ be a morphism locally of finite type, x a point of X. If f is flat and formally unramified at the point x, then, in every factorisation $f|U : U \xrightarrow{j} X' \xrightarrow{h} Y$, where U is an open neighbourhood of x, j a closed immersion and h an étale morphism, the homomorphism $\mathscr{O}_{X',j(x)} \to \mathscr{O}_{X,x}$ corresponding to j is bijective (in particular $\mathscr{O}_{X,x}$ is an essentially étale $\mathscr{O}_{Y,f(x)}$-algebra (18.6.1)).*

(ii) For a morphism $f : X \to Y$ to be étale, it is necessary and sufficient that it be locally of finite type, formally unramified and flat.

Proof. (i) Since the homomorphism $\mathscr{O}_{X',j(x)} \to \mathscr{O}_{X,x}$ is surjective, it suffices to prove that it is injective, and for that it suffices to prove that $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{X',j(x)}$-module *faithfully flat*, or merely *flat* ((**0**, 6.5.1) and (6.6.2)); in other words, it amounts to showing that j is a flat morphism at the point x; but since $h \circ j = f$ is flat at the point x by hypothesis and h is étale, this follows from (18.4.10) and (18.4.11), (i). IV-4 | 124

(ii) It suffices to prove the sufficiency of the stated conditions. For every $x \in X$, we have in a neighbourhood open U of x a factorisation of $f|U$ having the properties stated in (i), and the fact that f is flat and formally non ramified at *all* the points of U. For every $x \in X$, the quasi-coherent Ideal of $\mathscr{O}_{X'}$ corresponding to the closed sub-prescheme X' associated to j, we have $\mathscr{J}_{j(x)} = 0$ for every $z \in U$, hence j is an open immersion, and f is hence étale at every point of X. □

Corollary (18.4.13). — *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that y admits an open neighbourhood which is a reduced prescheme and has only a finite number of irreducible components. Then, for f to be étale at the point x, it is necessary and sufficient that f be flat and formally non ramified at the point x.*

Proof. There is only the sufficiency of the condition to prove. The question being local on X and Y, one can suppose (18.4.7) that f factorises as $X \xrightarrow{j} X' \xrightarrow{h} Y$ where h is étale and j a closed immersion. Then X' is reduced (17.5.7) and, replacing X' by an open neighbourhood of $j(x)$, one can suppose that X' has only a finite number of irreducible components. Then the maximal points of X' are above the maximal points of Y (2.3.4) and since they are finite, their number is finite. It all amounts to showing, with the notations of the proof of (18.4.12), that $\mathscr{J}_{x'} = 0$ for all points x' in a neighbourhood of $j(x)$ in X', knowing that $\mathscr{J}_{j(x)} = 0$. Now, replacing X' by an affine neighbourhood of $j(x)$, we can suppose that all the irreducible components of X' contain $j(x)$; if $X' = \operatorname{Spec}(A')$, and if $\mathfrak{p}'$ is the prime ideal of A' corresponding to the point $j(x)$, the morphism $\operatorname{Spec}(A'_{\mathfrak{p}'}) \to \operatorname{Spec}(A')$ is dominant, hence the corresponding homomorphism $A' \to A'_{\mathfrak{p}'}$ is *injective* since A' is reduced (**I**, 1.2.7). If $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A', $\mathfrak{J}$ identifies with an ideal of A', $\mathfrak{J}_{\mathfrak{p}'} = 0$ by hypothesis, hence $\mathfrak{J} = 0$. □

Corollary (18.4.14). — *Let A be a local ring with maximal ideal $\mathfrak{m}$, residue field k, B a finite A-algebra.*

(i) For B to be a formally unramified A-algebra, it is necessary and sufficient that $B \otimes_A k$ be a k-algebra étale.

(ii) For B to be a finite étale A-algebra, it is necessary and sufficient that $B \otimes_A k$ be an étale k-algebra and that B be a flat A-module (which is equivalent to saying ((**0**, 6.3.3)) *that for every maximal ideal $\mathfrak{n}$ of B (necessarily above $\mathfrak{m}$), $B_{\mathfrak{n}}$ is a flat A-module).*

Proof. It is clear that (ii) follows from (i) and (18.4.12), (ii), taking into account (17.1.2), (i). To prove (i), let us note that if $B \otimes_A k$ is an étale k-algebra, we have $\Omega^1_{B\otimes_A k/k} = 0$ (17.2.1). Since $\Omega^1_{B\otimes_A k/k} = \Omega^1_{B/A} \otimes_B k$ ((**0**, 20.5.5)) and since B is an A-algebra of finite type, $\Omega^1_{B/A}$ is a B-module of finite type ((**0**, 20.4.7)). But since B is a finite A-algebra, $\mathfrak{m}B$ is contained in the radical of B (Bourbaki, *Alg. comm.*, ch. V, § 2, no. 1, prop. 1), hence the lemma of Nakayama proves that $\Omega^1_{B/A} = 0$, and hence B is a formally unramified A-algebra (17.2.1). □ IV-4 | 125

18.5. Henselian local rings.

(18.5.1). Let X be a prescheme, $\mathscr{E}$ a locally free $\mathscr{O}_X$-Module of finite rank; the dual $\check{\mathscr{E}} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{O}_X)$ is therefore a locally free $\mathscr{O}_X$-Module whose rank at every point is equal to that of $\mathscr{E}$ at that point, and the canonical homomorphism $\mathscr{E} \to \mathscr{H}om_{\mathscr{O}_X}(\check{\mathscr{E}}, \mathscr{O}_X) = (\check{\mathscr{E}})^\vee$ is *bijective*. For every morphism $X' \to X$, set $\mathscr{E}_{(X')} = \mathscr{E} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$ which is a locally free $\mathscr{O}_{X'}$-Module, and consider the *set* $\Gamma(X', \mathscr{E}_{(X')})$ of sections of this $\mathscr{O}_{X'}$-Module over X'. We shall see that this defines a *contravariant representable functor*

$$(18.5.1.1) \qquad {}^t V_{\mathscr{E}} : X' \longmapsto \Gamma(X', \mathscr{E}_{(X')})$$

from the category of X-preschemes to the category of *sets* ((**0**, 8.1.8)). IV-4 | 126

First of all, this is indeed a well-defined functor, for if $f : X'' \to X'$ is an X-morphism, we have $\mathscr{E}_{(X'')} = f^*(\mathscr{E}_{(X')})$, whence ((**0**, 4.3.2)) a map $\Gamma(X', \mathscr{E}_{(X')}) \to \Gamma(X'', \mathscr{E}_{(X'')})$ which completes the definition of the functor ${}^t V$. Let us then show that the X-prescheme $\mathbf{V}(\check{\mathscr{E}})$ ((**II**, 1.7.8)) represents the functor ${}^t V$. It is immediate that $(\mathscr{E}_{(X')})^{\vee} = (\check{\mathscr{E}})_{(X')}$, hence $\mathbf{V}(\check{\mathscr{E}}) \times_X X' = \mathbf{V}((\check{\mathscr{E}})_{(X')})$; taking into account (**I**, 3.3.14), we are reduced to defining a bijection $\Gamma(\mathbf{V}(\check{\mathscr{E}}_{(X')})/X') \xrightarrow{\sim} \Gamma(X', \mathscr{E})$, and to verify that the bijection $\Gamma(\mathbf{V}(\check{\mathscr{E}})/X) \xrightarrow{\sim} \Gamma(X, \mathscr{E})$ is functorial in X'. Now, on the one hand a section $u \in \Gamma(X, \mathscr{E})$ identifies canonically with an $\mathscr{O}_X$-homomorphism $u : \mathscr{O}_X \to \mathscr{E}$, to which corresponds by transposition an $\mathscr{O}_X$-homomorphism ${}^t u : \check{\mathscr{E}} \to \mathscr{O}_X$, and hence an $\mathscr{O}_X$-Algebra homomorphism $v : \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \to \mathscr{O}_X$. Let $\mathrm{Al}(X, \mathscr{E}, \mathscr{J})$ denote the set of $u \in \Gamma(X, \mathscr{E})$ such that $\mathscr{J}$ is contained in the kernel of v; it follows immediately from these definitions that

$$X' \longmapsto \mathrm{Al}(X', \mathscr{E}_{(X')}, \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'})$$

is a representable functor by $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$. If $X' = \mathrm{Spec}(A)$ is affine and such that $\mathscr{E}_{(X')}$ is isomorphic to $\mathscr{O}^n_{X'}$, $\Gamma(X', \mathscr{E}_{(X')})$ can be identified with the set A^n; in an obvious way, we can say that the object $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ represents "the algebraic subset of the twisted affine space over X defined by $\mathscr{E}$". We write this X-prescheme $\mathbf{Al}(\mathscr{E}, \mathscr{J})$. We note also that if $\mathscr{J}$ is a type-finite Ideal of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, $\mathbf{Al}(\mathscr{E}, \mathscr{J})$ is an X-prescheme of finite presentation, since $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$ is an $\mathscr{O}_X$-Algebra of finite presentation.

(18.5.2). Recall ((**II**, 1.7.8)) that by definition $\mathbf{V}(\check{\mathscr{E}}) = \mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}))$. Let $\mathscr{J}$ be a quasi-coherent Ideal of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, so that $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ is a closed sub-prescheme of $\mathbf{V}(\check{\mathscr{E}})$; we will interpret it as *representing a functor* from the category of X-preschemes to that of sets. Let us note for this that a section $u \in \Gamma(X, \mathscr{E})$ identifies canonically with an $\mathscr{O}_X$-homomorphism $u : \mathscr{O}_X \to \mathscr{E}$, to which corresponds by transposition an $\mathscr{O}_X$-Algebra homomorphism $v : \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \to \mathscr{O}_X$. Let $\mathrm{Al}(X, \mathscr{E}, \mathscr{J})$ denote the set of $u \in \Gamma(X, \mathscr{E})$ such that $\mathscr{J}$ is *contained in the kernel* of v; if $X' = \mathrm{Spec}(A)$ is affine and such that $\mathscr{E}_{(X')}$ is isomorphic to $\mathscr{O}^n_{X'}$ and $\mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$ is a $\mathscr{O}_{X'}$-Module of the form $\widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of the polynomial algebra $C = A[\mathrm{T}_1, \dots, \mathrm{T}_n]$; the set $\mathrm{Al}(X', \mathscr{E}_{(X')}, \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'})$ then identifies with the subset of A^n consisting of points $(t_1, \dots, t_n)$ such that $\mathrm{P}(t_1, \dots, t_n) = 0$ for all polynomials $\mathrm{P} \in \mathfrak{J}$; thus we can say that the object $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ represents "the algebraic subset of the twisted affine space over X defined by $\mathscr{E}$". We also write this X-prescheme $\mathbf{Al}(\mathscr{E}, \mathscr{J})$. We note that if $\mathscr{J}$ is a type-finite Ideal of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, $\mathbf{Al}(\mathscr{E}, \mathscr{J})$ is an X-prescheme of finite presentation, since $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$ is an $\mathscr{O}_X$-Algebra of finite presentation.

Lemma (18.5.3). — *Let S be a prescheme, $f : X \to S$ a finite and locally free morphism. Consider the contravariant functor from the category of S-preschemes to the category of sets*

$$S' \longmapsto \mathrm{Of}(X \times_S S') \tag{18.5.3.1}$$

where $\mathrm{Of}(X \times_S S')$ *denotes the set of subsets that are both open and closed of the underlying space of* $X \times_S S'$*. Then this functor is representable by an S-prescheme* $\mathbf{Of}(X)$*, which is affine, étale and of finite presentation over S.*

Proof. We have by hypothesis $X = \mathrm{Spec}(\mathscr{B})$, where $\mathscr{B} = f_*(\mathscr{O}_X)$ is a finite $\mathscr{O}_S$-Algebra and locally **IV-4** | 127
free. Set $X' = X \times_S S'$, $f' = f_{(S')}$, $\mathscr{B}' = \mathscr{B} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} = f'_*(\mathscr{O}_{X'})$, so that $X' = \mathrm{Spec}(\mathscr{B}')$; there is then a canonical bijection, functorial in S', from the set $\mathrm{Of}(X')$ to the set $\mathrm{Id}(\mathscr{B}')$ of *idempotents* of the ring $\Gamma(S', \mathscr{B}') = \Gamma(X', \mathscr{O}_X)$. Indeed, by virtue of the equivalence of the category of affine S'-schemes and the opposite category of quasi-coherent $\mathscr{O}_{S'}$-Algebras ((**II**, 1.2.7) and (1.3.1)), there is a canonical bijective correspondence between the decompositions of X' into a *sum* $X'_1 \sqcup X'_2$ of two open subsets induced by open subsets of X' and the decompositions of $\mathscr{B}'$ into a *direct composite* of two Ideals $\mathscr{B}'_1$, $\mathscr{B}'_2$; these latter in turn form a set in canonical bijective correspondence with $\mathrm{Id}(\mathscr{B}')$. It suffices therefore to prove the lemma for the functor $S' \mapsto \mathrm{Id}(\mathscr{B}')$.

For this, we shall show that there exists a type-finite Ideal $\mathscr{J}$ of $\mathbf{S}_{\mathscr{O}_S}(\mathscr{B})$ such that $\mathrm{Id}(\mathscr{B}')$ is of the form $\mathrm{Al}(S', \check{\mathscr{B}}, \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'})$ (18.5.2). Note to this end that since $\mathscr{B}$ is an $\mathscr{O}_S$-Algebra, its inverse image $\mathscr{B}_{\mathbf{V}(\check{\mathscr{B}})}$ is an $\mathscr{O}_{\mathbf{V}(\check{\mathscr{B}})}$-Algebra, and we can form the "square" c^2, in this Algebra, of the canonical section c (18.5.1); it corresponds canonically to a homomorphism $u^{(2)} : \mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}}) \to \mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}}) \otimes_{\mathscr{O}_S} \mathscr{B}$, and one verifies immediately that for every affine open subset W of S, with the notations of (18.5.1), $u^{(2)}$ corresponds to the C-module homomorphism such that $u^{(2)}(1) = \sum_k (\sum_{i,j} c_{ijk} \mathrm{T}_i \mathrm{T}_j) \otimes e_k$, where (c_{ijk}) is the table of multiplication of the algebra $\Gamma(W, \mathscr{B})$. Let us show that the Ideal $\mathscr{J}$ of $\mathbf{S}_{\mathscr{O}_S}(\mathscr{B})$

generated by the kernel of the homomorphism $u - u^{(2)}$ answers the question. Indeed, it suffices to note that the ideal $\Gamma(W, \mathscr{J})$ of C is generated by the polynomials $\mathrm{P}_k(\mathrm{T}_1, \ldots, \mathrm{T}_n) = \mathrm{T}_k - \sum_{i,j} c_{ijk}\mathrm{T}_i\mathrm{T}_j$ and that the idempotents of $\Gamma(W, \mathscr{B})$ are precisely the elements $\sum_i t_i e_i$ of this algebra such that $(t_1, \ldots, t_n)$ annihilates all the polynomials P_k $(1 \leqslant i \leqslant n)$. We deduce from (18.5.2) that the S-prescheme affine $\mathbf{Of}(X) = \mathbf{Al}(\mathscr{B}, \mathscr{J})$ does represent the functor (18.5.3.1). It remains to show that $\mathbf{Of}(X)$ is étale over S, or equivalently, that it is *formally étale* over S. But if S' is an S-prescheme, S'_0 a closed sub-prescheme of S' defined by a locally nilpotent Ideal (and having the same underlying space as S'), it is clear that $X' = X \times_S S'$ and $X'_0 = X \times_S S'_0$ have the same underlying space, hence the canonical map $\mathrm{Of}(X') \to \mathrm{Of}(X'_0)$ is *bijective*, which completes the proof (17.1.1). □

Proposition (18.5.4). — *Let S be a prescheme, S_0 a closed sub-prescheme of S; consider the following properties:* **IV-4** | 128

a) *For every finite morphism $g : S' \to S$, the canonical map*

(18.5.4.1) $$\mathrm{Of}(S') \longrightarrow \mathrm{Of}(S' \times_S S_0)$$

is bijective (*cf.* (18.5.3)).

a′) *For every finite and locally free morphism $g : S' \to S$, the map* (18.5.4.1) *is bijective.*

b) *For every étale and separated morphism $g : S' \to S$, the canonical map*

(18.5.4.2) $$\Gamma(S'/S) \longrightarrow \Gamma(S' \times_S S_0/S_0)$$

is bijective.

Condition b) implies a′); if moreover S is quasi-compact and quasi-separated, condition a) implies b).

Proof. Let us first prove that b) implies a'). Suppose therefore b) satisfied, and let $g : S' \to S$ be a finite and locally free morphism; set $S'_0 = S' \times_S S_0$, so that $g_0 = g_{(S_0)} : S'_0 \to S_0$ is finite and locally free. Then it follows from (18.5.3) that $\mathrm{P} = \mathbf{Of}(S')$ is an S-prescheme *étale and separated*; moreover, by the definition of the functor Of, we see immediately that if we set $\mathrm{P}_0 = \mathbf{Of}(S'_0)$ (for the category of S_0-preschemes), we have $\mathrm{P}_0 = \mathrm{P} \times_S S_0$. This being so, by definition we have the commutative diagram

$$\begin{tikzcd}
\Gamma(\mathrm{P}/S) \arrow[r] \arrow[d, "l"] & \Gamma(\mathrm{P}_0/S_0) \arrow[d, "l"] \\
\mathrm{Of}(S') \arrow[r] & \mathrm{Of}(S'_0)
\end{tikzcd}$$

where the vertical arrows are the bijections. Since hypothesis b), applied to the morphism $\mathrm{P} \to S$, implies that the first row is a bijection, the same holds for the second, which establishes our assertion.

Before proving that a) implies b) when S is quasi-compact and quasi-separated, we will establish the following lemma.

Lemma (18.5.4.3). — *If S and S_0 satisfy condition a) of* (18.5.4)*, then, for every finite morphism $g : S' \to S$, S' is the unique open and closed neighbourhood of $S'_0 = g^{-1}(S_0) = S' \times_S S_0$ in S'.*

Proof. Indeed, it amounts to saying that if T' is a closed subset of S' such that $T' \cap S'_0 = \varnothing$, then $T' = \varnothing$. Now, if we also denote by T' the closed sub-prescheme of S' having T' as underlying space, the composite morphism $h : T' \to S' \xrightarrow{g} S$ is finite and $h^{-1}(S_0)$ is empty; condition a), applied to the morphism h, implies that T' is necessarily empty. □

This lemma being established, let us first prove that, under hypothesis a), the map (18.5.4.2) is *injective*. Indeed, if u', u'' are two S-sections of S', the fact that the morphism $S' \to S$ is non ramified implies that the prescheme of coincidences of u' and u'' is induced on an open subset U of S (17.4.6). If the restrictions to S_0 of u' and u'' are equal, then U contains S_0, hence is equal to S by virtue of the lemma (18.5.4.3) applied to the case where $S' = S$.

It remains to show that, under hypothesis a), the map (18.5.4.2) is *surjective* (when S is quasi-compact and quasi-separated). Let therefore $u_0 : S_0 \to S'_0$ be an S_0-section of S'_0; $u_0(S_0)$ being quasi-compact in S', can be covered by a finite number of open affine subsets V_i such that the restriction **IV-4** | 129

$g|V_i$ is a morphism of finite type; if V is the union of the V_i, we conclude that $g|V : V \to S$ is a morphism *of finite presentation*, being separated and locally of finite presentation (1.6.1). Replacing S' by V, one can hence suppose that g is of finite presentation. Since S is quasi-compact and quasi-separated (g quasi-finite and separated) (17.6.1), it follows from the "Main theorem" (8.12.6) that g factorises as $S' \xrightarrow{j} S'' \xrightarrow{f} S$, where j is an open immersion and f a *finite* morphism. Set $S''_0 = S'' \times_S S_0$, $j_0 = j_{(S_0)} : S'_0 \to S''_0$, which is an open immersion and $f_0 = f_{(S_0)} : S''_0 \to S_0$, which is also an S_0-section of S''_0. Then u_0 is an open immersion of S_0 into S'_0 (17.4.1), hence $g_0 : S'_0 \to S_0$ is étale, hence an open immersion of S_0 into S'_0, hence $X_0 = u_0(S_0)$ is *open and closed* in S''_0. In virtue of condition a), there exists an open and closed subset X of S'' such that $X \cap S''_0 = X_0$. Let us show that the morphism $f : S'' \to S$ is *étale at the points of* X: indeed, the set of points of X where $f|X$ is étale is open and contains $X_0 \subset S'_0$. But the lemma (18.5.4.3) applied to the finite morphism $f|X$ proves that $U = X$. On the other hand, $S' \cap X$ is open in X and contains X_0, hence by the same reasoning proves that $S' \cap X = X$, i.e. $X \subset S'$. It remains to show that, for every $s \in S$, the geometric number $n(s)$ of points of $X \cap f^{-1}(s)$ is equal to 1, for it will follow that $f|X$ is an isomorphism of the open subset $X \subset S'$ onto S, whose inverse will be the S-section sought extending u_0. But since $f|X$ is étale and finite, $s \mapsto n(s)$ is continuous on S (18.2.8), and since $X \cap S'_0 = X_0$, we have $n(s) = 1$ in S_0; the set of points $s \in S$ such that $n(s) = 1$ being open in S and containing S_0, is equal to S by (18.5.4.3). □

Remark (18.5.4.4). — One can show that the statement (18.5.4) remains valid when, in condition b), one supposes merely the morphism g étale (but not necessarily separated) [**?**, exp. XII].

Definition (18.5.5). — We say that a prescheme S and a closed sub-prescheme S_0 of S form a *Henselian couple* if they satisfy condition a) of (18.5.4).

Taking into account (**I**, 5.1.8), it amounts to the same to say that (S, S_0) is a Henselian couple or that $(S_{\mathrm{red}}, (S_0)_{\mathrm{red}})$ is one.

Proposition (18.5.6). — *(i) If (S, S_0) is a Henselian couple, then, for every finite morphism $f : S' \to S$, if S'_0 is the closed sub-prescheme $f^{-1}(S_0)$ of S', the couple (S', S'_0) is Henselian.*

(ii) Let $S = \coprod_\alpha S^{(\alpha)}$ be a sum of preschemes, S_0 a closed sub-prescheme of S, the sum of sub-preschemes $S_0^{(\alpha)}$ of the $S^{(\alpha)}$. For the couple (S, S_0) to be Henselian, it is necessary and sufficient that each of the couples $(S^{(\alpha)}, S_0^{(\alpha)})$ be Henselian.

Proof. Assertion (i) is an immediate consequence of the definition, since for every finite morphism **IV-4** | 130
$g : S'' \to S'$, $f \circ g : S'' \to S$ is a finite morphism. Similarly, under the conditions of (ii), for every morphism $g : S' \to S$ to be finite, it is necessary and sufficient that each of its restrictions $g^{(\alpha)} : S'^{(\alpha)} \to S^{(\alpha)}$ be finite, and the conclusion follows from (18.5.6), (ii) and the remark (18.5.7). □

Remark (18.5.7). — Let $S = \mathrm{Spec}(A)$ be an affine scheme, S_0 a closed sub-prescheme defined by an ideal $\mathfrak{J}$ of A. Then, if the couple (S, S_0) is Henselian, the ideal $\mathfrak{J}$ is necessarily contained in the *radical* of A. Indeed, if $\mathfrak{m}$ is a maximal ideal of A, $\mathfrak{m}$ must belong to $V(\mathfrak{J}) = S_0$, by virtue of (18.5.4.3), whence the conclusion. In particular, suppose that S_0 is *reduced to a point*, then $\mathfrak{J}$ must be the radical of A, so A must be a *local* ring, and S_0 the unique closed point of $\mathrm{Spec}(A)$.

Definition (18.5.8). — We say that a ring A is *Henselian* if it is semi-local and if, denoting by $\mathfrak{r}$ the radical of A, the couple $(\mathrm{Spec}(A), \mathrm{Spec}(A/\mathfrak{r}))$ is Henselian. We call a *Henselian local scheme* a scheme isomorphic to the spectrum of a Henselian local ring.

Proposition (18.5.9). — *(i) For a semi-local ring A to be Henselian, it is necessary and sufficient that it be a direct composite of Henselian local rings.*

(ii) For a local ring A to be Henselian, it is necessary and sufficient that every finite A-algebra B be isomorphic to a product of local rings.

Proof. (i) Indeed, the definition, applied to $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/\mathfrak{r})$, shows that S_0 is a finite discrete set, closed in S, and S is the union of a finite number of open and closed subsets S_i ($1 \leqslant i \leqslant n$), pairwise disjoint, each containing exactly one of the maximal ideals $\mathfrak{m}_i$ of A; the conclusion follows from (18.5.6), (ii) and the remark (18.5.7).

(ii) Every finite morphism $S' \to \operatorname{Spec}(A)$ is of the form $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$, where B is a finite A-algebra. If k is the residue field of A, $\operatorname{Spec}(B \otimes_A k)$ is a finite Artinian spectrum, hence finite and discrete. To say that the couple $(\operatorname{Spec}(A), \operatorname{Spec}(k))$ is Henselian means that B is a direct composite of rings A_i such that $\operatorname{Spec}(A_i \otimes_A k)$ is reduced to a point, i.e. that each A_i (which is a finite A-algebra) must have only a single maximal ideal (Bourbaki, *Alg. comm.*, ch. V, § 2, no. 1, prop. 1). $\square$

The study of Henselian rings is thus essentially reduced to that of Henselian *local* rings.

Proposition (18.5.10). — *If A is a Henselian ring, every finite A-algebra B is a Henselian ring (hence a direct composite of Henselian local rings (18.5.9)).*

Proof. Indeed, if $\mathfrak{r}'$ is the radical of B, the inverse image of $\mathfrak{r}'$ in A is the radical $\mathfrak{r}$ of A, and every maximal ideal of B is a maximal ideal of B, hence the set $V(\mathfrak{r}')$ in $\operatorname{Spec}(B)$ is the inverse image of the set $V(\mathfrak{r})$ in $\operatorname{Spec}(A)$. The proposition is then a consequence of (18.5.6), (i). $\square$

Theorem (18.5.11). — *Let A be a local ring, $\mathfrak{m}$ its maximal ideal. The following conditions are equivalent:*

a) *A is Henselian, in other words, every finite A-algebra B is isomorphic to a product of local rings.*

a') *Condition a) is satisfied for all A-algebras B of the form $A[\mathrm{T}]/F.A[\mathrm{T}]$, where $F \in A[\mathrm{T}]$ is a monic polynomial.*

b) *Let $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(k)$. For every étale morphism $g : S' \to S$, if we set $S'_0 = S' \otimes_A k$, every S_0-section u_0 of S'_0 is the restriction of an S-section u of S'.*

c) *For every morphism locally of finite type $f : X \to S$, separated, and every point $x \in X$ such that $f(x)$ is equal to the closed point s of S and that f is quasi-finite at the point x ((**III**, Err.20)), X is the sum of two preschemes X', X'' such that $X' = \operatorname{Spec}(\mathscr{O}_{X,x})$ and that $f|X' : X' \to S$ is a finite morphism.*

c') *For every morphism locally of finite type $f : X \to S$, and every point $x \in X$ such that f be quasi-finite at the point x and $f(x)$ equal to the closed point s of S, $\mathscr{O}_{X,x}$ is a finite algebra over $\mathscr{O}_{S,s} = A$.*

c'') *For every morphism locally of finite presentation $f : X \to S$, and every point $x \in X$ such that f be quasi-finite at the point x and $f(x)$ equal to the closed point s of S, $\mathscr{O}_{X,x}$ is a finite algebra and of finite presentation over $\mathscr{O}_{S,s} = A$.*

Proof. Let us first note that c') (resp. c'') is equivalent to the same condition where moreover f is supposed *separated*, the question being local on X. Similarly, condition b) is equivalent to the same condition where moreover g is supposed *separated*: indeed, it suffices to apply the latter to the restriction of g to an open neighbourhood affine of the point $u_0(S_0)$ in S'. Let us henceforth restrict to the case where, in b), c') and c'', the morphisms are separated. **IV-4** | 131

The fact that a) implies b) and that b) implies a') is a particular case of (18.5.4). Let us moreover show that a') implies a). It is a matter of proving that if e_0 is an idempotent of $C = B \otimes_A k$, there is an idempotent $e \in B$ whose image is e_0 in C. If b is an element of B whose image in C is e_0, the sub-A-algebra $A[b] = B'$ of B is finite, and the canonical image C' of $B' \otimes_A k$ in C contains e_0. Now, $B' \otimes_A k$ is a k-algebra, hence a direct composite of local k-algebras, and consequently e_0 is the image of an idempotent e'_0 of $B' \otimes_A k$. We are thus reduced to the case where B is monogenic, and hence by a'), a quotient of an algebra of the form $A[\mathrm{T}]/F.A[\mathrm{T}]$; now, by a'), $A[\mathrm{T}]/F.A[\mathrm{T}]$ is a direct composite of local rings, hence so is its quotient, which proves the existence of the idempotent e.

It is immediate that c) implies c'), and that c') implies c'') when B is of finite presentation. Conversely, let us show that c'') implies a'). For if $B = A[\mathrm{T}]/F.A[\mathrm{T}]$, the morphism $X = \operatorname{Spec}(B) \to \operatorname{Spec}(A) = Y$ is finite and of finite presentation, and the preceding demonstration shows that B is a direct composite of finite A-algebras $\mathscr{O}_{X,x_i}$, where the x_i are the points of the fibre of the closed point of $\operatorname{Spec}(A)$.

It is trivial that c) implies c'), by virtue of the remark at the beginning. Let us prove that c') implies c''). Suppose c') satisfied and let us prove that under the conditions of c''), the set $Z = \operatorname{Spec}(\mathscr{O}_{X,x})$ identifies with an open and closed subset of X, which will establish c'') (**I**, 2.4.2). In the first place, the composite morphism $Z \xrightarrow{j} X \xrightarrow{f} S$, where j is the canonical map (**I**, 2.4.1), is finite and of finite presentation by hypothesis, and since f is separated and locally of finite presentation, j is also a finite morphism closed (**II**, 6.1.10), hence this proves that Z is closed in X. Moreover, since j is a closed immersion (**I**, 4.2.2) and ((**I**, 4.2.2)), j is a closed immersion. But if $\mathscr{J}$ is the Ideal of

$\mathscr{O}_X$ defining Z, we also have $\mathscr{J}_z = 0$ for every point z in a neighbourhood open V of x in X, since by hypothesis $\mathscr{J}$ is a quasi-coherent $\mathscr{O}_X$-Module of finite type ((1.4.7)) and ((**0**, 5.2.2)). Now, in an open neighbourhood V containing Z, hence Z is open in X.

Finally, $c'')$ implies $a')$: indeed, if $B = A[\mathrm{T}]/F.A[\mathrm{T}]$, the morphism $X = \mathrm{Spec}(B) \to \mathrm{Spec}(A) = Y$ **IV-4** | 132
is finite and of finite presentation, and the preceding argument shows that B is a direct composite of finite A-algebras $\mathscr{O}_{X,x_i}$, where the x_i are the points of the fibre of the closed point of $\mathrm{Spec}(A)$. □

Corollary (18.5.12). — *Let A be a semi-local ring, $\mathfrak{r}$ its radical; set $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A/\mathfrak{r})$. For A to be Henselian, it is necessary and sufficient that, for every finite morphism $f : X \to S$ and every étale and separated morphism $g : Y \to S$, if we set $X_0 = X \times_S S_0$ and $Y_0 = Y \times_S S_0$, the canonical map*

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S_0}(X_0, Y_0)$$

is bijective.

Proof. The sufficiency of the condition follows from the equivalence of $a)$ and $b)$ in (18.5.11), by applying this condition to the case where $f = 1_S$. To see that the condition is necessary, note that if A is Henselian, the couple (X, X_0) is Henselian by (18.5.6), (i); moreover $\mathrm{Hom}_S(X, Y) = \Gamma(X \times_S Y/X)$ and $\mathrm{Hom}_{S_0}(X_0, Y_0) = \Gamma(X_0 \times_{S_0} Y_0/X_0)$; since $X \times_S Y$ is étale and separated over X, the conclusion follows from the fact that $a)$ implies $b)$ in (18.5.4). □

Remarks (18.5.13). — The equivalent conditions $a)$, $a')$, $b)$ of Theorem (18.5.11) are also equivalent to the following ("Hensel's lemma"):

a$''$) For every monic polynomial $F \in A[\mathrm{T}]$, with canonical image $F_0 \in k[\mathrm{T}]$, and every decomposition $F_0 = G_0 H_0$ of F_0 into a product of two monic polynomials that are coprime G_0, H_0 in $k[\mathrm{T}]$, there exists a unique couple (G, H) of monic polynomials of $A[\mathrm{T}]$ possessing the following properties: G_0 and H_0 are the canonical images respectively of G and H, we have $F = GH$, and the ideal of $A[\mathrm{T}]$ generated by G and H is equal to $A[\mathrm{T}]$.

Lemma (18.5.13.1). — *Let A be a local ring with residue field k, $F \in A[\mathrm{T}]$ a monic polynomial, B the A-algebra $A[\mathrm{T}]/F.A[\mathrm{T}]$. There exists a canonical bijective correspondence between the decompositions of B into direct composite of two A-algebra quotients B', B'' and the decompositions $F = GH$ of F into a product of two monic polynomials G, H of $A[\mathrm{T}]$, such that the ideal generated by G and H is equal to $A[\mathrm{T}]$; the quotient algebras B', B'' corresponding to such a couple of polynomials G, H are respectively $A[\mathrm{T}]/H.A[\mathrm{T}]$ and $A[\mathrm{T}]/G.A[\mathrm{T}]$.*

Proof. If $F = GH$ and G and H generate the ideal $A[\mathrm{T}]$, there are two polynomials P, Q of $A[\mathrm{T}]$ such **IV-4** | 133
that $1 = PG + QH$. We deduce that the intersection of the principal ideals $\mathfrak{a} = G.A[\mathrm{T}]$ and $\mathfrak{b} = H.A[\mathrm{T}]$ is equal to $\mathfrak{c} = F.A[\mathrm{T}]$: indeed, if $R \in G.A[\mathrm{T}] \cap H.A[\mathrm{T}]$, we can write $R = PRG + QRH$; since RG (resp. RH) is a multiple of F since R is a multiple of H (resp. G), therefore $R \in F.A[\mathrm{T}]$. Since $A[\mathrm{T}] = \mathfrak{a} + \mathfrak{b}$, $A[\mathrm{T}]/\mathfrak{c}$ is the direct sum of ideals $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$ and $\mathfrak{b}/(\mathfrak{a} \cap \mathfrak{b})$, canonically isomorphic respectively to $A[\mathrm{T}]/\mathfrak{b}$ and $A[\mathrm{T}]/\mathfrak{a}$.

Conversely, suppose given a decomposition of B into a direct composite of two A-algebras B', B'', which identify canonically with two ideals $e'B$, $e''B$ of B, corresponding to a decomposition $1 = e' + e''$ into orthogonal idempotents e', e'' of B. Set further $B_0 = B \otimes_A k = k[\mathrm{T}]/F_0.k[\mathrm{T}]$, where F_0 is the canonical image of F in $k[\mathrm{T}]$, of the same degree n as F; if e'_0, e''_0 are the canonical images of e', e'' in B_0, these are two orthogonal idempotents such that $1 = e'_0 + e''_0$, and B_0 is a direct composite $B'_0 = e'_0 B_0$ and $B''_0 = e''_0 B_0$. Let t and t_0 be the canonical images of T in B and B_0; B'_0 (resp. B''_0) is a k-algebra generated by $t'_0 = e'_0 t_0$ (resp. $t''_0 = e''_0 t_0$), and it admits a base of the form $\{e'_0, t'_0, t'^2_0, \ldots, t'^{s-1}_0\}$ (resp. $\{e''_0, t''_0, t''^2_0, \ldots, t''^{r-1}_0\}$) with $r + s = n$. On the other hand, B being a free A-module (of base $\{1, t, \ldots, t^{n-1}\}$), B' and B'' are projective A-modules, hence free since A is a local ring (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 3, cor. de la prop. 5); it follows from the above and from Bourbaki, *Alg. comm.*, ch. II, § 3, no. 3, prop. 5, that if we set $t' = e't$, $t'' = e''t$, $\{e', t', \ldots, t'^{s-1}\}$ (resp. $\{e'', t'', \ldots, t''^{r-1}\}$) is a base of the A-module B' (resp. B''). There is hence a monic polynomial H (resp. G) of degree s (resp. r) in $A[\mathrm{T}]$ such that $e'H(t') = 0$ and $e''G(t'') = 0$; as $t^h = e't'^h + e''t''^h$, $t''^h = e''t''^h$ for every integer $h \geqslant 1$, and as $t'^h = e't'^h$, we also have $G(t) = e'G'(t')$ and $H(t) = e''H(t'')$, whence $G(t)H(t) = 0$; we conclude that the polynomial $G(T)H(T)$ is divisible by $F(T)$; but since these two monic polynomials have the same degree, we have $GH = F$. Moreover, B' (resp.

B'') is isomorphic to $A[\mathrm{T}]/H.A[\mathrm{T}]$ (resp. $A[\mathrm{T}]/G.A[\mathrm{T}]$). Finally, there are two polynomials R, S in $A[\mathrm{T}]$ such that $e' = R(t)$ and $e'' = S(t)$; as $e'S(t') = 0$ and also $e'S(t'') = 0$, we have necessarily $e'R(t') = 0$ and $e''R(t'') = 0$, so that, by definition of G and H, $R = QH$ and $S = PG$, where P, Q belong to $A[\mathrm{T}]$; the relation $1 = R(t) + S(t)$ in B gives $PG + QH = 1 + LF$ for some polynomial $L \in A[\mathrm{T}]$, and since $F = GH$, this proves that the ideal generated by G and H is equal to $A[\mathrm{T}]$, and completes the proof of the lemma. □

This lemma being established, it suffices to apply it, to the ring A on the one hand, to the field k on the other, to see immediately that conditions a') and a'') are equivalent.

Proposition (18.5.14). — *Every semi-local ring, separated and complete for the $\mathfrak{r}$-preadic topology (where $\mathfrak{r}$ is the radical of A) is Henselian.*

Proof. Indeed, A is a direct composite of local rings separated and complete (Bourbaki, *Alg. comm.*, ch. III, § 2, no. 13, cor. of prop. 19), so we are reduced to the case where A is a local ring. Let us verify criterion a') of (18.5.11). Since B is an A-module libre of finite type, B is separated and complete for the $\mathfrak{J}$-preadic topology ((**0**, 7.3.6)). Replacing B by A and $\mathfrak{J}B$ by $\mathfrak{J}$, it all amounts to seeing that the map that, to every idempotent of A, associates its class mod $\mathfrak{J}$, gives a correspondence with its image, and one verifies immediately that for every affine open subset W of S with the notations of (18.5.1), is *bijective*. Let us write $\mathrm{Idem}(A)$ for the set of idempotents of A, and for every homomorphism of rings $\varphi : A \to B$, let $\mathrm{Idem}(\varphi)$ denote the map $\mathrm{Idem}(A) \to \mathrm{Idem}(B)$ restriction of φ; it follows from the definition of the projective limit that $\mathrm{Idem}(A) = \varprojlim \mathrm{Idem}(A/\mathfrak{J}^n)$, under the maps $\psi_{nm} : \mathrm{Idem}(A/\mathfrak{J}^m) \to \mathrm{Idem}(A/\mathfrak{J}^n)$ restriction of the canonical maps $A/\mathfrak{J}^m \to A/\mathfrak{J}^n$. But since $\mathrm{Spec}(A/\mathfrak{J}^m) \to \mathrm{Spec}(A/\mathfrak{J}^n)$ is a homeomorphism, the ψ_{nm} are bijections (as seen in the proof of (18.5.3)); this proves our assertion. □

Proposition (18.5.15). — *Let A be a Henselian ring, $\mathfrak{r}$ its radical. Then the functor $B \mapsto B/\mathfrak{r}B$ is an equivalence of the category of finite étale A-algebras with the category of finite étale $(A/\mathfrak{r})$-algebras.* **IV-4** | 134

Proof. The fact that the functor is fully faithful is a particular case of (18.5.12). To show that this functor is an equivalence of categories, we may reduce to the case where A is local; it suffices then to apply (18.1.2.thm) (for étale morphisms), and $S_0 = \mathrm{Spec}(A/\mathfrak{r})$, reduced to a single point. □

Remarks (18.5.16). —

(i) We do not know if Proposition (18.5.15) generalises to a Henselian couple (S, S_0), even when S is affine and Noetherian.

(ii) Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is *separated and complete* for the $\mathfrak{J}$-preadic topology. Then the couple $(\mathrm{Spec}(A), \mathrm{Spec}(A/\mathfrak{J}))$ is Henselian: indeed, for every finite A-algebra B, B is separated and complete for the $\mathfrak{J}$-preadic topology ((**0**, 7.3.6)). Replacing B by A and $\mathfrak{J}B$ by $\mathfrak{J}$, everything amounts to seeing that the map that to every idempotent of A associates its class mod $\mathfrak{J}$, is *bijective*. Let us write $\mathrm{Idem}(A)$ for the set of idempotents of A, and $\mathrm{Idem}(A) = \varprojlim \mathrm{Idem}(A/\mathfrak{J}^n)$ for the projective limit; but since $\mathrm{Spec}(A/\mathfrak{J}^n) \to \mathrm{Spec}(A/\mathfrak{J}^m)$ is a homeomorphism, the ψ_{nm} are bijections (as seen in the proof of (18.5.3)); this proves our assertion.

Theorem (18.5.17). — *Let A be a Henselian local ring, $S = \mathrm{Spec}(A)$, s the closed point. For every smooth morphism $f : X \to S$, the canonical map*

$$\Gamma(X/S) \longrightarrow \Gamma(X_s/\kappa(s))$$

(where $X_s = f^{-1}(s)$) is surjective.

Proof. The datum of a $\kappa(s)$-section of X_s is equivalent to that of a point $x \in X$ above s whose residue field $\kappa(x)$ is isomorphic to $\kappa(s)$, and it is a matter of proving that there exists an S-section $u : S \to X$ such that $u(s) = x$. Taking into account (17.16.3), (i), we can suppose that f is étale. Then the conclusion follows from the criterion (18.5.11), b). (The reader will note that by virtue of this criterion, the validity of (18.5.17) for a local ring is necessary and sufficient for A to be Henselian.) □

Remark (18.5.18). — The conditions of (18.5.11) are also equivalent to the following:

d) For every morphism locally of finite type $f : X \to S$ and every point $x \in X$ such that $f(x)$ be equal to the closed point s of S and that $\mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$ be a field canonically isomorphic to $k = \kappa(s)$, there exists an open neighbourhood U of x in X such that $f|U$ is a closed immersion.

Proof. It is immediate that d) implies condition b) of (18.5.11) since a closed immersion étale is an open immersion (18.9.1). Conversely, suppose the conditions of (18.5.11) satisfied, and let us prove d). The hypothesis of d) implies that f is quasi-finite at the point x ((**III**, Err.20)), hence by virtue of condition c) of (18.5.11), $\mathscr{O}_{X,x}$ is a finite A-algebra and $\mathrm{Spec}(\mathscr{O}_{X,x})$ a neighbourhood open U of x in X. Moreover, since $\mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$ is isomorphic to $k = A/\mathfrak{m}_s$, the Nakayama lemma proves that the homomorphism $A \to \mathscr{O}_{X,x}$ is surjective, hence $f|U$ is a closed immersion. □

Proposition (18.5.19). — *Let A be a Noetherian local ring and Henselian, $S = \mathrm{Spec}(A)$, s the closed point of S, $f : X \to S$ a proper morphism; set $X_s = f^{-1}(s)$. Then the map $Y \to Y \cap X_s$ is a bijection from the set of connected components of X to the set of connected components of X_s.*

Proof. By considering a closed sub-prescheme of X having for underlying space a connected component of X, we are reduced to showing that if X is *connected and non-empty*, then X_s is *connected and non-empty*. The fact that X_s is non-empty follows from ((**II**, 7.2.1)); to show that X_s is connected, we argue by contradiction, considering the Stein factorisation $f : X \xrightarrow{f'} S' \xrightarrow{g} S$ of the proper morphism f ((**III**, 4.3.3)); by hypothesis the finite discrete set $g^{-1}(s)$ would contain at least two points. Since A is Henselian and g separated and of finite type, it would then follow from (18.5.11), c) that S' would be the sum of two non-empty preschemes S'_1, S'_2 (since the intersection of any one of them with $g^{-1}(s)$ is reduced to a single point). Since f' is surjective ((**III**, 4.3.1)), we would conclude that X is a sum of two non-empty preschemes, contradicting the hypothesis. □

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(18.6.1). Given a local ring A, there exists a *set* $\mathfrak{E}$ of local A-algebras that are essentially étale such that every A-algebra belonging to $\mathfrak{E}$ is A-isomorphic to an algebra belonging to $\mathfrak{E}$. It suffices indeed to observe that there exists a set $\mathfrak{F}$ of A-algebras of finite type such that every A-algebra of finite type is isomorphic to an algebra belonging to $\mathfrak{F}$; one can take $\mathfrak{F}$ to be the set of quotients of polynomial algebras $\mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_n]$ ($n \in \mathbf{N}$).

In what follows, if A and B are two local rings, we denote by $\mathrm{Hom.loc}(\mathrm{A}, \mathrm{B})$ the set of *local* homomorphisms from A to B.

Lemma (18.6.2). — *Let* A, A′ *be two local rings, k, k' their residue fields, $\phi : \mathrm{A} \to \mathrm{A}'$ a local homomorphism making* A′ *an A-algebra essentially étale* (18.6.1) *and such that the corresponding homomorphism $k \to k'$ is bijective. Then, for every local Henselian ring* B, *the canonical map*

$$\mathrm{Hom}(\phi, 1_{\mathrm{B}}) : \mathrm{Hom.loc}(\mathrm{A}', \mathrm{B}) \longrightarrow \mathrm{Hom.loc}(\mathrm{A}, \mathrm{B})$$

is bijective.

Proof. Let $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$, and let s and y be the closed points of S and Y respectively. By hypothesis, A′ is isomorphic to a local ring $\mathscr{O}_{\mathrm{X},x}$ of an S-scheme X étale over S, at a point $x \in \mathrm{X}$ above s. Suppose given a local homomorphism $\psi : \mathrm{A} \to \mathrm{B}$, making Y an S-scheme; it amounts to showing that there exists *one and only one* S-morphism $f : \mathrm{Y} \to \mathrm{X}$ such that $f(y) = x$. Let $\mathrm{X}' = \mathrm{X} \times_{\mathrm{S}} \mathrm{Y}$, and note that since $k(x) = k(s)$, there exists a unique point $x' \in \mathrm{X}$ above x and above y, and that $k(x') = k(y)$. It remains to show that there exists a unique Y-section f' of X′ such that $f'(y) = x'$. Now, the morphism $g : \mathrm{X}' \to \mathrm{Y}$ is étale and separated, and the fibre $\mathrm{X}'_0 = g^{-1}(y)$ at the point x' has the residue field $k(x') = k(y)$. If we set $\mathrm{Y}_0 = \mathrm{Spec}(k(y))$, there exists a unique Y_0-section f'_0 of X'_0 such that $f'_0(y) = x'$, and the conclusion follows from the fact that B is assumed Henselian and from (18.5.11), b). □

We say that a local A-algebra A′ satisfying the conditions of (18.6.2) is *strictly essentially étale*. Note that the criterion (18.5.11), b) means that, *for A to be Henselian, it is necessary and sufficient that every strictly essentially étale A-algebra be A-isomorphic to* A.

Lemma (18.6.3). — *Let* A *be a local ring,* A_1, A_2 *two A-algebras that are local and strictly essentially étale.*

i) *There exists at most one A-homomorphism (necessarily local) from* A_1 *to* A_2.
ii) *There exists a local A-algebra* A_3 *that is strictly essentially étale, and two A-homomorphisms* $A_1 \to A_3$, $A_2 \to A_3$.

Proof. Let $S = \mathrm{Spec}(A)$; by hypothesis there are two S-schemes étale over S, X_1, X_2, and two points $x_1 \in X_1$, $x_2 \in X_2$ above the closed point s of S and such that $A_1 = \mathscr{O}_{X_1,x_1}$, $A_2 = \mathscr{O}_{X_2,x_2}$; the hypotheses $k(x_1) = k(x_2) = k(s)$ (**I**, 3.4.9) imply that there exists a unique point $x_3 \in X_3$ above x_1 and x_2, and that $k(x_3) = k(s)$. Let $X_3 = X_1 \times_S X_2$; the hypotheses $k(x_1) = k(x_2) = k(s)$ (**I**, 3.4.9) imply that there exists a unique point $x_3 \in X_3$ above x_1 and x_2, and that $k(x_3) = k(s)$. Furthermore, X_3 is étale over S (17.3.3), so $A_3 = \mathscr{O}_{X_3,x_3}$ satisfies the conditions of (ii). Moreover, as X'_2 is connected, **IV-4** | 137
the uniqueness of f' follows from (17.4.9), whence (i). □

(18.6.4). Let us denote by $\mathfrak{S}$ the subset of the set $\mathfrak{E}$ defined in (18.6.1) formed by the A-algebras that are strictly essentially étale belonging to $\mathfrak{E}$. It follows from (18.6.3) that the relation "there exists an A-homomorphism from A_1 to A_2" is a *preorder* relation on $\mathfrak{S}$, and makes $\mathfrak{S}$ a *filtered increasing* set. Indexing $\mathfrak{S}$ by itself by means of the identity map, it follows from (18.6.3), (i) that if $\lambda \leqslant \mu$ in $\mathfrak{S}$, there exists a *unique* $\phi_{\mu\lambda} : A_\lambda \to A_\mu$ and $(A_\lambda, \phi_{\mu\lambda})$ is clearly an *inductive system* of local A-algebras, the $\phi_{\mu\lambda}$ being local homomorphisms. Furthermore, it follows from (17.3.5) that for $\lambda \leqslant \mu$, A_μ is an A_λ-algebra that is strictly essentially étale.

Definition (18.6.5). — The *Henselization* of a local ring A is the inductive limit of the inductive system $(A_\lambda, \phi_{\mu\lambda})$, denoted ${}^hA$, defined in (18.6.4).

This definition does not depend on the choice of $\mathfrak{E}$; if $\mathfrak{E}'$ is another set having the same property as $\mathfrak{E}$, and $\mathfrak{S}'$ the subset of $\mathfrak{E}'$ consisting of the strictly essentially étale A-algebras, it follows from (18.6.3), (ii) that, for the preorder relations considered, $\mathfrak{S}$ and $\mathfrak{S}'$ are *cofinal* in $\mathfrak{S}'' = \mathfrak{S} \cup \mathfrak{S}'$, so they give the same inductive limit up to an isomorphism. We shall moreover see below (18.6.6), (i) and (ii) that ${}^hA$ and the canonical homomorphism $A \to {}^hA$ are the solution of a universal problem, and are therefore determined up to a unique isomorphism.

One will observe that if A is a field, then evidently ${}^hA = A$. In the general case, ${}^hA$ is also the Henselization of all the A-algebras that are strictly essentially étale A_λ ($\lambda \in \mathfrak{S}$), since if B is an A_λ-algebra that is strictly essentially étale, B is also an A-algebra that is strictly essentially étale (18.6.1), so (up to an A-isomorphism) one of the A_μ for $\mu \geqslant \lambda$, and the A_μ such that $\mu \geqslant \lambda$ and that A_μ is an A_λ-algebra that is strictly essentially étale form a cofinal set in the preordered set of the A_λ, by virtue of (18.6.1) and (18.6.3).

Theorem (18.6.6). — *Let* A *be a local ring,* ${}^hA$ *its Henselization.*

i) ${}^hA$ *is a Henselian local ring and the structural homomorphism* $A \to {}^hA$ *is local.*
ii) *For every Henselian local ring* B*, the canonical map*

$$\mathrm{Hom.loc}({}^hA, B) \longrightarrow \mathrm{Hom.loc}(A, B)$$

is bijective.

iii) ${}^hA$ *is a faithfully flat A-module, and if* $\mathfrak{m}$ *is the maximal ideal of* A, $\mathfrak{m}.{}^hA$ *is the maximal ideal of* ${}^hA$, *and the homomorphism* $A/\mathfrak{m}A \to {}^hA/\mathfrak{m}.{}^hA$ *of residue fields is bijective.*
iv) *If* $\widehat{A}$ *and* $({}^hA)^\wedge$ *are the separated completions of the local rings* A *and* ${}^hA$, *the homomorphism* $\widehat{A} \to ({}^hA)^\wedge$ *deduced from the structural homomorphism* $A \to {}^hA$ *by completion is bijective.*
v) *For* ${}^hA$ *to be Noetherian, it is necessary and sufficient that A be so.*
vi) *If A is Henselian, the canonical homomorphism* $A \to {}^hA$ *is bijective.*

Proof. Let $\mathfrak{m}_\lambda$ be the maximal ideal of A_λ; the fact that, in (17.6.1), *a*) implies *c′*), that for $\lambda \leqslant \mu$, on **IV-4** | 138
has $\mathfrak{m}_\mu = \mathfrak{m}_\lambda A_\mu$, the homomorphism $A_\lambda/\mathfrak{m}_\lambda \to A_\mu/\mathfrak{m}_\mu$ is bijective and A_μ is a flat A_λ-module. The fact that ${}^hA$ is local, that the homomorphism $A \to {}^hA$ is local, assertion (iii) and the sufficiency of (v) follow therefore from (**0**, 10.3.1.3); the necessity of (v) follows from the fact that ${}^hA$ is a flat A-module (6.5.2).

To prove that ${}^hA$ is Henselian, let us apply the criterion (18.5.11), *b*). Let $S = \mathrm{Spec}({}^hA)$, $S_0 = \mathrm{Spec}({}^hA/\mathfrak{m}.{}^hA)$, and let $g : S' \to S$ be an étale morphism; reasoning as in (18.5.4), one can suppose that g is of finite presentation. It follows then from (8.8.2) and (17.7.5) that there exists an index

$\lambda \in \mathrm{S}$, an étale morphism $g_\lambda : \mathrm{S}^{(\lambda)} \to \mathrm{Spec}(\mathrm{A}_\lambda)$ and, if one sets $\mathrm{S}_0^{(\lambda)} = g^{-1}(\mathrm{S}_0^{(\lambda)})$, an $\mathrm{S}_0^{(\lambda)}$-section $f_0^{(\lambda)} : \mathrm{S}_0^{(\lambda)} \to \mathrm{S}'^{(\lambda)}$ such that $\mathrm{S}' = \mathrm{S}'^{(\lambda)} \times_{g^{(\lambda)}} \mathrm{S}$, $g = g^{(\lambda)} \times 1$ and $f_0 = f_0^{(\lambda)} \times 1$. Let s be the closed point of S, $x = f_0(s)$, let s_λ be the closed point of $\mathrm{S}^{(\lambda)}$; as x_λ is above the closed point s_λ of $\mathrm{S}^{(\lambda)}$, the local ring C_λ of $\mathrm{S}'^{(\lambda)}$ at the point x_λ is an A_λ-algebra that is strictly essentially étale; furthermore, as $f_0^{(\lambda)}(s_\lambda) = x_\lambda$, in other words, C_λ is an A_λ-algebra that is local and strictly essentially étale, and by (18.6.1) is an A-algebra A_μ with $\mu \geqslant \lambda$. There is therefore an A_λ-homomorphism $\mathrm{C}_\lambda \to {}^h\mathrm{A}$, that is to say there exists a well-determined S-morphism $f : \mathrm{S} \to \mathrm{S}'$ of S′; by (18.5.4), there exists a neighbourhood of $\mathfrak{n}_\lambda$ in $\mathrm{Spec}(\mathrm{C}_\lambda)$ isomorphic to $\mathrm{Spec}(\mathrm{A}_\lambda)$ (**I**, 6.5.4); one thus concludes that there is a neighbourhood of $\mathfrak{n}$ in $\mathrm{Spec}(\mathrm{C})$ isomorphic to $\mathrm{Spec}(\mathrm{A}')$, hence $\mathrm{C}_\mathfrak{n}$ is isomorphic to A', which proves that A' is Henselian (18.6.2) and completes the proof.

To prove (ii), it suffices to note that $\mathrm{Hom.loc}({}^h\mathrm{A}, \mathrm{B}) \cong \varprojlim \mathrm{Hom.loc}(\mathrm{A}_\lambda, \mathrm{B})$ and that by (18.6.2) the canonical homomorphisms $\mathrm{Hom.loc}(\mathrm{A}_\lambda, \mathrm{B}) \leftarrow \mathrm{Hom.loc}(\mathrm{A}, \mathrm{B})$ are bijective.

To prove (iv), note that for every λ and every integer $n > 0$, $\mathfrak{m}_\lambda^n = \mathfrak{m}^n\mathrm{A}_\lambda$, and $(\mathfrak{m}.{}^h\mathrm{A})^n = \mathfrak{m}^n.{}^h\mathrm{A} = \varinjlim(\mathrm{A}_\lambda/\mathfrak{m}_\lambda^n)$, and by the exactness of the functor $\varinjlim$ in the category of A-modules, that ${}^h\mathrm{A}/(\mathfrak{m}.{}^h\mathrm{A})^n = \varinjlim(\mathrm{A}_\lambda/\mathfrak{m}_\lambda^n)$, and it therefore suffices to show that, for every n and every index λ, the homomorphism $\mathrm{A}/\mathfrak{m}^n \to \mathrm{A}_\lambda/\mathfrak{m}_\lambda^n$ is bijective. On the other hand, since A_λ is a flat A-module, one has

$$\mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1} = (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_\mathrm{A} \mathrm{A}_\lambda = (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_{\mathrm{A}/\mathfrak{m}} (\mathrm{A}_\lambda/\mathfrak{m}\mathrm{A}_\lambda)$$

and as $\mathrm{A}/\mathfrak{m} \to \mathrm{A}_\lambda/\mathfrak{m}\mathrm{A}_\lambda$ is bijective, the homomorphism $\mathfrak{m}^n/\mathfrak{m}^{n+1} \to \mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1}$ is also bijective; the conclusion then follows from Bourbaki, *Alg. comm.*, chap. III, § 2, no 8, cor. 3 du th. 1.

Finally, if A is Henselian, it follows from the remark preceding (18.6.3) that the canonical homomorphisms $\mathrm{A} \to \mathrm{A}_\lambda$ are bijective, which proves (vi) by definition of ${}^h\mathrm{A}$. □

(18.6.7). Let A be a *semi-local* ring, $\mathfrak{m}_i$ ($1 \leqslant i \leqslant r$) its maximal ideals. The *Henselization* of A is the ring denoted again ${}^h\mathrm{A}$, defined as the *product* $\prod_i {}^h(\mathrm{A}_{\mathfrak{m}_i})$ of the Henselizations of the local rings $\mathrm{A}_{\mathfrak{m}_i}$. It is a faithfully flat A-module and a semi-local A-algebra, whose maximal ideals are the $\mathfrak{m}_i.{}^h\mathrm{A}$ by virtue of (18.6.6), (iii); furthermore, if $\mathfrak{r} = \bigcap \mathfrak{m}_i$ is the radical of A, it follows from the above that $\mathfrak{r}.{}^h\mathrm{A}$ is the radical of ${}^h\mathrm{A}$ and that the canonical map $\mathrm{A}/\mathfrak{r} \to {}^h\mathrm{A}/\mathfrak{r}.{}^h\mathrm{A}$ is bijective. As the separated completion $\widehat{\mathrm{A}}$ of A for the $\mathfrak{r}$-preadic topology is the product of the separated completions $\widehat{\mathrm{A}}_{\mathfrak{m}_i}$ (Bourbaki, *Alg. comm.*, chap. III, § 2, no 13, prop. 18), the canonical homomorphism $\widehat{\mathrm{A}} \to ({}^h\mathrm{A})^\wedge$ is bijective by (18.6.6), (iv), and it is clear by (18.6.6), (v) that for ${}^h\mathrm{A}$ to be Noetherian, it is necessary and sufficient that A be so. **IV-4 | 138** **IV-4 | 139**

To obtain the analogue of the universal property (18.6.6), (ii), when A and B are two *semi-local* rings of radicals $\mathfrak{r}, \mathfrak{s}$, let us agree to call a *semi-local homomorphism* ϕ from A to B any homomorphism such that $\phi(\mathfrak{r}) \subset \mathfrak{s}$. In the particular case where B is a *product* of local rings B_j ($1 \leqslant j \leqslant n$), the data of such a homomorphism amounts to the data of its projections $\phi_j : \mathrm{A} \to \mathrm{B}_j$, which are homomorphisms subject only to the condition that $\phi_j(\mathfrak{r}) \subset \mathfrak{n}_j$, where $\mathfrak{n}_j$ is the maximal ideal of B_j. Moreover, since $\mathfrak{m}_i$ ($1 \leqslant i \leqslant m$) are the maximal ideals of A, the prime ideal $\phi_j^{-1}(\mathfrak{n}_j)$ must contain one of the $\mathfrak{m}_i$, since it must contain their intersection $\mathfrak{r}$, hence it is equal to one of the $\mathfrak{m}_i$ and ϕ_j factorises as $\mathrm{A} \to \mathrm{A}_{\mathfrak{m}_i} \xrightarrow{\psi_{ij}} \mathrm{B}_j$, where ψ_{ij} is a *local* homomorphism. If $\mathrm{Hom.sloc}(\mathrm{A}, \mathrm{B})$ denotes the set of semi-local homomorphisms from A to B, one can thus identify this set with $\prod_j \left(\bigcup_i \mathrm{Hom.loc}(\mathrm{A}_{\mathfrak{m}_i}, \mathrm{B}_j)\right)$; in the case considered, since a semi-local Henselian ring is a direct product of Henselian local rings, one sees that the universal property (18.6.6), (ii) is still valid for a semi-local ring A, provided one replaces "local homomorphisms" by "semi-local homomorphisms" (18.6.7).

Proposition (18.6.8). — *Let* A *be a semi-local ring,* B *an A-algebra that is semi-local and entire over* A*. Then* $\mathrm{B} \otimes_\mathrm{A} ({}^h\mathrm{A})$ *is a semi-local B-algebra isomorphic to* ${}^h\mathrm{B}$*.*

Proof. As $\mathrm{B} \otimes_\mathrm{A} ({}^h\mathrm{A})$ is entire over ${}^h\mathrm{A}$, each of its maximal ideals lies above one of the maximal ideals of ${}^h\mathrm{A}$, hence above a maximal ideal of A, and since B is entire over A, the prime ideals of B lying above a maximal ideal of A are maximal ideals of $\mathrm{B} \otimes_\mathrm{A} ({}^h\mathrm{A})$. Taking account of (18.6.6), (iii) and (**I**, 3.4.9), one concludes that the ring $\mathrm{B} \otimes_\mathrm{A} ({}^h\mathrm{A})$ is semi-local. Furthermore, for every Henselian local ring C, $\mathrm{Hom.sloc}(\mathrm{B} \otimes_\mathrm{A} ({}^h\mathrm{A}), \mathrm{C})$ is in bijection with the set of pairs (ϕ, ψ) of homomorphisms

$\psi : {}^h\mathrm{A} \to \mathrm{C}$, $\phi : \mathrm{B} \to \mathrm{C}$, $\mathrm{A} \to {}^h\mathrm{A} \xrightarrow{\psi} \mathrm{C}$ such that the composites $\mathrm{A} \to \mathrm{B} \xrightarrow{\phi} \mathrm{C}$ and $\mathrm{A} \to {}^h\mathrm{A} \xrightarrow{\psi} \mathrm{C}$ are equal. But by virtue of the bijection between $\mathrm{Hom.sloc}(\mathrm{A}, \mathrm{C})$ and $\mathrm{Hom.sloc}({}^h\mathrm{A}, \mathrm{C})$, one sees that for every $\phi \in \mathrm{Hom.sloc}(\mathrm{B}, \mathrm{C})$ there exists a unique ψ having the preceding property, hence the map $\mathrm{Hom.sloc}(\mathrm{B} \otimes_{\mathrm{A}} ({}^h\mathrm{A}), \mathrm{C}) \to \mathrm{Hom.sloc}(\mathrm{B}, \mathrm{C})$ is bijective, which proves the proposition by virtue of the uniqueness of the solution of a universal problem. □

Theorem (18.6.9). — *Let* A *be a semi-local ring.*

i) *For* ${}^h\mathrm{A}$ *to be reduced (resp. normal), it is necessary and sufficient that A be so.*
ii) *Suppose A Noetherian. Then, for every prime ideal* $\mathfrak{p}$ *of A, the ring* $({}^h\mathrm{A})_{\mathfrak{p}}/\mathfrak{p}.({}^h\mathrm{A})_{\mathfrak{p}}$ *is a composed direct product of a finite number of separable algebraic extensions of* $k(\mathfrak{p})$ *(which implies that the fibres of the canonical morphism* $\mathrm{Spec}({}^h\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$ *are* geometrically regular (18.6.9) *and are* discrete *spaces).*

Proof. (i) As ${}^h\mathrm{A}$ is a faithfully flat A-module, it follows from (2.1.13) that if ${}^h\mathrm{A}$ is reduced (resp. **IV-4** | 140
normal), so is A. To prove the converse, one can restrict to the case where A is local, so that, with the notations of (18.6.4), one has ${}^h\mathrm{A} = \varinjlim \mathrm{A}_\alpha$. Now, it follows from the definition of the A_α and from (17.5.7) that if A is reduced (resp. integral and integrally closed), so is each of the A_α; furthermore, the homomorphisms $\mathrm{A}_\alpha \to \mathrm{A}_\beta$ for $\alpha \leqslant \beta$ are injective by virtue of (**0**, 6.5.1) since A_β is a flat A_α-module; hence the morphisms $\mathrm{Spec}(\mathrm{A}_\beta) \to \mathrm{Spec}(\mathrm{A}_\alpha)$ are dominant, and one concludes from (5.13.2) (resp. (5.13.4)) that ${}^h\mathrm{A}$ is reduced (resp. integral and integrally closed).

(ii) One can restrict to the case where A is local. By virtue of the fact that the functor $\varinjlim$ commutes with tensor products, the fibre of the morphism $\mathrm{Spec}({}^h\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$ at a point $\mathfrak{p}$ is the projective limit of the fibres of the morphisms $\mathrm{Spec}(\mathrm{A}_\alpha) \to \mathrm{Spec}(\mathrm{A})$ at this point. As ${}^h\mathrm{A}$ is Noetherian, one is reduced to proving the following lemma: □

Lemma (18.6.9.1). — *Let* (B_α) *be an inductive system of Artinian rings,* $\mathrm{B} = \varinjlim \mathrm{B}_\alpha$ *its inductive limit. If* B *is Noetherian, then* B *is Artinian; if moreover, for* $\alpha \leqslant \beta$*, the homomorphisms* $\phi_{\beta\alpha} : \mathrm{B}_\alpha \to \mathrm{B}_\beta$ *are injective, then there exists* λ *such that, for* $\alpha \geqslant \lambda$*, the number of local components* $\mathrm{B}_\alpha^{(i)}$ *of* B_α *is constant, that* $\phi_{\beta\alpha}(\mathrm{B}_\alpha^{(i)}) \subset \mathrm{B}_\beta^{(i)}$ *for every i and that the* $\mathrm{B}^{(i)} = \varinjlim \mathrm{B}_\alpha^{(i)}$ *are the local components of* B*.*

Proof. Indeed $\mathrm{Y}_\alpha = \mathrm{Spec}(\mathrm{B}_\alpha)$, $\mathrm{Y} = \mathrm{Spec}(\mathrm{B})$, which, as a topological space, is equal to $\varprojlim \mathrm{Y}_\alpha$ (8.2.10). The spaces Y_α are finite and discrete; furthermore, if the $\phi_{\beta\alpha}$ are injective, the canonical morphisms $\mathrm{Y}_\beta \to \mathrm{Y}_\alpha$ are dominant, hence surjective. The lemma then follows from the following purely topological lemma: □

Lemma (18.6.9.2). — *Let* $(\mathrm{Y}_\alpha, \psi_{\alpha\beta})$ *be a projective system of finite topological spaces, and let Y be discrete. If* $\mathrm{Y} = \varprojlim \mathrm{Y}_\alpha$ *is Noetherian, Y is finite and discrete. If moreover the* $\psi_{\alpha\beta}$ *are surjective, then there exists* λ *such that for* $\alpha \geqslant \lambda$*, the number of elements of* Y_α *is constant and equal to the number of elements of* Y*.*

Proof. Indeed, Y being compact and Noetherian, every subset of Y is closed, hence compact, hence finite since it is compact. If moreover the $\psi_{\alpha\beta}$ are surjective, the same holds for $\psi_\alpha : \mathrm{Y} \to \mathrm{Y}_\alpha$, so $\mathrm{Card}(\mathrm{Y}) \geqslant \mathrm{Card}(\mathrm{Y}_\alpha)$, and as $\mathrm{Card}(\mathrm{Y}_\alpha)$ is an increasing function of α, there exists λ such that for $\alpha \geqslant \lambda$, $\mathrm{Card}(\mathrm{Y}_\alpha) = \mathrm{Card}(\mathrm{Y}_\lambda)$; the $\psi_{\alpha\beta}$ are then bijective for $\alpha \geqslant \lambda$ and one therefore has $\mathrm{Card}(\mathrm{Y}) = \mathrm{Card}(\mathrm{Y}_\lambda)$. □

Corollary (18.6.10). — *Let* A *be a local Noetherian ring. For* ${}^h\mathrm{A}$ *to possess one of the following properties:*

a) *being a Cohen-Macaulay ring* (**0**, 16.5.3)*;*
b) *satisfying the property* (S_n) (5.7.3)*;*
c) *being regular;*
d) *satisfying the property* (R_n) (5.8.2)*;*

it is necessary and sufficient that A possess the same property.

Proof. It follows immediately from the fact that the fibres of the morphism $\mathrm{Spec}({}^h\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$ **IV-4** | 141
are geometrically regular (18.6.9) and from (6.4.1), (6.5.1) and (6.5.3). □

Corollary (18.6.11). — *Let* A *be a semi-local Noetherian ring. For a prime ideal* $\mathfrak{p}$ *of* ${}^h\mathrm{A}$ *to belong to* $\mathrm{Ass}({}^h\mathrm{A})$*, it is necessary and sufficient that* $\mathfrak{p} \cap \mathrm{A}$ *belong to* $\mathrm{Ass}(\mathrm{A})$*.*

Proof. Taking account of the description of the fibres of the morphism $\mathrm{Spec}({}^h\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$, this also follows immediately from (3.3.1). ☐

Proposition (18.6.12). — *Let* A *be a local ring; the following properties are equivalent:*

a) *A is unibranch* (**0**, 23.2.1) *(resp. reduced and unibranch).*
b) *For every A-algebra* A' *that is local and strictly essentially étale,* $\mathrm{Spec}(\mathrm{A}')$ *is irreducible (resp. integral).*
c) $\mathrm{Spec}({}^h\mathrm{A})$ *is irreducible (resp. integral).*

Proof. Note first that ${}^h(\mathrm{A}_{\mathrm{red}}) = ({}^h\mathrm{A})_{\mathrm{red}}$, since ${}^h(\mathrm{A}_{\mathrm{red}}) = ({}^h\mathrm{A}) \otimes_\mathrm{A} \mathrm{A}_{\mathrm{red}}$ (18.6.8), and ${}^h(\mathrm{A}_{\mathrm{red}})$ is reduced (18.6.9). This allows, in the proof, to consider only the case where A is reduced, hence also A' (17.5.7) and ${}^h\mathrm{A}$ (18.6.9).

With the notations of (18.6.4), the condition *b*) means that all the A_α are integral; one therefore concludes that *b*) implies *c*) (5.13.3), hence *b*) implies *c*).

It is clear that if ${}^h\mathrm{A}$ is integral, so are each of the $\mathrm{A}_\alpha \subset {}^h\mathrm{A}$, hence *c*) implies *b*).

Let us show that *c*) implies *a*). Note first that $\mathrm{A} \subset {}^h\mathrm{A}$ is integral; let K and L be the fields of fractions of A and ${}^h\mathrm{A}$ respectively. It suffices to show that for every *finite* A-algebra B, $\mathrm{B} \subset \mathrm{K}$, B is an anneau local, whose Henselization is therefore $\mathrm{B} \otimes_\mathrm{A} ({}^h\mathrm{A})$ (18.6.8). Now B is a semi-local ring, whose Henselization is therefore $\mathrm{B} \otimes_\mathrm{A} ({}^h\mathrm{A})$ (18.6.8). By flatness, $\mathrm{B} \otimes_\mathrm{A} ({}^h\mathrm{A})$ identifies with a subring of L, hence ${}^h\mathrm{B}$ is integral. But as ${}^h\mathrm{B}$ is a semi-local Henselian ring, it is a composed direct product of local rings (18.5.9), hence it can only be integral if B is local, which shows that B is local.

Let us finally prove that *a*) implies *c*). Let C be the integral closure of the integral ring A. As C is a local ring by hypothesis, $\mathrm{C} \otimes_\mathrm{A} ({}^h\mathrm{A})$ is the Henselization of C (18.6.8), hence is integral and integrally closed (18.6.9). As A is a subring of C, hence ${}^h\mathrm{A}$ identifies with a subring of ${}^h\mathrm{C}$ by flatness, hence is also integral, which completes the proof. ☐

Corollary (18.6.13). — *Let* A *be a local Henselian ring. Then, for A to be unibranch, it is necessary and sufficient that* $\mathrm{A}_{\mathrm{red}}$ *be integral.*

Proposition (18.6.14). — i) *Every filtered inductive limit of Henselian local rings (where the transition homomorphisms are local) is a Henselian local ring.*

ii) *The functor* $\mathrm{A} \rightsquigarrow {}^h\mathrm{A}$ *in the category of local rings (where the morphisms are local homomorphisms) commutes with filtered inductive limits.*

iii) *Every Henselian local ring A is an inductive limit of an inductive filtered system of Henselian and Noetherian local rings* A_λ *(the transition homomorphisms* $\mathrm{A}_\lambda \to \mathrm{A}_\mu$ *being local).*

Proof. (i) Let (A_λ) be a filtered inductive system of Henselian local rings, where the transition homomorphisms are local. Let us show that $\mathrm{A}' = \varinjlim \mathrm{A}_\lambda$, which is local (**0**, 10.3.1.3), is Henselian. **IV-4** | 141
Now, let C be an A'-algebra étale, $\mathfrak{n}$ a prime ideal of C above the maximal ideal $\mathfrak{m}'$ of A', such that $\mathrm{C}_\mathfrak{n}$ is strictly essentially étale over A' (18.6.1), and in virtue of (8.8.2), (ii) and (17.7.8), there exist an index λ and an A_λ-algebra C_λ étale such that $\mathrm{C} = \mathrm{C}_\lambda \otimes_{\mathrm{A}_\lambda} \mathrm{A}'$ up to an isomorphism. Furthermore, if $\mathfrak{n}_\lambda$ is the inverse image of $\mathfrak{n}$ in C_λ, and $\mathfrak{m}_\lambda$ the maximal ideal of A_λ, one can, by virtue of (8.8.2.4) and of the transitivity of fibres (**I**, 3.6.4), suppose that $(\mathrm{C}_\lambda)_{\mathfrak{n}_\lambda}$ is strictly essentially étale over A_λ; since A_λ is Henselian, $(\mathrm{C}_\lambda)_{\mathfrak{n}_\lambda}$ is necessarily isomorphic to A_λ (18.5.11), *b*); there exists therefore a neighbourhood of $\mathfrak{n}_\lambda$ in $\mathrm{Spec}(\mathrm{C}_\lambda)$ isomorphic to $\mathrm{Spec}(\mathrm{A}_\lambda)$ (**I**, 6.5.4); hence $\mathrm{C}_\mathfrak{n}$ is isomorphic to A', which proves that A' is Henselian (18.6.2) and completes the proof. **IV-4** | 142

(ii) Suppose that $\mathrm{B} = \varinjlim \mathrm{B}_\lambda$, where the B_λ are local rings, the transition morphisms $\mathrm{B}_\lambda \to \mathrm{B}_\mu$ ($\lambda \leqslant \mu$) being local. Set $\mathrm{A}_\lambda = {}^h\mathrm{B}_\lambda$; the A_λ are Henselian local rings and Noetherian local rings (18.6.6), (v), and by virtue of (18.6.6), (vi), (ii) the transition morphisms $\mathrm{B}_\lambda \to \mathrm{B}_\mu$ uniquely determine local homomorphisms $\mathrm{A}_\lambda \to \mathrm{A}_\mu$, so that (A_λ) is an inductive system. It amounts to showing that $\mathrm{A}' = \varinjlim \mathrm{A}_\lambda$ is canonically isomorphic to $\mathrm{A} = {}^h\mathrm{B}$. By virtue of (18.6.6), (ii) and of the definition of inductive limits, one has, for every Henselian local ring E,

$$\mathrm{Hom.loc}(\mathrm{A}', \mathrm{E}) = \varprojlim \mathrm{Hom.loc}(\mathrm{A}_\lambda, \mathrm{E}) = \varprojlim \mathrm{Hom.loc}(\mathrm{B}_\lambda, \mathrm{E}) = \mathrm{Hom.loc}(\mathrm{B}, \mathrm{E}).$$

But as A' is Henselian by (i), this proves our assertion.

(iii) The ring A is a filtered inductive limit of its sub-local Noetherian rings B_λ, the transition homomorphisms being local (5.13.3), (iii). As the ${}^h B_\lambda$ are Henselian and Noetherian local rings (18.6.6), (v) and that $A = {}^h A$ (18.6.6), (vi), it suffices to apply (ii) to the inductive system (B_λ). □

Corollary (18.6.15). — *Let* A *be a Henselian local ring,* X *an A-prescheme of finite presentation over* A*. Then there exists a Noetherian Henselian local ring* A_0*, a local homomorphism* $A_0 \to A$*, an* A_0*-prescheme* X_0 *of finite type over* A_0 *and an A-isomorphism* $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.

Proof. This follows from (18.6.14) and (8.8.2), (ii). □

18.7. Henselization and excellent rings.

(18.7.1). In this paragraph we denote by $\boldsymbol{P}(Z, k)$ a property of the form considered in (7.3.1), where we further suppose that the property $\boldsymbol{Q}(A, k)$ satisfies the following condition: **IV-4** | 142

For every separable algebraic extension k' of k, and every k'-local algebra that is Noetherian, the property $\boldsymbol{Q}(A, k)$ is *equivalent* to $\boldsymbol{Q}(A, k')$.

It is immediate that all the properties $\boldsymbol{P}$ considered in (7.3.8) satisfy the preceding condition, since if K is a finite extension of k, $k' \otimes_k K$ is a composed direct product of a finite number of separable algebraic extensions of K.

Proposition (18.7.2). — *Let* A *be a local Noetherian ring. For* ${}^h A$ *to be a* $\boldsymbol{P}$*-ring* (7.3.13)*, it is necessary and sufficient that A be so.*

Proof. Indeed, by virtue of (18.6.6), (iv), the completion $\widehat{A}$ of A is also the completion of ${}^h A$ and one therefore has $A \subset {}^h A \subset \widehat{A}$. By (18.6.9), for every $x \in \mathrm{Spec}(A)$, the fibre at x of the morphism $\mathrm{Spec}({}^h A) \to \mathrm{Spec}(A)$ is discrete, finite, and each of the points y_i of this fibre is a separable algebraic extension of $k(x)$. The formal fibre of A at the prescheme *sum* of the formal fibres of ${}^h A$ at the points y_i. The conclusion then follows from the hypothesis on $\boldsymbol{P}$ in (18.7.1). □ **IV-4** | 143

Corollary (18.7.3). — *Let* A *be a local Noetherian ring. For* ${}^h A$ *to be universally Japanese, it is necessary and sufficient that A be so.*

Proof. Indeed, for a Noetherian local ring B, it amounts to the same to say that B is universally Japanese, or that the formal fibres of B are geometrically reduced (7.6.4) and (7.7.1); the conclusion therefore follows from (18.7.2). □

Corollary (18.7.4). — *Let* A *be a local Noetherian ring. For the formal fibres of* ${}^h A$ *to be geometrically regular, it is necessary and sufficient that those of A be so.*

Proof. This is a particular case of (18.7.2). □

Proposition (18.7.5). — *Let* A *be a local Noetherian ring. If A is universally catenary* (5.6.2) *(resp. formally catenary* (7.1.9)*, resp. strictly formally catenary* (7.2.6)*), so is* ${}^h A$.

Proof. With the notations of (18.6.4), if A is universally catenary, the same holds for the A_α, which are A-algebras essentially of finite type (5.6.3). To prove that in this case ${}^h A$ is universally catenary, it will therefore suffice to establish the following lemma: □

Lemma (18.7.5.1). — *Let* $(B_\alpha, \phi_{\beta\alpha})$ *be an inductive system of Noetherian rings and suppose that, for* $\alpha \leqslant \beta$*,* $\phi_{\beta\alpha}$ *makes* B_β *a flat* B_α*-module and that* $B = \varinjlim B_\alpha$ *is Noetherian. Then, if the* B_α *are catenary (resp. universally catenary), the same holds for* B.

Proof. To say that B is universally catenary is equivalent to saying that $B[T_1, \dots, T_n]$ is catenary for all n (5.6.2), and it is immediate that the inductive system $(B_\alpha[T_1, \dots, T_n])$ satisfies the same conditions as (B_α); it therefore suffices to prove that the B_α are catenary. If $\mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_r$ is a saturated chain of prime ideals, the inverse images $\mathfrak{p}_{0\alpha} \supset \mathfrak{p}_{1\alpha} \supset \cdots \supset \mathfrak{p}_{r\alpha}$ of these ideals in B_α form a chain of prime ideals for each α sufficiently large, and it suffices to show that for α sufficiently large, this chain is saturated. Now, each $\mathfrak{p}_i$ is an ideal of finite type, hence there exists λ such that, for $\alpha \geqslant \lambda$, on has $\mathfrak{p}_i = \mathfrak{p}_{i\alpha}.B$. On has therefore $1 = \mathrm{codim}(V(\mathfrak{p}_i), V(\mathfrak{p}_{i+1})) = \mathrm{codim}(V(\mathfrak{p}_{i\alpha}), V(\mathfrak{p}_{i+1,\alpha}))$ since B is a flat B_α-module (6.1.4) and (**0**, 6.2.3), which completes the proof of the lemma.

Suppose now A formally catenary. Let $\mathfrak{p}$ be a prime ideal of ${}^h\mathrm{A}$, and $\mathfrak{q}$ its inverse image in A, so that ${}^h\mathrm{A}/\mathfrak{q}.{}^h\mathrm{A}$ is a quotient ring of ${}^h\mathrm{A}$, and as ${}^h\mathrm{A}/\mathfrak{q}.{}^h\mathrm{A} = {}^h(\mathrm{A}/\mathfrak{q})$ by (18.6.8), it suffices to show that if A is *integral* and formally catenary, ${}^h\mathrm{A}$ is formally *equidimensional*. But as the completion of ${}^h\mathrm{A}$ is equal to $\widehat{\mathrm{A}}$, this follows from the hypothesis that A is formally catenary. □

Corollary (18.7.6). — *If a local Noetherian ring A is excellent* (7.8.2)*, so is* ${}^h\mathrm{A}$.

Remark (18.7.7). — It can happen that ${}^h\mathrm{A}$ is an excellent local ring without A being universally catenary (nor *a fortiori* excellent). To see this, let us take the example of A from (5.6.11), with the same notations. The ring E constructed in (5.6.11) is excellent when k_0 is of characteristic 0 (7.8.3), (ii) and (iii); consider two rings E_1, E_2 isomorphic to E, and "glue" $\mathrm{Spec}(\mathrm{E}_1)$ and $\mathrm{Spec}(\mathrm{E}_2)$ such that if $\mathfrak{m}_i$ and $\mathfrak{m}'_i$ are the maximal ideals of E_i $(i = 1, 2)$, corresponding to $\mathfrak{m}$ and $\mathfrak{m}'$, the point $\mathfrak{m}_1$ is "glued" to $\mathfrak{m}'_2$ and the point $\mathfrak{m}_2$ to $\mathfrak{m}'_1$; to be precise, if ε_i and ε'_i are the homomorphisms from E_i on $k(\mathfrak{m}_i)$ and $k(\mathfrak{m}'_i)$ $(i = 1, 2)$, the schema obtained is $\mathrm{Spec}(\mathrm{R})$, where R is the subring of $\mathrm{E}_1 \times \mathrm{E}_2$ formed by the pairs (x_1, x_2) such that $\varepsilon'_1(x_1) = \sigma(\varepsilon_2(x_2))$ and $\varepsilon_1(x_1) = \sigma(\varepsilon'_2(x_2))$. One verifies easily (for example with the help of (17.6.3)) that R is a finite C-algebra that is étale, and that there are two maximal ideals $\mathfrak{r}_1$, $\mathfrak{r}_2$ of R above the maximal ideal $\mathfrak{n}$ of C, with completions of $\mathrm{R}_{\mathfrak{r}_1}$ and $\mathrm{R}_{\mathfrak{r}_2}$, and that the Henselization is equal to ${}^h\mathrm{A}$, having $\mathrm{A} = \mathrm{C}_{\mathfrak{n}}$; $\mathrm{R}_{\mathfrak{r}_i}$ is therefore an A-algebra that is essentially étale, and its Henselization is also equal to ${}^h\mathrm{A}$. Furthermore, one has, by (7.8.4), (ii) that the formal fibres of A are geometrically regular, hence R is universally catenary. Finally, the ring R is universally catenary, and the same holds for the quotients by its two minimal prime ideals, whence the conclusion by (5.6.3), (iii). The ring $\mathrm{R}_{\mathfrak{r}_i}$ is therefore excellent, and by the same token ${}^h\mathrm{A}$ (18.7.6), while A is not universally catenary.

18.8. Strictly local rings and strict Henselization. **IV-4** | 144

Proposition (18.8.1). — *Let* A *be a local ring. The following conditions are equivalent:*

a) *A is a Henselian ring and its residue field k is separably closed.*
b) *A is a Henselian ring and every étale covering of* $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$ *is trivial.*
c) *For every étale morphism* $f : \mathrm{X} \to \mathrm{S}$ *and every point* $x \in \mathrm{X}$ *such that* $f(x)$ *is the closed point s of S, there exists an S-section* $u : \mathrm{S} \to \mathrm{X}$ *of X such that* $u(s) = x$.

Proof. The hypothesis c) implies that for every étale morphism $f : \mathrm{X} \to \mathrm{S}$, the residue fields at the points of $f^{-1}(s)$ are all isomorphic to k, and moreover the condition b) of (18.5.11) is satisfied. Hence A is Henselian, and the fact that every étale covering of S is trivial follows from (18.2.10), (ii); hence c) implies b). As étale coverings of $\mathrm{Spec}(k)$ are trivial if and only if k is separably closed, b) implies a) by virtue of (18.5.11), b). Finally, it follows also from (18.5.11), b) that a) implies c). □

Definition (18.8.2). — A local ring is said to be *strictly local* if it satisfies the equivalent conditions of (18.8.1). A scheme isomorphic to the spectrum of a strictly local ring is called a *strictly local scheme*.

Remark (18.8.3). — The conditions of (18.8.1) are also equivalent to the following:

d) For every morphism locally of finite type $f : \mathrm{X} \to \mathrm{S}$ and every point $x \in \mathrm{X}$ such that $f(x)$ is the closed point s of S and that $\mathscr{O}_{\mathrm{X},x}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X},x}$ is a field, a finite separable extension of $k(s) = k$, there exists an open neighbourhood U of x in X such that $f|U$ is a closed immersion.

(One will note that the hypothesis of d) is satisfied if f is *formally unramified* at the point x (17.4.1.2).)

We leave the reader the proof, which is substantially the same as that of (18.5.18).

Lemma (18.8.4). — *Let* A, A′ *be two local rings,* $\phi : \mathrm{A} \to \mathrm{A}'$ *a local homomorphism making* A′ *an A-algebra essentially étale* (18.6.1)*. Then, for every strictly local ring* B*, every local homomorphism* $\psi : \mathrm{A} \to \mathrm{B}$ *and every homomorphism of* $k(\mathrm{A})$*-algebras* $\alpha : k(\mathrm{A}') \to k(\mathrm{B})$*, there exists one and only one local homomorphism* $\psi' : \mathrm{A}' \to \mathrm{B}$ *such that* $\psi = \psi' \circ \phi$ *and that* α *is equal to the homomorphism* $\overline{\psi}'$ *deduced from* ψ' *by passage to quotients.* **IV-4** | 145

Proof. With the notations of (18.6.2), the homomorphism α corresponds to a $k(s)$-morphism $\mathrm{Spec}(k(y)) \to \mathrm{Spec}(k(x))$, hence to a point x' well determined in X′ above y. The existence of the Y-section f' of X′ such that $f'(y) = x'$ then follows from (18.8.1), c) since f' is étale and B strictly local, and its uniqueness follows from this and from (17.4.9). □

Lemma (18.8.5). — *Let* A *be a local ring,* A_1*,* A_2 *two A-algebras that are local and essentially étale, K a field that is an extension of* $k(A)$*.*

i) *For every* $k(A)$*-homomorphism* $\gamma : k(A_1) \to k(A_2)$*, there exists at most one A-homomorphism (necessarily local)* $\psi : A_1 \to A_2$ *such that* γ *is the homomorphism* $\overline{\psi}$ *deduced from* ψ *by passage to quotients.*

ii) *For every pair of local A-homomorphisms* $\beta_1 : A_1 \to K$*,* $\beta_2 : A_2 \to K$ *there exists a local A-algebra* A_3 *that is essentially étale, two A-homomorphisms* $\phi_1 : A_1 \to A_3$*,* $\phi_2 : A_2 \to A_3$ *and an A-homomorphism* $\beta : A_3 \to K$ *such that* $\beta_1 = \beta \circ \phi_1$ *and* $\beta_2 = \beta \circ \phi_2$*.*

Proof. With the notations of (18.6.3), the homomorphisms β_1, β_2 in (ii) correspond to two S-morphisms $\mathrm{Spec}(K) \to X_1$, $\mathrm{Spec}(K) \to X_2$, with respective images x_1, x_2; one thus deduces a well-determined $\mathrm{Spec}(K) \to X_3$ (**I**, 3.2.1) whose image x_3 lies above x_1 and x_2; the A-algebra $A_3 = \mathscr{O}_{X_3,x_3}$ and the homomorphism $A_3 \to K$ corresponding to these data then satisfy the conditions of (ii). On the other hand, the data of γ in (i) corresponds to an S-morphism $\mathrm{Spec}(k(x_2)) \to \mathrm{Spec}(k(x_1))$ or also to a point x'_3 of X_3 above x_2; the uniqueness of ψ follows from this and from (17.4.9). □

(18.8.6). Let A be a local ring, $\mathfrak{m}$ its maximal ideal, k its residue field, Ω a separable closure of k. Consider the set $\mathfrak{E}$ of A-algebras essentially étale defined in (18.6.1), and denote by L the set of A-*homomorphisms* $A' \to \Omega$, where A' ranges over $\mathfrak{E}$; one will note that such a homomorphism factorises as $A \to k(A') \xrightarrow{\omega_\lambda} \Omega$, where $\omega_\lambda : k(A') \to \Omega$ is a k-homomorphism (since $\mathfrak{m}A'$ is the maximal ideal of A'); conversely the data of a homomorphism ω_λ uniquely determines a homomorphism $\lambda \in L$. For every $\lambda \in L$, we denote by A_λ the A-algebra essentially étale belonging to $\mathfrak{E}$ on which λ is defined. The set L is *preordered* by the relation "there exists a homomorphism $\phi_{\mu\lambda} : A_\lambda \to A_\mu$ such that $\lambda = \mu \circ \phi_{\mu\lambda}$", and it follows from (18.8.5), (i) that this homomorphism $\phi_{\mu\lambda}$ is *unique*. Furthermore, L is *filtered increasing* for the preceding preorder relation (which one also denotes $\lambda \leqslant \mu$), by virtue of (18.8.5), (ii). It is clear that $(A_\lambda, \phi_{\mu\lambda})$ is an *inductive system* of **IV-4** | 146
local A-algebras, the $\phi_{\mu\lambda}$ being local homomorphisms; furthermore, it follows from (17.3.5) that for $\lambda \leqslant \mu$, A_μ is an A_λ-algebra that is essentially étale. If $\overline{\phi}_{\mu\lambda} : k(A_\lambda) \to k(A_\mu)$ is the k-homomorphism deduced from $\phi_{\mu\lambda}$ by passage to quotients, $(k(A_\lambda), \overline{\phi}_{\mu\lambda})$ is an inductive system of separable algebraic extensions of k, and the $\omega_\lambda : k(A_\lambda) \to \Omega$ form an inductive system of homomorphisms of k. Furthermore, the inductive limit

$$\omega : \varinjlim k(A_\lambda) \longrightarrow \Omega \tag{18.8.6.1}$$

is a k-*isomorphism*. It suffices indeed to prove that for every finite separable extension k' of k, there exists an A-algebra A' such that k' is the residue field of A' and that the homomorphism $k \to k'$ is deduced from $A \to A'$ by passage to quotients. But this follows from (18.1.1) applied to étale morphisms, since $\mathrm{Spec}(A)$ is the unique neighbourhood of its closed point.

Note now that if $\mathfrak{m}_\lambda$ is the maximal ideal of A_λ, it follows that $\mathfrak{m}_\mu = \mathfrak{m}_\lambda A_\mu$ and that A_μ is a flat A_λ-module. Hence it follows from (**0**, 10.3.1.3) that the ring $\varinjlim A_\lambda$ is *local*, that the homomorphism $w : A \to \varinjlim A_\lambda$ is local, and that one has a k-isomorphism

$$\varinjlim k(A_\lambda) \xrightarrow{\sim} k(\varinjlim A_\lambda); \tag{18.8.6.2}$$

identifying these two fields, one deduces a k-isomorphism

$$\omega^{-1} : \Omega \xrightarrow{\sim} k(\varinjlim A_\lambda).$$

Definition (18.8.7). — Let A be a local ring, $k = k(A)$ its residue field, $i : k \to \Omega$ a homomorphism from k into a separable closure Ω of k; the *strict Henselization* of A relative to i, denoted ${}^{hs}A_{(i)}$ (or ${}^{hs}A$ when this causes no confusion), is the inductive limit of the inductive system $(A_\lambda, \phi_{\mu\lambda})$ defined in (18.8.6).

As one has a k-isomorphism $\omega^{-1} : \Omega \xrightarrow{\sim} k({}^{hs}A_{(i)})$, Ω is endowed canonically with the structure of a ${}^{hs}A_{(i)}$-algebra by the local homomorphism

$${}^{hs}A_{(i)} \longrightarrow k({}^{hs}A_{(i)}) \xrightarrow{\omega} \Omega.$$

As in (18.6.5) one sees that the definition given in (18.8.7) does not depend on the choice of $\mathfrak{E}$; we shall moreover see below (18.8.8), (i) and (ii) that ${}^{hs}\mathrm{A}_{(i)}$ is an object representing a functor entirely defined by the data of A and i (**0**, 8.1.8), and is hence defined as such up to an isomorphism.

Proposition (18.8.8). — *Let* A *be a local ring, k its residue field, $i : k \to \Omega$ a homomorphism from k into a separable closure Ω of k, ${}^{hs}\mathrm{A}_{(i)}$ the strict Henselization of* A *corresponding to i.*

i) *${}^{hs}\mathrm{A}_{(i)}$ is a strictly local ring (18.8.2) and the structural homomorphism $w : \mathrm{A} \to {}^{hs}\mathrm{A}_{(i)}$ is local.*
ii) *For every local homomorphism $u : \mathrm{A} \to \mathrm{B}$, where* B *is a strictly local ring, and every k-homomorphism $\psi : \Omega \to k(\mathrm{B})$, there exists one and only one A-homomorphism $v : {}^{hs}\mathrm{A}_{(i)} \to \mathrm{B}$ such that ψ factorises as $\Omega \xrightarrow{\omega^{-1}} k({}^{hs}\mathrm{A}_{(i)}) \xrightarrow{\overline{v}} k(\mathrm{B})$, where $\overline{v}$ is deduced from v by passage to quotients.*
iii) *${}^{hs}\mathrm{A}_{(i)}$ is a faithfully flat A-module, and if $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}.{}^{hs}\mathrm{A}_{(i)}$ is the maximal ideal of ${}^{hs}\mathrm{A}_{(i)}$, and $k({}^{hs}\mathrm{A}_{(i)})$ is a separable closure of k.*
iv) *For ${}^{hs}\mathrm{A}_{(i)}$ to be Noetherian, it is necessary and sufficient that A be so.*
v) *If A is strictly local (hence $\Omega = k$), one has ${}^{hs}\mathrm{A} \cong \mathrm{A}$.*
vi) *If $i' : k \to \Omega'$ is a second homomorphism from k into a separable closure of k, for every k-isomorphism $\sigma : \Omega \xrightarrow{\sim} \Omega'$, the corresponding A-homomorphism (by (ii)) $v_\sigma : {}^{hs}\mathrm{A}_{(i)} \to {}^{hs}\mathrm{A}_{(i')}$ is an isomorphism.*

Proof. We have already seen above that ${}^{hs}\mathrm{A}_{(i)}$ is a local ring and that w is a homomorphism local; the fact that ${}^{hs}\mathrm{A}_{(i)}$ is Henselian (and hence strictly local) is demonstrated as in (18.6.6). The assertions of (iii) also follow from (**0**, 10.3.1.3); the necessity of the condition (iv); the necessity of condition results from (**0**, 6.5.2). **IV-4** | 147

To prove (ii), remark, with the notations of (18.8.6), that for every $\lambda \in \mathrm{L}$, there exists one and only one local homomorphism $v_\lambda : \mathrm{A}_\lambda \to \mathrm{B}$, such that the composite $k(\mathrm{A}_\lambda) \xrightarrow{\omega_\lambda} \Omega \xrightarrow{\psi} k(\mathrm{B})$ is deduced from v_λ by passage to quotients, by virtue of (18.8.4) and the hypothesis that B is strictly local. The uniqueness of v_λ implies moreover that the v_λ form an inductive system, whence the existence of the homomorphism v satisfying the conditions of (ii); uniqueness results from (vi) asserting the composites $v_\alpha \circ v_{\alpha-1}$ and $v_{\alpha-1} \circ v_\alpha$. Finally, if A is strictly local, and $\mathrm{A}' = \mathrm{B}_n$ is an A-algebra local and essentially étale, where B is an A-algebra étale, it follows from (18.8.1) that there exists a Spec(A)-section of Spec(B) taking the value $\mathfrak{n}$ at the closed point of Spec(A), hence by (17.4.1) that the homomorphism $\mathrm{A} \to \mathrm{A}'$ is bijective, which proves (v). □

(18.8.8.1). Let $\mathscr{C}$ be the category of strictly local rings, with local homomorphisms as morphisms; for every $\mathrm{B} \in \mathscr{C}$, denote by F(B) the set of pairs (u, ψ) consisting of a local homomorphism $u : \mathrm{A} \to \mathrm{B}$ and a k-isomorphism $\psi : k = k(\mathrm{A}) = k \xrightarrow{\psi} \Omega \xrightarrow{\psi} k(\mathrm{B})$ such that the composite $k(\mathrm{A}) = k \xrightarrow{i} \Omega \xrightarrow{\psi} k(\mathrm{B})$ is deduced from u by passage to quotients. One can say that the object ${}^{hs}\mathrm{A}_{(i)}$ *represents* the covariant functor $\mathrm{F} : \mathscr{C} \to \mathtt{Ens}$ (**0**, 8.1.11).

One will remark that by virtue of (18.8.8), (vi), the strict Henselizations ${}^{hs}\mathrm{A}_{(i)}$ are all A-isomorphic (for the various k-homomorphisms i of k into separable closures of k), but for given i and i', there is in general an infinity of A-isomorphisms ${}^{hs}\mathrm{A}_{(i)} \xrightarrow{\sim} {}^{hs}\mathrm{A}_{(i')}$. To be precise, the group of A-automorphisms of ${}^{hs}\mathrm{A}_{(i)}$ is isomorphic to the Galois group of Ω over k.

(18.8.9). Let now A be a *semi-local* ring, $\mathfrak{m}_j$ $(1 \leqslant j \leqslant r)$ its maximal ideals, and for every j, consider a homomorphism $i_j : k(\mathrm{A}_{\mathfrak{m}_j}) \to \Omega_j$ of the residue field into a separable closure. The *strict Henselization* of A relative to the i_j is the product $\prod_{j=1}^{r}({}^{hs}(\mathrm{A}_{\mathfrak{m}_j}))$, denoted ${}^{hs}\mathrm{A}$ (when there is no risk of confusion). It is therefore a faithfully flat A-module and a semi-local A-algebra, whose maximal ideals are the $\mathfrak{m}_j.{}^{hs}\mathrm{A}$ and the radical $\mathfrak{r}.{}^{hs}\mathrm{A}$ (if $\mathfrak{r}$ denotes the radical of A). Finally, the universal property (18.8.8), (ii) still holds when replacing "local homomorphism" by "semi-local homomorphism" (18.6.7), and replacing Ω by $\prod_j \Omega_j$. **IV-4** | 148

Proposition (18.8.10). — *Let* A *be a semi-local ring,* B *a finite A-algebra. Let $\mathfrak{n}_j$ $(1 \leqslant j \leqslant r)$ be the maximal ideals of* B*. Then there is, for every j, a maximal ideal $\mathfrak{q}_j$ of $\mathrm{B} \otimes_\mathrm{A} ({}^{hs}\mathrm{A})$ above $\mathfrak{n}_j$, such that ${}^{hs}\mathrm{B}$ is isomorphic to the composed direct product of local rings $(\mathrm{B} \otimes_\mathrm{A} ({}^{hs}\mathrm{A}))_{\mathfrak{q}_j}$ $(1 \leqslant j \leqslant r)$.*

Proof. One can evidently restrict to the case where A is local, replacing A by $A_{\mathfrak{m}_j}$. On the other hand, let $C = B_{\mathfrak{n}_j}$; by virtue of the definition of ${}^{hs}B$ (18.8.1), everything amounts to seeing that ${}^{hs}C$ is C-isomorphic to $(C \otimes_A ({}^{hs}A))_{\mathfrak{q}}$, where $\mathfrak{q}$ is one of the maximal ideals of $C \otimes_A ({}^{hs}A)$. Let k and K be the residue fields of A and C, Ω the separable closure of k, and let us suppose K and Ω contained in the same closure of k; as K is a finite extension of k, it is immediate that $K \otimes_k \Omega$ is a composed direct product of a finite number of fields, all isomorphic to the separable closure $K\Omega$ of K. On the other hand, the ring ${}^{hs}A$ being Henselian, $C \otimes_A ({}^{hs}A)$ is a composed direct product of local rings $(C \otimes_A ({}^{hs}A))_{\mathfrak{q}_i}$, where $\mathfrak{q}_i$ $(1 \leqslant i \leqslant s)$ ranges over the maximal ideals of $C \otimes_A ({}^{hs}A)$, and these local rings are Henselian (18.5.10) and have $K\Omega$ for residue field, hence they are strictly local (18.8.1). Let $C' = C \otimes_A ({}^{hs}A)$ and $C'_i = C'_{\mathfrak{q}_i}$, which is therefore a quotient ring of C'; let $f : C \to C'$, $g : {}^{hs}A \to C'$, $\phi_i : C' \to C'_i$ be the canonical maps.

$$\begin{tikzcd}
 & C'_i \arrow[r, "w"] & {}^{hs}C \\
C \arrow[r, "f"] & C' \arrow[u, "\phi_i"] & \\
A \arrow[u] \arrow[r] & {}^{hs}A \arrow[u, "g"'] &
\end{tikzcd}$$

Let $v : C \to {}^{hs}C$ be the canonical homomorphism local; there exists a unique $u : {}^{hs}A \to {}^{hs}C$ which, by passage to quotients, gives the canonical injection $\Omega \to K\Omega$ and such that the composite $A \to {}^{hs}A \xrightarrow{u} {}^{hs}C$ is equal to the composite $A \to {}^{hs}A \xrightarrow{u} {}^{hs}C$ (18.8.8); hence there exists a unique local homomorphism $w_0 : C' \to {}^{hs}C$ such that $w_0 \circ g = u$, $w_0 \circ f = v$, and since ${}^{hs}C$ is a local ring, this homomorphism factorises as $C' \to C'_i \xrightarrow{w} {}^{hs}C$ for a well-determined index i, w being a local homomorphism. On the other hand, the ring C'_i being strictly local, there exists a local homomorphism $w' : {}^{hs}C \to C'_i$ which, by passage to quotients, gives the identity automorphism of $K\Omega$ and which is such that $w' \circ v = \phi_i \circ f$ (18.8.8). One deduces first that $w \circ w'$ is an endomorphism of ${}^{hs}C$ which, by passage to quotients, gives the identity automorphism of $K\Omega$, hence is the identity **IV-4** | 149
(18.8.8). On the other hand, $w' \circ w \circ \phi_i \circ f = w' \circ v = \phi_i \circ f$ and $w' \circ w \circ \phi_i \circ g = w' \circ u = \phi_i \circ g$; whence $w' \circ w \circ \phi_i = \phi_i$, which is to say $w' \circ w$ is the identity automorphism of C'_i. □

Remark (18.8.11). — The proof of (18.8.10) uses the fact that B is a *finite* A-algebra, namely to establish that $K \otimes_k \Omega$ is a composed direct product of a finite number of fields. However, this last property is still satisfied when one supposes that B is an *entire* semi-local A-algebra whose maximal ideals $\mathfrak{n}_j$ have residue fields of finite separable degree $[k(\mathfrak{n}_j) : k]_s$ finite.

Proposition (18.8.12). — *Let* A *be a semi-local ring.*

i) *For* ${}^{hs}A$ *to be reduced (resp. normal), it is necessary and sufficient that A be so.*
ii) *Suppose A Noetherian. Then, for every prime ideal* $\mathfrak{p}$ *of A, the ring* $({}^{hs}A)_{\mathfrak{p}}/\mathfrak{p}.({}^{hs}A)_{\mathfrak{p}}$ *is a composed direct product of a finite number of separable algebraic extensions of* $k(\mathfrak{p})$ *(which implies that the fibres of the canonical morphism* $\mathrm{Spec}({}^{hs}A) \to \mathrm{Spec}(A)$ *are* geometrically regular *and are* discrete *spaces).*

Proof. The proofs are the same as those of (18.6.9), (18.6.10) and (18.6.11) respectively. □

Corollary (18.8.13). — *Let* A *be a local ring. For* ${}^{hs}A$ *to possess one of the following properties:*

a) *being a Cohen-Macaulay ring* (**0**, 16.5.3);
b) *satisfying the condition* (S_n) (5.7.3);
c) *being regular;*
d) *satisfying the condition* (R_n) (5.8.2);

it is necessary and sufficient that A possess the same property.

Corollary (18.8.14). — *Let* A *be a semi-local Noetherian ring. For a prime ideal* $\mathfrak{p}$ *of* ${}^{hs}A$ *to belong to* $\mathrm{Ass}({}^{hs}A)$*, it is necessary and sufficient that* $\mathfrak{p} \cap A$ *belong to* $\mathrm{Ass}(A)$.

Proof. The proofs are the same as those of (18.6.9), (18.6.10) and (18.6.11) respectively. □

Proposition (18.8.15). — *Let* A *be a local ring; the following properties are equivalent:*

a) *A is geometrically unibranch (resp. reduced and geometrically unibranch).*
b) *For every A-algebra* A' *that is local and essentially étale,* $\mathrm{Spec}(A')$ *is irreducible (resp. integral).*
c) $\mathrm{Spec}({}^{hs}A)$ *is irreducible (resp. integral).*

Proof. One reduces to the case where A is reduced, using the relation ${}^{hs}(A_{\mathrm{red}}) = ({}^{hs}A) \otimes_A A_{\mathrm{red}}$ (18.8.11), and the equivalence of $b)$ and $c)$ is proved as in (18.6.12); the same holds, the fact that $a)$ implies $c)$, taking into account the definition of a geometrically unibranch integral local ring (18.8.11). Finally, to prove that $c)$ implies $a)$, using the same notations as in (18.6.12), it suffices to see that ${}^{hs}B$ is integral. But, by (18.8.10), ${}^{hs}B$ is a localised ring of the semi-local ring $B \otimes_A ({}^{hs}A)$, which identifies by flatness with a subring of L, hence is integral; it follows that ${}^{hs}B$ is integral, hence also ${}^{hs}B$, which completes the proof. ☐

Corollary (18.8.16). — *Let* A *be a local Henselian ring (resp. strictly local). For A to be unibranch (resp. geometrically unibranch), it is necessary and sufficient that* $\mathrm{Spec}(A)$ *be irreducible.*

Proposition (18.8.17). — *Let* A *be a local Noetherian ring. If A is universally catenary, the same holds for* ${}^{hs}A$.

Proof. The proof is the same as that of (18.7.5). ☐

Proposition (18.8.18). — i) *Every filtered inductive limit of strictly local rings (where the transition homomorphisms are local) is a strictly local ring.*
ii) *The functor* $A \rightsquigarrow {}^{hs}A$ *in the category of local rings (where the morphisms are local homomorphisms) commutes with filtered inductive limits.*
iii) *Every strictly local ring A is an inductive limit of a filtered inductive system of Noetherian strictly local rings (the transition homomorphisms being local).*

Proof. The proofs are modelled on those of (18.6.14). ☐

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18.9. Formal fibres of Noetherian Henselian rings.

Theorem (18.9.1). — *Let* A *be a local Noetherian Henselian ring, whose formal fibres are geometrically normal. Then all the fibres of* A *are geometrically connected (hence geometrically integral).*

Proof. The two assertions stated are in fact equivalent, a prescheme that is locally Noetherian connected and normal being integral. As every finite integral A-algebra is a Henselian local ring (18.5.9) and (18.5.10), it follows from (7.3.16.2) that one is reduced to proving that if one further supposes A *integral*, then the fibre of the morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ at the generic point of $\mathrm{Spec}(A)$ is *integral*, which follows from the ☐

Corollary (18.9.2). — *Let* A *be a local Noetherian Henselian integral ring, whose formal fibres are geometrically normal. Then* $\widehat{A}$ *is integral.*

Proof. Indeed, it follows from (18.8.16) that A is unibranch, hence the theorem follows from the hypothesis made on the formal fibres of A and from (7.6.3). ☐

Corollary (18.9.3). — *Under the hypotheses of (18.9.2), the field of fractions L of* $\widehat{A}$ *is algebraically closed in L.*

Proof. This follows from the fact that the fibre of the morphism $\mathrm{Spec}(\widehat{A}) \to \mathrm{Spec}(A)$ at the generic point is geometrically integral and from (4.3.2) and (4.3.5). ☐

Corollary (18.9.4). — *Let* A *be a local Noetherian ring, Henselian, and whose formal fibres are geometrically normal (for example a Noetherian Henselian local ring that is excellent), and let* $A' = \widehat{A}$. *Let* X *be an A-prescheme,* $X' = X \otimes_A A'$, $g : X' \to X$ *the canonical projection. Then the map* $U \mapsto g^{-1}(U)$ *is a bijection from the set of subsets of* X *that are both open and closed to the set of subsets of* X' *that are both open and closed.*

Proof. As g is a faithfully flat and quasi-compact morphism (2.3.12), one knows that the topology of X is the *quotient* of that of X′ by the equivalence relation defined by f. It therefore amounts to seeing that an open and closed subset U′ of X′ is *saturated* for this relation. Now, as the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ has its fibres geometrically connected (18.9.1), the fibres of g are connected, whence our assertion immediately follows.

In particular, if X is locally connected (which will be the case if X is locally Noetherian), then, for every connected component U of X (which is open and closed), $g^{-1}(\mathrm{U})$ is connected (Bourbaki, *Top. gén.*, chap. I[er], 3[e] éd., § 11, no 3, prop. 7). If one denotes by $\pi_0(\mathrm{X})$ the set of connected components of X, the map $\pi_0(\mathrm{X}') \to \pi_0(\mathrm{X})$ canonically deduced from g is therefore *bijective*. □

Corollary (18.9.5). — *Under the hypotheses of (18.9.4), the functor*

(18.9.5.1) $$\mathrm{Z} \longmapsto \mathrm{Z} \otimes_{\mathrm{A}} \mathrm{A}'$$

from the category of étale preschemes over X, *to that of étale preschemes over* X′, *is fully faithful.*

Proof. Let Z_1, Z_2 be two étale preschemes over X, and set $\mathrm{Z}'_i = \mathrm{Z}_i \otimes_{\mathrm{A}} \mathrm{A}'$ ($i = 1, 2$); it is a matter of proving that every X′-morphism $\mathrm{Z}'_1 \to \mathrm{Z}'_2$ comes, by base change, from a unique X-morphism $\mathrm{Z}_1 \to \mathrm{Z}_2$. Suppose first that Z_2 is *separated* over X; then, as $\mathrm{Hom}_{\mathrm{X}}(\mathrm{Z}_1, \mathrm{Z}_2)$ identifies with $\Gamma((\mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2)/\mathrm{Z}_1)$, and that $\mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2$ is étale and separated over Z_1, by virtue of (17.9.3), to the set of open and closed subsets U of $\mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2$ such that the restriction to U of the projection $p : \mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2 \to \mathrm{Z}_1$ is a surjective and radicial morphism. The assertion thus follows from (18.9.4) and the fact that $\mathrm{Z}'_1 \times_{\mathrm{X}'} \mathrm{Z}'_2 = (\mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2) \otimes_{\mathrm{A}} \mathrm{A}'$. **IV-4** | 151

Let us pass to the general case: it will suffice to prove that the graph Γ of an X′-morphism $\mathrm{Z}'_1 \to \mathrm{Z}'_2$ in $\mathrm{Z}'_1 \times_{\mathrm{X}'} \mathrm{Z}'_2$ is of the form $\Gamma \otimes_{\mathrm{A}} \mathrm{A}'$, where Γ is induced on an *open* subset of $\mathrm{Z} = \mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2$. Indeed, the restriction to Γ of the projection $p : \mathrm{Z} \to \mathrm{Z}_1$ is an isomorphism; by base change $\mathrm{A} \to \mathrm{A}'$, this restriction becomes the isomorphism restriction $p' : \mathrm{Z}' \to \mathrm{Z}'_1$ (with $\mathrm{Z}' = \mathrm{Z} \otimes_{\mathrm{A}} \mathrm{A}'$), and it suffices to apply (2.7.1), (viii); the conclusion then follows from the characterisation of graphs (**I**, 5.3).

If $q : \mathrm{Z}' \to \mathrm{Z}$ is the canonical projection, to prove Z′ is of the form $q^{-1}(\Gamma)$, where Γ is open in Z, it suffices, since q is a faithfully flat and quasi-compact morphism, to prove that $q^{-1}(\Gamma) = q^{-1}(\mathrm{U})$ for an open $\mathrm{U} \subset \mathrm{Z}$. Let $\mathrm{Z}'' = \mathrm{Z}' \times_{\mathrm{Z}} \mathrm{Z}' = \mathrm{Z} \times_{\mathrm{X}} \mathrm{X}''$ where $\mathrm{X}'' = \mathrm{X}' \times_{\mathrm{X}} \mathrm{X}'$, and let $q_1 : \mathrm{Z}'' \to \mathrm{Z}'$, $q_2 : \mathrm{Z}'' \to \mathrm{Z}'$ be the canonical projections. Applying (4.5.19.1), it suffices to show that $q_1^{-1}(\Gamma') = q_2^{-1}(\Gamma')$. But this is a property which is true if and only if it is true after each base change $\mathrm{Spec}(k(x)) \to \mathrm{X}$, where x ranges over X. In other words, it suffices to prove the corollary when X is the *spectrum of a field*; but as every X-prescheme that is étale is automatically *separated* over X (17.6.2), c'), one is reduced to the case considered at the beginning of the proof. □

Remarks (18.9.6). — *i)* It is possible that, if the residue field of A is of characteristic 0, the functor (18.9.5.1) induces even an *equivalence* of the category of étale coverings of X and the category of étale coverings of X′. One can indeed prove this when A is *excellent*, using the resolution of singularities of Hironaka (M. Artin). It is plausible that an analogous statement is still true without restriction on the characteristic of the residue field, provided one restricts to the étale coverings whose "principal Galois groups" have order prime to the residual characteristic (cf. [41]).

ii) One does not know whether the formal fibres of A are geometrically connected (or even geometrically irreducible) when A is any Henselian local ring. It would suffice that, for every Noetherian Henselian local ring, Henselian and *integral* A, $\mathrm{Spec}(\widehat{\mathrm{A}})$ be irreducible. One does not know if this is always the case.

iii) One does not know whether, when A is an excellent local ring, its strict Henselization ${}^{hs}\mathrm{A}$ has geometrically regular formal fibres. One can indeed elucidate this question, one can reduce to the case where $\mathrm{A} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_n]]$, k being a field of characteristic $p > 0$. Taking account of (18.8.17), the question is whether the formal fibres of ${}^{hs}\mathrm{A}$ are geometrically regular. The answer is affirmative when $n = 1$, but is not known for $n = 2$.

The connectedness assertion made in (18.9.1) generalises in the following way:

Theorem (18.9.7). — *Let* A, B *be two local Noetherian rings,* $\rho : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* $f :$ **IV-4** | 152
$\mathrm{Spec}(\mathrm{B}) = \mathrm{X} \to \mathrm{Spec}(\mathrm{A}) = \mathrm{Y}$ *the corresponding morphism. Suppose the following conditions satisfied:*

i) *f is flat and all the fibres $X_y = f^{-1}(y)$ ($y \in Y$) are geometrically reduced* (4.6.2).
ii) *Either the ring* A *is geometrically unibranch* (**0**, 23.2.1)*, or else the ring A is unibranch* (**0**, 23.2.1) *and $k(B)$ is a primary extension* (4.3.1) *of $k(A)$.*

Then, if η is the generic point of Y*, X_η is connected, and for every closed subset $T \subset f^{-1}(y)$, $X - T$ is connected.*

Proof. The two conclusions stated are in fact equivalent; indeed, the hypothesis that $f(T)$ is rare in Y, T does not contain any maximal point of X (2.3.4), hence is rare in X; as X is a local scheme, hence connected, to prove that $X - T$ is connected, it suffices, by virtue of (15.5.6.1), to show that, for every $x \in f^{-1}(y)$, $\mathrm{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. One can therefore restrict to the case where $X = \mathrm{Spec}(B)$, B being a local Noetherian ring and $A \to B$ the corresponding local homomorphism; let $U = Y - \{y\}$, and note that $V = f^{-1}(U)$ is dense in X (2.3.10); it suffices therefore to prove that V is *connected*. Keeping the notations of (18.9.7.5), we thus have the commutative diagram

$$\begin{tikzcd} X & X_1 \arrow[l, "g_1"'] \\ Y \arrow[u, "f"] & Y_1 \arrow[l, "g"] \arrow[u, "f_1"'] \end{tikzcd}$$

□

Remark (18.9.7.1). — When the ring A is *Japanese*, one can, in the case preceding (18.9.7.2), take $A'' = A'$ by definition (**0**, 23.1.1); one sees therefore that in this case one would even be able to prove the theorem when A is *integrally closed*.

(18.9.7.2). Suppose therefore now that Y is integral and geometrically unibranch. Note that if T is a closed subset of X such that $f(T)$ is rare in Y, T contains no maximal point of X (2.3.4), hence is rare in X; as X is a local scheme, hence connected, to prove that $X - T$ is connected, it suffices, by virtue of (15.5.6.1), to show that for every $x \in f^{-1}(y)$, $\mathrm{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. One can therefore restrict to the case where $X = \mathrm{Spec}(B)$, B being a local Noetherian ring and the homomorphism $A \to B$ being local; let $U = Y - \{y\}$, and note that $V = f^{-1}(U)$ is dense in X (2.3.10); it suffices therefore to prove that V is *connected*.

Lemma (18.9.7.3). — *Let* A, B *be two local Noetherian rings,* $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $\rho : A \to B$ *a local homomorphism,* $f : X \to Y$ *the corresponding morphism. Suppose the following conditions satisfied:*

i) *f is flat and all the fibres X_y ($y \in Y$) are geometrically reduced.*
ii) *A is integral, geometrically unibranch and* $\dim(A) \geqslant 1$.

Then, if b is the closed point of X*, $X - \{b\}$ is connected.*

Proof. Note first that by virtue of the theorem of Hartshorne (5.10.7), $X - \{b\}$ is connected if **IV-4** | 153
$\mathrm{prof}(B) \geqslant 2$. Now on has (6.3.1)

$$\mathrm{prof}(B) = \mathrm{prof}(A) + \mathrm{prof}(B \otimes_A k) \tag{18.9.7.4}$$

where k denotes the residue field of A. On the other hand, since A is integral and $\dim(A) \geqslant 1$, one has $\mathrm{prof}(A) \geqslant 1$, hence $\mathrm{prof}(B) \geqslant 2$ *unless* $\mathrm{prof}(B \otimes_A k) = 0$; moreover, $B \otimes_A k$ is reduced by (i), hence the relation $\mathrm{prof}(B \otimes_A k) = 0$ means that $B \otimes_A k$ is a *field* (**0**, 16.4.7). We shall henceforth suppose that this last condition is satisfied. We shall begin by treating a case where the proof is very simple.

A) *Case where A is integrally closed.* Then, if $\dim(A) \geqslant 2$, one has $\mathrm{prof}(A) \geqslant 2$ (**0**, 16.5.1), hence $\mathrm{prof}(B) \geqslant 2$. One therefore only has to consider the case where $\dim(A) = 1$ and where $B \otimes_A k$ is a field; A is then a discrete valuation ring, hence regular, and since $B \otimes_A k$ is a field, B is also a discrete valuation ring (**0**, 17.3.3); but in addition (6.1.1.1), one has $\dim(B) = \dim(A) + \dim(B \otimes_A k) = \dim(A) = 1$, hence B is also a discrete valuation ring, hence $X - \{b\}$ is reduced to a single point, whence the lemma (18.9.7.3) in this case.

One will observe that this proves the statement of the theorem (18.9.7) when one supposes the ring A *Japanese*, for by virtue of the remark (18.9.7.1), one is reduced to the case where A is *integrally* *closed*, and then, in the reduction (18.9.7.3), the ring $\mathrm{A}_1 = \mathscr{O}_{\mathrm{X},y}$ is also integrally closed and one can apply the result just proved.

B) *Case where* $\dim(\mathrm{A}) \geqslant 2$. The result of (18.9.7.3) will follow from the two following lemmas. □

Lemma (18.9.7.5). — *Let* A *be a semi-local Noetherian ring, reduced,* A' *its integral closure in its total ring of fractions,* $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$; *let S be the set of closed points of* X *and* $\mathrm{U} = \mathrm{X} - \mathrm{S}$. *Suppose that for every point* $x' \in \mathrm{X}'$ *above a point of* U, $\dim(\mathscr{O}_{\mathrm{X}',x'}) \geqslant 2$; *then* $\mathrm{A}^{(1)} = \Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}})$ *is entire over* A *and is a sub-A-algebra isomorphic to* A'.

Proof. Recall that the $\mathfrak{p}_i$ ($1 \leqslant i \leqslant m$) are the minimal prime ideals of A, A' is the composed direct product of the integral closures $\mathrm{A}_i = \mathrm{A}/\mathfrak{p}_i$, so that X' is the sum of the schemes $\mathrm{X}'_i = \mathrm{Spec}(\mathrm{A}'_i)$ ($1 \leqslant i \leqslant m$). If X_0 is the sum of the schemes $\mathrm{X}_i = \mathrm{Spec}(\mathrm{A}_i)$ ($1 \leqslant i \leqslant m$) and U_0 the inverse image of U in X_0, the canonical homomorphism $\Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}}) \to \Gamma(\mathrm{U}_0, \mathscr{O}_{\mathrm{X}_0})$ is injective since X is reduced and the morphism $\mathrm{X}' \to \mathrm{X}$ surjective. As $\Gamma(\mathrm{U}_0, \mathscr{O}_{\mathrm{X}_0})$ is the composed direct product of the $\Gamma(\mathrm{U}_i, \mathscr{O}_{\mathrm{X}_i})$ (where U_i is the inverse image of U in X_i), it suffices to prove that for each i, $\Gamma(\mathrm{U}_i, \mathscr{O}_{\mathrm{X}_i})$ is isomorphic to a sub-A_i-algebra of A'_i. As X is reduced and the morphism $\mathrm{X}' \to \mathrm{X}$ surjective, the homomorphism $\Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}}) \to \Gamma(\mathrm{U}', \mathscr{O}_{\mathrm{X}'})$ (where U' is the inverse image of U in X') is injective, and everything amounts to verifying that the canonical homomorphism $\mathrm{A}' = \Gamma(\mathrm{X}', \mathscr{O}_{\mathrm{X}'}) \to \Gamma(\mathrm{U}', \mathscr{O}_{\mathrm{X}'})$ is *bijective*. Now (**I**, 8.2.1.1), $\Gamma(\mathrm{U}', \mathscr{O}_{\mathrm{X}'})$ is the intersection of the local rings $\mathrm{A}'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over U', and by virtue of the hypothesis, among these local rings are included all those of height 1. But the reasoning of (**0**, 23.2.7) applies also to a semi-local Noetherian integral ring, and is therefore a *semi-local Krull ring*, and is therefore the intersection of the local rings $\mathrm{A}'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over the set of prime ideals of height 1 of A' (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 6, th. 4); *a fortiori*, A' is the intersection of the $\mathrm{A}'_{\mathfrak{p}'}$ for $\mathfrak{p}' \in \mathrm{U}'$, which completes the proof of the lemma. □ **IV-4** | 154

Remark (18.9.7.6). — We know (**0**, 23.2.5) that there exists a finite sub-A-algebra A'' of K, the field of fractions of A, such that the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A}'')$ is *radicial*; as this morphism is also entire and dominant, hence surjective (**II**, 6.1.10), it is a *homeomorphism* (2.4.5). By virtue of the geometric interpretation of $\dim(\mathscr{O}_{\mathrm{X}',x'})$ (5.1.2), one has therefore, for every point x'' of $\mathrm{X}'' = \mathrm{Spec}(\mathrm{A}'')$ above a point $x \in \mathrm{X}$, $\dim(\mathscr{O}_{\mathrm{X}'',x''}) = \dim(\mathscr{O}_{\mathrm{X}',x'})$, whence $\dim(\mathscr{O}_{\mathrm{X}'',x'}) \geqslant 2$ if x' is the unique point of X' above x. The same conclusion holds if $\mathscr{O}_{\mathrm{X},x}$ is *geometrically unibranch*, for the fibre of the morphism $\mathrm{X}' \to \mathrm{X}$ at the point x is then reduced to a single point, and $\mathscr{O}_{\mathrm{X}',x'}$ is an *entire* $\mathscr{O}_{\mathrm{X},x}$-algebra; it suffices then to apply (**0**, 16.1.5). **IV-4** | 155

Lemma (18.9.7.7). — *Let* A *be a local Noetherian ring, of dimension* $\geqslant 2$, *integral and geometrically unibranch,* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, y *the closed point of* Y. *Let* X *be a locally Noetherian prescheme and* $f : \mathrm{X} \to \mathrm{Y}$ *a flat morphism. Then, for every closed subset* $\mathrm{T} \subset f^{-1}(y)$, $\mathrm{X} - \mathrm{T}$ *is connected.*

Proof. The same reasoning as at the beginning of (18.9.7.2) reduces to proving that, for every point $x \in f^{-1}(y)$, $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) - \{x\}$ is connected. One can therefore restrict to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, B being a local Noetherian ring and the homomorphism $\mathrm{A} \to \mathrm{B}$ being local; let $\mathrm{U} = \mathrm{Y} - \{y\}$, and note that $\mathrm{V} = f^{-1}(\mathrm{U})$ is dense in X (2.3.10); it suffices therefore to prove that V is *connected*. Keeping the notations of (18.9.7.5), and setting $\mathrm{A}^{(1)} = \Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}})$, $\mathrm{X}_1 = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}_1 = \mathrm{Spec}(\mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}^{(1)})$, we have the commutative diagram **IV-4** | 155

$$\begin{tikzcd} \mathrm{X} & \mathrm{X}_1 \arrow[l, "g_1"'] \\ \mathrm{Y} \arrow[u, "f"] & \mathrm{Y}_1 \arrow[l, "g"] \arrow[u, "f_1"'] \end{tikzcd}$$

By virtue of (2.3.1) and of the definition of $\mathrm{A}^{(1)}$, one has a canonical isomorphism $\Gamma(\mathrm{V}, \mathscr{O}_{\mathrm{X}}) \xrightarrow{\sim} \mathrm{B} \otimes_{\mathrm{A}} \mathrm{A}^{(1)}$, and it suffices therefore to prove that this last ring is *local*, such a ring being a product of two nonzero rings (**III**, 7.8.6.1). But we have seen (18.9.7.5) that $\mathrm{A}^{(1)}$ is a sub-A-algebra of the integral closure A' of A, hence $\dim(\mathrm{A}) \geqslant 2$. The hypothesis that A is geometrically unibranch implies that the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ is radicial at the point y (6.15.3); *a fortiori* g_1 is radicial at the closed point of X; furthermore g_1 is an entire morphism, hence the closed points of X_1 are above x, hence there is only one closed point of X_1, which completes the proof of (18.9.7.7).

It is clear that (18.9.7.7) proves (18.9.7.3) when $\dim(A) \geqslant 2$ (even without supposing the fibres geometrically reduced).

C) *Case where* $\dim(A) = 1$. One has seen at the beginning of the proof of (18.9.7.3) that one can suppose $\dim(B \otimes_A k) = 0$, and by flatness (6.1.1.1), one therefore has

$$\dim(B) = \dim(A) + \dim(B \otimes_A k) = 1.$$

Since A is integral, Y consists of two points, the generic point η and the closed point a. Furthermore, since f is flat, every irreducible component of X dominates Y (2.3.4), hence $X - \{b\}$, which is the set of maximal points ξ_i of X, is equal to the underlying set of $f^{-1}(\eta)$, and the fibre $f^{-1}(\eta)$ is the sum **IV-4** | 156
of the $\mathrm{Spec}(k(\xi_i))$. The hypothesis on the X_y and the fact that these fibres are of dimension 0, show that they are *geometrically regular* (6.7.6), and *a fortiori* geometrically normal, in other words f is a *normal morphism*. But since A is integral and geometrically unibranch, it follows then from (6.15.10) that B is also integral and geometrically unibranch, hence $f^{-1}(\eta) = X - \{b\}$ is reduced to a single point, and *a fortiori* connected; this concludes the proof of (18.9.7.3) and of (18.9.7). ☐

Remark (18.9.7.8). — In case C) of the proof of (18.9.7.3), one can avoid making appeal to the delicate result (6.15.10) by reasoning as follows: since A is integral and unibranch of dimension 1, it follows from the theorem of Krull-Akizuki (Bourbaki, *Alg. comm.*, chap. VII, § 2, no 5, prop. 5) that its integral closure A' is a *Noetherian* ring. The same reasoning as at the beginning of the proof of (18.9.7) shows that $B' = B \otimes_A A'$ is a local ring; furthermore, by flatness, B' is contained in the total ring of fractions R of the reduced B (3.3.5). Now, the theorem of Krull-Akizuki (Bourbaki, *loc. cit.*) also applies to a local Noetherian *reduced* ring of dimension 1, and shows that for every such ring B, everything contained between B and its total ring of fractions is *Noetherian*. The ring B' being a local Noetherian ring and the morphism $\mathrm{Spec}(B') \to \mathrm{Spec}(A')$ being normal, B' is an integral ring and integrally closed (6.5.4), and one concludes as at the beginning of the proof of (18.9.7).

Corollary (18.9.8). — *With the notations of* (18.9.7), *suppose verified the condition (i) of* (18.9.7) *and one of the two conditions:*

(ii′) *A is a strictly local ring* (18.8.2).

(ii″) *A is a Henselian local ring and* $k(B)$ *is a primary extension of* $k(A)$.

Then all the fibres X_y *(*$y \in Y$*) are geometrically connected.*

Proof. Let $\mathfrak{p}$ be the prime ideal of A corresponding to the point y, and set $A' = A/\mathfrak{p}$ and $B' = B \otimes_A A' = B/\mathfrak{p}B$, which are evidently local Noetherian rings; it is clear that the morphism $f' : \mathrm{Spec}(B') \to \mathrm{Spec}(A')$ satisfies condition (i) of (18.9.7); and as y is the generic point of $\mathrm{Spec}(A')$ and $k(A') = k(A)$, $k(B') = k(B)$, one sees that it suffices to verify that, in the case (ii′) (resp. (ii″)), A' is geometrically unibranch (resp. unibranch). But A' is integral, and in the case (ii′) (resp. (ii″)) it is strictly local (resp. Henselian) by virtue of (18.5.10). One therefore concludes that A' is geometrically unibranch (resp. unibranch) by virtue of (18.8.15) (resp. (18.6.12)). ☐

Corollary (18.9.9). — *Let* A *be a local Noetherian ring, Henselian, universally Japanese (i.e.* (7.6.4) *and* (7.7.2)*), whose formal fibres* (7.3.13) *are geometrically reduced; then all the fibres formelles of A are geometrically connected.*

Proof. It suffices to apply (18.9.8) in the case where $B = \widehat{A}$, since B is an A-module that is flat and that $k(B) = k(A)$. ☐

Remark (18.9.10). — The proofs of (18.9.4) and (18.9.5) do not use the fact that the formal fibres of A are geometrically connected. The conclusions of these two propositions are therefore still valid **IV-4** | 157
when one supposes the local ring A Noetherian, Henselian and universally Japanese.

Corollary (18.9.11). — *Let* X, Y *be two locally Noetherian preschemes,* Y *being supposed geometrically unibranch and X connected. Let* $f : X \to Y$ *be a reduced morphism (i.e.* (6.8.1)*), flat and with geometrically reduced fibres. Then, for every closed subset* T *of* X *such that* $f(T)$ *is rare in* Y, $X - T$ *is connected.*

Proof. As f is flat and $f(T)$ is rare, T cannot contain any maximal point of X (2.3.4), hence is rare in X. Utilising (15.5.6.1), it suffices to show that, for every $x \in T$, $\mathrm{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. Setting $y = f(x)$, as the ring $\mathscr{O}_{Y,y}$ is geometrically unibranch, one can apply (18.9.7) to the morphism

$$f_1 : \mathrm{Spec}(\mathscr{O}_{X,x}) \longrightarrow \mathrm{Spec}(\mathscr{O}_{Y,y}),$$

which satisfies conditions (i) and (ii) of this theorem, and to the closed subset $\{x\}$ of $\mathrm{Spec}(\mathscr{O}_{X,x})$, since $f_1(\{x\}) = \{y\}$ is rare in $\mathrm{Spec}(\mathscr{O}_{Y,y})$. □

18.10. Étale preschemes over a geometrically unibranch or normal prescheme

Theorem (18.10.1). — *Let $f : X \to Y$ be a locally of finite presentation morphism, x a point of X, $y = f(x)$. Suppose that Y is integral and geometrically unibranch at the point y. Then, for f to be étale at the point x, it is necessary and sufficient that the following two conditions be satisfied:*

i) *f is formally unramified at the point x;*
ii) *the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective.*

Moreover, if this is the case, X is integral and geometrically unibranch at the point x.
Finally, if $\mathscr{O}_{Y,y}$ is Noetherian, one can, in the preceding criterion, replace the condition (ii) *by the condition* (ii bis) $\dim \mathscr{O}_{Y,y} \leqslant \dim \mathscr{O}_{X,x}$.

Proof. The fact that, if f is étale, X is integral and geometrically unibranch at the point x is a partic- **IV-4** | 157
ular case of (17.5.7). It is clear on the other hand that the conditions (i) and (ii) are necessary for f to be étale at the point x, since $\mathscr{O}_{X,x}$ is then an $\mathscr{O}_{Y,y}$-module faithfully flat (17.6.1). Let us prove that these conditions are sufficient.

Set $\mathrm{A} = \mathscr{O}_{Y,y}$, $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$ (18.8.7), $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$. Let y' be the closed point of Y', x' a point of X' above x and y'; since the morphism $g : \mathrm{Y}' \to \mathrm{Y}$ is *flat* (18.8.8), it amounts to showing that f' is étale at the point x' (17.7.1),(ii)). The hypothesis (ii) implies that the morphism $\mathrm{Spec}(\mathscr{O}_{X,x}) \to \mathrm{Spec}(\mathscr{O}_{Y,y})$ is dominant (**I**, 1.2.7); as $\mathscr{O}_{Y,y}$ is integral, $\mathrm{Spec}(\mathscr{O}_{Y,y})$ has a single generic point y_1, and there is a generization x_1 of x in X above y_1. The projection morphism $\mathrm{X}' \to \mathrm{X}$ being flat, there exists a generization x_1' of x' above x_1 (2.3.4); set $y_1' = f'(x_1')$. Now, the hypothesis on y implies that A' is *integral* (18.8.15), hence Y' has a single generic point η, necessarily above y_1 since g is flat (2.3.4); on the other hand, η is also the generic point of $g^{-1}(y_1)$ **IV-4** | 158
(**0**, 2.1.8), and one knows on the other hand that the fibres of g are discrete spaces (18.8.12), hence $g^{-1}(y_1) = \{\eta\}$, and consequently $y_1' = \eta$. The morphism $\mathrm{Spec}(\mathscr{O}_{X',x'}) \to \mathrm{Spec}(\mathscr{O}_{Y',y'})$, restriction of f', is therefore dominant, and since $\mathscr{O}_{Y',y'} = \mathrm{A}'$ is integral, the corresponding homomorphism $\mathscr{O}_{Y',y'} \to \mathscr{O}_{X',x'}$ is injective (**I**, 1.2.7). On the other hand, f' is not ramified at the point x', (hence (18.8.3), *d*)), there exists a neighbourhood U of x' in X' such that $f'|\mathrm{U}$ is a *closed immersion* (18.8.3); *a fortiori* the homomorphism $\mathscr{O}_{Y',y'} \to \mathscr{O}_{X',x'}$ is surjective, and since it is injective, it is therefore *bijective*. But then $\mathscr{O}_{X,x'}$ is an $\mathscr{O}_{Y',y'}$-module *flat*, and f' is therefore étale at the point x' (17.6.1).

Finally, when $\mathscr{O}_{Y,y}$ is Noetherian, we know that if f is étale at the point x, we have $\dim \mathscr{O}_{Y,y} = \dim \mathscr{O}_{X,x}$ (17.6.4). Conversely, suppose the conditions (i) and (ii bis) satisfied, and let us show that they imply conditions (i) and (ii). Let $\mathfrak{J}$ be the kernel of the canonical homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$; restricting Y as needed to a neighbourhood of y, we may suppose that $\mathfrak{J} = \mathscr{J}_y$, where $\mathscr{J}$ is an Ideal of $\mathscr{O}_Y$; if Y_1 is the closed sub-prescheme of Y defined by $\mathscr{J}$, one can, restricting further X and Y to neighbourhoods of x and y respectively, suppose that f factorises as $\mathrm{X} \xrightarrow{h} \mathrm{Y}_1 \to \mathrm{Y}$ (**I**, 6.5.1). It is clear that f_1 is still not ramified at the point x, hence (17.4.1) quasi-finite at this point; the demonstration of (5.4.1), (i)) then shows that $\dim(\mathscr{O}_{X,x}) \leqslant \dim(\mathscr{O}_{Y,y}/\mathfrak{J})$. But if one had $\mathfrak{J} \neq 0$, one would conclude that $\dim(\mathscr{O}_{Y,y}/\mathfrak{J}) < \dim(\mathscr{O}_{Y,y})$ since Y is integral (**0**, 16.1.2.2); one would then have $\dim \mathscr{O}_{X,x} < \dim \mathscr{O}_{Y,y}$, contradicting the hypothesis, which proves (ii). □

Remark (18.10.2). —

(i) When $\mathrm{A} = \mathscr{O}_{Y,y}$ is Noetherian, and one knows that its completion $\widehat{\mathrm{A}}$ is integral (for example if A is regular), one can, in the preceding proof, replace A' by $\widehat{\mathrm{A}}$.

(ii) When Y is locally Noetherian, one can give a faster proof of (18.10.1), without using strict henselisation, but invoking the rather delicate results of §§ 14 and 15. As f is quasi-finite at the point x by hypothesis (17.4.1), one can, replacing X by an open neighbourhood of x, suppose that $f^{-1}(y) = \{x\}$. The hypothesis (ii) implies, as we have seen, the existence of an irreducible component Z of X containing x and dominating the unique irreducible component Y_0 of Y containing y. The morphism $f|\mathrm{Z}$ being quasi-finite at the point x, hence equidimensional at this point (13.2.2), it follows from the hypothesis on Y and from the Chevalley criterion (14.4.4) that $g = f|\mathrm{Z}$ is universally open at the point x. Furthermore since f (and hence also g) is not ramified at the point x, $g^{-1}(y)$ is geometrically reduced on $k(y)$ (17.4.1); as $\mathscr{O}_{Y,y}$ is integral, one concludes from (15.2.3) that g is *flat* at the point x, hence f is *étale* at this point (18.4.9).

(iii) With the same notations and hypotheses on Y as in (18.10.1), suppose only that f is *locally of finite type* (and not necessarily locally of finite presentation). Then, for $\mathscr{O}_{X,x}$ to be an $\mathscr{O}_{Y,y}$-algebra *essentially étale* (18.6.1), it is necessary and sufficient that f satisfy the two following conditions:

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1° f is formally unramified at the point x;
2° the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective.

There is nothing more than proving the sufficiency of these conditions. Taking into account (18.4.12), it suffices to show that f is *flat* at the point x. Now, by going back to the proof of (18.10.1) and the notations used in this proof, it suffices to show that f' is flat at the point x' (2.5.1); by (17.1.3) furthermore f' is formally unramified at the point x'. One can therefore restrict to carrying out the proof when $Y = \mathrm{Spec}(A)$, where A is a *strictly local* local ring and $y = f(x)$ the closed point of Y. Using then (18.8.3) we see that $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is surjective, hence *bijective* by virtue of the hypothesis, and this suffices to show the flatness of f at the point x.

Suppose furthermore that Y is *locally integral* at the point y (**I**, 2.1.8). Then the conditions 1° and 2° above (together with the fact that Y is geometrically unibranch at the point y and f locally of finite type) already imply that f is *étale at the point* x. Indeed, this follows from what precedes and from (18.4.13).

Corollary (18.10.3). — *Let* Y *be an integral prescheme, integral and geometrically unibranch, η its generic point,* X *a connected prescheme, $f : X \to Y$ a locally of finite type morphism, and such that $f^{-1}(\eta)$ is non-empty. Then, if f is formally unramified, f is étale, and* X *is integral and geometrically unibranch.*

Proof. It follows from the hypothesis and from (18.4.13) that f is étale at all the points where it is flat, and in particular at the points of $f^{-1}(\eta)$, since $\mathscr{O}_{Y,\eta} = k(\eta)$ is a field. The open set U of points of X where f is étale is therefore *non-empty*. Let us show on the other hand that for all $x \in U$, the homomorphism $\mathscr{O}_{Y,f(x)} \to \mathscr{O}_{X,x}$ is *injective*. Indeed let s be a section of $\mathscr{O}_Y$ above an open neighbourhood of $f(x)$, that one can always suppose to be X and Y themselves (the question being local on X and Y), and suppose that its image $t \in \Gamma(Y, f_*(\mathscr{O}_X)) = \Gamma(X, \mathscr{O}_X)$ has zero germ at the point x, hence vanishes in a neighbourhood V of x; then the restriction of t to the open set $U \cap V \neq \varnothing$ is zero; but the restriction $f|(U \cap V)$ is étale, hence $W = f(U \cap V)$ is open in Y and contains by construction the generic point η; the homomorphism $\mathscr{O}_{Y,\eta} \to \mathscr{O}_{X,z}$ being injective for all points $z \in f^{-1}(\eta)$, the hypothesis that $t|(U \cap V) = 0$ implies that $s_\eta = 0$, and as Y is integral, this implies $s = 0$, whence our assertion. Applying now (18.10.2), (iii)), we conclude that f is *flat* at the point x, in other words $x \in U$, hence U is both open and closed in X; as it is non-empty and X is connected, we have $U = X$.

Note now that since f is étale, hence locally quasi-finite and Y is irreducible, the maximal points of X belong to $f^{-1}(\eta)$ (2.3.4) and for all $x \in X$, there is a neighbourhood of x that contains only a finite number of these maximal points; in other words, the set of irreducible components of X is locally finite. But by (18.10.1) X is integral and geometrically unibranch at every point, hence every point of X belongs to only a single irreducible component, and since these latter form a locally finite set of open (and closed) subsets in X; since X is connected, it is integral. □

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Remark (18.10.3.1). — If $f : X \to Y$ is locally of finite presentation, dominant and quasi-compact, the fibre $f^{-1}(\eta)$ is non-empty by virtue of (**I**, 1.1.5). On the other hand, if one does not suppose f quasi-compact, f *can be non-ramified and dominant without f being étale* (always assuming the hypotheses of (18.10.3) verified for X and Y). This is what the following example shows: one takes for Y the affine plane $\mathrm{Spec}(\mathbf{C}[T, U])$; one considers on the other hand a family $(X_i)_{i \in \mathbf{Z}}$ of preschemes isomorphic to the affine line $\mathrm{Spec}(\mathbf{C}[T])$, and one forms the prescheme obtained by “gluing together” X_{2j} and X_{2j+1} at the point -1 and X_{2j+1} and X_{2j+2} at the point $+1$. Let us define on the other hand $f : X \to Y$ as being equal on X_{2j} to the closed immersion whose image is the line of equation $y = 2j$, transforming the point -1 to $(2j-1, 2j)$ and the point 1 to $(2j+1, 2j)$; on X_{2j+1}, f_j is the closed immersion whose image is the line of equation $x = 2j+1$, transforming the point -1 to $(2j+1, 2j)$ and the point 1 to $(2j+1, 2j+2)$. It is clear that f is a local immersion (hence is not ramified) and is dominant but the fibre of the generic point of Y is empty and f is not étale.

Corollary (18.10.4). — *Let $g : X \to S$, $h : X \to S$ be two locally of finite presentation morphisms, $f : X \to Y$ an S-morphism, x a point of X, $y = f(x)$, $s = g(y) = h(x)$. Suppose that h is flat at the point x and $g^{-1}(s)$ normal at the point y. Then, for f to be étale at the point x, it is necessary and sufficient that f be formally unramified at the point x and that $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$.*

Proof. The conditions are necessary by virtue of the fact that if f is étale at the point x, it follows from this that f_s is also flat and quasi-finite at the point x (6.1.2). Conversely, if the conditions of the statement are satisfied, in order to see that f is étale at the point x, it suffices, by virtue of (17.8.3), to see that f_s is so. One can therefore restrict to the case where S is the spectrum of a field. It then follows from the hypothesis $\dim_x X = \dim_y Y$ and the fact that f is not ramified at the point x that one has (by (5.2.3) and (17.4.1)) $\dim \mathscr{O}_{X,x} = \dim \mathscr{O}_{Y,y}$ integral, and we deduce from (**0**, 7.4.4) and (**0**, 16.3.10) that the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective; it then suffices to apply (18.10.1). □

Corollary (18.10.5). — *With the notations of (18.10.4), suppose that h is flat and that $g^{-1}(s)$ is normal for all $s \in S$ (which will be the case for example if g is smooth). For f to be an open immersion, it is necessary and sufficient that f be a monomorphism and that, for all $x \in X$, $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$, where $y = f(x)$ and $s = g(y) = h(x)$.*

Proof. One needs only to prove the sufficiency of the conditions; now, it follows from (17.1.3) that a monomorphism locally of finite presentation is formally unramified, hence the hypotheses imply that f is étale (18.10.4); on the other hand, by (17.9.1), we conclude from these and from (18.10.4). □

Lemma (18.10.6). — *Let $f : X \to Y$ be a flat and locally quasi-finite morphism.*

(i) The set $\mathrm{Max}(X)$ of maximal points of X is in canonical bijection with the set $\coprod_{y \in \mathrm{Max}(Y)} f^{-1}(y)$.

(ii) If Y is irreducible with generic point η, the set of irreducible components of X is in canonical bijection with $f^{-1}(\eta)$, and is in particular finite if and only if $f^{-1}(\eta)$ is finite. **IV-4** | 161

(iii) If the set of irreducible components of Y is locally finite, so is the set of irreducible components of X, and in particular X is locally connected.

Proof. The assertion (i) follows immediately from (2.3.4) and (**0**, 2.1.6) and from the fact that the fibres $f^{-1}(y)$ are discrete sets. It is clear that (ii) is a particular case of (i). Finally, to prove (iii), one can, by virtue of (i), restrict to the case where f is quasi-finite and where Y is irreducible (**0**, 2.1.6) and the conclusion follows from (ii) and from the fact that the fibres of f are finite sets (**II**, 6.2.2). □

Proposition (18.10.7). — *Let Y be a geometrically unibranch and irreducible prescheme (resp. integral), $f : X \to Y$ an étale morphism. Then X is isomorphic to a sum of preschemes irreducible (resp. integral) X_λ, where λ ranges over the fibre $f^{-1}(\eta)$ of the generic point η of Y. The X_λ are geometrically unibranch; if Y is normal, the X_λ are normal.*

Proof. Indeed, X is geometrically unibranch (17.5.7), hence every point of X belongs to only a single irreducible component; moreover, by virtue of (18.10.6) and (17.6.1) the set of irreducible components of X is locally finite, hence (**0**, 2.1.6) these are the connected components X_λ of X ($\lambda \in f^{-1}(\eta)$), and X is the sum of the X_λ; by virtue of (17.5.7), if Y is integral (resp. normal), the X_λ are integral (resp. normal). □

Proposition (18.10.8). — *Let Y be a normal and integral prescheme, of generic point η, $K = k(\eta)$ its field of rational functions, $f : X \to Y$ an étale morphism. Suppose that $f^{-1}(\eta)$ is finite and non-empty, so that the K-scheme $f^{-1}(\eta)$ is the spectrum of a finite separable K-algebra L, composed as a direct product of a finite number of fields K_i ($1 \leqslant i \leqslant n$), finite separable extensions of K (17.6.2). Let Y′ be the integral closure of Y in L, isomorphic to the sum of the integral closures Y_i of Y in K_i (**II**, 6.3.6). Then the morphism f factorises uniquely as $X \xrightarrow{f'} Y' \xrightarrow{g} Y$, such that the restriction $f'_\eta : f^{-1}(\eta) \to g^{-1}(\eta)$ is the canonical isomorphism. Moreover f' is a local isomorphism; for f' to be an open immersion, it is necessary and sufficient that f be separated.*

Proof. By virtue of (18.10.7), one can restrict to the case where X is integral and normal. The existence and uniqueness of the factorisation considered for f follow from (**II**, 6.3.9). As f' is birational and locally quasi-finite, the final assertions follow from (8.12.10). □

Corollary (18.10.9). — *Under the hypotheses of (18.10.8), for f to be a finite morphism (in other words, for X to be an étale covering of Y), it is necessary and sufficient that X be isomorphic to the integral closure Y' of Y in L.*

Proof. If f is finite, the canonical injection $j : \mathrm{X} \to \mathrm{Y}'$ (which is an open immersion) is also a finite morphism (**II**, 6.1.5), (v)), hence a closed morphism (**II**, 6.1.10), and as $j(\mathrm{X})$ is dense in Y' by (18.10.8), we have $j(\mathrm{X}) = \mathrm{Y}'$. Conversely, as Y is normal and L is a composed direct product of finite separable extensions of K, Y' is finite over Y (Bourbaki, *Alg. comm.*, chap. V, § 1, nº 6, cor. 1 of prop. 18), whence the corollary. □

(18.10.10). Let Y be a normal and integral prescheme, $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ its field of rational functions. We say that a K-algebra of finite rank L is *non-ramified over* Y if: 1° L is a separable K-algebra, hence composed as a direct product of finite separable extensions K_i of K ($1 \leqslant i \leqslant n$); 2° the integral closure Y' of Y in L (sum of the preschemes that are integral closures of Y in the K_i) is *non-ramified over* Y (which, by (18.10.3), is equivalent to saying that Y' is étale over Y). When $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, **IV-4** | 162
where A is an integral ring, integrally closed, K its field of fractions, one says also (by a slight abuse of language, and when there is no confusion to be feared with the terminology of (17.3.2), (ii)) that L is non-ramified over A instead of "non-ramified over Y".

Remark (18.10.11). — Instead of saying that L is non-ramified over Y, certain authors say also that L is "non-ramified over K"; we shall not follow this usage, which lends itself to confusions.

One can then express (18.10.9) in the following form:

Corollary (18.10.12). — *Let Y be a normal and integral prescheme, $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ its field of rational functions. Then the functor $\mathrm{X} \mapsto \mathrm{R}(\mathrm{X})$ is an equivalence of the category of étale coverings of Y and of the category of finite K-algebras that are non-ramified over Y. One obtains a quasi-inverse functor by making correspond to every finite étale K-algebra L, non-ramified over Y, the integral closure Y' of Y in L.*

Proposition (18.10.13). — *Let Y be a normal and integral prescheme, $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ its field of rational functions.*

- *(i) The field K is a K-algebra non-ramified over Y.*
- *(ii) Let L be a finite extension of K, non-ramified over Y, and let Z be the integral closure of Y in L. Then, if M is an L-algebra non-ramified over Z, M is also a K-algebra non-ramified over Y ("transitivity" of non-ramification).*
- *(iii) Let Y' be a normal and integral prescheme, K' its field of rational functions, $g : \mathrm{Y}' \to \mathrm{Y}$ a dominant morphism. If L is a K-algebra non-ramified over Y, then $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$ is a K'-algebra non-ramified over Y' ("translation" property). Furthermore if $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ are affine, and if C is the integral closure of A in L, then $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ is the integral closure of A' in $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$.*

Proof. The assertion (i) is immediate, Y being its own normalisation in K. To prove (ii), let Z' be the integral closure of Z in M, which is by hypothesis an étale covering of Z. As Z is étale and separated over Y, so is Z' (17.3.3), and it is clearly the integral closure of Y in M. As M is a K-algebra, finite and separable, it follows from (18.10.10) that M is non-ramified over Y. Finally, to prove (iii), let us note that if Z is the integral closure of Y in L, $\mathrm{Z} \times_{\mathrm{Y}} \mathrm{Y}'$ is étale and separated over Y' (17.3.3), finite over Y' since Z is finite over Y, and admits for ring of rational functions $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$ (**I**, 3.4.9); as Y' is normal, it follows from (17.5.7), which finishes the proof. □

Corollary (18.10.14). — *Let Y and Y' be two normal preschemes and integral, K, K' their respective fields of rational functions, $g : \mathrm{Y}' \to \mathrm{Y}$ a dominant morphism, so that K' is an extension of K. Let L be a finite extension of K non-ramified over Y, L_1 a composite extension (Bourbaki,* Alg.*, chap. VIII, § 8, def. 1) of L and K'. Then L_1 is an extension of K' non-ramified over Y'. Furthermore if $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ are affine, and if C is the integral closure of A in L, then the subring $\mathrm{C}_1 = \mathrm{A}[\mathrm{C}, \mathrm{A}']$ of L_1 is the integral closure of A' in L_1.*

Proof. Indeed, $L \otimes_K K'$ is composed as a direct product of finite separable extensions of K', and L_1 is one of these extensions; hence the integral closure of Y in L is the sum of preschemes, one of which is the integral closure of Y in L_1. The corollary therefore follows from (18.10.14). **IV-4** | 163

When for example A is the ring of integers of an algebraic number field K, K' and L algebraic extensions of K, there are classical examples showing that the relation $C_1 = A[C, A']$ does not hold when L is not ramified over A[ ]. □

(18.10.15). Suppose that $Y = \mathrm{Spec}(A)$ is affine, A being integral and integrally closed; let K be its field of fractions, L a finite separable extension of K, and we suppose that the integral closure C of A in L is a *projective* A-module of finite type (which will be the case for example if A is a Dedekind ring (Bourbaki, *Alg. comm.*, chap. VII, § 4, n° 10, prop. 22)). To say that L is non-ramified over A means that C is étale (18.10.10) and by virtue of (18.2.7), (ii)), this is equivalent to saying that the *discriminant* $d_{C/A}$ of Spec(C) over Spec(A) is *invertible* in A. In the particular case where C is a *free* A-module and $(x_i)_{1 \leqslant i \leqslant n}$ a basis of C over A, this means that $\det(\mathrm{Tr}_{C/A}(x_i x_j))$ is invertible in A.

Theorem (18.10.16). — *Let* Y *be an integral prescheme, η its generic point, $f : X \to Y$ a separated and locally of finite type morphism. Suppose that every irreducible component of* X *dominates* Y *and that the generic fibre $X_\eta = f^{-1}(\eta)$ is a finite prescheme over $K = k(\eta)$, hence $(X_\eta)_{\mathrm{red}}$ is the spectrum of a finite separable* K*-algebra* L*, where $L = \prod_i L_i$ is a product of finite extensions L_i of* K*. Set*

$$n = [L : K] = \sum_i [L_i : K], \quad n_s = \sum_i [L_i : K]_s \quad \textit{(sum of separable degrees of the } L_i\textit{)}.$$

Let y be a geometrically unibranch point of Y*, and denote by $n(y)$ the sum of the separable degrees over $k(y)$ of the residue fields of the isolated points of $f^{-1}(y)$. Then:*

(i) One has $n(y) \leqslant n_s \leqslant n$.

(ii) Suppose X *reduced and the point y normal in* Y*. For $n(y) = n$, it is necessary and sufficient that there exist a neighbourhood* U *of y in* Y *such that the restriction $f^{-1}(U) \to U$ of f is an étale and finite morphism.*

Proof. A) Reduction to the case where f is quasi-finite. — One knows (13.1.4) that the set Z of isolated points $x \in X$ of $f^{-1}(f(x))$ is open in X and by hypothesis Z contains $f^{-1}(\eta)$. Since $f^{-1}(\eta)$ is finite, Z is an increasing filtered union of quasi-compact open sets V_λ containing $f^{-1}(\eta)$ (dense in X by hypothesis). If $n_\lambda(y)$ is the sum of the separable degrees over $k(y)$ of the residue fields of the points of $V_\lambda \cap f^{-1}(y)$, we have $n(y) = \sup n_\lambda(y)$; to prove (i), it will therefore suffice to show that $n_\lambda(y) \leqslant n_s$. On the other hand, if $n_\lambda(y) = n$, suppose that we have shown that there exists a neighbourhood U of y in Y such that the restriction $V_\lambda \cap f^{-1}(U) \to U$ of f is an étale and finite morphism; since f is separated, the canonical injection $V_\lambda \cap f^{-1}(U) \to f^{-1}(U)$ is then also a finite morphism (**II**, 6.1.5), (v)), hence $V_\lambda \cap f^{-1}(U)$ is closed in $f^{-1}(U)$; but being dense everywhere in $f^{-1}(U)$, it is necessarily equal to it.

B) Reduction to the case where X *is reduced, f finite, and $Y = \mathrm{Spec}(\mathcal{O}_{Y,y})$.* — We note that the hypothesis on f implies that it is also satisfied for f_{red} and that the conclusion of (i) does not concern f_{red}. One can therefore replace f everywhere by f_{red} and thus suppose X reduced.

On the other hand, by virtue of (18.12.13) [1], one can (after having replaced Y by an open affine **IV-4** | 164 neighbourhood V of y and f by its restriction $f^{-1}(V) \to V$) factorise f as $X \xrightarrow{j} X' \xrightarrow{g} Y$, where g is *finite* and j an open immersion. Since X is reduced, one can replace X' by the reduced closed sub-prescheme with underlying space $\overline{j(X)}$ (**I**, 5.2.2), hence suppose X' reduced and such that every irreducible component of X' dominates Y. Moreover, $j(X)$ is an open dense set in X' and has only a finite number of irreducible components by virtue of the hypothesis; hence (**0**, 1.2.7) the irreducible components of X' are the closures of those of $j(X)$, and by construction the fibre $g^{-1}(\eta)$ identifies with $f^{-1}(\eta)$. If one supposes the theorem proved for f finite, one can apply it to g, and as $f^{-1}(y)$ identifies with a sub-prescheme of $g^{-1}(y)$, the relation (i) for g implies the same relation for f. Suppose on the other hand that $n(y) = n$; then, by virtue of what precedes, one can apply the conclusion of (ii) to the morphism g, and replacing Y as needed by a neighbourhood of y, one can suppose g étale; it follows in addition, on the one hand that $f^{-1}(y) = g^{-1}(y)$. As the morphism g is closed and $j(X)$ is open in X', there exists a neighbourhood U of y in Y such that $j(X) \cap g^{-1}(U) = j(f^{-1}(U))$, which proves that (ii) is also true for f.

We are thus reduced to the case where f is moreover a *finite* morphism. It is immediate that, to prove (i), one can restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$. Let us show that the same is true for proving (ii). Indeed, suppose we have shown that the condition $n(y) = n$ implies that f is flat and formally unramified at the points of $f^{-1}(y)$, and that Y is *integral*, we conclude from (18.4.13) that f is *étale at the points* of $f^{-1}(y)$, hence also in a neighbourhood V of $f^{-1}(y)$ in X; but since f is a finite morphism, hence closed, V contains an open set of the form $f^{-1}(\mathrm{U})$, where U is a neighbourhood of y in Y.

C) Reduction to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$ *and* $\mathscr{O}_{\mathrm{Y},y}$ *is strictly local.* — Set $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$, and let $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$ be the strict henselisation of A (18.8.7), which is a geometrically unibranch local integral ring (18.8.15). Let $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, and let y' and η' be the closed point and the generic point of Y'; y' is above y and since the morphism $g : \mathrm{Y}' \to \mathrm{Y}$ is flat (18.8.8), η' is above η (2.3.4). Set $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$; f' is finite and since g is flat, every irreducible component of X' dominates Y' (2.3.7). If $n'(y')$, n' and n'_s are the numbers defined for f' in the same way as $n(y)$, n and n_s for f, we have $n'_s = n_s$ and $n'(y') = n(y)$ (**I**, 6.4.8), and it therefore amounts to proving the assertion (i) for f or for f'. Moreover, if X is reduced and y a normal point of Y (hence A integrally closed (18.8.12)). On the other hand, it follows from the definition (18.8.7), from the remark (8.1.2), *a*)) and from the theorem of the double inductive limit, that A' is the inductive limit of rings A_α such that the morphisms $\mathrm{Spec}(\mathrm{A}_\alpha) \to \mathrm{Spec}(\mathrm{A}) = \mathrm{Y}$ are étale; consequently X' is the projective limit of the $\mathrm{X}_\alpha = \mathrm{X} \otimes_{\mathrm{A}} \mathrm{A}_\alpha$, which are étale over X, hence reduced since X is (17.5.7); by passage to the limit,
we deduce that X' is reduced (8.7.1). This being so, if $n(y) = n$, on has $n_s = n$ and consequently **IV-4** | 165
the residue fields at the points of X_η are necessarily separable over $k(\eta)$; the residue fields at the points of $\mathrm{X}'_{\eta'}$ are then algebraic and separable over $k(\eta')$ (4.6.1), in other words $n'_s = n' = n$. The morphism f' has therefore the same properties as f, and if one proves that f' is étale, it will follow by (17.7.1).

D) End of the proof. — Suppose therefore A strictly local. Since A is henselian and the morphism f finite, it follows from (18.5.11) that if x_j ($1 \leqslant j \leqslant m$) are the distinct points of $f^{-1}(y)$, X is the *sum* of open sub-preschemes $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j}) = \mathrm{X}_j$, which are the connected components of X. By hypothesis, each of them dominates Y, hence every irreducible component of X meets $f^{-1}(\eta)$, and so the number of points of $f^{-1}(\eta)$ is $\geqslant m$; *a fortiori* we have $n_s \geqslant m$. Suppose now the hypotheses of (ii) satisfied; the relations $n(y) = n_s = n$ imply first of all, since every irreducible component of X dominates Y, that each of the X_j is *irreducible*; moreover, as the X_j are *reduced*, they are *integral*; finally the relations $m = n_s = n$ imply that for the generic point ξ_j of X_j, $k(\xi_j)$ is a separable extension of $k(\eta)$ of degree 1, hence isomorphic to $k(\eta)$; in other words, the restriction $f|\mathrm{X}_j : \mathrm{X}_j \to \mathrm{Y}$ of f is *finite and birational*; moreover, X_j is *integral* and by hypothesis Y is *normal*; hence (8.12.10.1) $f|\mathrm{X}_j$ is an *isomorphism*, which proves that f is étale, and finishes the proof of (18.10.16). □

Remark (18.10.17). — The method of "étale localisation" used in the proof of (18.10.16) also allows one to improve the results of (15.5.1) by eliminating the Noetherian hypothesis. One has first of all the following result which generalises (15.5.2):

(18.10.17.1). Let $f : \mathrm{X} \to \mathrm{Y}$ be a separated and quasi-finite morphism, y a point of Y such that f is universally open at the points of $f^{-1}(y)$. Then, for every generisation y' of y, we have $n(y) \leqslant n(y')$.

Proof. One can replace Y by the reduced closed sub-prescheme $\overline{\{y'\}}$ and X by $f^{-1}(\mathrm{Z})$, hence suppose Y integral with generic point y'. Using (18.12.13) (as in (18.10.16), B)), one can suppose that X is an open dense subset of a prescheme X' and that f is the restriction to X of a finite morphism $g : \mathrm{X}' \to \mathrm{Y}$. Let us show that $g^{-1}(y') = f^{-1}(y')$. Indeed, let $i : g^{-1}(y') \to \mathrm{X}'$ be the canonical injection, which is a flat morphism since y' is the generic point of Y (**I**, 3.6.5); one can write $f^{-1}(y') = i^{-1}(\mathrm{X})$. On the other hand, the canonical injection $j : \mathrm{X} \to \mathrm{X}'$ is a morphism of finite type since the composite $g \circ j = f$ is of finite type and g is separated (1.5.4); hence X is a retrocompact open subset of X', and consequently pro-constructible in X' (**I**, 1.9.5), (v)). One concludes therefore from (2.3.10) that $i^{-1}(\overline{\mathrm{X}}) = \overline{i^{-1}(\mathrm{X})}$, in other words $f^{-1}(y')$ is dense in $g^{-1}(y')$; but as $g^{-1}(y')$ is discrete, we necessarily have $f^{-1}(y') = g^{-1}(y')$.

[5](1) The reader will verify that (18.10.16) is not used in the proof of (18.12.13). If f is supposed locally of finite presentation (resp. quasi-projective), one can replace (18.12.13) by (8.12.8) (resp. (8.12.6)).

We are thus reduced to proving that the sum of the numbers $[k(x) : k(y)]_s$ where x ranges over the points of $g^{-1}(y)$ where g is universally open, is at most equal to $n(y')$ (for the morphism g). **IV-4** | 166
Using then the fact that the property of being universally open at a point is preserved by base change, one shows, as in (18.10.16), C)), that one can reduce to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, with $\mathscr{O}_{\mathrm{Y},y}$ *strictly local*. Applying (18.5.11), *c*)), we see that if x_j ($1 \leqslant j \leqslant n$) are the points of $f^{-1}(y)$, X is the *sum* of n sub-preschemes $\mathrm{X}_j = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j})$; the hypothesis that g is open in each of the points x_j implies that the sub-prescheme X_j corresponding dominates Y (1.10.3); one concludes as in (18.10.16), D)).

We then deduce from (18.10.17.1) that *the conclusions of* (15.5.1) *are still valid when one no longer supposes* Y *locally Noetherian, but only that f is separated, quasi-finite, and of finite presentation*. □

Indeed, to prove assertion (i) of (15.5.1), one remarks that, since f is of finite presentation, the set E of points $z \in \mathrm{Y}$ such that $n(z) \geqslant n(y)$ is locally constructible (9.7.9); to show that y is interior to E, it suffices, by virtue of (1.10.1), to see that every generisation y' of y belongs to E, which is nothing other than (18.10.17.1).

To prove assertion (ii) of (15.5.1), one remarks that, since f is of finite presentation, using the method of (8.10.5), (xii)) applied following (8.1.2), *a*)), and one can therefore suppose that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, and the ring $\mathscr{O}_{\mathrm{Y},y}$ is *strictly local*. Applying (18.5.11), *c*)), we see then that if x_j ($1 \leqslant j \leqslant n$) are the points of $f^{-1}(y)$, X is the *sum* of n open sub-preschemes $\mathrm{X}_j = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j})$ which are *finite* over Y, and of an open prescheme X''. If one proves that $\mathrm{X}'' = \varnothing$, we will have shown that f is a finite morphism, and the conclusion follows from the fact that, since X is open in each of the points x_j, the restriction of f to X_j, being a finite morphism (**II**, 6.1.10), is *surjective*; the hypothesis that $z \mapsto n(z)$ is constant implies (17.4.1), hence one terminates as in (18.10.16), D)).

Finally, the proof of (iii) reproduces without change that given in (15.5.1).

(18.10.17.2). By analogous methods using strict henselisation, one can, in most of the results of §§ 14 and 15, eliminate the Noetherian hypotheses.

Lemma (18.10.18). — *Let* X, Y *be two preschemes, $f : \mathrm{X} \to \mathrm{Y}$ a birational morphism* ([1])*, x a point of* X*. For f to be an isomorphism local at the point x, it is necessary and sufficient that f be étale at the point x. For f to be an open immersion, it is necessary and sufficient that f be étale and separated.*

Proof. The conditions stated being trivially necessary, all that is needed is to see that they are sufficient. For the first assertion, the question is local on X and Y, hence one can suppose X and Y affine, hence f *separated*, and we are thus reduced to proving the second assertion. By virtue of (17.9.1), it is a question of proving that f is *radicial*. Let $y \in \mathrm{Y}$, and let us prove that $f^{-1}(y)$ is radicial on $k(y)$. The base change $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}) \to \mathrm{Y}$ not changing the fact that f is étale, separated, and birational, one can restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, with $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$. Let $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$, which is a local ring **IV-4** | 167
and an $\mathscr{O}_{\mathrm{Y},y}$-module faithful flat. Set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$, so that f' is étale and separated; moreover, as the morphism $\mathrm{Y}' \to \mathrm{Y}$ is flat, the reasoning of (6.15.4.1) shows that f' is also birational.

Suppose therefore $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, A strictly local, f separated, étale and birational, and let us show that $f^{-1}(y)$ can contain only a single point. Indeed, if $f^{-1}(y)$ contained two distinct points x_1, x_2, there would exist two Y-sections u_1, u_2 of f such that $u_1(y) = x_1$ and $u_2(y) = x_2$ (18.8.1); but since f is étale and separated, u_1 and u_2 would be isomorphisms of Y onto two connected components (open) of X (17.9.4) and there would then be two maximal points of X at least above every maximal point of Y, which contradicts the hypothesis that f is *birational*. □

Proposition (18.10.19). — *Let* Y *be a reduced prescheme whose set of irreducible components is locally finite, $f : \mathrm{X} \to \mathrm{Y}$ a separated morphism of finite type and formally unramified, g a* Y*-rational section of f (i.e. a* Y*-rational map from* Y *to* X*),* U *the domain of definition of g* (**I**, 7.2.1)*, and let* Z *be the closure of $g(\mathrm{U})$ in* X*. Then, for all points $y \in \mathrm{Y} - \mathrm{U}$ such that y is geometrically unibranch, $\mathrm{Z}_y = \mathrm{Z} \cap f^{-1}(y) = \varnothing$. In particular, if* Y *is geometrically unibranch, $g(\mathrm{U})$ is a subset of* X *that is both open and closed. If, for every closed irreducible* T *of* X*, $f(\mathrm{T})$ is closed in* Y*, g is defined at every geometrically unibranch point of* Y*.*

5([1]) We understand by this the notion defined in (6.15.4), but where we do not suppose X and Y reduced.

Proof. Since Y is reduced and f separated, it follows from (**I**, 7.2.2) that there exists a U-section u of $f^{-1}(\mathrm{U})$ belonging to the class g; by (17.4.1.2), moreover, f is formally unramified, hence u is an isomorphism of U onto the prescheme induced on the open set $u(\mathrm{U})$ of X. If one denotes also by Z the reduced sub-prescheme of X having Z for underlying space, $g(\mathrm{U}) = u(\mathrm{U})$, being reduced, is induced also by Z, and by construction $f' = f|\mathrm{Z}$ is a *birational* morphism of Z to Y, moreover formally unramified since f is (17.1.3). The question being local over Y at the point y, one can restrict (replacing Y by a neighbourhood of y) to the case where Y is *integral*; then $u(\mathrm{U})$ is irreducible, hence also Z, and by construction Z is also integral. Then (**I**, 8.2.7) since f' is dominant, the homomorphism $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{Z},x}$ is injective for all $x \in \mathrm{Z}_y$, and one concludes from (18.10.2), (iii)) that f' is an étale isomorphism local at the point x; but this would imply that the U-section u would be extendable to an open set distinct from U, contrary to the definition of the latter. We therefore necessarily have $\mathrm{Z}_y = \varnothing$, which establishes the first assertion; the second is an obvious consequence. Finally, under the last hypotheses of the statement, as Z is closed and irreducible, $f(\mathrm{Z})$ is closed in Y, hence equal to Y, i.e. $\mathrm{Z}_y \neq \varnothing$ for all $y \in \mathrm{Y}$, which finishes the proof. ☐

Remark (18.10.20). — Let Y be a locally Noetherian prescheme. One says that a morphism $f : \mathrm{X} \to \mathrm{Y}$ is *essentially proper* if it is locally of finite type and if, for every base change $\mathrm{Y}'' \to \mathrm{Y}$, where Y″ is *locally Noetherian*, the image by $f_{(\mathrm{Y}'')} : \mathrm{X}_{(\mathrm{Y}'')} \to \mathrm{Y}''$ of every closed irreducible subset of X″ is *closed*. To say that f is essentially proper for every Y-scheme $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A})$ where A is **IV-4** | 168
a discrete valuation ring, the canonical map (**II**, 7.3.2.2) relative to the Y′-prescheme $\mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, is bijective. The reasoning of (**II**, 7.3.8) proves that this is the same as saying that f (supposed locally of finite type) is *separated* and that for every base change $\mathrm{Y}'' \to \mathrm{Y}$, where Y″ is *locally Noetherian*, the image by $f_{(\mathrm{Y}'')} : \mathrm{X}_{(\mathrm{Y}'')} \to \mathrm{Y}''$ of every closed irreducible subset of X″ is *closed*. To say that f is *proper* (for Y locally Noetherian) means therefore that f is *essentially proper and of finite type* (**II**, 7.3.8); but one encounters important examples of essentially proper morphisms that are not of finite type (for example certain "Picard preschemes" or certain "preschemes of Néron-Severi" (chap. VI)). It evidently follows from (18.10.19) that if Y is *locally Noetherian, reduced and geometrically unibranch*, and if $f : \mathrm{X} \to \mathrm{Y}$ is *essentially proper and not ramified*, then every Y-rational section of f is *defined everywhere*.

Proposition (18.10.21). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a locally of finite presentation morphism, x a point of* X*, $y = f(x)$, and suppose the ring $\mathscr{O}_{\mathrm{Y},y}$ integral and geometrically unibranch. Let r be the minimum number of generators of the $\mathscr{O}_{\mathrm{X},x}$-module $(\Omega^1_{\mathrm{X}/\mathrm{Y}})_x$ (equal to the rank of the $k(x)$-vector space $(\Omega^1_{\mathrm{X}/\mathrm{Y}})_x \otimes_{\mathscr{O}_{\mathrm{X},x}} k(x)$). Then, for f to be smooth at the point x, it is necessary and sufficient that, if y' is the generic point of the unique irreducible component of* Y *containing y, there exists a generisation x' of x such that $f(x') = y'$ and* $\dim_{x'} f^{-1}(y') \geqslant r$.

Proof. If f is smooth at the point x, it is also so in an open neighbourhood U of x, hence in every generisation x' of x, and $\dim_{x'}(f^{-1}(f(x'))) = r$ in each of these points, by virtue of (17.10.2). As f is moreover flat in U, there exists a generisation of x above every generisation of y (2.3.4), which proves that the condition is necessary. To show that it is sufficient, let us first note that by (17.5.1), it suffices to prove that the morphism deduced from f by the base change $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}) \to \mathrm{Y}$ is smooth at the point x; one can therefore suppose that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$.

The question is local on X, hence (16.4.22) and (**0**, 5.2.2)) one can suppose that X is affine, and that there exist r sections s_i $(1 \leqslant i \leqslant r)$ of $\mathscr{O}_{\mathrm{X}}$ above X such that the sections $d_{\mathrm{X}/\mathrm{Y}}(s_i)$ generate $\Omega^1_{\mathrm{X}/\mathrm{Y}}$. Let

$$g : \mathrm{X} \longrightarrow \mathrm{Z} = \mathrm{Y}[\mathrm{T}_1, \dots, \mathrm{T}_r] = \mathbf{V}_{\mathrm{Y}}$$

be the Y-morphism corresponding to the homomorphism $\mathscr{O}^r_{\mathrm{Y}} \to f_*(\mathscr{O}_{\mathrm{X}})$ of $\mathscr{O}_{\mathrm{Y}}$-Modules defined by the s_i (considered as sections of $f_*(\mathscr{O}_{\mathrm{X}})$ above Y) (**II**, 1.2.7). We have the exact sequence

$$g^*(\Omega^1_{\mathrm{Z}/\mathrm{Y}}) \longrightarrow \Omega^1_{\mathrm{X}/\mathrm{Y}} \longrightarrow \Omega^1_{\mathrm{X}/\mathrm{Z}} \longrightarrow 0.$$

As the $d\mathrm{T}_i$ $(1 \leqslant i \leqslant r)$ generate $\Omega^1_{\mathrm{Z}/\mathrm{Y}}$, it follows from the definition of g that the homomorphism $g^*(\Omega^1_{\mathrm{Z}/\mathrm{Y}}) \to \Omega^1_{\mathrm{X}/\mathrm{Y}}$ is surjective, hence $\Omega^1_{\mathrm{X}/\mathrm{Z}} = 0$, and g is consequently *formally unramified* (17.4.1). We shall now see that g is *étale* at the point x, and taking into account (17.3.8), this will prove that f is smooth at the point x. As $\mathscr{O}_{\mathrm{Y},y}$ is supposed integral and geometrically unibranch, so is $\mathscr{O}_{\mathrm{Z},z}$ **IV-4** | 169 by (17.5.7) where $z = g(x)$, since the structural morphism $\mathrm{Z} \to \mathrm{Y}$ is smooth (17.3.8). By virtue of (18.10.1), it suffices to show that the homomorphism $\mathscr{O}_{\mathrm{Z},z} \to \mathscr{O}_{\mathrm{X},x}$ is injective; it amounts to the same thing to see that the morphism $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) \to \mathrm{Spec}(\mathscr{O}_{\mathrm{Z},z})$ corresponding is *dominant*, since $\mathscr{O}_{\mathrm{Z},z}$ is integral (**I**, 1.2.7). Let y' be the generic point of $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$ and let z' be the generic point of $\mathrm{Z} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}[\mathrm{T}_1, \dots, \mathrm{T}_r])$ which is integral; if $h : \mathrm{Z} \to \mathrm{Y}$ is the structural morphism, on has $h(z') = y'$. It suffices to prove that there exists in $f^{-1}(y')$ a generisation x' of x such that the image of x' by the morphism $g_{y'} : f^{-1}(y') \to h^{-1}(y')$ equals z'. Moreover z' is also the generic point of $h^{-1}(y') = \mathrm{Spec}(k(y')[\mathrm{T}_1, \dots, \mathrm{T}_r])$; as the morphism $g_{y'}$ is not ramified, hence quasi-finite (17.4.2), it follows from (5.4.1) that the restriction of $g_{y'}$ to every irreducible component of *dimension* $\geqslant r$ of $f^{-1}(y')$ is a dominant morphism. Now, there exists by hypothesis a component containing a generisation x' of x; *a fortiori* its generic point is a generisation of x, whence the conclusion. □

18.11. Application to Noetherian local algebras complete over a field The following lemma generalises (**0**, 21.9.1) and (**0**, 21.9.2):

Lemma (18.11.1). — *Let k be a field,* A *a Noetherian complete local ring which is a k-algebra,* K *the residue field of* A.

- *(i) If the* K*-vector space $\Omega^1_{\mathrm{K}/k}$ is of finite rank, the* A*-module $\widehat{\Omega}^1_{\mathrm{A}/k}$ is of finite type.*
- *(ii) If moreover* A *is a k-algebra formally smooth (for the adic topology), $\widehat{\Omega}^1_{\mathrm{A}/k}$ is a free* A*-module of rank equal to*

$$\dim(\mathrm{A}) + \mathrm{rg}_{\mathrm{K}}(\Omega^1_{\mathrm{K}/k}) - \mathrm{rg}_{\mathrm{K}}(\Upsilon_{\mathrm{K}/k}).$$

Furthermore, for every subfield k_0 of k such that $\Omega^1_{\mathrm{K}/k_0}$ is a k-vector space of finite rank, $\widehat{\Omega}^1_{\mathrm{A}/k_0}$ is a free A*-module of rank equal to*

$$\dim(\mathrm{A}) + \mathrm{rg}_{\mathrm{K}}(\Omega^1_{\mathrm{K}/k_0}) - \mathrm{rg}_{\mathrm{K}}(\Upsilon_{\mathrm{K}/k_0}) + \mathrm{rg}_{\mathrm{K}}(\Omega^1_{k/k_0}).$$

Proof. The assertion (i) is noted as a reminder, having been proved in (**0**, 20.7.15). To prove (ii), note that this assertion has been in fact established by the demonstration of (**0**, 21.9.2), making only intervene the finiteness of the transcendence degree of K over k (by way of the Cartier equality (**0**, 21.7.1)). □

Lemma (18.11.2). — *Let k be a field of characteristic $p > 0$,* C *a k-algebra, local Noetherian, complete, whose residue field is a finite extension of k, and which is integral; let n be its dimension,* L *its field of fractions. Then $\Omega^1_{\mathrm{L}/k}$ and $\Upsilon_{\mathrm{L}/k}$ are* L*-vector spaces of finite rank, and one has*

(18.11.2.1) $$\mathrm{rg}_{\mathrm{L}}(\Omega^1_{\mathrm{L}/k}) - \mathrm{rg}_{\mathrm{L}}(\Upsilon_{\mathrm{L}/k}) \geqslant n.$$

Suppose furthermore that $[k : k^p] < +\infty$. Then one has the equality **IV-4** | 170

(18.11.2.2) $$\mathrm{rg}_{\mathrm{L}}(\Omega^1_{\mathrm{L}/k}) - \mathrm{rg}_{\mathrm{L}}(\Upsilon_{\mathrm{L}/k}) = n.$$

Furthermore, $\Omega^1_{\mathrm{C}/k}$ is then isomorphic to $\widehat{\Omega}^1_{\mathrm{C}/k}$, hence (18.11.1) is a C*-module of finite type.*

Proof. We know in fact (**0**, 19.8.9) that there exists a sub-k-algebra C_0 of C isomorphic to a formal power series algebra in n variables $k[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ and such that C is a C_0-algebra finite. Hence, L is a *finite* extension of the field of fractions $\mathrm{L}_0 = k((\mathrm{T}_1, \dots, \mathrm{T}_n))$ of C_0. Now, one has the exact sequence of L-vector spaces obtained by applying (**0**, 20.6.17.1) to the three fields $k \subset \mathrm{L}_0 \subset \mathrm{L}$ (and taking into account (**0**, 20.4.13)):

$$0 \longrightarrow \Upsilon_{\mathrm{L}/k} \otimes_{\mathrm{L}_0} \mathrm{L} \longrightarrow \Upsilon_{\mathrm{L}/k} \longrightarrow \Upsilon_{\mathrm{L}/\mathrm{L}_0} \longrightarrow \Omega^1_{\mathrm{L}/k} \otimes_{\mathrm{L}_0} \mathrm{L} \longrightarrow \Omega^1_{\mathrm{L}/k} \longrightarrow \Omega^1_{\mathrm{L}/\mathrm{L}_0} \longrightarrow 0.$$

Since L is a finite extension of L_0, $\Omega^1_{\mathrm{L}/\mathrm{L}_0}$ and $\Upsilon_{\mathrm{L}/\mathrm{L}_0}$ are L-vector spaces of finite rank having the same rank, by the Cartier equality (**0**, 21.7.1); from the preceding exact sequence we therefore already deduce that

$$\mathrm{rg}_{\mathrm{L}}(\Omega^1_{\mathrm{L}/k}) - \mathrm{rg}_{\mathrm{L}}(\Upsilon_{\mathrm{L}/k}) = \mathrm{rg}_{\mathrm{L}_0}(\Omega^1_{\mathrm{L}_0/k}) - \mathrm{rg}_{\mathrm{L}_0}(\Upsilon_{\mathrm{L}_0/k}).$$

To prove (18.11.2.1) or (18.11.2.2), one can therefore restrict to proving these relations by replacing C and L by C_0 and L_0. As L_0 is separable over k (**0**, 21.7.1), we have $\Upsilon_{\mathrm{L}_0/k} = 0$ (**0**, 20.6.3); one deduces hence already that in all cases $\mathrm{rg}_{\mathrm{L}_0}(\Omega^1_{\mathrm{L}_0/k}) \geqslant n$, with equality when $[k : k^p] < +\infty$; this already proves (18.11.2.1) and (18.11.2.2).

Finally, to see that $\Omega^1_{C/k}$ is isomorphic to $\widehat{\Omega}^1_{C/k}$ when $[k : k^p] < +\infty$, it suffices to prove that $\Omega^1_{C/k}$ is a C-module of finite type, since C is a Noetherian complete local ring (**0**, 7.3.5), taking into account (**0**, 20.4.5). Since $\Omega^1_{C/k}$ is isomorphic to $\widehat{\Omega}^1_{C/k}$ and that $k[C_0^p] \subset k[C^p]$, the result follows that $\Omega^1_{C/k}$ is a C-module of finite type and C_0 a $k[C_0^p]$-module of finite type in the hypothesis $[k : k^p] < +\infty$. □

Lemma (18.11.3). — *Let* A *be a Noetherian local ring,* $\mathfrak{p}$ *a prime ideal of* A *such that* $A_\mathfrak{p}$ *is a Cohen-Macaulay ring. For every system of parameters* $(t_i)_{1\leqslant i\leqslant s}$ *of* $A_\mathfrak{p}$*, there exists a sequence* $(x_i)_{1\leqslant i\leqslant s}$ *of elements of* $\mathfrak{p}$ *such that the* x_i *form part of a system of parameters of* A *and that, for all* i*,* $x_i/1$ *is congruent to* t_i *modulo* **IV-4** | 172
the ideal $t_1A_\mathfrak{p} + \ldots + t_{i-1}A_\mathfrak{p}$*. In particular, if* $A_\mathfrak{p}$ *is regular and if the* t_i *form a regular system of parameters of* $A_\mathfrak{p}$*, so it is also the case of the* $x_i/1$*.*

Proof. This lemma itself will be a consequence of the following: □

Lemma (18.11.3.4). — *Let* A *be a Noetherian local ring,* $\mathfrak{p}$ *a prime ideal of* A*,* x *an element of* A*; then there exists an element* $x' \in A$ *such that* $x'/1 = x/1$ *in* $A_\mathfrak{p}$ *and that* x' *does not belong to any of the minimal prime ideals of* A *not contained in* $\mathfrak{p}$*.*

Proof. Let us first show how (18.11.3.4) implies (18.11.3). Multiplying the t_i if necessary by invertible elements of $A_\mathfrak{p}$, one can suppose that they are of the form $x_i/1$, with $x_i \in \mathfrak{p}$ for $1 \leqslant i \leqslant s$. We reason by induction on s. As t_1 forms part of a system of parameters of $A_\mathfrak{p}$ (**0**, 16.5.5), there is no $x'_1 \in A$ such that $x'_1/1 = t_1$ that can belong to a minimal prime ideal of $A_\mathfrak{p}$, hence it cannot belong to any of the minimal prime ideals of A contained in $\mathfrak{p}$. By virtue of (18.11.3.4), one can moreover choose x'_1 such that $x'_1/1 = t_1$ (hence $x'_1 \in \mathfrak{p}$) and that x'_1 does not belong to *any* minimal prime ideal of A, hence x'_1 forms part of a system of parameters of A (**0**, 16.3.4) and (16.3.7). One reasons then by induction, considering the ring $A' = A/x'_1A$ and the prime ideal $\mathfrak{p}' = \mathfrak{p}/x'_1A$; as $A'_{\mathfrak{p}'} \cong A_\mathfrak{p}/t_1A_\mathfrak{p}$, $A'_{\mathfrak{p}'}$ is also a Cohen-Macaulay ring (**0**, 16.5.5), and the images t'_i $(2 \leqslant i \leqslant s)$ of the t_i in $A'_{\mathfrak{p}'}$ form a system of parameters of this ring; it suffices to apply A' and to the t'_i $(2 \leqslant i \leqslant s)$ the induction hypothesis. The last assertion of (18.11.3) follows from the fact that in $A_\mathfrak{p}$, a system of parameters that generates the maximal ideal is a regular system of parameters (**0**, 17.1.1). □

Proof of (18.11.3.4). Let $(p'_k)_{1\leqslant k\leqslant r}$ be the suite of minimal prime ideals of A not contained in $\mathfrak{p}$, and let $(\mathfrak{p}_j)_{1\leqslant j\leqslant n}$ be the suite of other minimal prime ideals of A; one can suppose that $r \geqslant 1$. As $\mathfrak{p}'_k$ does not contain $\bigcap_{1\leqslant i\leqslant n}\mathfrak{p}_j$, there exists an element $y \in \bigcap_{1\leqslant i\leqslant n}\mathfrak{p}_j$ that does not belong to any of the $\mathfrak{p}'_k$ (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2); moreover $y/1$ belongs to all the minimal prime ideals of $A_\mathfrak{p}$, hence is nilpotent; if $(y/1)^h = 0$, the element $x' = x + y^h$ will satisfy the requirement, since $x'/1 = x/1$ by definition of the $\mathfrak{p}'_k$, on has $x' \notin \mathfrak{p}'_k$ for $1 \leqslant k \leqslant r$, and on the other hand, if $\mathfrak{p}_j$ is a prime ideal of A not contained in $\mathfrak{p}$ but such that $x \notin \mathfrak{p}_j$, we also have $x' \notin \mathfrak{p}_j$ since $y \in \mathfrak{p}_j$. □

Proof of (18.11.3)*, continued.* We return to the proof of (18.11.3) when $p = 1$. By virtue of (18.11.3.4), there exists a regular system of parameters $(t_i)_{1\leqslant i\leqslant s}$ of $A_\mathfrak{p}$ where the x_i $(1 \leqslant i \leqslant s)$ belong to $\mathfrak{p}$ and form part of a system of parameters $(x_j)_{1\leqslant j\leqslant n}$ of A. Set $A_0 = k[[T_1, \ldots, T_n]]$; as A is a k-algebra complete and the x_j belong to the maximal ideal $\mathfrak{m}$ of A, there exists a local k-homomorphism $u : A_0 \to A$ such that $u(T_j) = x_j$ for $1 \leqslant j \leqslant n$ (Bourbaki, *Alg. comm.*, chap. III, § 4, n° 5, prop. 6); if n is the ideal of A generated by the x_j $(1 \leqslant j \leqslant n)$, it is by hypothesis an ideal of definition of A; one deduces therefore from (**0**, 7.4.4) and (7.4.3) that u makes A an A_0-algebra *finite*.

Set $\mathfrak{p}_0 = \sum_{j=1}^s A_0T_j$, $B_0 = (A_0)_{\mathfrak{p}_0}$ and $B = A \otimes_{A_0} B_0$, so that $u_{\mathfrak{p}_0} : B_0 \to B$ makes B a B_0-algebra **IV-4** | 173
finite; moreover, if $\mathfrak{p}'$ is the ideal of B generated by $\mathfrak{p}$, on has $B_{\mathfrak{p}'} = A_\mathfrak{p}$; as $\mathfrak{p}'$ contains $\mathfrak{p}_0B_0$ by construction, it is above the maximal ideal $\mathfrak{p}_0B_0$ of B_0. Let us show that the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(B_0)$ is *not ramified* at the point $\mathfrak{p}'$: indeed this follows from the fact that $k(\mathfrak{p}')$ is a finite extension of the field $k(\mathfrak{p}_0)$ of characteristic 0, hence is *necessarily separable*, and that $B_{\mathfrak{p}'}/\mathfrak{p}_0B_{\mathfrak{p}'} = k(\mathfrak{p}')$ by virtue of the choice of the x_i for $1 \leqslant j \leqslant s$ (17.4.1).

Let us now note the following lemma:

Lemma (18.11.3.5). — *Let* k *be a field,* R, S *two Noetherian complete local* k*-algebras, such that, if* K *is the residue field of* S*,* $\Omega^1_{K/k}$ *is of finite rank over* K*; let* $u : R \to S$ *be a* k*-homomorphism making* S *an* R*-algebra finite; then* $\widehat{\Omega}^1_{S/R} = \Omega^1_{S/R}$*, and the sequence*

(18.11.3.6)
$$\widehat{\Omega}^1_{R/k} \otimes_R S \xrightarrow{v} \widehat{\Omega}^1_{S/k} \xrightarrow{w} \Omega^1_{S/R} \longrightarrow 0$$

(cf. (**0**, 20.7.17.3)*) is exact.*

Proof. The first assertion follows from the fact that $\Omega^1_{S/R}$ is an S-module of finite type (**0**, 20.4.7) and that w is surjective; but it follows from (18.11.1) that $\widehat{\Omega}^1_{S/k}$ is an S-module of finite type; on concludes that every sub-S-module of $\widehat{\Omega}^1_{S/k}$ is closed (**0**, 7.3.5), whence the lemma. ☐

Applying this lemma to the case where $R = A_0$, $S = A$, and noting that $\widehat{\Omega}^1_{A_0/k}$ is an A_0-module free of rank n (**0**, 21.9.3); hence on a $\widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A = \widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A$ and this is an A-module free of rank n. Since Spec(B) is not ramified on Spec(B_0) at the point $\mathfrak{p}'$, on a $(\Omega^1_{A/A_0})_\mathfrak{p} = \Omega^1_{B_{\mathfrak{p}'}/B_0} = 0$ (16.4.15) and (17.4.1). If one localises the exact sequence (18.11.3.6) (applied to A_0 and A) at $\mathfrak{p}$, one obtains therefore a surjective homomorphism $(\widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A)_\mathfrak{p} \to (\widehat{\Omega}^1_{A/k})_\mathfrak{p}$, and the conclusion follows that $\widehat{\Omega}^1_{A/k}$ is a B-module of rank n, and the ring being complete and integral by the Cohen theorem (**0**, 19.8.8), (ii)) and the fact that the field of fractions of A is of characteristic 0. Hence the $A_\mathfrak{p}$-module $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$ admits a system of n generators. But the $A_\mathfrak{p}$-module $\widehat{\Omega}^1_{A/k}$ is of *rank* n, by virtue of (**0**, 21.9.5), which is applicable to the ring being complete and integral by the Cohen theorem (**0**, 19.8.8), (ii)) and the fact that the field of fractions of A is of characteristic 0. Hence the $A_\mathfrak{p}$-module $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$ admits also a system of n generators, and since it is of rank n, this quotient is necessarily *free*, whence the conclusion. ☐

Lemma (18.11.4). — *Let k be a field, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k. Let k' be an arbitrary extension of k, and set $A' = A\widehat{\otimes}_k k'$. Then:*

(i) A' is a semi-local Noetherian ring, complete, composed as a direct product of complete local rings A'_i $(1 \leqslant i \leqslant r)$ which are A-modules faithfully flat, and whose residue fields are finite extensions of k'; if $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}A'$ is an ideal of definition of A'; and $\dim(A'_i) = \dim(A)$ for all i.

(ii) For all i, $\widehat{\Omega}^1_{A/k}$ is canonically isomorphic to $\widehat{\Omega}^1_{A/k} \otimes_A A'_i$.

Proof. The first assertions follow immediately from (7.5.5) and (**0**, 6.6.2); the fact that $\mathfrak{m}A'$ is an ideal of definition of A' also follows from (7.5.5), since K is the residue field of A, $K \otimes_k k'$ is a finite k'-algebra. Finally $A'_i/\mathfrak{m}A'_i$ is one of the Artinian local rings whose direct product is $A' \otimes_k k'$; hence as A'_i is an A-module flat, the equality of dimensions of A and A'_i follows from (6.1.2). **IV-4** | 174

(ii) As $\widehat{\Omega}^1_{A/k}$ is, by the hypothesis on K, an A-module of finite type (18.11.1), $\widehat{\Omega}^1_{A/k} \otimes_A A'$ is complete and identifies with the completed tensor product $\widehat{\Omega}^1_{A/k} \widehat{\otimes}_A A'$; by virtue of the associativity of the completed tensor product (**0**, 7.7.4), $\widehat{\Omega}^1_{A/k} \otimes_A A' = \widehat{\Omega}^1_{A/k} \otimes_A (A\widehat{\otimes}_k k')$ identifies with $\widehat{\Omega}^1_{A/k} \widehat{\otimes}_k k'$. But $\Omega^1_{A/k} \otimes_k k'$ identifies with $\Omega^1_{A''/k'}$, where $A'' = A \otimes_k k'$ (**0**, 20.5.5), and as A' is by definition the separated completion of A'', the separated completion $\widehat{\Omega}^1_{A''/k'}$ identifies by construction to $\widehat{\Omega}^1_{A'/k'}$ (**0**, 20.7.4); furthermore, if $\mathfrak{r}$ is the radical of A', the $\mathfrak{r}$-preadique topology on $\Omega^1_{A'/k'}$ identifies with the product of the $\mathfrak{m}'_i$-preadique topologies on the $\Omega^1_{A'_i/k'}$ (where $\mathfrak{m}'_i$ is the maximal ideal of A'_i); finally, it follows that the separated complete $\widehat{\Omega}^1_{A'/k'}$ for the $\mathfrak{r}$-preadique topology identifies with the product of the separated completions $\widehat{\Omega}^1_{A'_i/k'}$ for the $\mathfrak{m}'_i$-preadique topologies, and it suffices to use (**0**, 20.4.5).

The conclusion of (ii) follows now from the fact that $\Omega^1_{A'/k'}$ is a direct sum of the $\Omega^1_{A'_i/k'}$ (**0**, 20.4.13), and if $\mathfrak{r}$ is the radical of A', the $\mathfrak{r}$-preadique topology on $\Omega^1_{A'/k'}$ identifies with the product of the $\mathfrak{m}'_i$-preadique topologies on the $\Omega^1_{A'_i/k'}$ (where $\mathfrak{m}'_i$ is the maximal ideal of A'_i); the separated complete $\widehat{\Omega}^1_{A'/k'}$ for the $\mathfrak{r}$-preadique topology identifies therefore with the product of the separated completions $\widehat{\Omega}^1_{A'_i/k'}$ for the $\mathfrak{m}'_i$-preadique topologies, hence one has a canonical isomorphism

$$\widehat{\Omega}^1_{A'/k'} \xrightarrow{\sim} \widehat{\Omega}^1_{A/k} \otimes_A A'.$$

☐

Proposition (18.11.5). — *Let k be a field, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of A distinct from the maximal ideal $\mathfrak{m}$, such that there exists a* *unique minimal prime ideal* $\mathfrak{q} \subset \mathfrak{p}$ *for which* $\dim(A/\mathfrak{q}) = \dim(A)$ *(which will in particular hold when* A *is equidimensional), m an integer* $\geqslant 0$*. The following conditions are equivalent:*

a) The $A_\mathfrak{p}$*-module* $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$ *admits a system of m generators.*
b) There exists a local k-homomorphism $u : B \to A$*, where* $B = k[[T_1, \dots, T_m]]$*, making* A *a* B*-algebra finite, and such that the corresponding morphism* $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ *is étale at the point* $\mathfrak{p}$*.*

Proof. To prove that *b*) implies *a*), let us note that by virtue of the lemma (18.11.3.5), on has the exact sequence

$$\widehat{\Omega}^1_{B/k} \otimes_B A \longrightarrow \widehat{\Omega}^1_{A/k} \longrightarrow \Omega^1_{A/B} \longrightarrow 0$$

since A is a B-algebra finite; localising at $\mathfrak{p}$ and noting that by hypothesis $(\Omega^1_{A/B})_\mathfrak{p} = 0$ (17.4.1), one obtains a surjective homomorphism $(\widehat{\Omega}^1_{B/k} \otimes_B A)_\mathfrak{p} \to (\widehat{\Omega}^1_{A/k})_\mathfrak{p}$, and the conclusion follows that **IV-4 | 175**
$\widehat{\Omega}^1_{B/k}$ is a B-module of rank m (**0**, 21.9.3).

To prove that *a*) implies *b*), let us first prove the following lemmas. □

Lemma (18.11.5.1). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k,* $\mathfrak{p}$ *a prime ideal of* A*,* $\mathfrak{q}$ *a minimal prime ideal of* A *contained in* $\mathfrak{p}$*. Then the minimum number of generators of the* $A_\mathfrak{p}$*-module* $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$ *is at least equal to* $\dim(A/\mathfrak{q})$*.*

Proof. Set $n = \dim(A/\mathfrak{q})$, and let m be the minimum number of generators of the $A_\mathfrak{p}$-module $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$, which is equal to $\mathrm{rg}_{k(\mathfrak{p})}((\widehat{\Omega}^1_{A/k})_\mathfrak{p} \otimes_{A_\mathfrak{p}} k(\mathfrak{p}))$ (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. 2 of prop. 4). Note that $(\widehat{\Omega}^1_{A/k})_\mathfrak{q} = (\widehat{\Omega}^1_{A/k})_\mathfrak{p} \otimes_{A_\mathfrak{p}} A_\mathfrak{q}$, hence the minimum number of generators of the $A_\mathfrak{q}$-module $(\widehat{\Omega}^1_{A/k})_\mathfrak{q}$ is *at most* equal to m. It suffices therefore to consider the case where $\mathfrak{p} = \mathfrak{q}$ is minimal. In second place, let us show one can restrict to the case where k is *algebraically closed*. Indeed, let k' be an algebraic closure of k, and set $A' = A\widehat{\otimes}_k k'$, which is composed as a direct product of k'-local algebras A'_i $(1 \leqslant i \leqslant r)$, whose residue fields are isomorphic to k', and which are A-modules faithfully flat (18.11.4). Applying (2.3.4) and (6.1.1), one sees that in a given A'_i, there exists a minimal prime ideal $\mathfrak{q}'_i$ above $\mathfrak{q}$ such that $n' = \dim(A'_i/\mathfrak{q}'_i) \geqslant \dim(A/\mathfrak{q}) = n$. On the other hand by (18.11.4), the minimum m' of generators of the $(A'_i)_{\mathfrak{q}'_i}$-module $(\widehat{\Omega}^1_{A'_i/k'})_{\mathfrak{q}'_i}$ is *at most* equal to m; whence our assertion.

Suppose therefore k algebraically closed and $\mathfrak{q} = 0$. In the case where $\mathfrak{q} = 0$, one can moreover restrict to the case where A is integral and K separable over the algebraically closed field k, hence A is a geometrically regular algebra (6.7.6); one can therefore apply (18.11.3) for $\mathfrak{p} = 0$ since $k = k^p$ and in this case $m = n$. □

Lemma (18.11.5.2). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k,* $\mathfrak{m}$ *its maximal ideal,* $\mathfrak{p} \subset \mathfrak{m}$ *a prime ideal of* A*. Let m be the minimum number of generators of the* $A_\mathfrak{p}$*-module* $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$*, and let* $(x_i)_{1 \leqslant i \leqslant m}$ *be a family of elements of* $\mathfrak{m}$ *not belonging to* $\mathfrak{p}$*. Then there exist m invertible elements* u_i $(1 \leqslant i \leqslant m)$ *of* A *such that if* $y_i = u_i x_i$*, the canonical images of the* dy_i *in* $(\widehat{\Omega}^1_{A/k})_\mathfrak{p}$ *generate this* $A_\mathfrak{p}$*-module.*

Proof. For all $x \in A$, denote by $\delta(x)$ the canonical image of dx $(= d_{A/k}x)$ in $(\widehat{\Omega}^1_{A/k})_\mathfrak{p} \otimes_{A_\mathfrak{p}} k(\mathfrak{p}) = \widehat{\Omega}^1_{A/k} \otimes_A k(\mathfrak{p})$; as this is a $k(\mathfrak{p})$-vector space of dimension m by hypothesis, it suffices (by virtue of the Nakayama lemma) to prove that one can determine the u_i such that the $\delta(u_i x_i)$ form a free system. We reason by induction and suppose for an integer $r < m$, we have determined the u_i $(1 \leqslant i \leqslant r)$ such that the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$ are linearly independent over $k(\mathfrak{p})$; if $\delta(x_{r+1})$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$, one takes $u_{r+1} = 1$ to continue the induction. In the contrary case, note that for $u \in A$, $\delta(u x_{r+1})$ is the canonical image of $(du)x_{r+1} + u(dx_{r+1})$; as $u(dx_{r+1})$ has a canonical image which is a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$, there **IV-4 | 176**
exists therefore $z \in A$ such that $\delta(z)$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$; if $z \notin \mathfrak{m}$, one takes $u = z$; otherwise, $1 + z$ is invertible and $d(1+z) = d(z)$ since $(du)x_{r+1} = \delta(z)$; one takes then $u = 1 + z$, which finishes the proof of (18.11.5.2). □

Proof of (18.11.5)*, continued.* We now return to the proof of the implication *a*) ⇒ *b*) in (18.11.5). Let us first note that since $\mathfrak{p} \neq \mathfrak{m}$, there exists a system of parameters $(x_i)_{1 \leqslant i \leqslant n}$ of A (with $n = \dim(A)$) such that $x_i \notin \mathfrak{p}$ for all i, without which $A/\mathfrak{p}$ would be of finite length, and $\mathfrak{p}$ would be maximal, contrary to the hypothesis. But if for example $x_1 \notin \mathfrak{p}$, it suffices, for each index i such that $x_i \in \mathfrak{p}$, to replace x_i by $x_1 + x_i$, to have a system of parameters none of whose elements belong to $\mathfrak{p}$. The relation $\dim(\mathrm{A}/\mathfrak{q}) = n$ implies, by virtue of (18.11.5.1), that $m \geqslant n$; one can therefore consider a family $(x_i)_{1\leqslant i\leqslant m}$ of elements $x_i \notin \mathfrak{p}$, whose n first form a system of parameters of A. Multiplying furthermore the x_i by invertible elements u_i of A, one can, by (18.11.5.2), suppose that the images of dx_i $(1 \leqslant i \leqslant m)$ in $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ generate this $\mathrm{A}_{\mathfrak{p}}$-module, and the multiplication by the u_i has not altered the fact that the x_i for $1 \leqslant i \leqslant n$ form a system of parameters. We shall consider the local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$ such that $u(\mathrm{T}_i) = x_i$ for $1 \leqslant i \leqslant m$ (Bourbaki, *Alg. comm.*, chap. III, § 4, n° 5, prop. 6); as the x_i generate an ideal of definition of A, it follows from (**0**, 7.4.4) and (7.4.3) that u makes A a B-algebra *finite*.

But the $d_{\mathrm{A}}x_i$ are the canonical images by v of the elements $d_{\mathrm{B}}\mathrm{T}_i \otimes 1$ (**0**, 20.5.2.6). If one localises the exact sequence preceding at $\mathfrak{p}$, we see therefore that $v_{\mathfrak{p}} : (\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}} \to (\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is *surjective*, and consequently $0 = (\Omega^1_{\mathrm{A}/\mathrm{B}})_{\mathfrak{p}} = \Omega^1_{\mathrm{A}_{\mathfrak{p}}/\mathrm{B}_{\mathfrak{r}}}$ (where $\mathfrak{r}$ is the inverse image of $\mathfrak{p}$ in A, cf. (16.4.15)), thus the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at $\mathfrak{p}$ by (17.4.1), and the property *b*) of (18.11.5) follows. □

Remark (18.11.6). — In the statement of (18.11.5), one can drop the hypothesis $\mathfrak{p} \neq \mathfrak{m}$. Indeed, for every local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$, where $\mathrm{B} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_r]]$ (r any integer) making A a B-algebra finite, $\mathfrak{m}$ is the only point of Spec(A) above the maximal ideal $\mathfrak{n}$ of B; the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ cannot be unramified at $\mathfrak{m}$ unless $\mathrm{A}/\mathfrak{n}\mathrm{A}$ is a field, an extension separable of k (17.4.1), which implies that the residue field K of A is a separable extension of k. If this condition is not satisfied, the conclusion of (18.11.5) can never be satisfied for $\mathfrak{p} = \mathfrak{m}$, whatever the integer m.

Corollary (18.11.7). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, equidimensional, whose residue field is a finite extension of k,* $\mathfrak{p}$ *a prime ideal of* A. *Then the minimum number of generators of the* $\mathrm{A}_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ *is at least equal to* dim(A).

Proof. It suffices to set $\mathfrak{p} = 0$ in (18.11.7). □

Corollary (18.11.8). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, integral, whose residue field is a finite extension of k. If* K *is the field of fractions of* A, *one has* $\mathrm{rg}_{\mathrm{K}}((\widehat{\Omega}^1_{\mathrm{A}/k}) \otimes_{\mathrm{A}} \mathrm{K}) \geqslant \dim(\mathrm{A})$.

Proof. It suffices to set $\mathfrak{p} = 0$ in (18.11.7). □

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Corollary (18.11.9). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, of dimension n, whose residue field is a finite extension of k. Let $\mathfrak{p}$ be a prime ideal of* A *distinct from the maximal ideal, and containing a minimal prime ideal $\mathfrak{q}$ such that* $\dim(\mathrm{A}/\mathfrak{q}) = n$. *Suppose that the* $\mathrm{A}_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ *admits n generators. Then:*

- *(i) The* $\mathrm{A}_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ *is free of rank n.*
- *(ii) There exists a local k-homomorphism* $u : \mathrm{B} \to \mathrm{A}$, *where* $\mathrm{B} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_n]]$, *making* A *a* B*-algebra finite, and such that the corresponding morphism* $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ *is étale at the point* $\mathfrak{p}$.
- *(iii) The k-algebra* $\mathrm{A}_{\mathfrak{p}}$ *is geometrically regular over k.*

Proof. Let us prove first of all (ii); by virtue of (18.11.5), there exists a local k-homomorphism $u : \mathrm{B} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_n]] \to \mathrm{A}$ making A a B-algebra finite, and such that the corresponding morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at the point $\mathfrak{p}$; set $\mathfrak{r} = u^{-1}(\mathfrak{p})$. The hypothesis on $\mathfrak{q} \subset \mathfrak{p}$ and the fact that A is a quotient ring of a regular ring (**0**, 19.8.8) imply that $\dim(\mathrm{A}_{\mathfrak{p}}) = n - \dim(\mathrm{A}/\mathfrak{p})$ (**0**, 16.5.12). Similarly $\dim(\mathrm{B}_{\mathfrak{r}}) = n - \dim(\mathrm{B}/\mathfrak{r})$. Finally, since the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at the point $\mathfrak{p}$, the fibre of this morphism at the point $\mathfrak{r}$ is of dimension 0, hence (**0**, 16.3.9) $\dim(\mathrm{A}/\mathfrak{p}) \leqslant \dim(\mathrm{B}/\mathfrak{r})$ and consequently $\dim(\mathrm{B}_{\mathfrak{r}}) \leqslant \dim(\mathrm{A}_{\mathfrak{p}})$. We conclude therefore from (18.10.1) that the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is étale at $\mathfrak{p}$.

The assertion (iii) follows from the fact that $\mathrm{A}_{\mathfrak{p}}$ is formally smooth and that $\mathrm{B}_{\mathfrak{r}}$ is formally smooth on k for their preadique topologies (**0**, 19.3.4) and (19.3.5)), hence $\mathrm{A}_{\mathfrak{p}}$ is formally smooth on k for its preadique topology, and consequently geometrically regular over k (**0**, 22.5.8).

Let us finally prove (i). Let k' be an algebraically closed extension of k, and let us consider the k'-algebra semi-local $A' = A\widehat{\otimes}_k k'$. As a B-module of finite type via u, A is considered as a B-algebra finite; since the morphism $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ is finite and étale at every point $\mathfrak{p}'$ above $\mathfrak{p}$ (17.3.3); furthermore, A' is composed as a direct product of local rings A'_i of dimension n (18.11.4) and $\mathfrak{p}'$ identifies with a prime ideal $\mathfrak{p}'_i$ of one of the A'_i. The same reasoning as above proves that $(A'_i)_{\mathfrak{p}'_i}$ is geometrically regular over k', and as k' is perfect, the fact d) implies b), hence by faithful flatness (18.11.4) and (2.5.2), $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module free of finite type. But $(\widehat{\Omega}^1_{A'_i/k'})_{\mathfrak{p}'_i} = (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} (A'_i)_{\mathfrak{p}'_i}$ (18.11.4) and $(A'_i)_{\mathfrak{p}'_i}$ is a faithfully flat $A_{\mathfrak{p}}$-module; one sees that $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module *free* of finite type (2.5.2); this finishes the proof. □

Theorem (18.11.10). — *Let k be a field of characteristic exponent p, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of A distinct from the maximal ideal. The following conditions are equivalent:* IV-4 | 178

- *a) For every extension k' of k, and every prime ideal $\mathfrak{p}'$ of $A' = A\widehat{\otimes}_k k'$ above $\mathfrak{p}$, $A_{\mathfrak{p}'}$ is a regular ring.*
- *a′) There exists a perfect extension k' of k and a prime ideal $\mathfrak{p}'$ of $A' = A\widehat{\otimes}_k k'$ above $\mathfrak{p}$, such that $A_{\mathfrak{p}'}$ is regular.*
- *b) Let n be the largest of the dimensions of the irreducible components of $\operatorname{Spec}(A)$ to which $\mathfrak{p}$ belongs; then $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module free of rank n.*

Suppose furthermore that $n = \dim(A)$ (which will be the case if A is equidimensional); then the preceding conditions are also equivalent to:

- *c) There exists a local k-homomorphism $u : B \to A$, where $B = k[[T_1, \dots, T_n]]$, making A a B-algebra finite, and such that the corresponding morphism $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ is étale at the point $\mathfrak{p}$.*
- *d) The ring $A_{\mathfrak{p}}$ is geometrically regular over k.*

Furthermore, if $[k : k^p] < +\infty$, the condition d) is equivalent to a′), b), and c).

Proof. The fact that d) implies b) when $[k : k^p] < +\infty$ is nothing other than (18.11.3). It is clear that a) implies a'). Let us show c) implies a) when $\dim(A) = n$. One says that A is an A-module flat (18.11.4) and $\dim(A') = \dim(A) = n$; it follows from (2.3.4) and (6.1.1) that there exists a minimal prime ideal $\mathfrak{q}'$ of A' contained in $\mathfrak{p}'$, above $\mathfrak{q}$ and such that $\dim(A'/\mathfrak{q}') = \dim(A/\mathfrak{q}) = n$. Furthermore, by (18.11.4), $(\widehat{\Omega}^1_{A'/k'})_{\mathfrak{p}'} = (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A'_{\mathfrak{p}'}$; as k' is perfect, the hypothesis implies that it is geometrically regular over k', one can apply to A' and $\mathfrak{p}'$ the fact d) implies b), hence $(\widehat{\Omega}^1_{A'/k'})_{\mathfrak{p}'}$ is an $A_{\mathfrak{p}'}$-module free of rank n; by faithful flatness (18.11.4) and (2.5.2), $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module free of rank n. This shows the equivalence of a), a'), b), and c) when $\dim(A) = n$. It remains to prove that a), a'), and b) are still equivalent in the general case. Let $\mathfrak{J}$ be the kernel of A, the noyau of the canonical homomorphism $A \to A_{\mathfrak{p}}$, and set $A_1 = A/\mathfrak{J}$; we necessarily have $\mathfrak{J} \subset \mathfrak{p}$, and if we set $\mathfrak{p}_1 = \mathfrak{p}/\mathfrak{J}$, the canonical homomorphism $A_{\mathfrak{p}} \to (A_1)_{\mathfrak{p}_1}$ is *bijective*; it follows from (**I**, 6.5.4) that the canonical injection $\operatorname{Spec}(A_1) \to \operatorname{Spec}(A)$ is an *isomorphism local* at the point $\mathfrak{p}_1$, and one has $\mathfrak{J}_{\mathfrak{p}} = 0$. One sees as in (18.11.3) that the sequence is exact, and that one concludes that the condition a) (resp. a'))) for the ring A and the prime ideal $\mathfrak{p}$ is *equivalent* to the condition a) (resp. a'))) for the ring A_1 and the prime ideal $\mathfrak{p}_1$. Now, all the minimal prime ideals of A contained in $\mathfrak{p}$ contain $\mathfrak{J}$ since $\operatorname{Spec}(A_1) \to \operatorname{Spec}(A)$ is an isomorphism local at $\mathfrak{p}_1$; on the other hand the $\operatorname{Ass}_A(A/\mathfrak{J})$ are the prime ideals of $\operatorname{Ass}(A)$ which are contained in $\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 2, prop. 6); hence all the minimal prime ideals of A_1 are contained in $\mathfrak{p}_1$, and consequently $\dim(A_1) = n$. It suffices then to apply to A_1 and $\mathfrak{p}_1$ which has been proved above. □

Remark (18.11.11). — IV-4 | 179

(i) The equivalence of conditions d) and b) in (18.11.10) is no longer valid when one no longer supposes that $[k : k^p] < +\infty$. Indeed, by the example of (**0**, 22.7.7), (ii)), the ring $B = A/\mathfrak{q}$ is integral and of dimension 1; on the other hand, the sequence

$$(\mathfrak{q}/\mathfrak{q}^2) \otimes_A L \xrightarrow{j} \widehat{\Omega}^1_{A/k} \otimes_A L \longrightarrow \widehat{\Omega}^1_{B/k} \otimes_B L \longrightarrow 0$$

is exact (same demonstration as in (18.11.3)), and we know that *j is not injective*, hence $j = 0$ since $(\mathfrak{q}/\mathfrak{q}^2) \otimes_A L$ is of rank 1. We conclude that $\operatorname{rg}_L(\widehat{\Omega}^1_{B/k} \otimes_B L) = 2$; as $B_{\mathfrak{p}}$ is a k-algebra geometrically regular, one sees that the condition d) does not imply here the condition b).

(ii) With the notations of (18.11.10) and supposing that $n = \dim(\mathrm{A})$ and $(x_i)_{1\leqslant i\leqslant n}$ a system of parameters of A not belonging to $\mathfrak{p}$; one deduces a local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$ such that $u(\mathrm{T}_i) = x_i$ for $1 \leqslant i \leqslant n$, making A a B-algebra finite. For the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ to be *étale* at the point $\mathfrak{p}$, it is necessary and sufficient that the images of $d_{\mathrm{A}}x_i$ in $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ form a *system of generators* of this $\mathrm{A}_{\mathfrak{p}}$-module. Indeed, as one has seen in the proof of (18.11.5) that if the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at $\mathfrak{p}$, the canonical homomorphism $(\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}} \to (\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is surjective and the images of the elements $d_{\mathrm{B}}\mathrm{T}_i \otimes 1$, which generate $(\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}}$, are the $d_{\mathrm{A}}x_i$; whence the *necessity* of the condition. Conversely, the same reasoning shows that if this condition is satisfied, the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at $\mathfrak{p}$, and the reasoning of (18.11.9) proves in fact that this morphism is étale at $\mathfrak{p}$.

Corollary (18.11.12). — *Let k be a field of characteristic exponent p such that $[k : k^p] < +\infty$, A a k-algebra, local Noetherian and complete and integral, which is not a field, and whose residue field is a finite extension of k. Then there exists a finite radicial extension k' of k such that, setting $\mathrm{A}' = \mathrm{A} \otimes_k k'$, the k'-algebra $\mathrm{A}'_{\mathrm{red}}$ and the prime ideal $\mathfrak{o}$ of this algebra satisfy the equivalent conditions a), a′), b), c), and d) of (18.11.10). In particular, if $n = \dim(\mathrm{A}) = \dim(\mathrm{A}')$, there exists a k'-homomorphism local $\mathrm{B}' = k'[[\mathrm{T}_1, \dots, \mathrm{T}_n]] \to \mathrm{A}'_{\mathrm{red}}$* **IV-4** | 180 *making $\mathrm{A}'_{\mathrm{red}}$ a B'-algebra finite, and such that the field of fractions K' of $\mathrm{A}'_{\mathrm{red}}$ is a finite separable extension of the field of fractions $k'((\mathrm{T}_1, \dots, \mathrm{T}_n))$ of B'.*

Proof. The morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ being complete, radicial, and finite, A' is a local ring, complete and its nilradical is the only prime ideal above $\mathfrak{o}$ of A; by flatness, A' identifies with a sub-ring of $\mathrm{K} \otimes_k k'$, which is also the total ring of fractions of A'; on has therefore $\mathrm{K}' = (\mathrm{K} \otimes_k k')_{\mathrm{red}}$. It is a question of proving, by the equivalence $d) \Leftrightarrow c)$, that there exists a finite radicial extension k' of k such that $(\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is a separable extension of k'. Or one knows (**0**, 19.8.9) that, under all the hypotheses made, there exists a sub-k-algebra $\mathrm{C} = k[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ of A such that A is a finite C-algebra; hence, K is a *finite* extension of the field of fractions $\mathrm{K}_1 = k((\mathrm{T}_1, \dots, \mathrm{T}_n))$ of C, which is *separable* on k (**0**, 21.9.6.4). If p is the characteristic exponent of k, one can therefore write $\mathrm{K} \otimes_k k^{p^{-\infty}} = \mathrm{K} \otimes_{k_1} (\mathrm{K}_1 \otimes_k k^{p^{-\infty}})$ and $\mathrm{K}_1 \otimes_k k^{p^{-\infty}}$ is a *field*, a radicial extension of K_1; on concludes that $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is a finite algebra over the field $\mathrm{K}_1 \otimes_k k^{p^{-\infty}}$, hence is *Artinian*. The conclusion follows therefore from the following more general lemma: □

Lemma (18.11.12.1). — *Let k be a field of characteristic $p > 0$, K an extension of k. The following conditions are equivalent:*

a) The ring $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is Artinian.
b) There exists an extension k' of k of finite type such that $(\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is a k'-algebra separable.
c) There exists a finite radicial extension k' of k such that $(\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is a separable extension of k'.

Proof. Let us first show that $c)$ implies $a)$. The ring $\mathrm{A} = \mathrm{K} \otimes_k k'$ is then a local Artinian ring, and if $\mathfrak{N}$ is nilradical, the residue field $\mathrm{L} = \mathrm{A}/\mathfrak{N}$ is separable over k'; hence $\mathrm{L} \otimes_{k'} k^{p^{-\infty}}$ is separable over k', and as it is equal to $(\mathrm{A} \otimes_{k'} k^{p^{-\infty}})/(\mathfrak{N} \otimes_{k'} k^{p^{-\infty}})$, one sees that $\mathfrak{N} \otimes_{k'} k^{p^{-\infty}}$ is the nilradical of $\mathrm{K} \otimes_k k^{p^{-\infty}}$; as $\mathfrak{N}$ is a nilpotent ideal of finite type, the nilradical of $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is of finite type, hence the ring $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is Artinian.

Conversely, let us prove that $a)$ implies $c)$. Let $\mathfrak{N}$ be the nilradical of the local Artinian ring $\mathrm{B} = \mathrm{K} \otimes_k k^{p^{-\infty}}$, which is by hypothesis generated by a finite number d of elements of the form $y_i = \sum_j x_{ij} \otimes \xi_{ij}$, $x_{ij} \in \mathrm{K}$, $\xi_{ij} \in k^{p^{-\infty}}$. Let k' be the finite radicial extension of k generated by the ξ_{ij}, $\mathfrak{N}_0$ the ideal of $\mathrm{B}_0 = \mathrm{K} \otimes_k k'$ generated by the y_i; it is clear that $\mathfrak{N}_0 \otimes_{k'} k^{p^{-\infty}} = \mathfrak{N}$, and hence $\mathfrak{N} \cap \mathrm{B}_0 = \mathfrak{N}_0$, and so it is equal to $\mathrm{B}_0/\mathfrak{N}_0 = (\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is separable over k (4.6.1).

It is clear that $c)$ implies $b)$. Conversely, suppose $b)$ is satisfied, and let us note that there exists a separable extension k_1 of k such that k' is a finite radicial extension of k_1; set $\mathrm{K}_1 = \mathrm{K} \otimes_k k_1$, which is a field. Applying the equivalence of $a)$ and $c)$ to the extension K_1, on sees that $\mathrm{K}_1 \otimes_{k_1} k_1^{p^{-\infty}}$ is **IV-4** | 181 also Artinian (Bourbaki, *Alg. comm.*, chap. I, § 3, nº 5, cor. of prop. 8); we have thus shown that $b)$ implies $a)$, which finishes the proof. □

18.12. Applications of étale localisation to quasi-finite morphisms The results of this section were communicated by P. Deligne.

Theorem (18.12.1). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a locally of finite type morphism, x a point of* X*, $y = f(x)$. Suppose that x is an isolated point of the space $f^{-1}(y)$. Then there exist an étale morphism $\mathrm{Y}' \to \mathrm{Y}$, a point x' of $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$ above x and a neighbourhood V' of x' in X' such that, if $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$, $f'|\mathrm{V}'$ is a finite morphism. Furthermore V' is both open and closed, and X' is therefore the sum of two sub-preschemes induced on open subsets of X', of which one is finite on Y' and contains x'.*

Proof. The last assertion follows from the fact that, if f is separated, it is the same for f'; hence, if $j' : \mathrm{V}' \to \mathrm{X}'$ is the injection canonical, and the fact that f' is finite implies that it is the same for j' (**II**, 6.1.5), and $\mathrm{V}' = j'(\mathrm{V}')$ is closed in X' (**II**, 6.1.10).

The question being local on X, one can suppose $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ and $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$ affine, B being an A-algebra of finite type, and one can then suppose $f^{-1}(y) = \{x\}$. Hence the question is local on Y, hence of the form $\mathrm{C}/\mathfrak{J}$, where $\mathrm{C} = \mathrm{A}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ and $\mathfrak{J}$ is an ideal of type finite of E; we then have $\mathrm{A}' = \mathrm{E}'/\mathfrak{J}\mathrm{E}'$, where $\mathrm{E}' = \mathrm{C}'[\mathrm{T}_1, \ldots, \mathrm{T}_r]$. Let $(\mathfrak{J}_\lambda)$ be the family of ideals of type fine of C contained in $\mathfrak{J}$, and denote the corresponding $\mathrm{X}_\lambda = \mathrm{Spec}(\mathrm{C}/\mathfrak{J}_\lambda)$ considered as closed sub-preschemes of $\mathrm{Z} = \mathrm{Spec}(\mathrm{C})$, so that $\mathfrak{J}$ is the filtered union of the $\mathfrak{J}_\lambda$. If X and the $\mathrm{X}_\lambda = \mathrm{Spec}(\mathrm{C}/\mathfrak{J}_\lambda)$ are considered as closed sub-preschemes, $\mathrm{X} = \bigcap \mathrm{X}_\lambda$; if $f_\lambda : \mathrm{X}_\lambda \to \mathrm{Y}$ is the structural morphism, and as the sets $f_\lambda^{-1}(y)$ are closed in the Noetherian space $\mathrm{Z}_y = \mathrm{Spec}(k(y)[\mathrm{T}_1, \ldots, \mathrm{T}_r])$, there exists an index λ such that $f_\lambda^{-1}(y) = f^{-1}(y) = \{x\}$. One can then suppose that for a given X_λ, the condition holds that X is at point x; if one proves the proposition for such an X_λ, it will follow for X, since x' is a point above x in $\mathrm{Z}' = \mathrm{Z} \times_{\mathrm{Y}} \mathrm{Y}'$, $\mathrm{X}'_\lambda = p^{-1}(\mathrm{X}_\lambda)$, where $p : \mathrm{Z}' \to \mathrm{Z}$ is the projection (from the restriction of f_λ to X_λ), the same will hold for the open set V'_λ of x' in X'_λ which will be finite over Y'. Furthermore V' and X' are sums of two open sub-preschemes, V' and X'_λ; one answers the question by taking V' as a sum of U' and V'. In particular, note that the set of isolated points of X in their fibre et open in X (13.1.4), hence f is *quasi-finite*. Set $\mathrm{A}'' = \mathscr{O}_{\mathrm{Y},y}$, and set $\mathrm{Y}'' = \mathrm{Spec}(\mathrm{A}'')$. Set $\mathrm{X}'' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}''$, $f'' = f_{(\mathrm{Y}'')} : \mathrm{X}'' \to \mathrm{Y}''$; $\mathrm{X}'' \to \mathrm{Y}''$ is *separated*, quasi-finite and of presentation finite; A'' is henselian, for every $x'' \in \mathrm{X}''$ above x an open and closed neighbourhood V'' of x'' in X'' which is finite on Y'' (18.5.11), *c*)); as the immersion $\mathrm{V}'' \to \mathrm{X}''$ is open and closed, it is also quasi-compact, hence $f''|\mathrm{V}''$ is of presentation finite since $\mathrm{V}'' \to \mathrm{X}''$ is open, and $\mathrm{X}'' \to \mathrm{Y}''$ is of presentation finite. f'' is finite and quasi-compact over Y'' (1.6.2), hence $f''|\mathrm{V}''$ is of presentation finite.

Since, by definition, A'' is the inductive limit of a filtered inductive family (A_λ) of $\mathscr{O}_{\mathrm{Y},y}$-algebras strictly étales ((18.6.5)) and (8.1.2), *a*))); also $\mathrm{B}_{\mathrm{M}_\lambda}$ can be assumed to be a $\mathscr{O}_{\mathrm{Y},y}$-algebra of presentation finite, and it follows from (18.6.1) and (8.1.2), *a*)) and the result of (17.7.8) that $\mathrm{B}_{\mathrm{M}_\lambda}$ is a $\mathscr{O}_{\mathrm{Y},y}$-algebra étale for all λ; Namely, Y'' is the inductive limit projective of a family (Y_α) of affine Y-schemes étale. If x''_α is the projection of x'' on $\mathrm{X}'_\alpha = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}_\alpha$, on can suppose that $\mathrm{X}'_\alpha \cap f'^{-1}(\mathrm{U})$ where V'_α is a neighbourhood of x''_α in X'_α, is finite on Y_α (8.2.11). Finally, V'' being of presentation finite over Y'', on concludes from (8.10.5), (x)) that for a suitable α, V'_α is finite over Y'_α. □

Remark (18.12.2). — In the preceding proof, one sees (taking into account (18.6.2)) that one has constructed Y' answering the question, and in addition such that if $y' = f'(\mathrm{X}')$, the homomorphism $k(y) \to k(y')$ is bijective.

Corollary (18.12.3). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a locally of finite type morphism and separated, y a point of* Y *such that the underlying space $f^{-1}(y)$ is finite and discrete; in particular, a proper and quasi-finite morphism. Then there exist an étale morphism $\mathrm{Y}' \to \mathrm{Y}$, a point $y' \in \mathrm{Y}'$ above y such that $k(y') = k(y)$, and a decomposition of $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$ as a sum of two sub-preschemes X'_1, X'_2 induced on open subsets of X', such that the restriction $f' = f_{(\mathrm{Y}')} : \mathrm{X}'_1 \to \mathrm{Y}'$ is a finite morphism and that $\mathrm{X}'_2 \cap f'^{-1}(y') = 0$.*

Proof. Indeed, one can apply the corollary (18.12.3) at any point whatsoever y of Y. Since f' is a **IV-4** | 182
closed morphism, hence, since X'_2 is closed in X', $f'(\mathrm{X}'_2)$ is closed in Y' and does not contain y'; hence there exists by construction an open neighbourhood U' of y' in Y' such that $f'^{-1}(\mathrm{U}') = \mathrm{X} \times_{\mathrm{Y}} \mathrm{U}'$ is finite on Y', and the morphism $\mathrm{U}' \to \mathrm{Y}$ is étale; the morphism $f'^{-1}(\mathrm{U}') \to \mathrm{U}'$ is faithfully flat and quasi-compact, hence the conclusion. □

Corollary (18.12.4). — *Let $f : X \to Y$ be a finite morphism, it is necessary and sufficient that it be separated, universally closed and, for all $y \in Y$, $f^{-1}(y)$ is a finite and discrete space; in particular, a proper and quasi-finite morphism.*

Proof. Indeed, one can apply the corollary (18.12.3) at any point whatsoever y of Y. Since f' is a closed morphism, hence, since X'_2 is closed in X', $f'(X'_2)$ is closed in Y' and does not contain y'; hence there exists by construction an open neighbourhood U' of y' in Y' such that $f'^{-1}(U') = X \times_Y U'$ is finite over Y', and the morphism $U' \to Y$ is étale; the morphism $U' \to Y$ is faithfully flat and quasi-compact, hence by (2.7.1), (xv)), the restriction of f to X_λ, being a finite morphism (**II**, 6.1.10), hence the conclusion results from (11.3.1), since f restricted is finite and quasi-compact. □

Remark (18.12.5). — One notes that in the proof of (18.12.4) one has not used the hypothesis (18.12.4) in its general form, but only that $f_{(Y')}$ is a universally closed morphism for every étale morphism $Y' \to Y$.

Corollary (18.12.6). — *Let $f : X \to Y$ be a morphism of preschemes. The following conditions are equivalent:*

a) *f is a closed immersion.*
b) *f is a proper monomorphism.*
c) *f is proper and, for all $y \in Y$, the $k(y)$-prescheme $X_y = f^{-1}(y)$ is radicial and geometrically reduced on $k(y)$ (i.e. empty or $k(y)$-isomorphic to $\mathrm{Spec}(k(y))$).*

Proof. This is deduced from (18.12.4) as (18.11.5) is deduced from (8.11.1). □

Proposition (18.12.7). — *Let $f : X \to Y$ be a locally of finite type morphism, y a point of Y. For there to exist a neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is a closed immersion, it is necessary and sufficient that X_y be a $k(y)$-prescheme radicial and geometrically reduced on $k(y)$ (i.e. empty or $k(y)$-isomorphic to $\mathrm{Spec}(k(y))$) and that there exist a neighbourhood V of y such that the restriction $f^{-1}(V) \to V$ of f is a universally closed morphism.*

Proof. The conditions being evidently necessary, let us prove that they are sufficient. One can, by the second condition, already suppose f universally closed. Let us show, moreover, hence separated; this follows from the following lemma: □

Lemma (18.12.7.1). — *Let $f : X \to Y$ be a closed morphism, $y \in Y$ a point such that every neighbourhood of X_y contains an affine neighbourhood of X_y (condition always satisfied when X_y is empty or reduced to a single point). Then there exists a neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is an affine morphism.*

Proof. Indeed, let U_0 be an affine open neighbourhood of y in Y and let V be a neighbourhood of X_y contained in $f^{-1}(U_0)$ such that $f^{-1}(U) \subset V$, which is affine. As f is closed and V open, there exists a neighbourhood $U \subset U_0$ of y such that $f^{-1}(U) \subset V$. As the restriction $f^{-1}(U) \to U$ is also an affine morphism (**II**, 1.2.5). □

Suppose therefore f affine and universally closed, and let us show that for all $y \in Y$ such that the restriction $f^{-1}(U) \to U$ of f is a closed immersion, it suffices to prove that f is quasi-finite (18.12.4), and since X_y is empty or reduced to a single point, f is quasi-finite (18.12.4), taking into account (18.12.6).

Proposition (18.12.8). — *Let $f : X \to Y$ be a morphism of preschemes. For f to be entire, it is necessary and sufficient that f be affine and universally closed.*

Proof. We can suppose $X = \mathrm{Spec}(B)$, and it suffices to prove that every element $b \in B$ is entire on A (**II**, 6.1.1). Let B' be the sub-A-algebra of B generated by b, which is an A-algebra of finite type; set $X' = \mathrm{Spec}(B')$, so that $f : X \to Y$ factorises as $X \xrightarrow{g} X' \xrightarrow{h} Y$, where h is of finite type, and g is *dominant* since the homomorphism $B' \to B$ is injective (**I**, 1.2.7). As h is separated and $h \circ g$ universally closed, g is universally closed (**II**, 5.4.3) and (5.4.9)); hence h is also universally closed, **IV-4** | 183
therefore proper since it is of finite type and separated. But then, for all $y \in Y$, the morphism $h^{-1}(y) \to \mathrm{Spec}(k(y))$ deduced from h is proper and affine, hence *finite* (**III**, 4.4.2), in other words

h is *quasi-finite*; one deduces therefore from (18.12.4) that h is finite, and consequently b is entire on A. ☐

Remark (18.12.9). — One can say that a morphism $f : X \to Y$ is entire when one supposes only separated, universally closed, and such that for all $y \in Y$, the morphism $f^{-1}(y) \to \operatorname{Spec}(k(y))$ deduced from f is entire. It would suffice to prove that these conditions imply that f is affine, or again that every fibre X_y is contained in an affine open neighbourhood.

Corollary (18.12.10). — *A morphism $f : X \to Y$ which is injective and universally closed is entire.*

Proof. It suffices, by (18.12.8), to prove that f is affine, which follows from the lemma (18.12.7.1) and the hypothesis. ☐

Corollary (18.12.11). — *Let $f : X \to Y$ be a morphism of preschemes (resp. a locally of finite type morphism). For f to be a universal homeomorphism* (2.4.2), *it is necessary and sufficient that f be entire (resp. finite), radicial, and surjective.*

Proof. One knows that the conditions are sufficient (2.4.5); they are necessary by virtue of (18.12.10) and (18.12.4). The last proposition improves (8.11.2). ☐

Proposition (18.12.12). — *Every morphism $f : X \to Y$ which is quasi-finite and separated is quasi-affine.*

Proof. Set $\mathscr{A} = f_*(\mathscr{O}_X)$, which is an $\mathscr{O}_Y$-Algebra quasi-coherent and separated (1.7.4); and $Z = \operatorname{Spec}(\mathscr{A})$ (**II**, 1.3.1), so that f factorises canonically as $X \xrightarrow{g} Z \xrightarrow{h} Y$, h being affine and g corresponding to the automorphism identical to $\mathscr{A}$ (**II**, 1.2.7); it suffices to prove that g is an *open immersion*, or, what comes to the same (17.9.1), that the morphism g is étale and *radicial*. It suffices evidently to prove that, for all $y \in Y$, the morphism g is étale and radicial at each point of $f^{-1}(y)$. One can apply the result to X, f the restriction and Y by X, Y, and the sections of f of f; hence f is quasi-compact and separated, and the fact that f is étale and radicial at each point of $f^{-1}(y)$ is part of (18.10.2), (iii)), quasi-separable, from X, f, and g as in (18.12.3), whose X' is the extension $Z' = Z \times_Y Y'$ a $\mathscr{O}_{Y'}$-Module $\mathscr{O}_{X'}/\mathfrak{J}E'$, so that as X and Y are both Y-schemes étale (17.3.2); hence by (**II**, 1.2.7) f factorises as $X \xrightarrow{g} Z \xrightarrow{h} Y$, where g is the canonical injection from the identity on $\mathscr{A}$, and g is an *open immersion*, and g is *radicial*, whence the result. ☐

Corollary (18.12.13). — *("Main Theorem" of Zariski). Let Y be a prescheme quasi-compact and quasi-separated. Then there exists a factorisation of f*

$$X \xrightarrow{g} Z \xrightarrow{u} Y$$

where g is an open immersion (necessarily quasi-compact) and u a finite morphism.

Proof. Indeed, it follows from (18.12.12) that f is quasi-affine, hence quasi-projective. One can therefore apply (8.12.8), where the hypothesis on $\mathscr{O}_Y$-Module ample can in fact be replaced by the sole hypothesis that Y is quasi-compact and quasi-separated: indeed, it follows from (8.12.3) that the existence of the factorisation announced is a *local* property on Y, and it suffices to prove it when Y is affine. ☐

Remark (18.12.14). — One can give a proof of (18.12.13) analogous to that of (18.12.12) (but not using (8.12.3), hence (18.5.11)), which itself uses the "Main Theorem" under its local form (8.12.9)). Keeping indeed the notations of the proof of (18.12.12), and let $\mathscr{B}$ be the $\mathscr{O}_Y$-Algebra quasi-coherente, integral closure of $\mathscr{O}_Y$ in $\mathscr{A}$ (**II**, 6.3.2). If one sets $T = \operatorname{Spec}(\mathscr{B})$, it suffices, by virtue of (8.12.3), to prove that $g : X \to T$ corresponding to the canonical injection $\mathscr{B} \to \mathscr{A}$ is an immersion, and radicial. With the notations of (18.12.12), and one can suppose that $Y = \operatorname{Spec}(C)$ and $Y' = \operatorname{Spec}(C')$ are affine, C' being a C-algebra étale (17.3.2); hence $\mathscr{A} = \overline{A}$, where A is a C-algebra, B the integral closure of C in A. The algebra A' is a C'-algebra, $A' = A \otimes_C C'$, and $\mathscr{B} = \overline{B}$, where B is the integral closure of C in A. As $Z' = Z \times_Y Y'$ and the X'_λ and X_λ are étale over X, the conclusion follows in this case.

When C is Noetherian, the demonstration of (6.14.4) no longer requires the use of (6.14.1). Indeed, the reductions successives of the proof of (6.14.4) lead back, without using (6.14.1), to the case where C is *integral*, A the field of fractions of C, and the morphism $\operatorname{Spec}(B') \to \operatorname{Spec}(B)$ is étale, it follows from (17.5.7) that B' is a normal ring, and the reasoning terminates by making appeal only to the easy lemma (6.14.1), and not to the difficult part of the proof of (6.14.1).

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Proposition (18.12.15). — *Let* G *be a ring,* C' *a* C*-algebra étale,* A *a* C*-algebra,* B *the integral closure of* C *in* A*; on sets* $\mathrm{A}' = \mathrm{A} \otimes_{\mathrm{C}} \mathrm{C}'$, $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{C}} \mathrm{C}'$, B' *identifying to a sub-algebra of* A'*; then* B' *is the integral closure of* C' *in* A'.

Proof. Considering the filtered increasing family of sub-C-algebras of type finite of A and using the fact that A is a C-algebra of type fine, hence of the form $\mathrm{E}/\mathfrak{J}$, where $\mathrm{E} = \mathrm{C}[\mathrm{T}_1, \ldots, \mathrm{T}_r]$ and $\mathfrak{J}$ is an ideal of type fine of E; we then have $\mathrm{A}' = \mathrm{E}'/\mathfrak{J}\mathrm{E}'$, where $\mathrm{E}' = \mathrm{C}'[\mathrm{T}_1, \ldots, \mathrm{T}_r]$. Let $(\mathfrak{J}_\lambda)$ be the increasing filtered family of the ideals of type fine of C contained in $\mathfrak{J}$ and denote B_λ the integral closure of C in $\mathrm{E}/\mathfrak{J}_\lambda$; if B_λ is the inductive limit of the integral closures B_λ, B is the inductive limit of the B_λ, as one sees by the reasoning of (5.13.4); same reasoning for C and the $\mathrm{B}_\lambda \otimes_{\mathrm{C}} \mathrm{C}'$ for $\beta \geqslant \alpha$. The same, A' is the inductive limit of the integral closures of C' in $\mathrm{E}'/\mathfrak{J}_\lambda\mathrm{E}'$, hence the same reasoning as the preceding shows that $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{C}} \mathrm{C}'$ is the integral closure of C' in A' once we have reduced to the case where C is Noetherian, the proposition becomes a particular case of (6.14.4). It is necessary however to observe that, when C' is supposed *étale* on C Noetherian, the demonstration of (6.14.4) no longer requires that one use (6.14.1). Indeed, the successive reductions in the proof of (6.14.4) lead back, *without using* (6.14.1), to the case where C is *integral*, A the field of fractions of C, hence B is a normal ring and the morphism $\mathrm{Spec}(\mathrm{B}') \to \mathrm{Spec}(\mathrm{B})$ is étale, it follows from (17.5.7) that B' is a normal ring, and the reasoning terminates by making appeal only to the elementary lemma (6.14.1), and not the difficult part of the proof of (6.14.1). □

Proposition (18.12.16). — *Let* $g : \mathrm{Y} \to \mathrm{S}$ *be a quasi-compact morphism,* $f : \mathrm{X} \to \mathrm{Y}$ *a separated and quasi-finite morphism. For every* $\mathscr{O}_{\mathrm{Y}}$*-Module invertible* $\mathscr{L}$*, ample relative to* S (**II**, 4.6.1)*, the* $\mathscr{O}_{\mathrm{X}}$*-Module* $f^*(\mathscr{L})$ *is ample relative to* S.

Proof. Indeed, by virtue of (18.12.12), the morphism f is quasi-affine, hence (**II**, 5.1.6) the $\mathscr{O}_{\mathrm{X}}$-Module $\mathscr{O}_{\mathrm{X}}$ is ample for f. We deduce from (**II**, 4.6.13), (iii)) that $\mathscr{O}_{\mathrm{X}} \otimes_{\mathscr{O}_{\mathrm{X}}} f^*(\mathscr{L}^{\otimes n}) = f^*(\mathscr{L}^{\otimes n})$ is ample relative to S for n large enough. But as $f^*(\mathscr{L}^{\otimes n}) = (f^*(\mathscr{L}))^{\otimes n}$, this implies that $f^*(\mathscr{L})$ is ample relative to S (**II**, 4.6.9), (i)). □

Corollary (18.12.17). — *Let* Z *be an affine scheme,* $h : \mathrm{X} \to \mathrm{Z}$ *a morphism of finite type,* $\mathscr{L}$ *an* $\mathscr{O}_{\mathrm{X}}$*-Module invertible. With the notations of* (**II**, 4.5.2)*, the conditions of* (**II**, 4.5.2) *(which are equivalent to saying that* $\mathscr{L}$ *is ample) are also equivalent to each of the following:*

b$''$) *h is separated, one has* $\mathrm{G}(\varepsilon) = \mathrm{X}$ *and the canonical morphism* $u : \mathrm{G}(\varepsilon) \to \mathrm{Proj}(\mathrm{S})$ *has finite and discrete fibres.*

b$'''$) *One has* $\mathrm{G}(\varepsilon) = \mathrm{X}$ *and the canonical morphism u is radicial.*

If $\mathrm{Z} = \mathrm{Spec}(\mathrm{A})$*, the ring* S *is canonically equipped with a structure of graded* A*-algebra, hence* $h = g \circ u$*. Since every radicial morphism is separated* (**I**, 1.8.7.1)*, b$'''$) implies b$''$). As the condition b) of* (**II**, 4.5.2) *implies evidently b$''$), it remains to see that b$''$) implies $\mathscr{L}$ ample. Since $h = g \circ u$ is of finite type by hypothesis, u is also quasi-finite and separated* (**I**, 6.3.4)*, hence by the hypothesis b$''$) implies that u is quasi-finite and separated* (**I**, 5.5.1)*. To prove that $\mathscr{L}$ is ample, we shall now apply* (18.12.16)*. Set* $\mathrm{Y} = \mathrm{Proj}(\mathrm{S})$*. Since* X *is quasi-compact, there exists an integer $n > 0$ and a finite number of elements $f_j \in \mathrm{S}_n$ such that the inverse images $u^{-1}(\mathrm{D}_+(f_j))$ cover* X*; one can therefore consider u as a quasi-finite and separated morphism of* X *in the open* $\mathrm{Y}' = \bigcup \mathrm{D}_+(f_j)$ *of* Y*. Let us consider then the* $\mathscr{O}_{\mathrm{Y}'}$*-Module invertible* $\mathscr{L}' = \mathscr{O}_{\mathrm{Y}}(n) \,|\, \mathrm{Y}'$ (**II**, 2.5.8)*; it is very ample relative to* Z (**II**, 4.4.3)*, and consequently it is the same for* $u^*(\mathscr{L}')$ (**II**, 4.4.3).

$\mathscr{L}' = \mathscr{O}_{\mathrm{Y}}(n) \,|\, \mathrm{Y}'$ (**II**, 2.5.8)*; it is very ample relative to* Z (**II**, 4.4.3)*, and as* $u^*(\mathscr{L}') = \mathscr{L}^{\otimes n}$ *by* (**II**, 4.4.3)*,* $\mathscr{L}^{\otimes n}$ *is thus ample relative to* S *(this amounts to saying that it is ample* (**II**, 4.6.6)*), and consequently the same is true of* $\mathscr{L}$ (**II**, 4.5.6)*, (i)).*

We leave it to the reader to state analogous conditions equivalent to those of (**II**, 4.6.3) *for the* $\mathscr{O}_{\mathrm{X}}$*-Modules relatively ample.*